

Transgenerational Hangovers: A Rat Model for Behavioural and
Epigenetic Changes Across Generations due to Paternal
Preconceptual Alcohol Consumption

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Abstract

Alcohol is the most harmful drug of abuse, making alcoholism a major economic and public health crisis. Unsurprisingly, this has led to the majority of neurobiological research on alcohol focussing on its direct effects on an individual. This includes those affected by foetal alcohol spectrum disorders (FASDs). However, research has shown that heavy paternal drinking predicts earlier and heavier adolescent drinking in the offspring, accompanied with other behavioural and molecular changes. While alcohol use disorder (AUD) is highly heritable, research on genetic variants alone does not sufficiently account for its risk and the FASDs like symptoms seen in the offspring of alcoholic fathers. Recently, there has been an increase in appreciation of the importance of epigenetic mechanisms of inheritance, which transfer changes due to parental experiences through the germline. This thesis aimed to assess both the inter and transgenerational impacts of preconceptual paternal alcohol consumption (PPAC) using a rat model. Sprague Dawley male rats (F0) underwent chronic voluntary ethanol intake and at the end of the drinking paradigm were kept for one spermatogenesis cycle before being mated with ethanol naïve females. The litters and matched controls were behaviourally assessed, and a cohort of F1 males mated to observe a F2 generation.

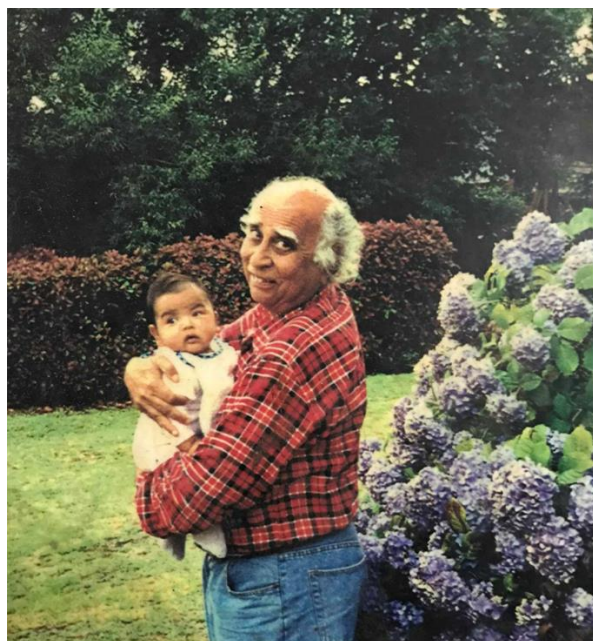
PPAC caused behavioural changes in both the F1 and F2 generations, including altering litter sizes and delaying development. The F1 also show a reduction in sensitivity to the motor impairing effects of ethanol compared to controls. Sexually dimorphic effects of PPAC were seen with female offspring having a reduced preference to ethanol in both the F1 and F2, while tolerance to ethanol in motor coordination was again seen in the F2 females but not F2 males. Likewise, F1 males presented reductions in locomotor activity but these effects did not persist in the F2.

Analysis of liver mitochondrial DNA copy number (mtDNA-CN) found that drinking F0 fathers had reduced mtDNA-CN at the end of the ethanol intake paradigm and 12 weeks post-drinking relative to controls. The livers of F1 showed lower mtDNA-CN in the offspring of ethanol sired rats compared to offspring of non-drinkers after undergoing ethanol consumption, suggesting differences in liver metabolism. Finally, livers of F0 drinkers had reduced expression of Peroxisome proliferator-activated receptor γ coactivator 1 α (*Pgc1a*), relating to mitochondrial biogenesis, and methyl-CpG-binding protein 2 (*Mecp2*), indicating putative changes in methylation signalling, together suggesting epigenetic inheritance of changes due to xenotoxic stress.

The findings show PPAC induces developmental delays, causes transgenerational changes in drinking behaviour, ethanol sensitivity and liver mitochondrial function in a sexually dimorphic manner. These changes may be protective to the female offspring of PPAC to modify their ethanol intake. The alterations in F1 liver mtDNA-CN demonstrate potential far-reaching consequences for the metabolism of xenotoxic substances extending beyond ethanol and provides evidence to support behavioural and epigenetic changes across generations due to PPAC.

Dedication

Dedicated to my grandparents, Nana and Dadi,



Thank you for being my greatest fans.

Wish we could celebrate this together.

Acknowledgements

I would like to begin by thanking those who are going to be examining this thesis because they are the only ones who will read this cover to cover. I feel markers do not get the credit they deserve, most New York times best sellers are tedious at best, so having to read the mumblings of underpaid, overworked, perpetually stressed humans must not be easy. I hope you find my work as interesting as I did, I look forward to talking about it with you, and please remember, literal generations of rats gave their life for this project so it's as much a family biography as it is a thesis. I am not sure what rat squeaks translate to "thank you" but I do owe them.

Considering that people who are going on to read this thesis have plenty to cover, they are excused from reading the rest of this section. However, I know most of you will probably just glance over here to look for your name. My mother already said this section is what she was most looking forward to. Thanks for the vote of confidence Mum. Anyway.... We made it! Twenty-three years of education has led to this moment, and I am pretty sure that I have made multiple errors here. Considering I have lived in four different countries; you guys can pick which education system is to blame for the errors. But for now, let us get onto the thanking.

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List of Abbreviations

5-HTTLPR	Serotonin-transporter-linked promoter region
5caC	5-carboxylcytosine
5fC	5-formylcytosine
5hmC	5-hydroxymethylcytosine
5mC	5-methylcytosine
acH4K5	acetyl-histone histone 4-lysine
ADH	Alcohol dehydrogenase
ADHD	Attention deficit hyperactivity disorder
ADT	Affective disorders test
ALDH	Aldehyde dehydrogenase
ANOVA	Analysis of Variance
ASD	Autism Spectrum Disorder
AUD	Alcohol use disorder
BD	Bipolar disorder
BDNF/Bdnf	Brain Derived Neurotropic Factor
BDNF-AS	BDNF antisense
CTCF	CCCTC-binding factor
DAT/ Slc6a3	Dopamine transporter
dlPFC	Dorsolateral PFC
DNA	Deoxyribonucleic acid
DNMT	DNA methyltransferase
FASD	Foetal alcohol spectrum disorders
gDNA	Genomic/nuclear DNA
H3K9ac	Acetylation of histone 3-lysine 9
H3K4me3	Trimethylation of histone H3 at lysine 4
HPA axis	Hypothalamic-Pituitary-Adrenal axis
Htr2c	5-Hydroxytryptamine Receptor 2C
IA2BC	Intermittent access two bottle choice paradigm
IG-DMR	Intergenic differentially methylated region
IGF2/Igf2	Insulin-like growth factor 2 gene
lncRNA	long non-coding RNA
M1A	N1-methyladenosine
m6A	N6-methyl adenosine
MBD1	Methyl-CpG binding domain protein 1
MDD	Major depressive disorder
MeCP2/Mecp2	Methyl-CpG-binding protein 2

mPFC	medial prefrontal cortex
miRNA	microRNA
mRNA	Messenger ribonucleic acid
mtDNA	Mitochondrial DNA
mtDNA-CN	Mitochondrial DNA copy number
ncRNA	Non-coding RNA
NGF/Ngf	Nerve Growth Factor
NAc	Nucleus accumbens
ONS	Office of National Statistics, United Kingdom
PatEE	Paternal Ethanol Exposure
PER2	Period Circadian Regulator 2
PFC	Prefrontal cortex
PGC1a/Pgc1a	Peroxisome proliferator-activated receptor γ coactivator 1 α
piRNA	piwi RNAs
PND	Postnatal day
POMC	Proopiomelanocortin
PPAC	Paternal Preconceptual Alcohol Consumption
RNA	Ribonucleic acid
ROS	Reactive oxygen species
SAM	S-adenyl methionine
SAT	Successive Alleys Test
Slc6a4	Solute Carrier Family 6 Member 4
sncRNA	small non-coding RNA
tRNA	Transfer RNA
tDR	tRNA-derived RNA
WHO	World Health Organisation
Ψ	Pseudouridines

Chapter 1: An Introduction to Paternal Preconceptual Alcohol Consumption

“They say so many people die because of alcohol. Ironically, we never talk about how many of them were born because of it”- Seen outside a bar in Glasgow, United Kingdom

The above quote points out some interesting aspects of many societies' relationship with alcohol. We know it causes harm and yet we often rely on its power as a social facilitator. While this is perhaps not the original intention of the quote, this thesis would like to oblige and talk about what they say is not spoken about, namely, births due to alcohol consumption. But we want to go a step further and talk about how drinking might affect those born due to it. If we know a night of heavy drinking can lead to a hangover the next day, can extended periods of drinking lead to hangover effects in the next generation? To answer this question, we still need to start with the first sentence of the quote.

Alcohol is the most widely used substance of abuse. The World Health Organization (WHO) reports approximately 3 million deaths every year resulting from harmful use of alcohol (WHO, 2022). This represents 5.3% of all deaths per year and 5.1% of the global burden of disease and injury is attributable to alcohol, as measured in disability-adjusted life years (WHO, 2022). A study in Scandinavian countries showed individuals with alcohol use disorder (AUD) show 24–28 years shorter life expectancy than the general population (Westman et al., 2015). Therefore, AUD (also referred to as alcoholism or alcohol abuse) represents a major economic and public health crisis. There is widespread agreement among experts in the field that alcohol is the most harmful addictive substance, particularly if one combines harm to the individual with

harm to society (Crossin et al., 2023; van Amsterdam et al., 2015). As a result, alcoholism places a significant burden on health and many national economies. Example, the recent New Zealand (Ministry of Health, 2022) report stated 1 in 5 New Zealanders show hazardous drinking habits with an even higher number in the age group of 18 – 34 years (up to 32 – 38%). Similar behavioural statistics are seen in other western nations like Australia (Australian Bureau of Statistics, 2021) the United Kingdom (ONS, 2022). This excessive alcohol use is directly related to reduced productivity and increased medical claims, road accidents, violence, and mortality. Reducing the damage caused by alcoholism will therefore not only benefit the patient and their family but can also substantially reduce the economic and judicial costs associated with hazardous drinking.

Data indicates national, cultural and gender differences in drinking behaviours with Europe and the Americas having the highest levels of drinking while the Eastern Mediterranean nations consume the lowest amounts of alcohol per capita (WHO, 2023). Differences in trends between regions across the world are evident too with drinking showing a decline in Europe while we see rises in South-East Asia and Western Pacific regions. There are also cultural differences within countries which are sometimes not mentioned. For New Zealand, adjusted for age and gender, Māori are the most likely to show hazardous drinking patterns while Asians show the lowest (Ministry of Health, 2022). In the Netherlands, the prevalence of drinking was substantially lower in all ethnic minority groups compared to the Dutch and minorities showed a lower risk of alcohol abuse (van Amsterdam et al., 2020). Lastly, men are overwhelmingly the highest consumers of alcohol with them drinking nearly four times more pure alcohol per capita than women across the world in 2019 (WHO, 2023).

1.1. Gene x Environment in Alcohol use

The development and severity of psychiatric disorders are characterized as being a product of both nature (family genetics) and nurture (the influence of ones surrounding environment across a lifetime) with these being termed gene x environment interactions (Ellenbroek & Yoon, 2016). Twin and adoption studies demonstrate a clear genetic component for the development of alcohol abuse. However, the estimated heritability ranges from 40% to 70% (Hottenga et al., 2018; Verhulst et al., 2015; Young-Wolff et al., 2011) which indicates an environmental influence in the development of these behaviours. Factors like availability of alcohol, childhood neglect, parental dinking and socio-economic status among others have been associated (Dasgupta, 2016). Importantly, the consequences to environmental factors seem to be influenced by genes associated with substance abuse. Examples of these include carriers of the s-allele of the 5-HTTLPR (serotonin-transporter-linked promoter region), which is associated with lower transcriptional efficiency, showing increased alcohol use especially after childhood maltreatment (Kaufman et al., 2007; van der Zwaluw et al., 2010) and the protective effect of the minor A allele of rs1229984, which increases alcohol dehydrogenase (ADH) activity and is associated with reduced drinking, is attenuated by peer drinking (Olfson et al., 2014). These indicate that the environment can moderate gene expression resulting in complex phenotypes. These are termed epigenetic changes and refer to the modification of gene expression by environmental factors without a change to the genetic code itself.

Until recently, virtually all research in the field of alcoholism has focused on the affected individual, trying to unravel the neurobiology of addictive behaviour, or identify genetic risk factors involved in the development of alcoholism. This includes research on foetal alcohol

spectrum disorders (FASDs), as this investigates the effects of alcohol exposure on the unborn foetus, as a consequence of maternal alcohol intake. Research indicates at least 50% of children with FASD symptoms also have an alcoholic father (Abel, 2004) and, as early as 1968, researchers have reported FASD symptoms in children with alcoholic fathers in cases where mothers do not consume alcohol (Lemoine et al., 1968 cited in (Calhoun & Warren, 2007) which suggests paternal drinking also plays a role in FASD symptoms. While overall research in this area is still limited there are a growing number of studies using human and animal models to understand the consequences of paternal alcohol exposure on the next generation. This chapter aims to summarize the work examining the effects of alcohol on subsequent generations. This means looking at the effects of alcohol which are transmitted through the germline due to preconceptual exposure to ethanol. Governments often post strict guidelines for women around drinking just before and during pregnancy (Armstrong, 2017). In contrast, not much focus is put on men during the period of conception. As drinking in men is more prevalent, it is important to assess the impact male drinking can have on future generations. Additionally, while maternal preconceptual drinking can have effects due to changes in in-utero conditions and maternal care (Rompala & Homanics, 2019) any changes seen due to paternal drinking would have to be directly through the germline or through social factors. As the social effects of paternal drinking have been well documented (Scholte et al., 2008; Thor et al., 2022) the rest of the chapter will primarily focus on intergenerational inheritance of phenotypes due to Paternal Preconceptual Alcohol Consumption (PPAC) and the potential mechanisms by which this occurs. As rodent models often use different methods of ethanol delivery, we will also be using the more general term of Paternal ethanol exposure (PatEE) for when methods including but not limited to ethanol consumption are used.

1.2. Effects of Preconceptual Alcohol Exposure on the First Generation

There is a growing list of research showing intergenerational effects of PatEE. While some of these are in humans, most of them use rodent models. A complicating factor in assessing the exact impact of PatEE is the large variety of experimental design, differing in mode, dose and duration of ethanol administration, species, strains, and other experimental variations. Nonetheless, some general conclusions can be drawn from the current literature.

1.2.1. Birth and early development

The effects of paternal ethanol exposure can be seen as early as during pregnancy with heavy PPAC shown to increase the risk of early pregnancy loss and stillbirths in humans (Henriksen et al., 2004). There are even mixed reports on the influence on maternal fertility, some studies note an increase in placental weights and gestation length in dams of litters sired by rodents chronically exposed to ethanol (Abel, 1995; Chang et al., 2019; Klassen & Persaud, 1976). Although Meek et al. (2007) reported the opposite with shorter gestation lengths for offspring of acute ethanol exposed sires. The study also showed that male mice with acute alcohol exposure before mating had lower numbers of pups with reduced birth weights and increased mortality after birth (Meek et al., 2007). Other studies looking at paternal ethanol exposure confirmed these effects on litter characteristics with reported alterations in litter sizes and pup sex ratios (Abel, 1989; Abel, 1995; Stockard & Papanicolaou, 1916a; Stockard & Papanicolaou, 1916b), fewer viable litters (Rathod et al., 2020), an increased number of runts (underweight births) and low weights in alcohol sired litters (Asimes, 2018; Bielawski & Abel, 1997; Bielawski et al., 2002; Ceccanti et al., 2016; Chang et al., 2017; Ledig et al., 1998). However, some experiments have found no effect of paternal drinking on the number of pups,

sex ratio, and mortality rate of litters (Abel & Lee, 1988; Abel & Tan, 1988; Liang et al., 2014; Nieto et al., 2022; Randall, C. L. et al., 1982; Rompala et al., 2016) which may be due to the differences in experimental designs. The studies mentioned above vary heavily in aspects like ethanol exposure duration (acute vs chronic), ethanol dosage (high vs low), method of exposure (vapour vs forced diet vs free choice), time between end of drinking and conception (immediately after exposure vs allowing for time between last exposure and mating) and age of the sires at ethanol exposure and mating. Notably, the inconsistency also extends to human studies where some groups have reported no effect of PPAC on birth weights or early motor development in infants (Hutchinson et al., 2019; McCormack et al., 2018; Passaro et al., 1998).

Rodent studies have reported differences in organ weights based on paternal ethanol exposure (Abel, 1993; Chang et al., 2017; Lee, H. J. et al., 2013) while studies in humans indicate potential differences in anogenital distance (Xia et al., 2018). Other reports on humans saw no differences due to paternal drinking and early foetal growth when maternal drinking was accounted for (Windham et al., 1995). Few studies of PatEE look at early developmental milestones and those that do show mixed results with one showing a delay in eye opening and achievement of surface righting abilities in ethanol-sired pups (Meek et al., 2007) while another study reports ethanol-sired pups developing the righting reflex earlier than controls (Jamerson et al., 2004). Another study found no difference in the achievement of developmental landmarks between offspring of ethanol exposed rats and offspring (Wozniak et al., 1991). Taken together, these highlight the heterogeneity in current studies of PatEE.

1.2.2. Behavioural impacts on Offspring Phenotypes

While birthweights and litter characteristics show the impact of paternal alcohol, most research done on paternal ethanol exposure has looked at cognitive, behavioural, and epigenetic changes seen in the offspring. Below we go through the potential effects of PatEE/PPAC on the development of a broad range of impairments in the next generation.

1.2.2.1. Cognition and Perception.

There is evidence demonstrating sons of alcoholics show delayed cognitive performance and deficits in memory, language processing and motor skills compared to sons of non-alcoholics (Abel & Tan, 1988; Aronson & Gilbert, 1963; Ervin et al., 1984). Additionally, preschool going children of alcoholic fathers show poorer visuospatial processing of information compared to controls (Schandler et al., 1988). Another study showed that sons of active alcoholics perform significantly worse on visuospatial, memory, and attentional tasks as well as general intellectual functioning than sons of controls as well as sons of social drinking fathers (Ozkaragoz et al., 1997)

Rodent models support these findings with ethanol-sired offspring showing reduced spatiotemporal learning (Wozniak et al., 1991) and memory impairments associated with changes in the hippocampus and prefrontal cortex (PFC) (Ledig et al., 1993). Learning and memory has been assessed using multiple different experimental designs, with poorer performances being seen by the offspring of ethanol sired rodents in the Passive Avoidance test (Abel & Lee, 1988), Radial Arm maze (Wozniak et al., 1991) and Morris water maze (Liang et al., 2014). PatEE mice have trouble completing T-maze performance tests indicative of reduced cognitive functioning (Hollander et al., 2019). All these studies had the sires undergo chronic ethanol exposure, although the time between final exposure and copulating ranged from 10 days

to 1 month. These indicate the changes which influenced learning and memory in the next generation are long lasting in the fathers. Other studies report increased immobility of offspring of ethanol exposed fathers in the forced swim test and reduced grooming behaviour despite no differences being seen in the fathers themselves (Abel, 1991a; Abel, 1991b). Lastly, there is evidence to suggest chronic ethanol exposure in mice has a dose-dependent effects on offspring's hearing loss (Liang et al., 2015).

1.2.2.2. Locomotor Activity and Motor Coordination.

PatEE has also been shown to affect motor skills of the offspring. Abel showed an increase in open-field locomotor activity in PPAC F1 offspring at Postnatal day (PND) 21, PND 42 and PND 90 (Abel, 1993) while similar effects of PatEE were seen in 1 year old rats (Ledig et al., 1998). There is some indication that these effects may be dose dependent as Abel showed an increased activity in offspring of sires subject to high ethanol intake but decreased activity in offspring of rats with a lower alcohol intake (Abel, 1989). There are also studies which have found no differences between the paternal drinking conditions and locomotor (Beeler et al., 2019; Nieto et al., 2022; Rompala et al., 2017). This heterogeneity of outcomes is also seen in tests of motor coordination, with poorer performances of PatEE offspring being observed on balance and sensorimotor tasks in some studies (Bottom et al., 2022; Conner et al., 2020) but not in others (Nieto et al., 2022).

Although these studies show differences in baseline locomotor activity and coordination, there is also firm evidence that PatEE offspring respond differently to the effect of alcohol on motor behaviour. Our lab has shown a reduced sensitivity to the motor-impairing effects of ethanol with PPAC offspring showing lower drops in performance on the accelerating rotarod

task compared to controls (Hussain et al., 2022). Other studies have found sex dependent effects when under the influence of ethanol with alcohol sired males showing attenuated effects (Finegersh & Homanics, 2014; Nieto et al., 2022). In females, PatEE has either shown no effects (Finegersh & Homanics, 2014) or an increase in sensitivity (Nieto et al., 2022) to the motor disruptive effects of ethanol. These differences may in part be due to differences in animal strain as Abel found that the reduction in activity due to PPAC was more prominent in Long Evans rats when compared to Sprague-Dawley rats (Abel, 1989). Taken together, while there remains some ambiguity of the effects of PatEE on motor skills at baseline levels, there are clear differences in response to ethanol sensitivity which may even be sex dependent.

1.2.2.3. Alcohol and reward.

One of the major areas of focus in drug abuse research is the changes it brings about in reward systems. In relation to the topic of the present chapter, research has indicated the exposure to ethanol by either parent pre-conception brings about changes in neural circuitry associated with reward processing. In a human study, offspring of parents with an ethanol abuse history show a heightened sensitivity to the rewarding effects of alcohol and preference to alcohol related cues (Orford & Velleman, 1990). Rodent studies have also found that PatEE induces a higher susceptibility to the rewarding effects of ethanol, affecting *BDNF* (Brain Derived Neurotropic Factor) and nerve growth factor (*NGF*) in the frontal cortex, striatum, hippocampus, and olfactory lobes (Ceccanti et al., 2016). *BDNF* and *NGF* are neurotrophic factors involved, among others, cell growth, differentiation, and maturation, and cell migration, metabolism, and apoptotic cell death and play a significant role in regulating structural and behavioural plasticity involved in drug addiction (Russo et al., 2009). Ceccanti et al. found

alterations in *NGF* and *BDNF* in reward related brain regions in offspring of mice exposed to alcohol (Ceccanti et al., 2016). These findings, particularly *BDNF* dysregulation in reward pathways has also been reported by other studies (Finegersh & Homanics, 2014; Rompala et al., 2017).

In terms of behaviour, one study reported a bidirectional effect on a conditioned place preference (CPP) task in male offspring depending on the paternal alcohol exposure dosage where a lower dose (0.5 g/kg) increased preference, but a high dose (1.5 g/kg) decreased preference (Ceccanti et al., 2016). An increasing number of studies over the past decade have looked at alcohol consumption in the offspring of PatEE. Finegersh and Homanics found that adult male mice exposed intermittently to alcohol vapour over 5 weeks sired male offspring with decreased alcohol drinking preference and consumption (Finegersh & Homanics, 2014). Those key findings were replicated with the male offspring of chronic ethanol exposed mice via both vapour (Rompala et al., 2017) and free choice ethanol drinking (Beeler et al., 2019) showing reduced ethanol preference and consumption. However, another study exposing mice to ethanol vapours found lower ethanol consumption in females rather than male offspring of the ethanol-sired mice (Rathod et al., 2020). Alternatively, one study found that paternal binge-alcohol exposure over 8 days increased alcohol intake by intraoral infusion in postnatal offspring (Hollander et al., 2019). While these studies looked at chronic ethanol exposure within 48-hours of conception, studies with longer delays between ethanol exposure and copulation have shown different effects on offspring drinking. Nieto and colleagues subjected Wistar rats to chronic ethanol exposure in vapour chambers for 6 weeks then mated them with ethanol naïve females 8 weeks after treatment and found alcohol-sired offspring consumed less alcohol in a self-

administration task without any differences in their response to food (Nieto & Kosten, 2023; Nieto et al., 2022).

Our earlier research looking at a serotonin transporter (SERT) knock-out rat model used an intermittent voluntary 2 bottle choice paradigm on their F0 generation for chronic ethanol intake with a one-month delay between exposure and mating (Hussain et al., 2022). We found overall reductions in ethanol drinking in ethanol sired offspring, an effect mediated by the SERT genotype of the offspring, although there were no clear effects of genotype on drinking behaviours in the fathers. As serotonin is known to play a role in reward circuitry (Fischer & Ullsperger, 2017) these results link well with other reports on the influence of PPAC on offspring reward. In one such study, analysis of gene expression showed serotonin transporter Solute Carrier Family 6 Member 4 (*Slc6a4*) and low-affinity pan-neurotrophins receptor (*p75^{ntr}*) elevation and decreased 5-Hydroxytryptamine Receptor 2C (*Htr2c*) in the brainstem of offspring of chronic alcohol-exposed sires (Ferraguti et al., 2020). Taken together with the behavioural findings, it appears PatEE may interfere with ethanol sensitivity in future generations by epigenetic regulations of pathways involved in mood, emotion and neurotrophins and provide some protection against the initial reinforcing effects of alcohol.

1.2.3. Psychiatric disorders

1.2.3.1. Depression and Anxiety

There are a host of studies which report anxiety and depression like behaviours in offspring of parents with alcohol exposure. Unfortunately, most of these studies looked at the maternal effects on offspring behaviour and it has been noted that preference in intergenerational studies is often given to conduct disorders (Easey & Sharp, 2021). Nevertheless, one systematic

review of human studies found significant links between fathers with AUD and depressive symptoms in their offspring (particularly males) later in life (Mahesh et al., 2021). Rodent studies parse this association further with mice of ethanol sired rats showing increased anxiety and an increase in depressive like behaviours, particularly in females (Kim, P. et al., 2014; Liang et al., 2014). Another study found no baseline differences in anxiety like behaviours but increases in the anxiolytic effects of ethanol in the male offspring of mice with chronic ethanol exposure (Finegersh & Homanics, 2014). Conversely, two studies one in mice and the other in rats looking at anxiety behaviours in the open field test showed that anxiety decreased in male progenies of ethanol-sired rats but not in female progenies (Beeler et al., 2019; Koabel et al., 2021). While not directly PPAC, a study did find offspring of male mice who were prenatally exposed to ethanol show increased depressive-like behaviours on the forced swim task indicating nongenomic inheritance of traits through the male germline (Bottom et al., 2022). These findings suggest that parental ethanol exposure has complex sex dependent effects on offspring anxiety and depression like behaviours, with these changes being dependent on multiple factors.

1.2.3.2. Attention deficits and hyperactivity disorders.

Conduct issues have also been associated with PPAC, a large population study found both paternal and maternal alcohol exposure being associated with attention deficit hyperactivity disorder (ADHD) especially when combined with tobacco use (Han, J. et al., 2015). In animal models, one acute dose of alcohol before mating may cause behavioural abnormalities such as significantly lower risk assessment behaviour and increased aggression along with fewer fearful behaviours as adults when compared to a control group (Meek et al., 2007). Even in chronic PPAC two months before gestation, offspring showed increased impulsivity in the novelty-

seeking tests, and an increased time spent in the light compartment during the light-dark test (Ledig et al., 1998). The paternal effect may be mediated by catecholaminergic activity with increased hyperactivity in PPAC offspring when compared to controls being altered in response to amphetamines (Abel, 1993). Extending these findings, another study found paternal ethanol exposure induced ADHD-like hyperactive, inattentive, and impulsive behaviours in offspring and found that both protein and messenger ribonucleic acid (mRNA) expression of dopamine transporter (*Slc6a3*) were significantly decreased in the cerebral cortex and striatum (Kim, P. et al., 2014). This implies that in addition to the behavioural differences, clear alterations in gene expression and development of cortical structures are induced by PPAC which likely impact development of offspring. Therefore, it is important to understand which mechanisms may underlie these PPAC induced changes in brain and behaviour.

1.3. Mechanisms of Preconception Environment Heritability

The concept of parental and ancestral environment playing a role in phenotypes of subsequent generations has been mentioned throughout history. Roman author Gellius observed that children of alcoholic men were “weak brained” (Ouko et al., 2009) while more scientific works in the 1800s by biologist Jean-Baptiste Lamarck mention ancestral environment as a driver of phenotypes but these views had previously been rejected for the singular genetic basis of inheritance (Skinner, 2016). However, over the years there is an increasing amount of evidence which supports the idea of nongenomic inheritance. A particular point of importance for the influence of the parental environment seems to be around conception with epigenetic changes being crucial at fertilization to drive early embryonic development (Greeson et al., 2023; Rompala & Homanics, 2019)

Some of the best human evidence of intergenerational effects comes from research done on food scarcity. The Dutch famine of 1944-45 showed an increase in cardiovascular, metabolic and mental health conditions in offspring of mothers who suffered from malnutrition during this time (Painter et al., 2006; Roseboom et al., 2011). Interestingly, the offspring of fathers exposed to this prenatal stress produced offspring with higher weights and body mass index (BMI) scores compare to offspring on non-exposed fathers (Veenendaal et al., 2013). While the famine would have affected both parents, the studies suggest different influences from each parent. Other analyses on this cohort have noted persistent epigenetic changes caused by this famine exposure particularly in the insulin-like growth factor 2 gene (*IGF2*) (Heijmans et al., 2008).

As reporting on paternal trends has increased over the past 10 years, numerous preclinical studies have emerged which support the evidence of environmental heritability in humans. Example, paternal experiences like diet, body mass index and smoking have been shown to affect deoxyribonucleic acid (DNA) methylation in their offspring (Billah et al., 2022; Merrill et al., 2021; Yin et al., 2022). Additionally, reviews on paternal and maternal trauma and post-traumatic stress disorder show strong transgenerational effects (Burchert et al., 2017; Harkness, 1993; Youssef et al., 2018). While generally controlled for, human studies come with a multitude of confounding factors which include socioeconomic conditions, influence of parents during child development, infections and genetic variations within sample populations to name a few.

As mentioned earlier, rodent sires are generally not involved in any of the stages of gestation and rearing which allows for better separation of social and biological factors. The range of the behavioural and neurochemical effects seen after PatEE in rodents described here

and reviewed by others (Rompala & Homanics, 2019; Terracina et al., 2022) clearly points to a biological role in the manifestation of these phenotypes.

1.3.1. Epigenetic mechanisms

Epigenetic changes can be described as cellular and physiological trait variations that are not caused by changes in the DNA sequence (Ellenbroek & Youn, 2016). This is unlike the qualitative change seen due to genetic mutations which cause changes in the nucleotide sequence of messenger ribonucleic acid (mRNA) and the amino-acid sequence of proteins produced. Additionally, epigenetic most changes are reversible and can last from anywhere between a few seconds to years. However, some epigenetic changes are known to last a lifetime and even influence future generations. There are multiple mechanisms through which this can occur, such as histone modification, DNA methylation and noncoding RNA (ribonucleic acid) interference. While the mechanism of each of these is different, all of them work to alter gene expression without altering the underlying DNA code. These epigenetic mechanisms are essential for controlling normal development and provide means for specific cells, and the organism, to react to environmental cues (Oppermann, 2013) and they also help explain the variances and plasticity seen in phenotypes of a disease (Feinberg, 2007).

DNA methylation is the most often investigated form of epigenetic change and is characterized by the transfer of a methyl group from S-adenyl methionine (SAM) to the fifth carbon of a cytosine residue to form 5-methylcytosine (5mC) (Moore, L. D. et al., 2013). This process is catalysed by a family of DNA methyltransferases (DNMTs) and occurs most often in CpG nucleotides that are distributed unevenly across the genome making its patterns cell and tissue specific allowing for complex patterns of gene expression. Subsequently the 5-

methylcytosine can be oxidized to 5-hydroxymethylcytosine (5hmC) (Figure 1.1) (Ellenbroek & Youn, 2016). The methylation of CpGs in the promoter region of a gene is typically associated with gene repression while hypomethylation in the promoter region and hypermethylation in the gene body is often associated with enhanced gene expression (Lanata et al., 2018). It is important to remember that altered methylation can cause region specific up and down regulations of activity, for example, lower genomic DNA methylation alterations in alcohol drinkers are associated with the upregulation of N-methyl-D-aspartate 2b receptor subtype (NR2B) which is associated with high levels of lifetime drinking and daily alcohol intake (Biermann et al., 2009) and elevated DNA hypermethylation in the promoter regions of the alpha synuclein gene leads to a downregulation of alpha synuclein expression in alcohol drinkers, followed by disturbed dopaminergic neurotransmission (Bönsch et al., 2006).

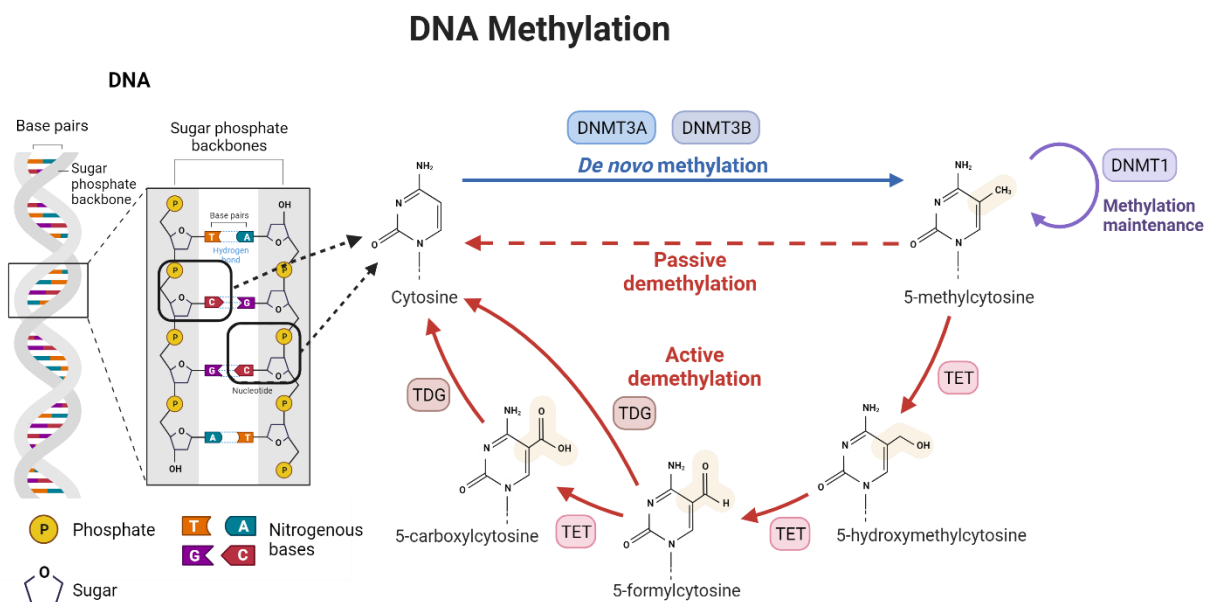


Figure 1.1. DNA methylation and demethylation. DNA methylation occurs on the fifth carbon of a cytosine base catalysed by DNA methyltransferases (DNMTs). Passive DNA demethylation can

occur when cells divide. A replication independent active demethylation occurs when ten-eleven translocation (TET) methyl cytosine dioxygenases enzymes mediate DNA demethylation by oxidizing 5-methylcytosine (5mC) in DNA to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC). This active DNA demethylation involves excision of 5fC and 5caC by the DNA repair enzyme thymine DNA glycosylase (TDG). Created with BioRender.com

In addition to DNA methylation, histone modifications can mediate the effects of environmental stressors on gene expression (Ciafrè et al., 2019). As eukaryotic DNA is tightly wrapped around histones, they play an integral role in maintaining the structure of DNA. The modification of histones can loosen the bonds with DNA allowing for an increase in gene expression while the opposite can occur through modification which tighten these bonds. While most modifications occur on histone tails, they can also occur on the globular portion and different residues on each of the histones can be modified (Figure 1.2). There are many identified histone modifications (Tweedie-Cullen et al., 2012), however, the three most common ones are histone acetylation and phosphorylation which are associated with transcriptional activation and histone methylation which can be associated with both transcriptional repression and activation (Ellenbroek & Youn, 2016).

Lastly, a host of different non-coding RNAs (ncRNAs) including long non-coding RNAs (lncRNAs), piwi-interacting RNAs (piRNA), small interfering RNAs (siRNA), transfer RNAs (tRNA) and micro RNAs (miRNAs) are all involved in epigenetic activity through modifying chromatin structure and silencing gene transcription as well as mediating the recruitment of histones and DNMTs (Wei et al., 2017). They even seem to form memories of these events through self-reinforcing epigenetic loops (Holoch & Moazed, 2015). The most abundant internal mRNA/lncRNA modification is made of N6-methyl adenosine (m6A) while some of the other

common modifications are 5mC, N1-methyladenosine (m1A) and pseudouridines (Ψ) (Liu, N. & Pan, 2015). M6A modifications in the m6A methyltransferase *METTL14*, which is important for cell development, is associated with alcoholism (Niu et al., 2013; Wang, X. et al., 2016). Similarly, Variants within introns of m6A demethylase gene *FTO* have been associated with risk of type 2 diabetes and childhood obesity (Bell, C. G. et al., 2010).

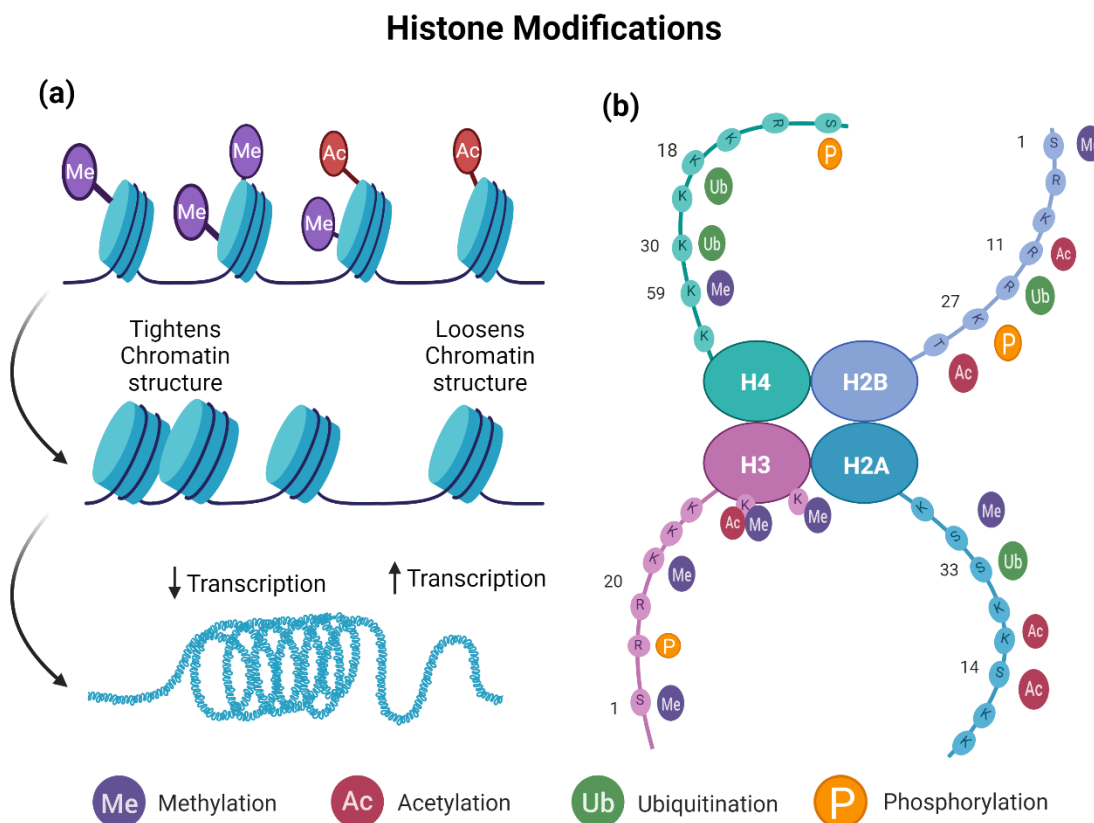


Figure 1.2. Depiction of histone modifications (a) Histone modifications like methylation are believed to tighten the chromatin structure by increasing binding between histones and thus reduces the amount of transcription from genes in that region while histone acetylation reduces binding between histones and increases transcription. (b) All 4 histones can be modified at different residues. Created with BioRender.com

1.3.2. Other possible mechanisms

While current research suggests epigenetic changes through the above-mentioned mechanisms appear to be the most likely way through which preconceptional environments are passed down, we should consider the other potential explanations for the phenotypes seen in offspring due to paternal ethanol exposure.

1.3.2.1. Genetic changes due to ethanol

First, is the idea that alcohol is causing direct mutations in sperm DNA which is causing the inherited differences. Interestingly, a lot of work assessing the genotoxic properties of ethanol were conducted in the 1980's and 90s and a detailed review on the genotoxic ability of ethanol found no significant evidence of it being a genotoxic hazard with the majority of assays looking at chromosome aberrations coming out negative (Phillips & Jenkinson, 2001). The review goes on to describe few studies looking at excessively high drinking which might give rise to some degree of genotoxicity (Phillips & Jenkinson, 2001). Indeed, studies looking at drinking have found some evidence of genotoxicity in peripheral blood due to alcohol (Ishikawa et al., 2007; Kotova et al., 2013). Again, *in vitro* research as well as *in vivo* work indicates alcohol can cause chromosomal damage and increase in DNA mutations in high concentrations (Garaycochea et al., 2018; Kayani & Parry, 2010; Obe & Anderson, 1987; Voordeckers et al., 2020).

When it comes to sperm cells, there have been multiple studies in humans on the effects of alcohol on overall sperm health and structure, with reports associating alcohol drinking to reduced sperm count, sperm motility, higher levels of aneuploidy and fragmentation (Joo et al., 2012; Ricci et al., 2017). However, there is no clear evidence of DNA mutations in sperm which can be passed down through generations being caused due to ethanol exposure. Even two recent

studies describe lower sperm motility, higher presence of immature sperm nuclei, abnormal proteins and altered protamine/histone ratios in high drinkers but show no significant differences in DNA sustainability or fragmentation (Blaseg et al., 2022; Trautman, 2023). Another important consideration is that sperm DNA fragmentation is associated with lower implantation rates, poor embryo development and high rates of pregnancy termination (spontaneous abortions) even in good quality oocytes (Alvarez Sedó et al., 2017; Borges et al., 2019; McQueen et al., 2019). Suggesting that in competition for successful pregnancies, fitter sperm beat out those with DNA damage. Taken together, the high doses required to induce a genetic mutation in sperm cells might not be easy to attain and even if they are reached, the damaged sperm would not be able to form viable offspring. These suggest that while direct mutations cannot be ruled out, epigenetic changes, which are subtler, might be what are responsible for the differences brought about by preconceptual ethanol exposure.

1.3.2.2. Changes in seminal fluids

As research focus has shifted away from gene mutations and towards epigenetics, there is also some recent evidence of the other components of sperm ejaculate playing a role. The seminal fluid which comprises a mixture of proteins, peptides, different types of small noncoding RNAs, and cell-free DNA fragments in the ejaculate appears to be involved in paternal epigenetic inheritance (Patlar, 2022). Seminal fluid compositions are highly influenced by the paternal environment such as diet, infection and exercise (Binder et al., 2015; Patlar et al., 2019; Ramm, 2020). Evidence of environmental differences in seminal fluid playing a role in offspring development comes from studies in neriid fly *Telostylinus angusticollis* where the seminal fluids seem to be associated with body size (Crean et al., 2014) and European whitefish *Coregonus*

lavaretus where seminal fluids appear to impact swimming performance (Kekäläinen et al., 2020). In addition, embryo viability appears to be partially dependent on seminal fluids (Simmons & Lovegrove, 2019).

However, due to the novelty of the field, the mechanisms by which seminal fluids play a role are currently not well understood (Patlar, 2022). One of the suggested mechanisms for seminal fluids are that they mediate sperm epigenetics. There is evidence which links seminal fluids to sperm RNA composition (Kasimanickam et al., 2019; Reilly et al., 2016; Vojtech et al., 2014) and on how protein/peptide concentrations can influence methylation status in both rodent and human sperm DNA (Åsenius et al., 2020; Poon et al., 2009). The other suggested route is through modulation of the female and egg epigenetics. This is backed by research which shows seminal fluids bringing about physiological and behavioural changes in females (Alvarez-Fernandez et al., 2019; Bromfield, 2018; Schjenken & Robertson, 2020) and direct contact with the egg may potentially cause epigenetic changes within the egg, example, in golden hamsters, disruption of seminal fluid producing organs altered histone modification in fertilized eggs and the DNA methylation pattern of embryos (Poon et al., 2009).

While we do not rule out these alternate mechanisms, due to the current evidence provided pointing against genetic mutations and the current mechanisms of seminal fluid being poorly understood but suggesting a role in mediation of sperm epigenetics. The remainder of this chapter focusses on the epigenetic alterations seen in both fathers and their offspring due to ethanol exposure.

1.4. Epigenetic Changes due to Alcohol Exposure

With regards to alcohol, research suggests ethanol consumption plays a role in DNA methylation and histone modifications thus influencing multiple epigenetic changes throughout the body (Ciafrè et al., 2019). Some of these changes have been seen in the germline (Chastain & Sarkar, 2017) and epigenetic changes have been identified in the subsequent generations (Finegersh et al., 2015). Below we detail the effects of ethanol on epigenetic mechanisms in the fathers and introduce the complexities of the patterns of alterations seen in the offspring.

1.4.1. Epigenetic changes in the fathers

1.4.1.1. Human studies

Most studies in humans looking at the effects of ethanol focus on DNA methylation by looking at blood samples. Genome-wide methylation analysis studies of mononuclear blood cells have found significant differences between AUD patients and controls in total DNA methylation levels (Philibert et al., 2018). With other reports seeing a 10% increase in genomic DNA methylation in blood cells in patients with AUD (Bönsch et al., 2004) . Studies which examine specific genes in individuals have seen an increase in DNA methylation and a reduction in gene expression of Period Circadian Regulator 2 (*PER2*) and Proopiomelanocortin (*POMC*) in heavy drinkers which were predictors of alcohol craving (Gangisetty et al., 2019; Muschler et al., 2010). Assessment of DNMTs show lowered DNMT-3b mRNA expression is associated with an increase of genomic DNA methylation in patients with alcoholism (Bönsch et al., 2006) and another study of DNMTs and methyl-CpG-binding domain proteins in the blood of individuals with AUD also show a downregulation of methyl-CpG binding domain protein 1 (*MBD1*) and methyl-CpG-binding protein 2 (*MECP2/Mecp2*) in the alcohol-dependent group (Krasteva et al., 2020). Interestingly, these differences in methylation seem to be related to patterns of

consumptions with one study finding significantly lower DNA methylation levels in the leukocytes of heavy drinkers compared to non-drinkers and moderate drinkers when looked at cross-sectionally (Maeda et al., 2022) and another reported differential blood DNA methylation in certain CpG islands in former drinkers compared to current drinkers in a longitudinal study (Dugué et al., 2021).

Epigenome wide analysis of alcohol dependence from human brain tissues identified differentially methylated regions in the dorsolateral PFC (dlPFC) and NAc with reduced methylation in drinkers (Meng et al., 2021). A study of postmortem brain tissue of 104 individuals showed a number of differentially methylated regions in glucocorticoid signalling and inflammation-related genes in the amygdala and PFC associated with AUD (Lohoff et al., 2021). However, others found no differences in DNA methylation patterns between alcoholics and matched controls in the PFC highlighting the tissue specific nature of methylation changes. Apart from methylation, epigenetic regulation in the brain has also been reported through alterations in lncRNAs and histone modifications (Manzardo et al., 2012). The lncRNA brain-derived neurotrophic factor-antisense (*BDNF-AS*) acts as an epigenetic regulator in the human amygdala only in early onset AUD but shows no increase in levels in late onset AUD suggesting the regulation is age dependent (Bohnsack et al., 2019). Lastly, histone modifications are also reported in the frontal cortex of individuals with AUD (Manzardo et al., 2012; Ponomarev et al., 2012).

While these studies show an effect of alcohol on epigenetic change, the way these changes would pass down the germline is through sperm cells and potentially the seminal fluid. Alcohol is linked with greater oxidative stress which in turn is associated with greater DNA

damage and increased global DNA hypomethylation in human sperm (Shen et al., 1999). To the best of our knowledge, only one study directly investigated sperm DNA methylation in humans. The authors found a reduction in DNA methylation of the regulatory gene *H19/IGF2* (insulin like growth factor 2) intergenic differentially methylated region (*IG-DMR*), where individuals who heavily consumed alcohol showed significant demethylation compared to those who did not drink alcohol (Ouko et al., 2009). The study showed a correlation between chronic alcohol use and demethylation of regions of sperm DNA that are typically hypermethylated, indicating epigenetic changes in specific imprinted genes that could be transmitted to the next generation. In addition, while no studies on humans have directly tested the effects of ethanol on ncRNAs in sperm, ethanol consumption is known to alter miRNA expression in the gut (Tang et al., 2008).

Lastly, it is important to mention that sperm DNA is different to mitotic DNA structure as it is much more tightly bound with most histones being replaced with protamines (Balhorn, 2007). This was thought to mean that epigenetic changes through histone modifications were not a way of paternal epigenetic inheritance, however, recent work shows a fraction of the genome escapes this remodelling and keep their histones which makes changes in chromatin a less likely but viable route for epigenetic inheritance (Sharma, U., 2019). Together, this means all major mechanisms of epigenetic changes can occur in epigenetic inheritance, however based on current knowledge, the most likely and best studied routes for the transmission of any epigenetic changes are through DNA methylation of select genes or small RNA modifications.

1.4.1.2. Animal studies

The amount of research looking at the effects of ethanol on epigenetic modifications in animal models outweigh the work done in humans. Much like the reports on blood samples and

brain region changes mentioned above, rodent studies also found changes in gene expression in the brain. Adolescent alcohol exposure was associated with dysregulation in DNA methylation mechanisms in the amygdala through altering DNMT activity during ethanol exposure and as well as in withdrawal (Sakharkar et al., 2019). The study also mentions the decrease in DNMT enzyme activity is accompanied by an increase in DNA-damage-inducible (*Gadd45g*) mRNA expression, which facilitates active DNA demethylation (Sakharkar et al., 2019). Rats exposed to ethanol also show increased hypermethylation of the PFC (Barbier et al., 2015; Otero et al., 2012) while ethanol treatments in adolescent rats leads to epigenetically encoded decrease in levels of BDNF protein and mRNA in the amygdala of adults (Kokare et al., 2017; Kyzar et al., 2017; Pandey et al., 2015). Additionally, rodent studies on ethanol exposure have identified modifications of multiple histones, examples of these are the increased levels of acetyl-histone histone 4-lysine 5 (*acH4K5*) in the medial prefrontal cortex (mPFC) after long term exposure (Montesinos et al., 2016) and acetylation of histone 3-lysine 9 (*H3K9ac*) in hypothalamic microglia after early life exposure to ethanol (Chastain et al., 2019). Lastly, the effects of alcohol on miRNAs are also seen with prolonged alcohol exposure affecting specific miRNA in the amygdala, hippocampus, and cerebellum (Barney et al., 2022; Kyzar et al., 2019; Rossetto et al., 2019).

Of greater interest to understand the intergenerational impact of alcohol drinking is the research linking ethanol to epigenetic modifications in the sperm. Two weeks of ethanol injections decreased sperm counts and plasma testosterone as well as increased oxidative stress in mice (Jana et al., 2010). Oxidative stress can deplete SAM multiple mechanisms, first, the transsulfuration pathway can regenerate glutathione under oxidative stress conditions from SAM,

depleting SAM in the process (Kreuz & Fischle, 2016). More directly, SAM is synthesised by methionine adenosyltransferase (MAT) and oxidation of MAT leads to its inactivation and again a reduction of SAM (Kreuz & Fischle, 2016). As ethanol induces oxidative stress, it may have a particularly robust effect on DNA methylation during spermatogenesis due to a negative regulation of SAM (Kruman & Fowler, 2014), the primary methyl donor for DNMTs. Studies have found reduced DNMT1 RNA levels in the sperm of chronic alcohol treated mice (Bielawski et al., 2002).

Importantly many genes are methylated from conception, and there are a substantial number of genes where only the maternal or the paternal copy of the gene is methylated (referred to as genomic imprinted genes). Disruption to the methylation status at known imprinting loci has been shown to have dramatic developmental effects (Monk et al., 2019). In this regard, studies have found reduced DNA methylation in imprinted gene loci *Snrpn*, *H19* and *Peg3* in sperm of chronic ethanol treated rodents (Knezovich & Ramsay, 2012; Liang et al., 2014) mirroring what was seen in humans (Ouko et al., 2009). A reduction in methylation in the *Bdnf* promoter in sperm of mice chronically exposed to ethanol vapour was also seen (Finegersh & Homanics, 2014). Another study observed an increase in methylation of dopamine transporter (DAT, *Slc6a3*) gene promoter in the sperm of sire mice, suggesting germline changes in the epigenetic methylation signature of DAT gene by ethanol exposure (Kim, P. et al., 2014). There is also evidence that supports changes in offspring methylation even in the absence of clear methylation changes in the paternal sperm. Trends but no significant DNA methylation changes in *H19* in the sperm but significant decreases in expression levels of *H19* in tail biopsies of ethanol-sired offspring have been noted (Knezovich & Ramsay, 2012). Similar patterns with the

paternally imprinted *Snrpn* gene found no methylation differences in the sperm but decreased methylation in the offspring cerebral cortex (Liang et al., 2014). Additionally, paternal voluntary alcohol consumption had no effect on sperm methylation but did alter hepatic fibrosis markers in offspring (Chang et al., 2017). Taken together these suggest paternal drinking may transmit heritable disruptions but the capacity of sperm-inherited changes in DNA methylation remains unclear.

With respect to other epigenetic mechanisms, ethanol alters the integrity of sperm chromatin and DNA (Rahimipour et al., 2013). In another recent study, ethanol exposure was shown to elevate the levels of H3K4me3 (trimethylation of histone H3 at lysine 4) in sperm leading to enhanced CTCF (CCCTC-binding factor) binding which was associated with altered developmental programming (Bedi et al., 2022). Lastly, a third study found male alcohol consumption significantly decreased sperm histone epigenetic mark H3K4me3 in the treated group (Cambiasso et al., 2022). The lack of research looking at sperm chromatin profile is suggested to be due to the unique properties and minimal histone content, however, acute ethanol does increase histone acetylation in rat testes, which is important in developing germ cells (Kim, J. & Shukla, 2006). Ethanol is also a modulator of histone-modifying enzymes in the brain and liver (Pandey et al., 2008; You et al., 2008). Researchers note that as sperm chromatin profile continues to be detailed, it would lead to improved understanding of any epigenetic mechanisms which would explain environmental inheritance (Rompala & Homanics, 2019).

RNA modifications are considered essential to models of RNA-mediated inheritance and underlie small non-coding RNA (sncRNA) stability (Rompala & Homanics, 2019). Multiple miRNAs and tDR (transfer RNA-derived RNA) in the sperm appear to be altered by chronic

ethanol exposure (Rompala et al., 2018). One of those noted to be altered was miR10a which is suggested to function as a posttranscriptional regulator for *BDNF* (Jiajie et al., 2017). The other prominent finding was the alteration of *tRNA-Gly-GCC* which is associated with controlling retrotransposon-regulated gene networks in the fertilized oocyte (Sharma, U. et al., 2016) and *tRNA-Glu-CTC* as it makes up about half of the small RNA content in the sperm (Peng et al., 2012). Rompala et al., also found 2 altered posttranscriptional RNA modifications specific to mitochondrial tRNAs which could be involved in sperm mitochondrial function (Rompala et al., 2018). Additionally, chronic alcohol use was shown to alter the ratio of tRNA-derived small RNAs to Piwi-interacting RNAs in sperm which was associated with late term foetal growth restriction (Bedi et al., 2022). While few studies have looked at alcohol, exposure of male mice to stress or paternal stress alters several sperm miRNA profiles (Rodgers et al., 2013; Rodgers et al., 2015). The same is seen with high fat diet-induced obese male mice showing altered miRNAs transcription profiles (McPherson et al., 2015). As there are over 150 identified RNA modifications (Boccaletto et al., 2022), most of which have not been assessed with regards to ethanol. Future studies will need to fully characterize the effects of ethanol on the epitranscriptome.

1.4.2 Gene expression changes in offspring

We earlier discussed that the epigenetic changes in the sperm of fathers who consume alcohol may play a critical role in the phenotypical differences seen in PatEE offspring when compared to non-drinkers. We also pointed out that the methylation of a particular gene in the sperm does not coincide with dysregulation of the same gene in the offspring. Liang et al (2014) found reduced DNA methylation in genes *H19* and *Peg3* in sperm of chronic ethanol treated

rodents but found no methylation differences of these in their offspring instead finding decreased *Snrpn* methylation in the cerebral cortex, a gene which was not differentially methylated in the sperm. Similarly, a study found no significant DNA methylation of *H19* in the sperm but saw reduction in *H19* expression in the offspring (Knezovich & Ramsay, 2012). Another study with comparable results found no changes in gene methylation in the sperm but found decreased expression of imprinted genes associated with foetal and placental growth in the offspring, specifically, *Gnas*, *Grb10* (males) and *Sgce* (females) in the placenta, as well genes *Cdkn1c*, *Dcn*, *Gnas* (males), and *Meg3* in the foetal liver which are associated with liver fibrosis (Chang et al., 2017).

One of the common set of gene with altered expression due to PatEE are neurotrophins with a reduction in DNA methylation levels of the *Bdnf* gene in the ventral tegmental area of male but not the female offspring of ethanol exposed sires (Finegersh & Homanics, 2014). The authors also reported no changes, in either sex, in *Bdnf* methylation in the mPFC. On the other hand, increased *Bdnf* and reduced *Ngf* expression in the PFC, striatum, and olfactory lobes of offspring of ethanol exposed mice as well as changes in Trk-A, which is a receptor for *Ngf*, in the hippocampus has been found (Ceccanti et al., 2016). Additionally, there was an increase in DAT levels in the olfactory bulbs of the ethanol-sired offspring (Ceccanti et al., 2016). A recent rat study of PatEE found that while *Bdnf* DNA methylation levels were lower in the NAc and greater in the mPFC of the ethanol sires, but the same patterns of DNA methylation were not seen in the offspring despite the offspring showing aberrant *Bdnf* DNA methylation patterns compared to control-sired offspring (Nieto et al., 2022).

Several other methylation patterns and gene expression changes as a result of PatEE have been noted. The offspring of ethanol-sires show increased methylation in the *Pomc* gene which can lead to HPA (Hypothalamic Pituitary Adrenal) axis dysregulation (Govorko et al., 2012). This possible HPA disruption is confirmed by another study which found lower corticosterone levels in offspring of preconception ethanol animals compared to control offspring (Asimes, 2018). The study also found lower mRNA expression of gonadotropin releasing hormone which is important in regulating the reproductive axis (Asimes, 2018). Chronic paternal ethanol exposure from late adolescence was found to decrease *Dnmt1* and *Mecp2* levels, as well as DAT expression in the mPFC and striatum of male offspring (Kim, P. et al., 2014). Another study detailed the effects of ethanol exposure before conception on the offspring's hypothalamus and found differences in expression of genes that mediate neurogenesis and synaptic plasticity during neurodevelopment (*Bmp1*, *Fgf13*, *Serpini1*, *Igf2r*, *Fgfr3*, *Pak*), genes important for directing chromatin remodelling (*Hdac3*, *Sirt2*, *Dnmt1*), posttranslational modifications or transcription regulation (*Egr2*, *Wbp4*, *Dicer1*) and genes involved in regulation of obesity (*ApoE*) and reproductive function (*GnRH*) (Przybycien-Szymanska et al., 2014). Likewise, genome-wide DNA methylation patterns in the hypothalamus of offspring who had either one or both parents being exposed to ethanol preconception found methylation patterns were different when only one parent was exposed to ethanol compared to both and the patterns again differed whether it was the mother or father who was exposed (Asimes et al., 2017). All three cases did have different DNA methylation patterns when compared to controls while the global methylations appeared to not be significantly different (Asimes et al., 2017).

To summarize, research has revealed a clear influence of paternal alcohol exposure on the expression levels of genes associated with neurotransmitter systems, synaptic plasticity, neurodevelopment, endocrine functions, transcription regulation and chromatin modifications. The studies also indicate the complexities of intergenerational epigenetic mechanisms with methylation not being a direct predictor of RNA expression levels and the high sex based and region-specific alterations seen thus far.

1.5. Transgenerational Impacts of PatEE

While most research focusses on intergenerational non-genomic inheritance, it is important to also consider the effects across multiple generations. However, this inheritance appears slightly different when coming from males or females. For males, a phenotype being inherited from the father (F0 generation) to the offspring (F1 generation) is considered intergenerational. If the same phenotype persists in the F2 or subsequent generations, it can then be called transgenerational. In females, as the F0 pregnant female already carries the F1 foetus and the already formed oocytes which will be used for the F2, a true transgenerational effect through the female germline would be reserved for the F3 generation (Perez & Lehner, 2019).

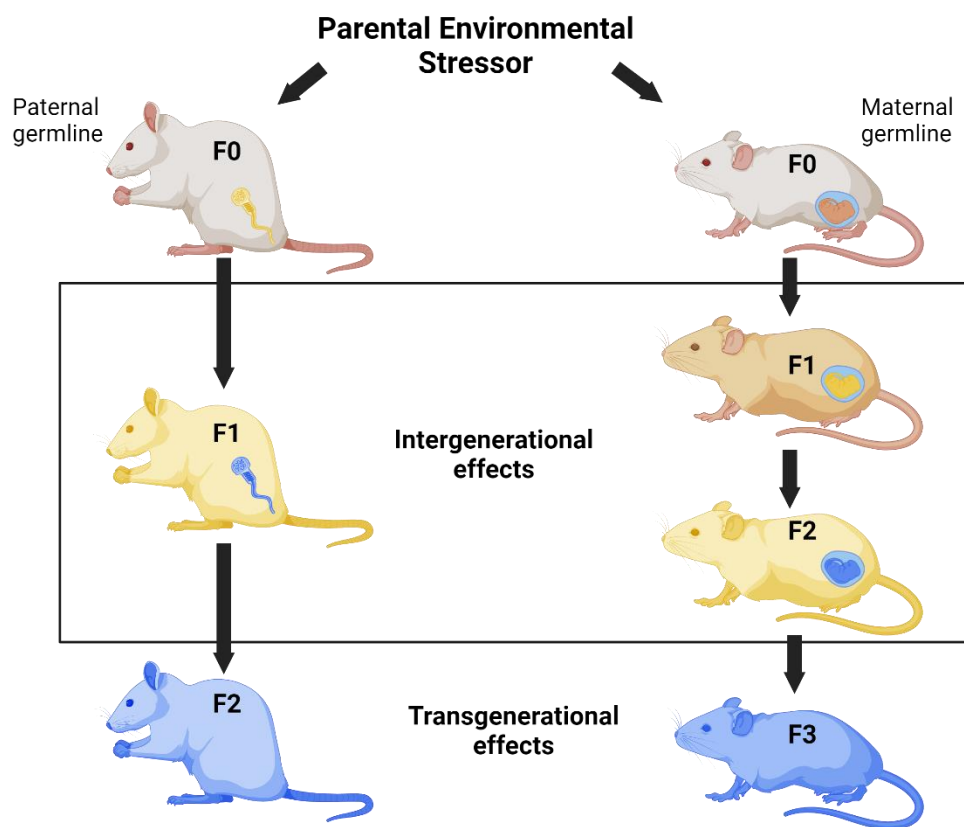


Figure 1.3. Difference between Intergenerational and Transgenerational Effects in male and female Germline. Intergenerational and transgenerational inheritance to environmental stressors like alcohol consumption. In paternal exposure, only the sperm cells for the F1 generation are present and directly exposed to the stressor (intergenerational inheritance), whereas phenotypes which persist into F2 and beyond are transgenerational (blue) due to not having any direct contact with the stressor. Maternal exposure exposes both her and the developing foetus in utero (F1, orange) as well as the developing germ cells in the foetus which later give rise to F2 (yellow), both of which are present at the time of environmental exposure (therefore they both are intergenerational). The F3 generation cells (blue) are not present at the time of environmental exposure; if they carry phenotypes, they do so by transgenerational inheritance. Created with BioRender.com

The Swedish Överkalix study showed an effect of the paternal grandfather's food availability and nutrition on the grandchildren (Bygren et al., 2001). A further analysis of these cohorts found sex dependent links with paternal grandfather's food supply being linked to the mortality of grandsons, while paternal grandmother's food supply was only associated with the

granddaughters' mortality (Pembrey et al., 2006). With regards to alcohol, a human study noted that ancestral exposure to alcohol can have enduring effects that impact epigenetic processes such as hypomethylation of *HRAS* and hypermethylation of *TP53* which are genes involved in cancer development (Hill et al., 2017).

True transgenerational effects of PatEE have rarely been reported in rodents. One early study on guinea pigs indicated lasting effects of paternal ethanol on offspring mortality and deformations lasting up to 4 generations (Stockard & Papanicolaou, 1916b). Other works have noted decreased testosterone levels and damage to sperm DNA in adult male rats sired by ethanol exposed rats which could lead to differences in future generations (Abel & Lee, 1988; Cambiasso et al., 2022), although this was not investigated. One study did find a 5% reduction in cortex methylation in the F2 of ethanol sired mice although the study did not see any malformations in the animals at 8 weeks (Liang et al., 2014). Indeed, to the best of our knowledge, no studies in rats have investigated transgenerational behavioural effects due to PPAC. However, studies which have looked at prenatal alcohol exposure have seen *Pomc* dysregulations caused by prenatal exposure to ethanol being passed down through the male germline for at least 2 further generations (Govorko et al., 2012; Sarkar, 2016). While these changes are passed down through the males, they appear to not do so through the female germline (Sarkar, 2016). Recently, another study looking at a transgenerational model of prenatal alcohol exposure saw abnormal sensorimotor processing, increased risk-taking behaviour, and increased depressive-like behaviours extending to the F3 generation in 20-day old mice (Bottom et al., 2022). The study however only saw differences in phenotypes and could not see any significant differences in cortical connectivity after the directly exposed F1 generation. This is in

line with two other studies which have observed maintenance of epigenetically encoded phenotypes despite loss of transmission of altered ncRNAs to offspring (Gapp et al., 2014; Radford et al., 2014). These studies clearly indicate that more research is needed to investigate whether the effects of PPAC extend beyond the F1 generation due to PPAC.

1.6. Discussion

1.6.1. Conclusions

The chapter presents an overview on the current knowledge surrounding the intergenerational impacts of alcohol with a primary focus on paternal drinking. Research, while being heterogenous, has clearly shown PatEE influences offspring behaviours and gene expression. Ethanol is shown to affect DNA methylation, ncRNAs and even histone modifications in germ cells and these contribute to developmental and behavioural changes in the offspring in ways ranging from birth defects and increased mortality to altered sensitivity to alcohol and cognitive impairments. Interestingly, these changes appear to have the potential to be transmitted to future generations and highlight the complexity of non-Mendelian inheritance. Current research has shown that these intergenerational changes appear to be highly dependent on a range of factors which include acute or chronic alcohol exposure, species of subject, sex of offspring and region analysed to name a few. Despite a clear effect of PPAC, the small number of studies and high variance within experiments indicates how early PPAC is in its research in terms of understanding the effects seen. Understanding these impacts is imperative for designing public health policy around alcohol use as it is increasingly showing to not only affect the health of the user but also any future offspring.

1.6.2 Current project

The current project focusses on paternal preconceptual alcohol exposure research. As mentioned, the large majority of alcohol research looks at direct harm to the individual and with an increasing amount of evidence indicating parental environment influencing offspring development. Of these preconception environments the project chose to focus on the paternal aspects as opposed to maternal for a few reasons. First, as mentioned, public health guidelines around alcohol during pregnancy and while trying to get pregnant are generally consistent for women with a strict emphasis on abstinence from alcohol drinking (Armstrong, 2017; Centers for Disease Control and Prevention, 2023; The American College of Obstetricians and Gynecologists, 2023). Conversely, such public health advice when trying to start a family is generally absent for men apart from in cases of low fertility (National Health Service, 2023). With the mounting evidence suggesting the impact of paternal preconceptual environment on offspring development, a project focussing on the effects of paternal drinking can further build a case for changing public perceptions. Second, our lab previously looked at preconceptual maternal drinking and found no significant differences in the offspring when compared to controls (unpublished data). Additionally, a large-scale study looking at 5,600 women found no effect of alcohol intake preconception (McCarthy, F. P. et al., 2013). While the absence of differences could be due to methodological limitations, this does encourage us to take a deeper dive into the paternal effects first before going back to look at the maternal effects of preconceptual alcohol exposure.

The project aims to assess both the inter and transgenerational impacts of preconceptual paternal alcohol consumption using a rat model. First, the project uses a method of voluntary ethanol consumption in the F0 males as opposed to forced exposure techniques to assess for any

variations in offspring seen due to sire drinking amounts. Next, while the animals are of the same outbred rat strain (Sprague Dawley rats) each animal in the F1 comes from a different litter and one male and female offspring from each litter will be a part of each behavioural test to account for litter effects and contamination seen by having one animal go through multiple behavioural tests. We will be looking at the behavioural performance of F2 generations of ethanol drinking F0s which means the project would be one of the first of its kind to look at true transgenerational impacts of PPAC. Lastly, the study hopes to identify changes seen in the F0, F1 and F2s due to ethanol consumption and check for similarities and any dose-response effects. To do this, sperm samples from all the sires used in mating will be collected, along with liver samples to assess metabolism changes due to ethanol and lastly brain samples from the sires and subsequent generations at different ages (PND 2- PND 180) to understand any temporal effects of PPAC. Our lab, through this project, aims to paint a comprehensive picture of paternal alcohol abuse and its effects across multiple generations. The current thesis comprises of a majority of the behavioural and developmental work and the preliminary work on liver function to assess alcohol damage and indications of potential epigenetic changes.

Chapter 2: Paternal Drinking and Mating Selection for Transgenerational studies

This study aimed to assess the effects of Paternal Preconceptual Alcohol Consumption (PPAC) in rats which voluntarily developed high ethanol drinking behaviours. Historically, it has been difficult to get standard laboratory rats to voluntarily consume considerable amounts of ethanol without initiation procedures. Such procedures include sucrose fading (Samson, 1986), food and water deprivation (Meisch & Thompson, 1973), pairing drinking with electrical brain stimulation (Wayner & Greenberg, 1972), and limiting access to ethanol (Sinclair et al., 1992). However, rats do not maintain intake at the same level once the initiation procedure is removed. Other attempts to increase ethanol consumption in rats have been made using different concentrations of ethanol (Kiefer et al., 1994; Wolffgramm & Heyne, 1995), or by selectively breeding animals to prefer alcohol (Bell, R. L. et al., 2006; Dyr & Kostowski, 2008; Myers et al., 1998). Most of the previous studies looking at paternal ethanol exposure have used such models or have used forced methods like ethanol vapour or injection (Abel & Lee, 1988; Abel, 1989; Abel, 1993; Conner et al., 2020; Finegersh & Homanics, 2014; Ledig et al., 1998; Nieto et al., 2022). This makes it difficult to design a preclinical model for behaviours associated with alcohol use disorder in humans and further affect the ability to assess associations with levels of ethanol intake and their transgenerational effects.

Studies show that intermittent access to ethanol induces high voluntary ethanol consumption (Hwa et al., 2011; Wayner & Greenberg, 1972; Wise, 1973). Simms et. al., (2008) developed an intermittent-access 20% ethanol 2-bottle-choice drinking paradigm (IA2BC) which showed potential as a preclinical animal model of drinking (Simms et al., 2008). The IA2BC

facilitates studies examining several stages of ethanol consumption including escalation, maintenance, and deprivation and has been successfully adapted to demonstrate individual differences in vulnerability for AUD (Spoelder et al., 2015; Spoelder et al., 2017). In addition, voluntary consumption is translationally relevant as it better mimics human drinking and it overcomes the stress confound that is associated with other forced exposure models (Rompala et al., 2018). Other studies, including from our own lab, have also successfully used similar procedures to induce elevated levels of voluntary ethanol intake (Beeler et al., 2019; Carnicella et al., 2014; Hussain et al., 2022; Sabino et al., 2013) and it therefore represents a suitable methodology to investigate the inter- and transgenerational effects of ethanol consumption.

2.0.1. Current study

The current study used an adapted version of the IA2BC design for voluntary ethanol intake starting from adolescence in the first generation (F0). Adolescence for the beginning of ethanol intake was selected due to studies showing adolescents are more ready to take up ethanol drinking and consume ethanol at faster rates than adults (Doremus et al., 2005; Vetter et al., 2007). These findings suggest starting the IA2BC in adolescence also has a practical advantage when trying to assess cross generational effects. Importantly, the decision also translates well to human research which shows people most likely to have drinking behaviours in adulthood start in adolescence (Bolland et al., 2016; Dawson et al., 2008; Spear, 2016).

After completion of IA2BC the F0 were then mated (after an ethanol free wash-out period) with ethanol naïve females to give birth to the next generation (F1). This is to make sure that the only contact the F1 generations had to ethanol was through the father's semen. A set of F1 rats were kept and mated with another set of ethanol naïve females so the only contact the F2

generation had was through the grandfather's semen. Below the IA2BC paradigm and selection criteria for the mating used in the experiment going forward are described.

2.1. Methods

2.1.1. Subjects

The first generation (F0) of Sprague-Dawley rats in this experiment comprised of 30 male rats which were bred at Victoria University of Wellington, New Zealand, vivarium. Animals were individually housed for the duration of the experiment and 20 rats were subjected to ethanol drinking through the IA2BC while the other 10 which were given ad libitum tap water under the same housing conditions acted as matched controls. Rats were postnatal day (PND) 30-33 at the start of the drinking paradigm. Eight weeks, to wait for the completion of one spermatogenesis cycle (Lara et al., 2016; van Haaster & de rooij, 1993), after the last drinking session, the 10 highest drinking males (D₀) and the non-drinking controls (ND₀) were mated with ethanol naïve females. Of the remaining 10 males, the 6 highest drinkers were culled for sample collection after the last day of drinking for molecular analysis (explained in Chapter 5). Samples were also collected from the 10 drinkers and 10 non-drinkers after mating was completed.

The housing room was temperature (20 °C) and humidity (60%) controlled. And was maintained on a 12-h reverse light/dark cycle (lights off at 7 am). Rats were provided food (Specialty Feeds, SF00-100, Australia) and water ad-libitum for the duration of the study. Animal care procedures and experimental protocols met with Victoria University ethics committee approval.

2.1.2. Procedures

2.1.2.1. Intermittent alcohol two bottle choice (IA2BC) paradigm

Rats were provided ethanol (20% v/v), made from diluting 99.8% ethanol solution (Purescience Ltd, Porirua New Zealand) with tap water, in their home cages using an adapted version of the intermittent access two bottle choice paradigm (IA2BC; Figure 2.1) (Simms et al., 2008). Rats were offered ethanol for 7 hours/day for three days/week (Monday, Wednesday, and Friday) for twelve sessions. On ethanol drinking days, rats were offered two plastic bottles with stainless steel, drip proof, double ball bearing drinking spouts containing 20% (v/v) ethanol or tap water. Bottles were inserted on either side of the cage front and were presented 30 minutes after lights off and remained until 7 hours had passed. The ethanol bottle was replaced with a second water bottle until the next session began. After the initial four weeks, the rats were given access to ethanol for 24 hours/day for three days/week (Monday, Wednesday, and Friday) for another 12 sessions. Ethanol was again presented 30 minutes after the lights went off and the bottle remained on for 24 hours after which it was replaced with water. Placement of ethanol bottles was randomized for each drinking session to control for side preferences. The ND (Non-Drinkers) group was kept under the same experimental conditions except they were only given water. Ethanol and water intake were recorded by weighing the bottles after each session to calculate alcohol preference. Rat bodyweight was measured weekly to calculate g/kg intake.

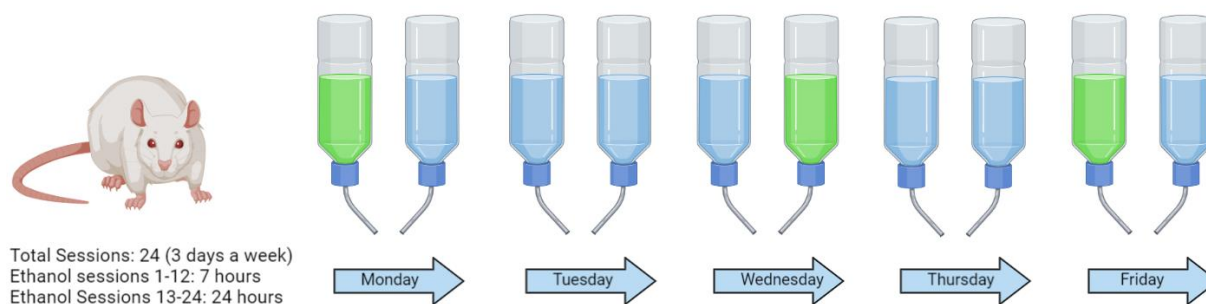


Figure 2.1. Schematic of the IA2BC drinking paradigm. A 20% (v/v) Ethanol solution was provided along with a bottle of water three times a week. Ethanol was provided for 7-hours for the first 12 sessions after which the rats were given ethanol access for 24-hours during remaining 12 session The ethanol bottle location was randomized to prevent side preference. Created with BioRender.com

2.1.3. Data analysis

Statistical analysis was carried out using IBM SPSS Statistics 25 (IBM; USA) and GraphPad Prism (version 8.4.1 for Windows, GraphPad Software; USA). A generalized linear mixed model analysis with an autoregressive covariance structure was used for ethanol consumption across the 24 sessions. This model was chosen to accommodate for days during which data of a subject was lost due to leaks in the drinking bottles (Wang, Z. & Goonewardene, 2004). Total fluid intake and ethanol preference were averaged for the 7-hour and 24-hour sessions and were analysed using a repeated measures analysis of variance (ANOVA).

2.2. Results and Discussion

2.2.1. Drinking acquisition

There were only 8 bottle leaks in total across the sessions with 1 rat being affected twice. All 20 rats in the study developed substantial alcohol drinking, confirming the effectiveness of the intermittent access paradigms for the acquisition of consistent drinking behaviours with a significant increase in drinking across days ($F(23, 391) = 16.90, p < .001$). However, we also see a large variance between rats with the ethanol consumption of the highest drinking rat at times being double that of the mean (Figure 2.2). The mean gram per kilogram (g/kg) ethanol intake across sessions was 3.60 g/kg ($SD = 1.10$ g/kg) with the minimum being 2.20 g/kg and maximum 6.05 g/kg. This variance within our selected mating cohort does allow us to look at a linear range of ethanol consumption which means the ability to look for dose dependent effects of PPAC.

F0 Ethanol drinking per day

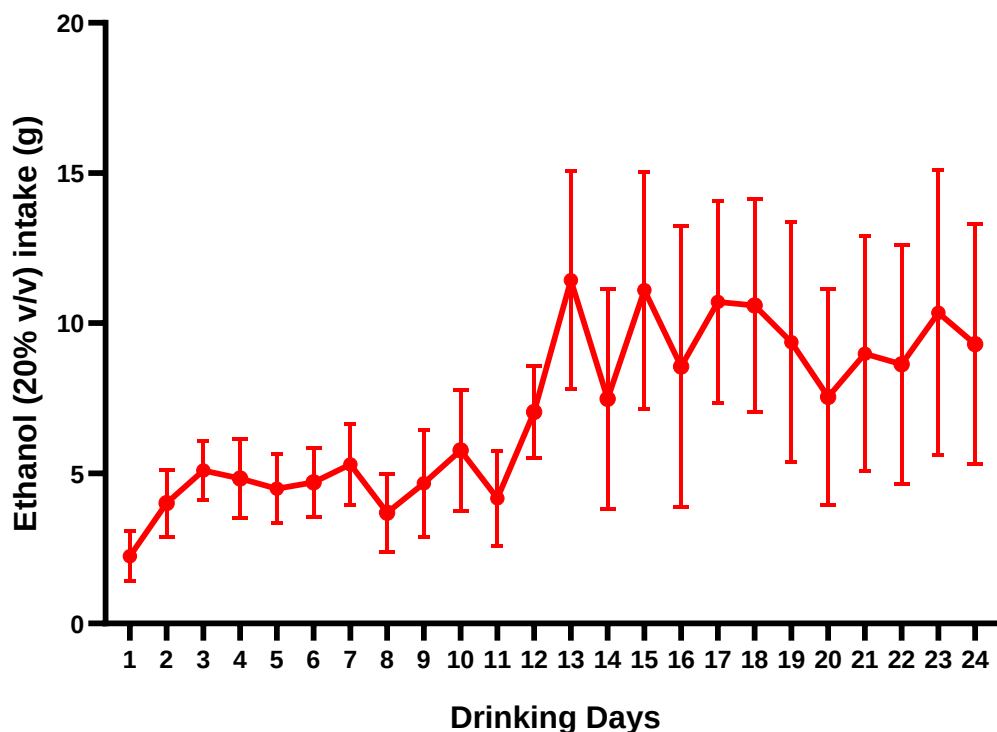


Figure 2.2. F0 Ethanol drinking. Daily ethanol intake for the 20 rats initially increased across sessions and then remained stable. There was a significant effect of day ($F(23, 391) = 16.90, p < .001$) which is shown to be due to the low drinking on the first two drinking sessions followed by the increase when sessions changes from 7-hours to 24-hours. Bars denote standard deviations.

Total fluid intake during the 7-hour and 24-hour drinking sessions was compared between the drinking animals and controls using a repeated measures ANOVA. As expected, there was an increase in fluid intake when we moved from 7-hour access to 24-hours ($F(1, 28) = 915.93, p < .001$) but there were no significant differences in fluid intake between the conditions ($F(1, 28) = 0.08, p = .786$; Figure 2.3). Alcohol preference averaged 29% ($SD = 8.18\%$) and varied with the rat with the highest preference to ethanol being 47.4% preference and the lowest preference being 16.9%. Like drinking, preference initially increased over the first few days, however it stagnates, there was no significant difference ($F(1, 19) = 1.07, p = .313$) in preference between the 7-hour sessions ($M = 31.5 \pm 6.71\%$) and the 24h-sessions ($M = 29.3 \pm 10.2\%$).

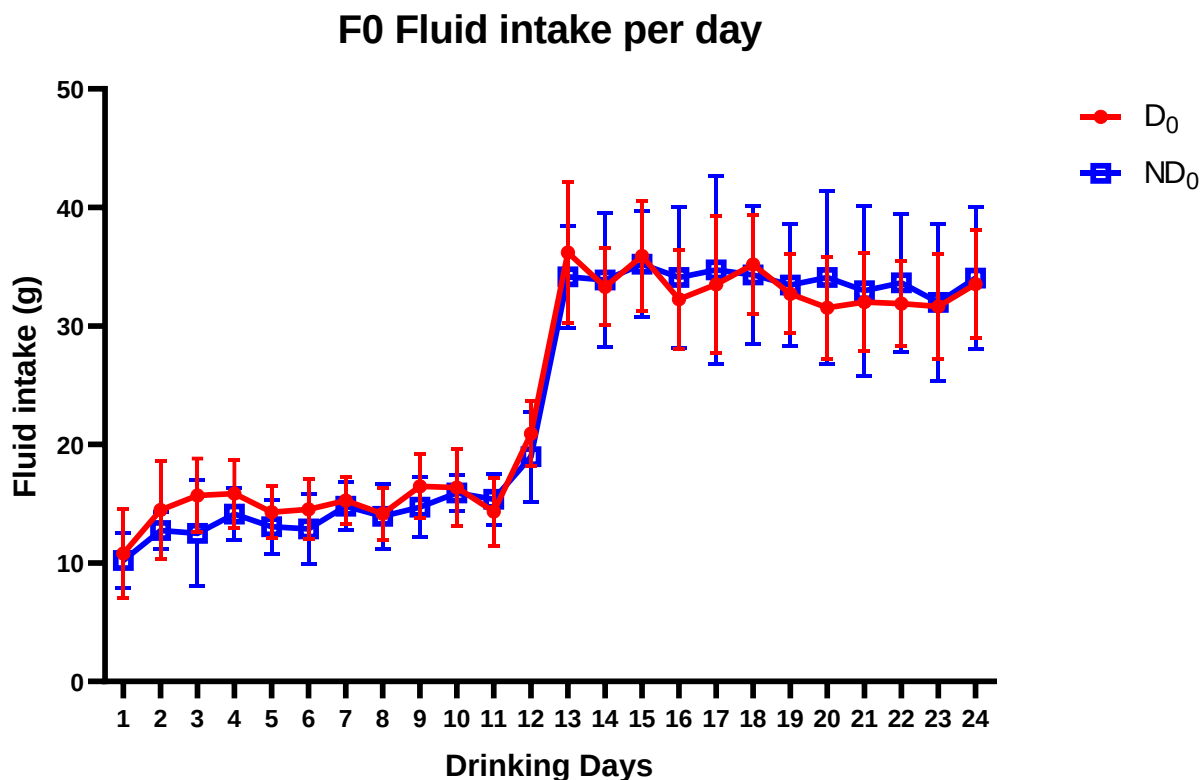


Figure 2.3. F0 Total fluid intake. Total liquid intake was compared between the controls and drinkers and there were no statistical differences between the two groups in fluid intake ($F(1, 28) = 0.08, p = .786$). The only difference seen was the expected jump in fluid intake between the 7-hour and 24-hour sessions $F(1, 28) = 915.93, p < .001$.

2.2.2. Power Analysis

A power analysis to estimate the required sample sizes for the study was done using the resource equation approach (Arifin & Zahiruddin, 2017; Charan & Kantharia, 2013). This method was selected as it is recommended for exploratory research and when it is hard to assume standard deviations and effect sizes (Festing & Altman, 2002) such as in this study where the transgenerational and intergenerational impacts of paternal ethanol consumption on behaviour are unclear (Chapter 1). The method is based on the use of Analysis of Variance (ANOVA) for analysis and suggests that the degrees of freedom (DF) for the error term used to test the effect of

the variable should be approximately 10 to 20 where over 20 leads to diminishing returns and under 10 is too low (Arifin & Zahiruddin, 2017; Festing & Altman, 2002). The formula used for calculation is:

$$n = DF/kr + 1$$

Where n is the sample size, k is the number of groups, r is the number of repeated measurements and DF is replaced by 10 for minimum and 20 for maximum.

Calculations for the F0 generation with just the drinkers and non-drinkers indicate a sample size between 6 and 11 for each condition in F0. Looking at F1 and F2, the 4 possible groups based on sire condition (D vs ND) and sex (male vs female) suggest a required sample size between 4 to 6 per condition for each experiment. Based on these numbers and accounting for potential unsuccessful mating pairs we decided to mate 10 F0 animals in each condition.

2.2.3. Study mating structure

The rats with an average drinking above the 50th percentile (3.30 g/kg) were selected to mate with ethanol naïve females. All 10 ND₀ rats were also mated. The non-drinkers were housed in a comparable way to the D₀ to form identical controls. This is because rats were housed in isolation for the period of drinking and isolation itself is shown to have effects on rat behaviour (Lesscher et al., 2015). These were then mated to form matched controls for the F1 and F2 generations.

The choosing of the highest 10 drinkers for mating was to make sure our sire rats had consistently consumed high amounts of alcohol. Those selected for mating were housed individually for the length of 1 spermatogenesis cycle (about 8 weeks) before being mated with

ethanol naïve females (Lara et al., 2016; van Haaster & de rooij, 1993). The pairs were kept together for 20 days after which they were separated, this occurred once and therefore if a sire was unable to impregnate a female in that time it produced no F1 litters. Each sire would produce a single litter. Of the litters produced, one male and female from each litter, numbers permitting, were used for different experiments. This is to reduce any litter effects which are often seen in animal studies. Litter effects are the effect where animals from the same litter show greater phenotypic similarities to each other than rodents from different litters of the same strain (Jiménez & Zylka, 2021). Males from each of the F1 litters were left alone to then be mated with a separate set of ethanol naïve females, in the same method as described above, to produce F2 litter. This is important to assess transgenerational effects as the F2 litters do not have any contact with alcohol apart from their F0 grandfathers (Figure 2.4).

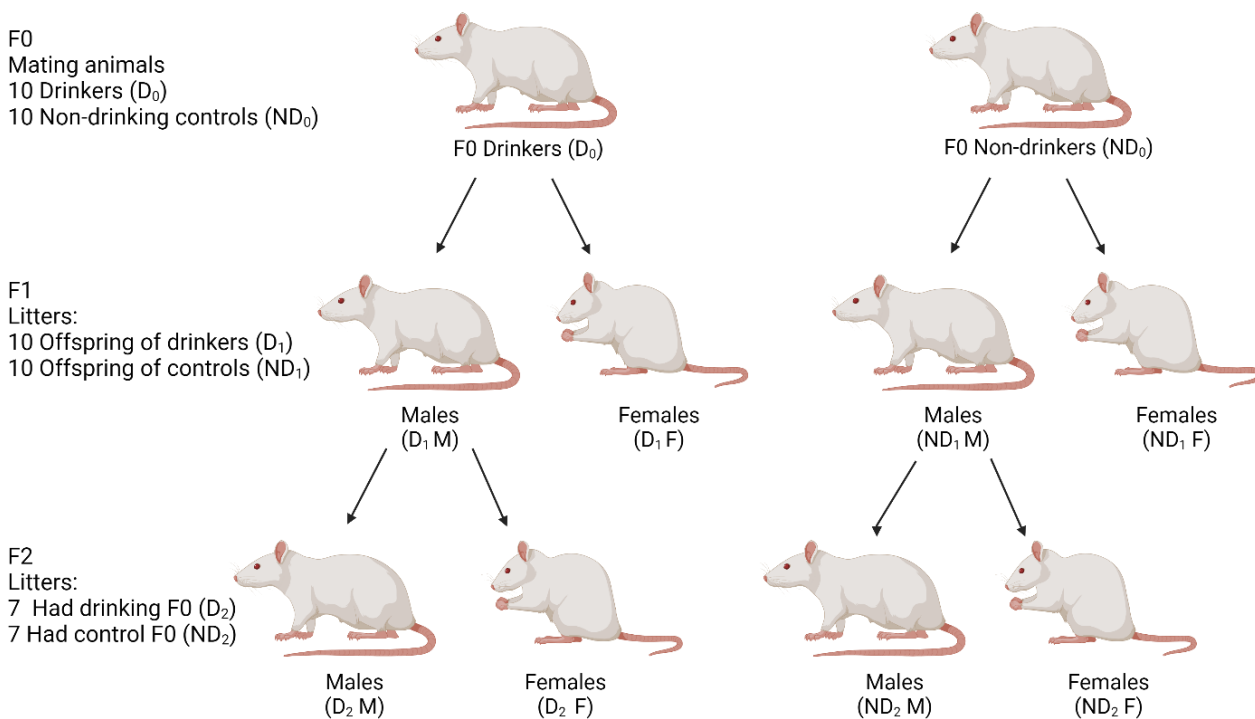


Figure 2.4. Schematic of animals in the study. Three generations of the male germline were assessed for PPAC. F1 rats used for mating were not exposed to ethanol making the D₀ drinkers the only mated animals exposed to ethanol. Created with BioRender.com

2.2.4. Thesis overview

As this thesis is the first part of a larger project looking at the transgenerational effects of paternal alcohol, the current thesis focusses on certain behavioural and biological aspects. The first set of experiments looks at early developmental milestone achievement in the F1 and F2s, while the next set focus on locomotor and drinking behaviours in adolescence and early adulthood. Lastly, we look for molecular differences in liver function and methylation in the F0 which could be involved in transgenerational effects of alcohol. After these, the thesis will

discuss its findings in context of other work done by our lab to present an overall picture of the project and its implications so far.

Chapter 3: Intergenerational and Transgenerational impacts of PPAC on Development

Early life stressors often play a significant role in the development of behavioural patterns later in life. Research has shown that early exposure to traumatic experiences and stress can induce neurodevelopmental abnormalities (Mesquita et al., 2007). A lot of the early brain development in rats begins in the second week of intrauterine life and completes by the first four weeks of postnatal life which is a lot faster than in humans, where the first two years of life are considered crucial (Barkur & Bairy, 2015). However, just like infants, rat pups show key behavioural and physical markers which underline early development. These markers include somatic markers (e.g., eye opening) and behaviours (e.g., walking and surface righting reflexes). Changes in the timing of these markers can be influenced by different events during development and often predict neurodevelopmental abnormalities later. Maternal stress during pregnancy has shown to cause delays in righting reflexes and achievement of some other developmental milestones (Weinstock et al., 1988) while prenatal alterations in immune function have been shown to induce delays in eye opening (Foley et al., 2014). Developmental marker delays are also associated with early life stress with delays in eye opening and incisor eruption reported in pups which had been overly handled by experimenters (Lee, M. & Williams, 1977) while maternal deprivation showed delays in surface righting reflexes, eye opening and acquisition of walking and rearing abilities (Ellenbroek et al., 2005; Mesquita et al., 2007). Sensory deprivation by whisker trimming is another early post-natal intervention which delays achievement of these developmental milestones (Smirnov & Sitnikova, 2019).

Research also shows that delays in developmental milestone achievement consistently act as predictors for larger behavioural abnormalities later in life (Ellenbroek et al., 2005; Foley et al., 2014; Smirnov & Sitnikova, 2019). However, in both pre-natal and post-natal cases, the stressor is acting directly upon the subjects being assessed. This opens questions into the effects of pre-conception stressors on the early development milestones of offspring. In this situation, neither the foetus nor the mother has been put directly under significant stress. Going a step further, what if the stressor was a generation removed? This chapter looks at the early developmental differences between offspring of rats sired by ethanol drinking fathers and controls. Additionally, we assess the transgenerational effects of ethanol drinking on early development in the second generation.

3.0.1. Development and PPAC

While research into the effects of PPAC is growing, most work has been done on behavioural and cognitive functions which are seen later in life (Chapter 1). The research which does speak to early developmental differences is scarce but informative. Early research report features of FASD in children born to alcoholic fathers (Lemoine et al., 2003) while other studies in humans indicate potential differences in anogenital distance (Xia et al., 2018), higher risk of childhood acute lymphoblastic leukaemia and brain tumours (Milne et al., 2013) and increased risk of congenital heart disease (Zhang, S. et al., 2020). Other studies find no effect of paternal drinking on birth weights or early motor development in infants (Hutchinson et al., 2019; McCormack et al., 2018; Passaro et al., 1998).

In rodents, as discussed previously, PatEE has shown effects on litter sizes, pup sex ratios, birth weights and litter viabilities (Abel, 1989; Abel, 1995; Chang et al., 2019; Emanuele

et al., 2001; Klassen & Persaud, 1976; Ledig et al., 1998; Meek et al., 2007; Rathod et al., 2020; Stockard & Papanicolaou, 1916a). Additionally, the few studies looking at paternal ethanol exposure on developmental milestones show mixed results (Jamerson et al., 2004; Meek et al., 2007). No studies we know of have looked at early developmental milestone differences one generation removed from paternal ethanol exposure. However, lasting differences have been reported which could influence effects in subsequent generations (Abel & Lee, 1988; Cambiasso et al., 2022; Govorko et al., 2012; Sarkar, 2016; Stockard & Papanicolaou, 1916a; Stockard & Papanicolaou, 1916b). Therefore, this chapter also looks at any potential transgenerational effects seen due to male alcohol drinking pre-conception in rats.

3.0.2. Current study

This chapter compares developmental markers of pups sired by rats exposed to ethanol and those sired by ethanol naïve rats (F1). The same markers were looked at in the next generation (F2) to assess the transgenerational effects. The measures were litter size, eye opening, pinnae development, fur coat growth, walking, self-grooming and unsupported rearing (Smirnov & Sitnikova, 2019). As these milestones are generally seen on specific days in normal rat development and can easily be assessed by observation in home cages, we can identify key milestones of development without disturbing/handling the animals and thus reducing the potential of stress effects on the litters which has shown to influence early development (Howe, 1983; Lee, M. & Williams, 1977). Animals were taken on PND 2 and PND 4 for other experiments in the project, and they were used to assess surface righting.

3.1. Methods

3.1.1. Subjects

3.1.1.1. Animals

The mating structure has already been described in the previous chapter (Figure 2.1). The 20 F0 pairs gave birth to 20 F1 litters, 10 offspring of drinking fathers (D₁) and 10 offspring of non-drinking fathers (ND₁). Litter sizes ranged from 3 to 17 for D₁ and 10 to 19 for ND₁. Larger litters were reduced to 10 animals on PND 2 to control for litter size. After weaning a male and a female from each litter was taken for each behavioural experiment. From the remaining animals, 1 male from each available litter was taken and kept to mate for the next generation (F2).

The F1 males kept for mating were paired with ethanol naïve females to produce the F2 generation. It is important to note that the F1 males used for mating were not exposed to ethanol themselves as it was to assess whether any effects of male drinking can carry over for multiple generations. Four F1 litters did not have any males for this part, which meant 16 F1 males (9 ND₁ and 7 D₁) were used and 14 pairs produced litters, 7 litters who had paternal grandfathers with a history of drinking (D₂) and 7 controls with no alcohol drinking history (ND₂). The litter sizes ranged from 4 to 17 for D₂ and 11 to 17 for ND₂. Again, larger litters were reduced to 10 on PND 2 and 2 animals from each litter were taken on PND 4. Like with the F1s, animals were selected to take part in different behavioural tests after weaning.

All subjects were bred at Victoria University of Wellington, New Zealand, vivarium.

3.1.1.2. Housing

Developmental milestone assessments were made pre-weaning and thus pups were in home cages with their mothers. The housing room was maintained on a 12-hour reverse light/dark cycle with temperature and humidity controlled (Chapter 2).

3.1.2. Procedures

3.1.2.1. Development milestones

The milestones assessed were based on criteria previously described in the literature (Barkur & Bairy, 2015; Geisler et al., 1993; Smirnov & Sitnikova, 2019). Milestones were measured in the dam home cages and were carried out observationally as to minimise any disturbance and stress on the litters. Litters were checked from PND 1 to weaning at PND 21 to check for the attainment of eye opening, pinnae detachment, coat growth, walking, rearing and self-grooming. The cages were viewed twice a day, once in the morning between 9 – 10 a.m. and another in the afternoon from 3 – 4 p.m. Litters were observed for approximately 5 minutes each and movement of cages from the housing rack to viewing table provided stimulation for if the pups were asleep/feeding at the time. A milestone was said to be achieved for the litter when more than two animals in the litter were found to meet the required criteria for the milestone. The milestones assessed here are all typically achieved between PND 8 and PND 20 which means they would be reached after the surface righting measurement. One male and female rat from each litter were taken on PND 2 and PND 4 to assess surface righting which meant the maximum litter size at the time of developmental milestone measures was 8. Those observing were blinded to the parental conditions of each of the litters.

3.1.2.1.1. Eyelid opening

Eyelid opening could be scored based on it being achieved or not on any given day. A "0" when both eyelids were closed and a "1" for when both eyes had been completely opened. The PND at which the eyelids began to open and when they were completely opened were taken for analysis.

3.1.2.1.2. Coat Growth

Rat pups are born without fur on their body and over the course of early development grow into their coats. Assessments were made as to when a full coat was developed by marking the days as “0” for no coat development, “.5” for the time which fur appeared to grow and finally “1” for when clear distinct coats were seen by virtue of no skin being visible to the naked eye.

3.1.2.1.3. Pinnae detachment

This is the outer ear separation in rats and the litters were marked as “0” for when there was no visible separation, “0.5” for when the ears began to separate and “1” for when the ears were completely separated.

3.1.2.1.4. Walking

Walking is determined by consistent and firm movements of all four paws when the belly did not touch the ground. It was not considered walking if the belly was in contact with the ground. When rats of a litter were seen to be walking, the date was marked as “1”

3.1.2.1.5. Self-grooming

Here we only looked at matured self-grooming. This is described as sequential grooming movements which began at the nose, continuing to the face and head, the body, the tail, and the genitals (Kalueff et al., 2016). The parameter was successful and marked as “1” when pups were able to groom.

3.1.2.1.6. Unsupported rearing/rearing

This is the ability to keep an upright position on the hind legs without any wall support. This was marked as "1" when a pup was able to stand steadily without support for more than 1 s.

3.1.2.2. Surface righting reflex

Here, one male and one female pup at PND 4 from each litter were taken for surface righting, after which they were culled for sample collection for other studies. In addition, the pups taken from litters with over 12 pups at PND 2 were also checked for surface righting before being culled.

Each pup was placed in a supine position (in abnormal position, back resting on the ground surface) and was left to change itself into a prone position (all the 4 feet in contact with surface). Each pup was given three trials two minutes apart. Each pup got a maximum of 1 minute per trial to be able to right itself. The shortest time to right was taken for analysis with that score acting as a representation for the remaining pups of that litter (Figure 3.1).

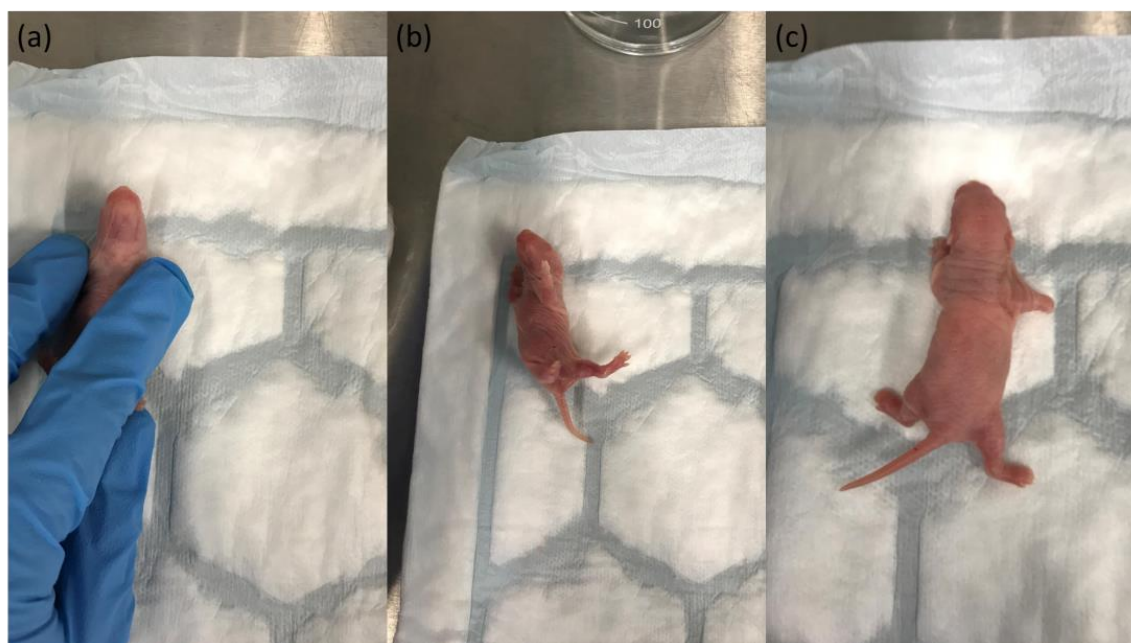


Figure 3.1. Surfacing righting procedure. (a) Placing the pup in the supine position, timer starts as soon as experimenter takes their hand off the pup. (b) Rat pup tries to right itself. (c) Timer stops as soon as pup rights itself in the prone position.

3.1.3. Data analysis

Statistical analysis was carried out using jamovi (Version 2.3.12) and GraphPad Prism (version 8.0.2 for Windows). Independent t-tests and ANOVAs (Analysis of Variance) were used to compare the developmental milestones and surface righting reflex across the conditions. Tukey's Honest Significant Difference (HSD) test was used as post hoc analysis where needed to assess the significance of differences between pairs of group means.

3.2. Results

3.2.1. Developmental milestones

As developmental milestones were observed in home cages, comparisons were just made between drinker and non-drinker litters in the F1 and F2. Litter size at PND 0, coat growth, eye opening, pinnae detachment, walking, self-grooming and unsupported rearing were compared using independent t-tests.

3.2.1.1. F1

All 10 drinkers and non-drinkers produced litters; the mean litter sizes were larger in the ND₁ ($M = 13.90 \pm 2.73$) compared to the D₁ ($M = 12.40 \pm 4.17$), but there was a greater variance in litter sizes in the D₁. One D₁ litter did not survive till the measurements of developmental milestones. D₁ litters also had 8 deaths between PND 0 and PND 4 unrelated to the culling while ND₁ litters only had 2 such occurrences. This difference was not statistically significant, but this could be because the 8 deaths came from two litters.

The only significant difference seen between the D₁ and the ND₁ litters was in pinnae detachment ($t(17) = 2.58, p = .019$) where D₁s showed slower development of complete pinnae detachment. The other measures were not significantly different between the groups, although the mean days of development for eye opening, walking and rearing were (slightly) higher for D₁ litters while those for assisted rearing and self-grooming were marginally higher in the ND₁ (Figure 3.2).

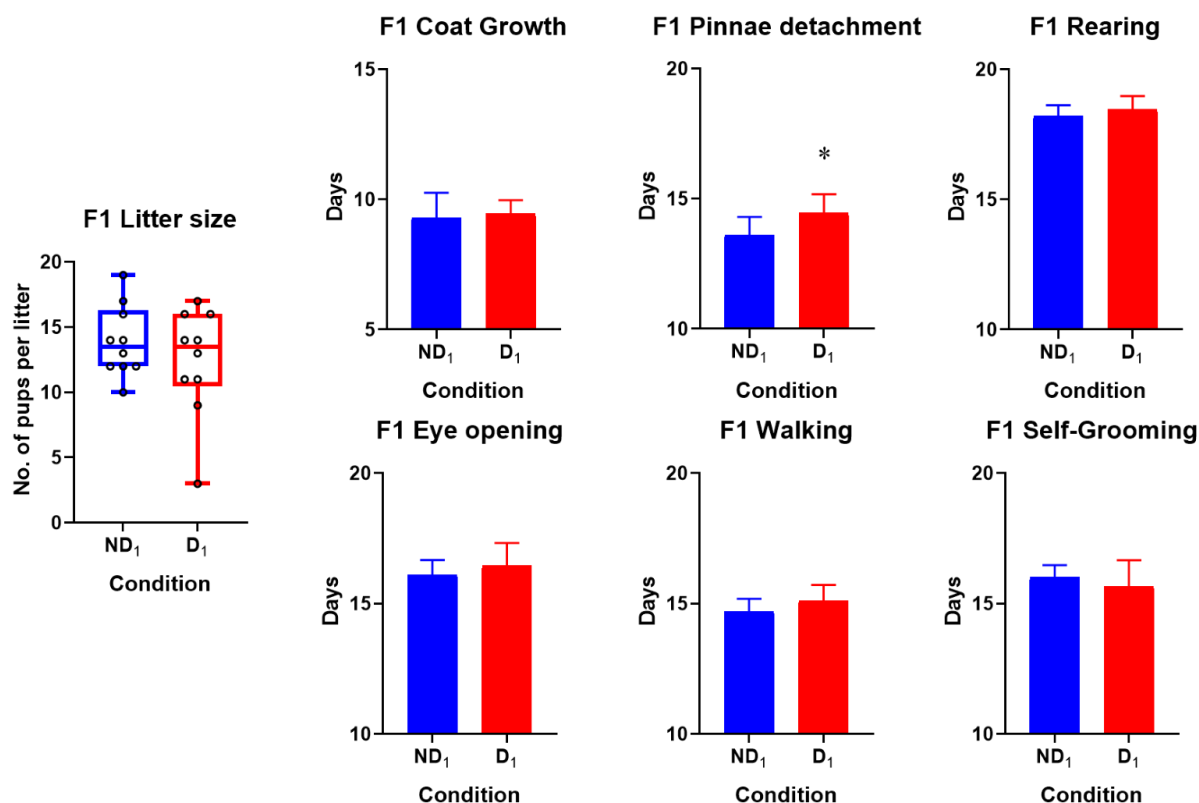


Figure 3.2. Litter sizes at birth and developmental milestones in F1 litters. The day at which milestone achieved on the Y-axes. Pinnae detachment showed a significant delay in offspring of drinkers compared to non-drinking sires ($t(17) = 2.58, p = .019$). Litter sizes showed high variance in D₁ and eye opening, walking and rearing were also slightly delayed in the D₁. Bars denote standard deviations, Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation. * Denotes significant difference.

3.2.1.2. F2

There were 7 litters for ND₂ and D₂ each, however, 2 D₂ litters did not survive long enough for measuring any developmental milestones. When looking at litters at PND 0, ND₂ - just like ND₁ - had a larger mean litter size and a smaller variance in litter numbers ($M = 13.00 \pm 1.92$) compared to D₂ litters ($M = 11.60 \pm 5.03$). Just like in the F1s, the D₂ litters had 5 deaths between PND 0 and PND 4 while only 1 was recorded in the ND₂ litters. Unlike the F1 comparison, this was statistically significant ($t(10) = 2.96, p = .014$).

When looking at developmental milestones, coat growth ($t(10) = 2.74, p = .021$) and eye opening ($t(10) = 2.76, p = .020$) were significantly different with them being delayed in D₂ litters. Ear separation, walking and rearing again had slightly higher means in the drinker litters, although these were not significant (Figure 3.3).

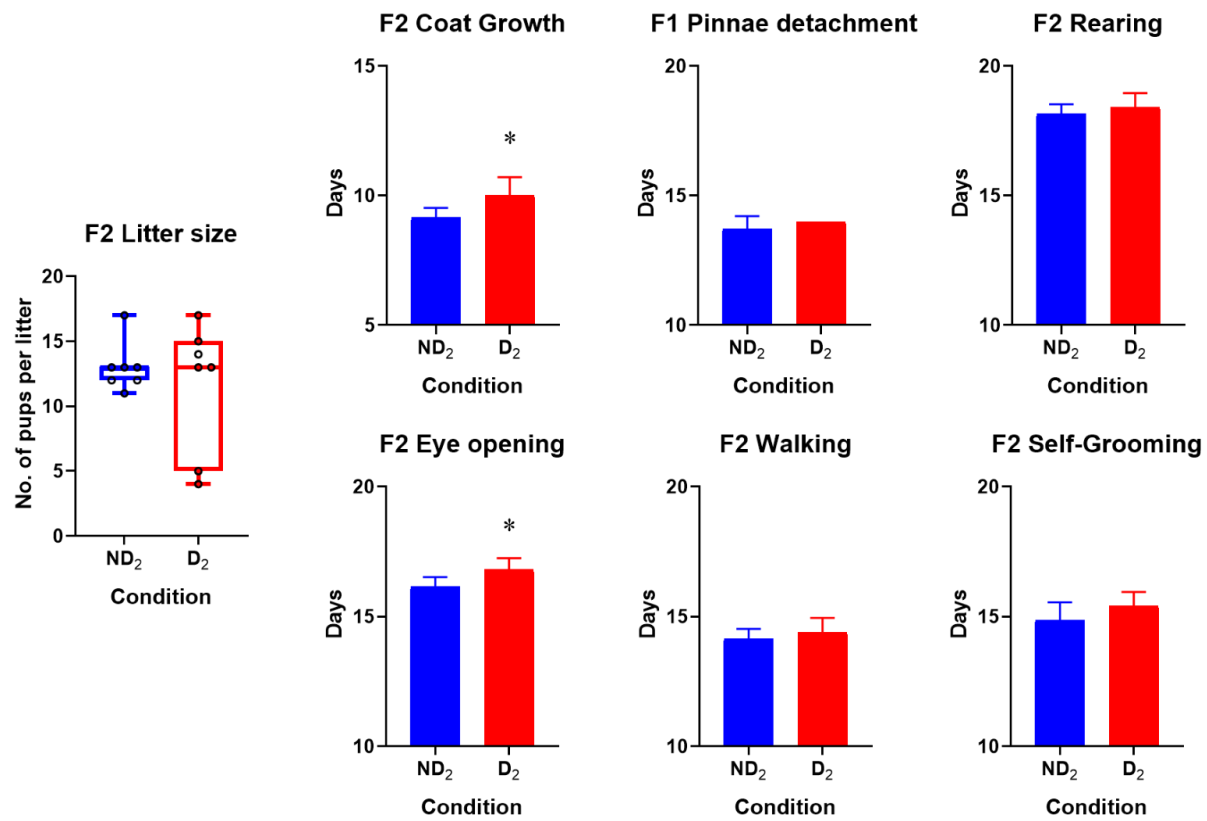


Figure 3.3. Litter sizes at birth and developmental milestones in F2 litters. The day at which milestone achieved on the Y-axes. Coat growth ($t(10) = 2.74, p = .021$) and eye opening ($t(10) = 2.76, p = .020$) showed a significant delay in offspring of drinkers compared to non-drinking sires. Again, the variance in litter size was high in the D₂ and all other milestones showed slight delays. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation. Bars denote standard deviations, * Denotes significant difference.

3.2.1.3. Crossgenerational comparisons

The F1s and F2s along with sire condition were compared to check for any differences between generations and to assess whether any differences seen above persisted using 2x2 ANOVAs for each of the milestones. There was no significant difference of litter size, coat growth and rearing. Eye opening ($F(1, 27) = 4.54, p = .042$) and pinnae detachment ($F(1, 27) = 6.19, p = .019$) showed a significant effect of sire condition and showed no differences between the two generations. Lastly, walking, and self-grooming had significant generation effects where F2 showed walking ($F(1, 27) = 11.13, p = .002$) and self-grooming ($F(1, 27) = 6.89, p = .014$) sooner than the F1 rats. Lastly, none of the conditions showed any significant sire condition x generation interaction.

3.2.2. Surface righting reflex

3.2.2.1. F1

Surface righting in F1s was measured in 37 PND 2 and 38 PND 4 pups. Pups taken out during surface righting gave the only opportunity to weigh rats before weaning. Differences for weight and surface righting ability were compared in ANOVAs by age (PND 2/PND 4) x sex (male/female) x sire drinking condition (D/ND).

Looking at weights, age ($F(1, 67) = 190.93, p < .001$) and sex ($F(1, 67) = 8.75, p = .004$) showed significant differences with PND 4 pups weighing more than PND 2 pups and

males weighing more than females (Table 3.1). There was no effect of sire condition on weights and no significant interactions. While looking at surface righting times, age again showed a significant effect ($F(1, 69) = 9.70, p = .003$) with latencies reducing from PND 2 to PND 4. Lastly, there was a significant three-way sire x sex x age interaction ($F(1, 69) = 4.42, p = .039$) where Tukey's post hoc analysis showed surface righting was slower in PND 2 ND₁ males compared to PND 4 ND₁M ($p = .018$), PND 4 D₁F ($p = .020$), PND 2 D₁M ($p = .043$) and PND 4 D₁M ($p = .018$) but the difference was not observed with the other conditions (Figure 3.4).

Sire Condition	Sex	Age	N	Weight (g)	Min time (sec)
Non-Drinker	Female	PND2	8	8.10 (1.29)	10.21 (5.14)
		PND4	10	11.83 (1.17)	6.16 (6.80)
	Male	PND2	12	8.15 (1.26)	18.04 (20.72)
		PND4	10	12.83 (1.43)	2.84 (1.71)
Drinker	Female	PND2	8	8.29 (1.03)	11.95 (13.49)
		PND4	9	11.41 (0.85)	3.44 (2.23)
	Male	PND2	9	9.14 (1.11)	4.16 (1.85)
		PND4	9	12.74 (1.11)	3.62 (2.00)

M (SD)

Table 3.1. Means and standard deviations for Surface righting times and weights of the F1.
Means and standard deviations for the weights and minimum surface righting times in each F1 group separated by sex, condition and age.

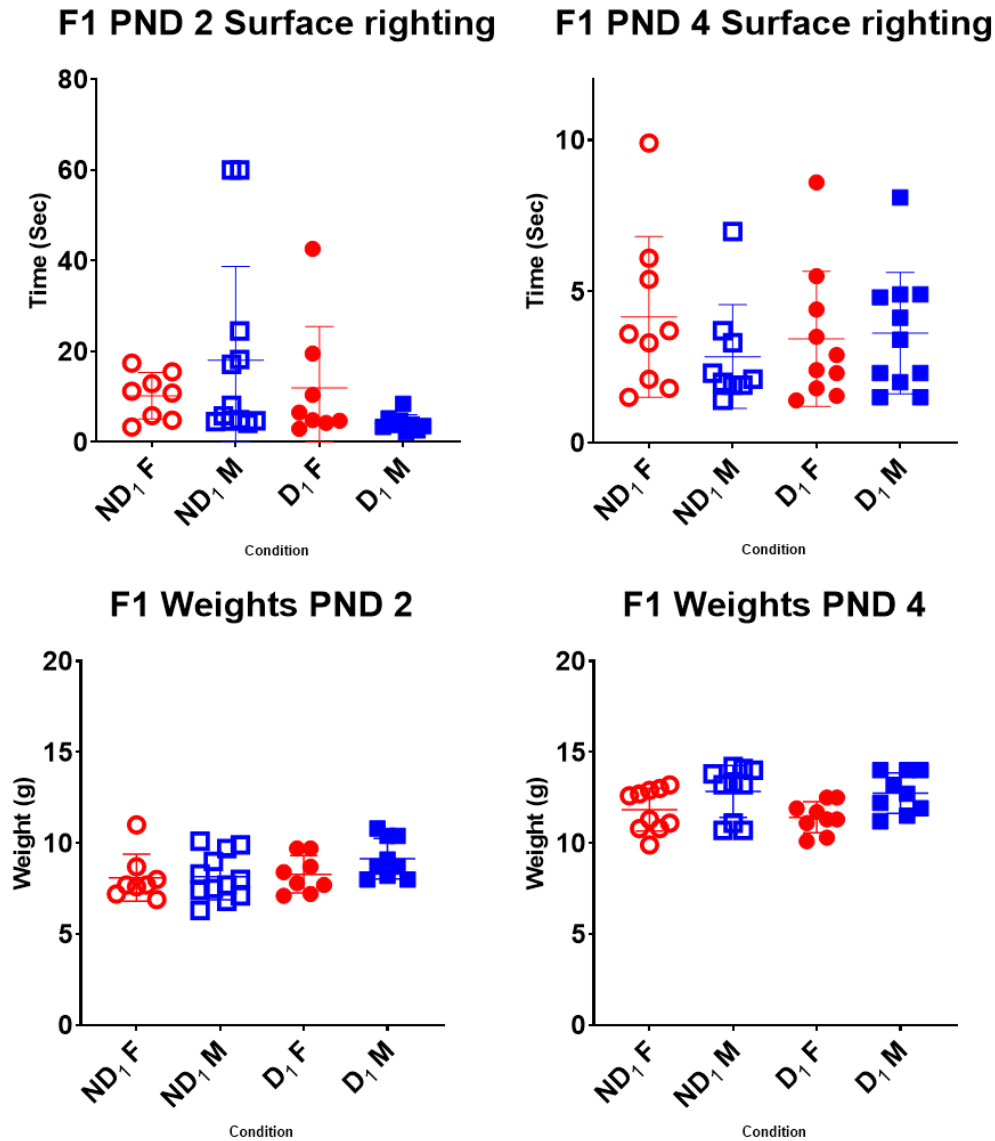


Figure 3.4. Comparison of surface righting times and weights in the F1. Age ($F(1, 67) = 190.93, p < .001$) and sex ($F(1, 67) = 8.75, p = .004$) showed significant differences with PND 4 pups weighing more than PND 2 pups and males weighing more than females. While looking at surface righting times, there was a significant three-way sire condition * sex * age interaction ($F(1, 69) = 4.42, p = .039$) where surface righting was slower in PND 2 ND₁ males compared to PND 4 ND₁M ($p = .018$), PND 4 D₁F ($p = .020$), PND 2 D₁M ($p = .043$) and PND 4 D₁M ($p = .018$) but the difference was not observed with others. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation. Bars denote standard deviations.

3.2.2.2. F2

Thirty-six PND 2 pups and 28 PND 4 F2 pups went through surface righting. The same 2x2x2 ANOVAs were used as in the F1s. Again, when looking at weights, age ($F(1, 54) = 113.68, p < .001$) and sex ($F(1, 54) = 6.00, p = .018$) showed significant differences with PND 4 pups weighing more than PND 2 pups and males weighing more than females (Table 3.2). There was no effect of parental condition on weights. Surface righting times were again significantly different when looking at age ($F(1, 54) = 9.87, p = .003$) with times reducing from PND 2 to PND 4. There were no other significant differences seen. The sire x sex x age interaction seen in F1s did not reappear here.

Sire Condition	Sex	Age	N	Weight (g)	Min time (sec)
Non-Drinker	Female	PND2	7	9.21 (1.18)	15.04 (18.38)
		PND4	7	12.59 (1.31)	4.28 (2.16)
	Male	PND2	9	9.64 (0.86)	8.04 (4.98)
		PND4	7	14.00 (0.86)	2.30 (0.76)
Drinker	Female	PND2	7	8.90 (0.74)	9.31 (8.46)
		PND4	7	13.70 (2.77)	5.20 (2.13)
	Male	PND2	11	9.77 (1.70)	8.97 (7.73)
		PND4	7	15.09 (2.45)	3.55 (2.48)

M(SD)

Table 3.2. Means and standard deviations for Surface righting times and weights of the F2. Means and standard deviations for the weights and minimum surface righting times in each F2 group separated by sex, condition and age.

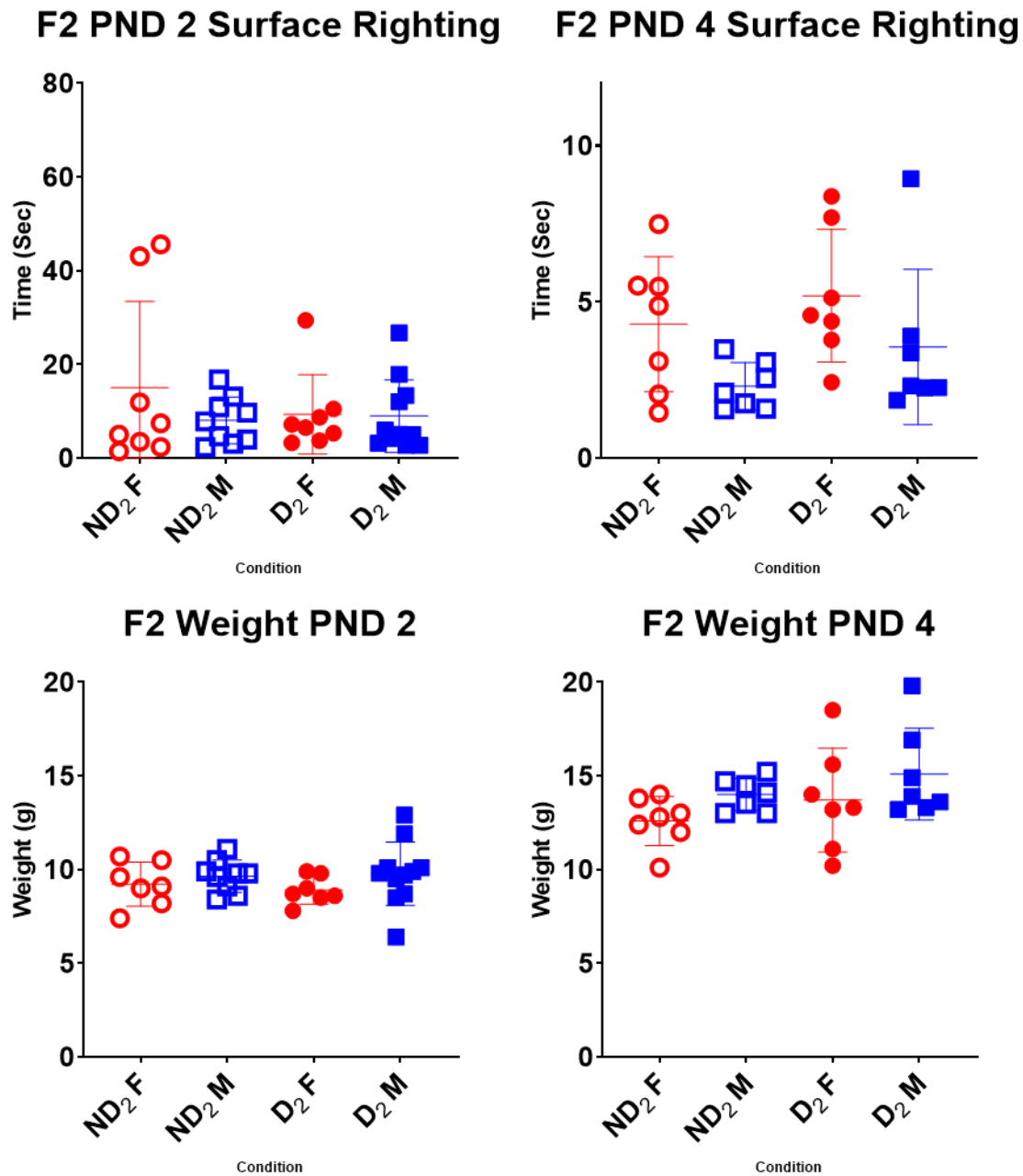


Figure 3.5. Comparison of surface righting times and weights in the F2. When looking at weights, age ($F(1, 54) = 113.68, p < .001$) and sex ($F(1, 54) = 6.00, p = .018$) showed significant differences with PND 4 pups weighing more than PND 2 pups and males weighing more than females. Surface righting improved with age ($F(1, 54) = 9.87, p = .003$) but there were no other significant effects. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation. Bars denote standard deviations.

3.2.2.3. Crossgenerational comparisons

Comparing the two generations, the same significant age ($F(1, 121) = 289.63, p < .001$) and sex ($F(1, 121) = 28.00, p < .001$) effects on weight and age effects ($F(1, 125) = 18.81, p < .001$) on surface righting persist when looking at both generations. There was a significant effect of generation on weight with the F2 litters being heavier than the F1s ($F(1, 121) = 56.23, p < .001$). However, when the rats from the two smallest litters, which were both D₂ litters and had the heaviest F2 animals were taken out, the difference disappeared (comparing Figures 3.4 and 3.5).

3.3. Discussion

The experiment's aim was to assess the differences in development between rats with a paternal drinking history and offspring of non-drinkers. Looking at the F1 rats, we saw delays in pinnae detachment in the D₁ while in the F2s we saw delays in the eye opening and coat growth in the D₂ compared to the ND₂. This suggests early development is influenced by paternal drinking history and even show transgenerational effects. There were no differences in weights between drinking conditions, however at PND 2 the ND₁ males performed worse than the other pups on the surface righting reflex. There were no differences in surface righting at PND 4.

3.3.1. Developmental milestones and litter size

The slower achievement of milestones in D₁ extends previous work showing parental stressors affect early development. The only significant delay in the F1 rats was in pinnae development but there was non-significant but directionally relevant delayed development of eye opening, walking and rearing which do support previous work on developmental delays (Meek et

al., 2007). However, while Meek and colleagues treated male rats with ethanol immediately prior to mating, our study indicates that developmental delays can still be seen in conditions where drinking has stopped up to 8 weeks (one spermatogenesis cycle) before mating. The differences in eye opening were significant in the F2s and along with the other developmental delay trends support the idea of changes due to PPAC being passed down through generations by the male germline (Sarkar, 2016; Stockard & Papanicolaou, 1916b). Taken together, the inter- and transgenerational developmental delays seen here are similar to those seen in rat models of schizophrenia (Ellenbroek et al., 2005) and autism spectrum disorder (ASD) (Foley et al., 2014). This should therefore fit well with research suggesting offspring of alcohol exposed sires are more likely to develop behavioural deficits in adulthood (Abel & Lee, 1988; Beeler et al., 2019; Ledig et al., 1998; Rompala et al., 2017).

Our study falls in line with other reports which do not see a change in litter size due to PAPC (Abel, 1995; Leichter, 1986; Rathod et al., 2020). However, it is important to mention that we did find a wider range of litter sizes in both the D1 and D2 which does suggest differences in fertility influenced by alcohol. While we did not measure litter weights for every animal in each litter to identify total ‘runts’ there were a larger number of deaths in the first three weeks in both the D1 and D2 litters as well as more instances of small litter sizes which is in line with earlier reports (Abel, 1989; Rathod et al., 2020; Stockard & Papanicolaou, 1916b). These findings might also be explained by studies which report changes in fertility and foetal reabsorption rates of females bred with alcohol treated males (Abel, 1995; Klassen & Persaud, 1976).

Finally, ethanol exposure is shown to reduce testosterone levels and increase testicular oxidative injury leading to fewer successful pregnancies, smaller litter sizes and sex ratio

changes, along with other developmental delays later (Cicero et al., 1990; Emanuele et al., 2001). Here we see similar differences in the F2 offspring to what we saw in the F1, and these could be associated with studies which show alcohol-sired rats also have reduced testosterone levels compared to controls (Abel & Lee, 1988; Abel, 1989). If reduced testosterone levels in ethanol exposed rats is one of the reasons for the variance, we see in the F1 rats, then they may also play a role in the differences we see in the F2 litters.

3.3.2. Surface righting

Consistent with previous work there were no differences in weights between the treatment conditions at PND 2 or at PND 4 (Abel & Tan, 1988; Abel, 1995; Leichter, 1986; Wozniak et al., 1991). We did not measure birthweights, which is where studies have previously reported differences (Bielawski et al., 2002; Chang et al., 2019; Chang et al., 2017; Ledig et al., 1998). However, weights in adolescence and adulthood are discussed in chapter 4 (behavioural experiments). The difference in weights between the F1 and F2 can be attributed to the two small D2 litters, the removal of these two litters from weight measurements also removed the differences seen between the two generations.

The surface righting reflex showed no difference on PND 4 in both the F1 and F2. Interestingly, in the F1s there was a sexually dimorphic difference with only the male ND₁ pups at PND 2 showing slower surface righting reflex attainment. This might fit with Jamerson and colleagues who found that pups born to sires who had stopped alcohol consumption prior to mating showed earlier attainment of surface righting than controls but also showed poorer balance (Jamerson et al., 2004). Sexually dimorphic behaviours in offspring of PatEE are common (Ledig et al., 1998; Rompala et al., 2017; Wozniak et al., 1991) but these are generally

associated with behaviours in adolescence and adulthood. This and the limitations in our method of assessing the righting reflex impedes any major interpretations we can make towards the faster righting in D1 males compared to controls. As this effect was not seen in the F2s, any differences in male surface righting seen here was not transgenerational unlike the other developmental milestones.

3.3.3. Limitations

The trade-off for minimizing disturbance for the developing litters was that we could not use many parameters while looking at early development. We already mentioned not measuring birthweights but other informative tests like incisor eruption, negative geotaxis and sound reflex were also left out. The developmental milestones were looked at as observations while in home cages which meant we did not look at individual pups and thus masks any intra-litter variability and sex differences in attaining these markers in turn limiting statistical power. The use of surface righting on separate animals for PND 2 and PND 4 also means the lack of ability to assess the development of a single animal over multiple days (generally PND 2 to PND 6) which would be hiding information which would have been parsed through repeated measures testing.

3.3.4. Conclusions

The chapter assessed early developmental differences due to PPAC over multiple generations. The variance in litter sizes and difference in survival rates along with delay in attainment of developmental milestones like pinnae detachment and eye opening indicates developmental changes in ethanol-sired offspring. These effects were seen to carry over into the next generation suggesting the effects of PPAC are passed through the male germline. Considering these initial changes, the next chapter looks at possible behavioural differences seen

in adolescence and adulthood in the F1 and F2 generations of male rats exposed to alcohol and provides insight into how these early developmental delays might manifest later in life.

Chapter 4: Intergenerational and Transgenerational impacts of PPAC on Behaviour

While early developmental milestones give us important clues with respect to animal models, most research looking at brain disorders focuses on behavioural and biological differences seen in adulthood. The last chapter demonstrated the potential effect of PPAC on the timing of developmental milestones on both the first (F1) and the next (F2) generation. This chapter aims to take the next step and investigate potential behavioural differences later in development and into early adulthood between the offspring of ND₀ (non-drinking) rats and D₀ (Drinking) rats. To do this, we consider locomotor activity, ethanol induced motor coordination and alcohol consumption behaviours in the F1 and F2 litters.

4.0.1. Locomotor activity and motor coordination

Locomotor activity is often used to analyse drugs, genes, and disease models. Measuring activity in an open chamber allows for a simple assessment of general activity levels and measures things like distance moved, vertical counts (rearing) and times spent in pre-defined areas. Similarly, the rotarod performance test is based on a rotating rod with forced motor activity being applied, by a rodent. The test measures parameters such as riding time (seconds) or endurance. The main functions of this test are to evaluate balance, grip strength and motor coordination of the subjects; especially in testing the effect of experimental drugs (Rustay et al., 2003) or after traumatic brain injury (Mouzon et al., 2012). The length of time that a given animal stays on this rotating rod is a measure of their balance, coordination, and physical condition. The advantage of these tests is that they create a discretely measurable, continuous

variable (length of time) that can be used to quantify the effects of different drugs, conditions, and procedures. For example, alcohol markedly impairs rodent performance in the rotarod test (Uzbay & Wallis, 1999), acute ethanol injections show dose dependent decreases in locomotor activity (Liljequist & Karcz-Kubicha, 1993; Overstreet et al., 2004; Randall, C. L. et al., 1975) and some aspects of locomotor activity are even increased under ethanol exposure (Correa et al., 2003; Frye & Breese, 1981).

Chapter 1 described the mixed reports on the effects of paternal alcohol exposure on locomotor activity in F1 offspring. Some have noted an increase in open-field locomotor activity at PND 21, PND 42, PND 90 and 1 year in ethanol sired rats while others suggest the effects depend on the amount of ethanol the sires have been exposed to (Abel, 1989; Abel, 1993; Ledig et al., 1998). On the other hand, several other studies found no differences between the paternal drinking conditions and locomotor activity (Beeler et al., 2019; Nieto et al., 2022; Rompala et al., 2017). Just like locomotor activity, the accelerating rotarod research has also been inconsistent, possibly due to methodological differences. Poorer baseline performances of PPAC offspring on motor coordination have been observed (Conner et al., 2020), while others found no difference during training between PatEE and non-drinker offspring (Finegersh & Homanics, 2014; Rompala et al., 2017). However, studies, including our lab, have found sex dependent effects when the offspring were treated with alcohol, with paternal ethanol exposure attenuating the effects of alcohol (Finegersh & Homanics, 2014; Hussain et al., 2022; Nieto et al., 2022). No such reports have been done in the F2 of paternal ethanol exposed offspring.

4.0.2. Alcohol drinking and PPAC

While ethanol drinking paradigms are often used for models of AUD, the effects of PPAC on compulsive drinking behaviour are not common, and as with the previous paradigms, the results are mixed. While some studies showed that male offspring of ethanol-sired mice had reduced ethanol consumption compared to controls, an effect not seen in females (Finegersh & Homanics, 2014; Rompala et al., 2017) another study showed the opposite effect: reduced drinking in ethanol-sired females but not males (Rathod et al., 2020). Beeler et al., also found a reduction in ethanol-sired male rats, though this was only observed in shorter drinking sessions in the dark (Beeler et al., 2019). Lastly, two self-administration studies show reduction in the rewarding effects of ethanol in offspring of ethanol exposed male rats with sex differences in behavioural patterns during the tasks (Nieto & Kosten, 2023; Nieto et al., 2022) . These suggest there may be important sex differences in ethanol reward and sensitivity due to paternal ethanol drinking.

4.0.3. Current study

Given the mixed results of the effects of PatEE on locomotor activity, motor coordination and ethanol consumption in F1, and the lack of studies investigating the F2 generation, this chapter assesses the effects of PPAC on locomotor activity, ethanol induced alterations in motor coordination and ethanol consumption in both the F1 and F2 generation. Just like previous studies where early development has led to the manifestation of behavioural differences later in life, we expect to see behavioural differences between the ND₁ and D₁ rats. In addition, the lack of research assessing true transgenerational effects of PPAC in rodents makes the behavioural work on the F2s intriguing.

4.1. Methods

4.1.1. Subjects

As mentioned previously, a single male and female from each litter was taken to run separate behavioural experiments. Here, for the IA2BC, there were 28 F1 rats (14 D₁ and 14 ND₁) which were again split equally by sex (7 males and 7 females each). In the F2s, 23 F2 rats underwent the IA2BC, 10 D₂ (5 males and 5 females) and 13 ND₂ rats (8 males and 5 females). Animals were PND 30-35 at the start of drinking and were housed individually during the paradigm and ethanol was given to them in their home cages. The results of the 20 F0 (male) rats put through the IA2BC are also compared to the F1 and F2s.

Locomotor activity was measured in a separate cohort from PND 22-24, just after weaning. This same cohort was used to measure rotarod activity beginning from when they were aged between PND 30 and PND 33. Thirty-six F1 rats 9 in each of the 4 groups ND1 females, ND1 males, D₁ females and D₁ males and 24 F2 rats, 14 ND₂ (7 females, 7 males) and 10 D₂ (5 females, 5 males) were used in this set.

Housing room conditions and food were the same as described in Chapter 2.

4.1.2. Procedures

4.1.2.1. Locomotor Activity

The chamber was a square arena mounted with two planes of infrared detectors in a specially designed Plexiglas chamber (Med Associates, W x 43.2 x 43.2, H x 30cm) kept inside a sound attenuating box. Rats were introduced to the testing room 10-15 minutes before to get habituated to the new surroundings. An animal was then placed into the centre of each chamber and activity was recorded for 60 minutes and processed by a computer program (Activity

Monitor Version 5 program: Med Associates Inc., St Albans, VT USA). Horizontal locomotor activity and vertical movements (rearing) were detected by rats breaking the infrared beams across the arena. The testing room was dimly lit by a red light and a white-noise generator was used to reduce extraneous noise disturbance. On completion, rats were placed back in their home cages and returned to the home room. The apparatus was then cleaned and dried in preparation for the next rat. Testing was done on 3 consecutive days.

4.1.2.2. Accelerating Rotarod

The experiments were performed using an accelerating Rotarod (Figure 4.1; Rotarod LE 8500 (76-0239), L x 50cm, H x 36cm, W x 24cm, Panlan, Barcelona, Spain). The apparatus is a four-station rotating roll (60mm diameter) mounted 10cm above four corresponding drop levers that register the time and speed in rotations per minute (rpm) at which an animal falls from the roll (figure). Rats are placed on the horizontally oriented, rotating cylinder (rod) suspended low enough not to injure the animal, but high enough to induce avoidance of fall. The speed of the rotarod is mechanically driven, and may either be held constant, or accelerated (Jones & Roberts, 1968).

Animals were weighed before each habituation and testing day. Testing occurred in a separate room under dimmed red light. A pseudo-randomized order was assigned for time of day and alcohol dose to counterbalance and evenly distribute conditions across the animals. The test had 3 days of training/habituation followed by 3 more days of testing with 48h in between each test day.

Rats were habituated to the testing room for 15min after which they were placed in separate lanes on the rod rotating at 5 rpm such that animals may walk forward to keep balance.

After 60 seconds on the rod, animals were returned to their home cage and the apparatus was cleaned between trials. This was repeated for three trials separated by 10min intertrial intervals. The last training day involved increasing the speed of the rotarod by 1 rpm (again starting at 5 rpm). The aim of this training day was to build up to 12 rpm over a 3-minute period. The minimum inclusion criterium for the testing days was that the animal could remain on the rotarod for at least 15 sec at 12 rpm on one trial on the final training day.

Testing started forty-eight hours after the last training session. After habituation to the room, rats were injected with a .9% saline, 1 g/kg ethanol or 2 g/kg ethanol intraperitoneal injection and placed on the rotarod 10- and 30-minutes post injection to allow for the drug to take effect. The apparatus was set to accelerate from 4 to 40rpm in 300s, and animals from the same cage were placed in separate lanes initially rotating at 4 rpm. The trial began when acceleration started and ended when the animal fell off or 300 seconds had passed. Latency to fall was recorded over two trials at each of these time points and the best performance was taken. There were no exclusion criteria for any of the test conditions and animals were given two days before being administered a different dose to allow prior dose to leave the body.



Figure 4.1. Rotarod apparatus. The accelerating Rotarod treadmill (Rotarod LE 8500 (76-0239), L x 50cm, H x 36cm, W x 24cm, Panlab, Barcelona, Spain).

4.1.2.3. Intermittent alcohol two bottle choice (IA2BC) paradigm

The IA2BC paradigm used was the same one used on the F0 animals and has been described previously (Chapter 2).

4.1.3. Solutions

Ethanol 20% (v/v) was made through dilution of 99.8% ethanol solution (Purescience Ltd, Porirua New Zealand) with distilled water. The rotarod and locomotor activity boxes were cleaned with diluted Virkon antiseptic (1:100) (Antec International, Sudbury, UK)

4.1.4 Data Analysis

Statistical analysis was carried out using IBM SPSS Statistics 25 (IBM; USA) and GraphPad Prism (version 8.4.1 for Windows, GraphPad Software; USA). Like previously, a generalized linear mixed model analysis with an autoregressive covariance structure was used for ethanol consumption. This model was chosen to accommodate for days during which data of a subject was lost due to leaks in the drinking bottles (Wang, Z. & Goonewardene, 2004). Ethanol preferences were averaged for the 7-hour and 24-hour session and like the ethanol-induced motor-coordination across 3 injections were measured using mixed factorial ANOVAs (analysis of variance), as no data points were missing. There was no difference in performance between the 10-min post-injection and the 30-min post injection trials and so the best time for each rat was taken. Mixed factorial ANOVAs were also used for locomotor activity across PND 22, 23 and 24. Tukey's HSD was used for post hoc analysis.

4.2. Results

4.2.1. Locomotor activity

Measures for total distance covered, ambulatory counts and vertical counts were looked at in ANOVAs with sex, F0 drinking condition and the 3 days of testing (PND 22, PND 23 and PND 24) being the independent variables. For the accelerating rotarod, the rats were measured

on their latency to fall and their percentage performance on the ethanol dose compared to saline. Here, the independent variables were sex, father drinking condition and ethanol dose.

4.2.1.1. F1

Measurement of locomotor activity in the F1 rats showed a significant change in distance travelled ($F(2, 64) = 11.53, p < .001$), ambulatory counts ($F(2, 64) = 13.86, p < .001$) and rearing ($F(2, 64) = 23.96, p < .001$) across the testing days with distance travelled and ambulatory counts decreasing from PND 22 to PND 24. The opposite was seen in rearing counts, with a steady increase from PND 22 to PND 24. There was also a significant effect of sex when looking at rears ($F(1, 32) = 4.92, p = .034$) with females showing more counts of rearing than males. In terms of sire differences, the ND₁ group on average had more rears than the D₁ but this, while close, did not reach statistical significance ($F(1, 32) = 3.94, p = .056$), however, the effect is almost entirely due to the low number of rears seen by the D₁ males (Figure 4.2). Lastly, females on average also travelled more distance than males but this was not statistically significant ($F(1, 32) = 2.94, p = .096$).

4.2.1.2. F2

In the F2s, the distance travelled ($F(2, 40) = 3.90, p = .028$), and the ambulatory counts ($F(2, 40) = 3.77, p = .032$), significantly decreased from PND 22 to PND 24. Also, like the F1s, there was a significant increase in rearing ($F(2, 40) = 13.15, p < .001$), from PND 22 to PND 24. There were no significant effects of either sex or F0 drinking condition in the F2s for locomotor activity.

4.2.1.3. Crossgenerational comparisons

When comparing the two generations, the reduction of distance travelled ($F(2, 104) = 12.83, p < .001$) and increase in rearing ($F(2, 104) = 33.34, p < .001$) across the days remained significant. There was a significant difference in distance travelled between generations ($F(1, 52) = 14.13, p < .001$) with the F1 rats travelling more than the F2 (Figure 4.2). There was no difference in rearing between the generations.

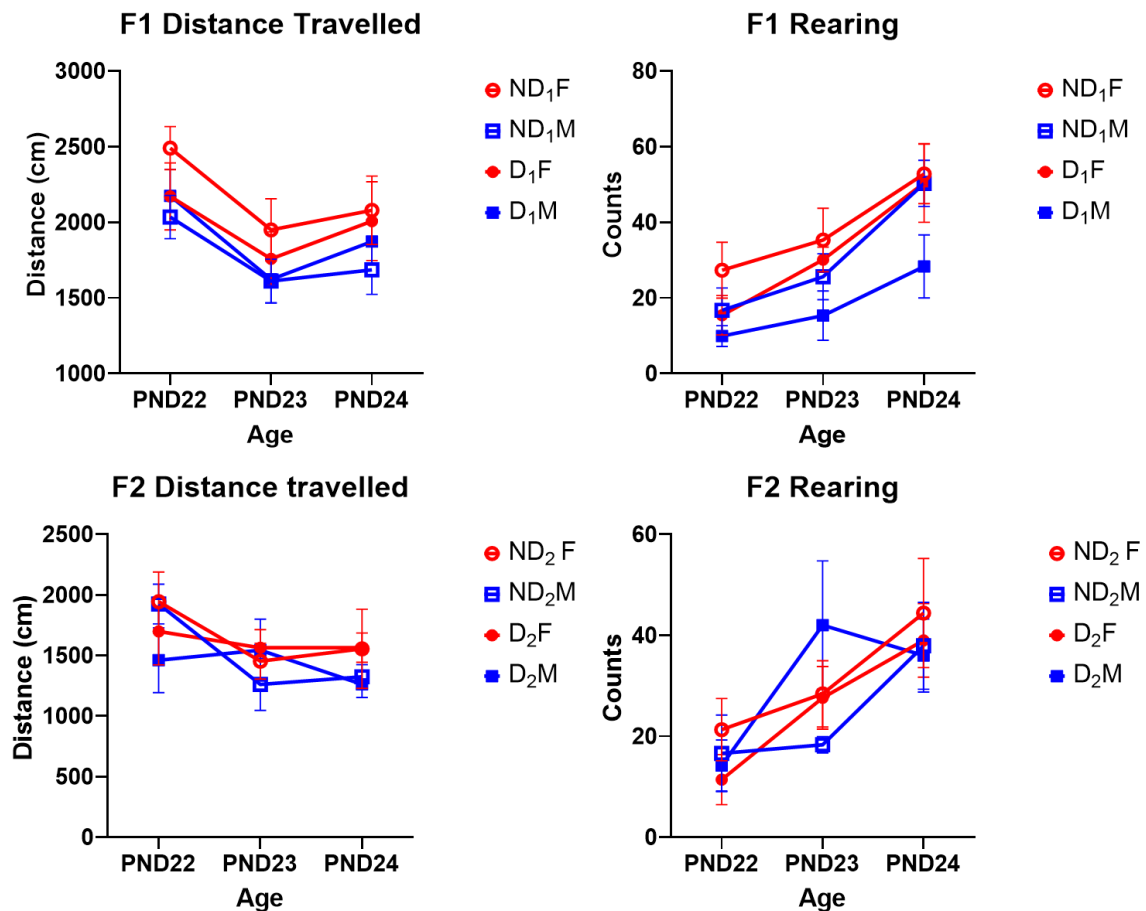


Figure 4.2. Distance travelled and rearing of the F1 and F2 rats in the locomotor activity chamber. Distance travelled reduced over the days while rearing behaviour increased. While the F1 travelled more than the F2 ($F(1, 52) = 14.13, p < .001$). F1 females travelled slightly more distance ($F(1, 32) = 2.94, p = .096$) and reared ($F(1, 32) = 4.92, p = .034$) more than the males on average. ND₁ showed more rears than D₁ ($F(1, 32) = 3.94, p = .056$) with this being due to

the D₁ males. F₂s showed no differences through either sex or parent. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation.

4.2.2. Accelerating Rotarod

For the accelerating rotarod, the rats were measured on their latency to fall and their percentage performance on the ethanol dose compared to saline. Here, the independent variables were sex, father drinking condition and ethanol dose.

4.2.2.1. F₁

When looking at the F₁'s latency to stay on the rotarod, there was a significant effect of dose ($F(2, 64) = 26.46, p < .001$) with a reduction in performance with increasing doses of ethanol. Tukey's post hoc analysis showed significant differences between saline and both 1 g/kg ($p = .008$) and 2 g/kg ($p < .001$) dose while there was also a significant difference between the 1 and 2 g/kg doses ($p < .001$). There was also a near significant effect of sex ($F(1, 32) = 4.12, p = .051$) with females performing better than males. There was a significant dose x sire drinking condition interaction ($F(2, 64) = 6.62, p = .002$) with the ND₁ animals performing better on the saline than the D₁ rats ($p = .007$), however this difference was no longer present at the 1g/kg ($p = .985$) and 2g/kg ($p = .989$) ethanol administration (Figure 4.3). Rotarod performance when injected with ethanol was also looked at as a percentage of the performance on saline. Here, there was again a significant effect of dose ($F(1, 32) = 27.83, p < .001$) with performance dropping when on the ethanol doses and being worst on 2 g/kg. There was no difference between males and females in terms of change in performance across doses but there was a significant main effect of father drinking condition ($F(1, 32) = 6.07, p = .019$) where the performance drop was blunted in the D₁ condition.

4.2.2.2. F2

Rotarod performance for F2s showed a significant effect of dose ($F(2, 40) = 53.70, p < .001$) just like the F1s. Again, Tukey's post hoc analysis showed significant differences between saline and both 1 g/kg ($p = .002$) and 2 g/kg ($p < .001$) dose while there was also a significant difference between the 1 and 2 g/kg doses ($p < .001$). The F2s also showed a significant effect of sex ($F(1, 20) = 9.30, p = .006$) with females performing better than males, a trend seen but not significant in the F1s. There was also a main effect of F0 drinking condition ($F(1, 20) = 12.27, p = .002$) where D₂ performed better than ND₂. There was also a significant condition x sex interaction ($F(1, 20) = 5.38, p = .031$), indicating that D₂ females show an attenuated response to the motor effects of ethanol compared to the D₂ males ($p = .011$), ND₂ males ($p = .003$) and ND₂ females ($p < .001$). Lastly there was a dose x sex interaction ($F(2, 40) = 4.48, p = .018$) where females performed better than males on 1g/kg ($p = .016$) but not on saline or the 2g/kg dose (Figure 4.3). This response again is moderated by the performance of the D₂ females. However, when looking at performance as a percentage of saline the only significant effect was that of dose ($F(1, 20) = 33.44, p < .001$). All the other effects seen when looking at just latency to fall were no longer significant.

4.2.2.3. Crossgenerational comparisons

Looking at latency to fall off the rotarod, the same dose effect was seen ($F(2, 104) = 80.62, p < .001$). There was a significant generation effect with the F2 performing better on the rotarod than the F1 ($F(1, 52) = 66.23, p < .001$). The females performed significantly better than males ($F(1, 52) = 15.82, p < .001$) and importantly the rats from F0 drinker origins stayed on the rotarod longer than the non-drinkers ($F(1, 52) = 7.33, p = .009$). Across the generations, there

was a sex x sire condition interaction ($F(1, 52) = 7.46, p = .009$) where females of the drinker sires significantly outperformed the males ($p < .001$) as well as of the males ($p < .001$) and females ($p = .002$) of the non-drinker sires. However, this difference is mainly carried by the performance of the D₂ females. Lastly, looking at percentage drop in performance from saline there was a significant difference between F1 and F2 ($F(1, 52) = 7.16, p = .010$) where the F1 rats perform better under the influence of ethanol than the F2s and offspring of drinkers performed better on ethanol than offspring of non-drinkers ($F(1, 52) = 4.21, p = .045$).

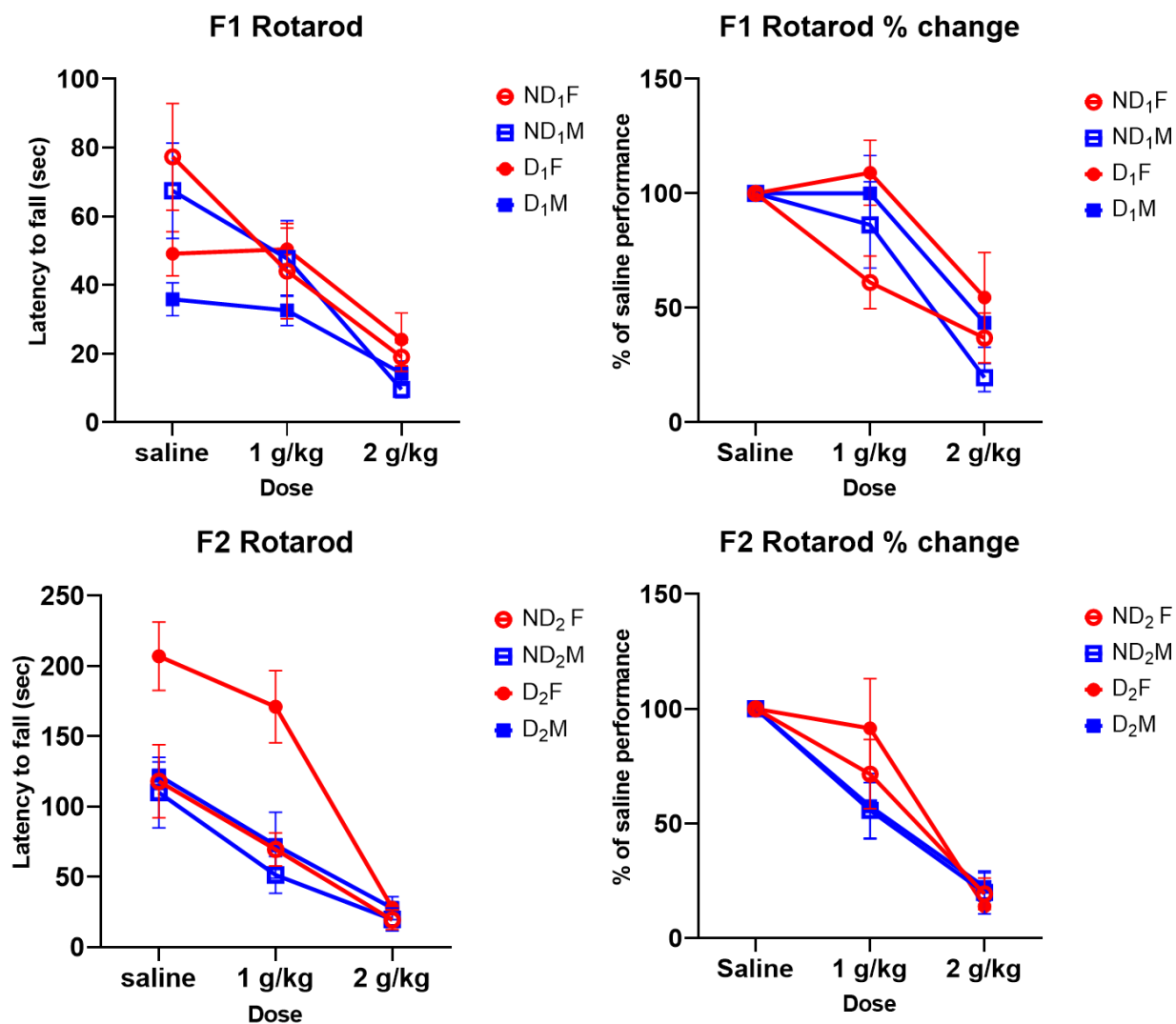


Figure 4.3. Average latency to fall (in seconds) and percentage drop in performance on the rotarod. A clear effect of dose is seen in both generations with performance dropping with increase in ethanol dose in F1 ($F(2, 64) = 26.46, p < .001$) and ($F(2, 40) = 53.70, p < .001$). Crossgenerational analysis showed females perform better than males in both the F1 and F2 rats ($F(1, 52) = 15.82, p < .001$). There was a sex x sire condition interaction ($F(1, 52) = 7.46, p = .009$) where D females significantly outperformed the D males ($p < .001$) as well as the ND males ($p < .001$) and ND females ($p = .002$). This is explained by the D₂ females performing significantly better than the other groups while the D₂ males performed similar to the non-drinkers indicating sexually dimorphic transgenerational effects ($F(1, 20) = 5.38, p = .031$). Looking at percentage drop in performance from saline. The F1 rats perform better under the influence of ethanol than the F2s ($F(1, 52) = 7.16, p = .010$) and offspring of drinkers performed better on ethanol than offspring of non-drinkers ($F(1, 52) = 4.21, p = .045$). Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation.

4.2.3. IA2BC

4.2.3.1. F1

First, a repeated measures ANOVA assessing total daily ethanol intake showed a significant ($F(23, 621) = 58.90, p < .001$) increase in total ethanol drinking across days (Figure 4.4). Next, a 2 (sex) x 2 (father drinking condition) x 24 (drinking days) linear mixed model analysis looking at g/kg ethanol consumption showed a significant effect of days ($F(23, 119.69) = 8.76, p < .001$), post hoc analysis indicates that drinking to bodyweight remained mostly stable with a few significant day differences but there was an increase from session 1 ($M = 3.02 \pm 1.28$ g/kg) to session 11 ($M = 4.12 \pm 1.46$ g/kg) and again a slight increase was seen in the 24 sessions from day 13 ($M = 4.96 \pm 2.29$ g/kg) to 24 ($M = 5.95 \pm 2.68$ g/kg; Figure 4.5). In addition, there were main effects of sex ($F(1, 81.452) = 60.95, p < .001$) with females consuming higher amounts of ethanol. There was a near significant main effect of sire condition ($F(1, 81.45) = 3.69, p = .058$) but no significant day x sire condition interaction ($F(23, 119.69) = 1.30, p = .182$). Importantly, there was a significant sire x sex interaction ($F(1, 81.45) = 10.96,$

$p = .001$) with post hoc analysis showing that ND₁ females consumed significantly ($p < .05$) higher amounts of g/kg ethanol ($M = 5.91 \pm 1.29$ g/kg) when compared to D₁ females ($M = 4.79 \pm 0.95$ g/kg) and the ND₁ ($M = 3.57 \pm 0.67$ g/kg) and D₁ ($M = 3.87 \pm 0.83$ g/kg) males (Figure 4.6). The other three conditions were not significantly different from each other.

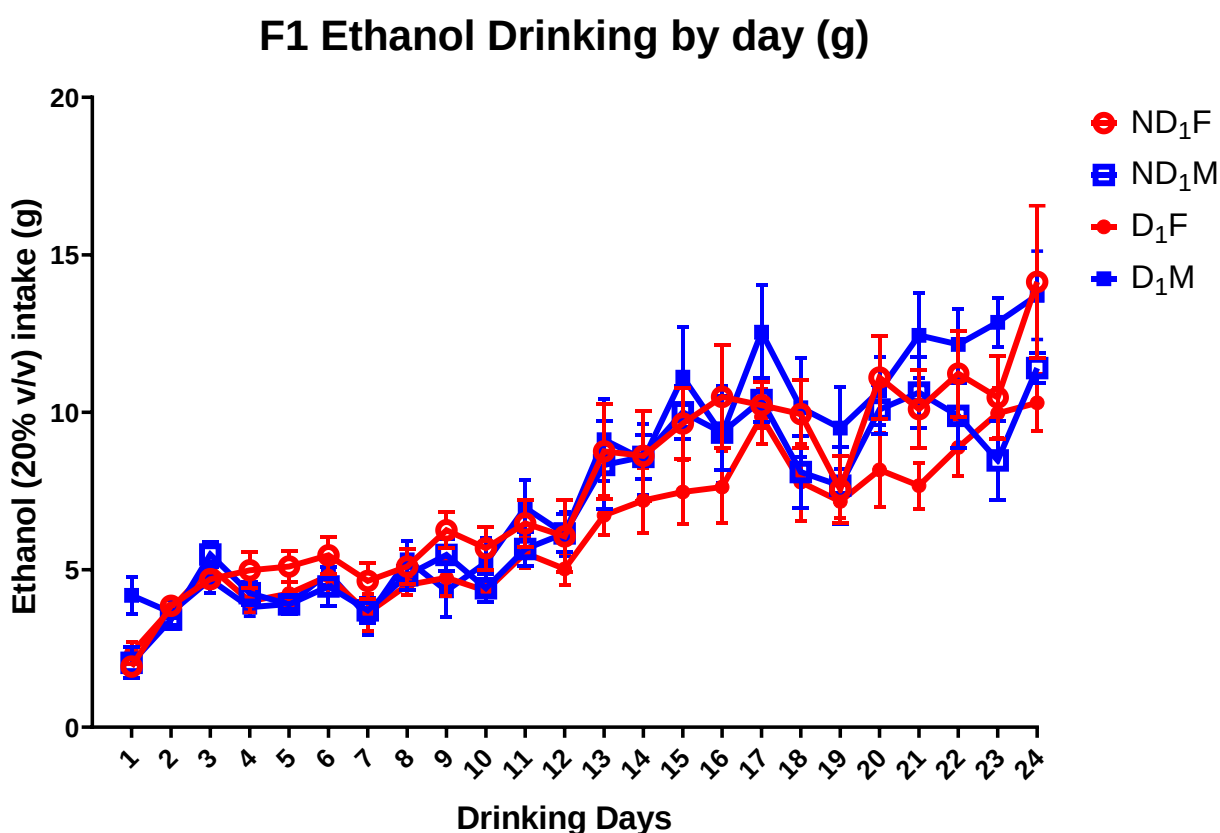


Figure 4.4. Daily ethanol intake of the F1 cohort. There was a clear increase of total ethanol intake across sessions ($F(23, 621) = 58.90, p < .001$). No differences between the cohorts were observed here. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation.

To assess ethanol preference, the preference in drinking days were averaged to create a single value for the preference for the 7-hour and for the 24-hour sessions. A 2 (sex) x 2

(drinking condition) x 2 (drinking sessions) mixed factorial ANOVA looking at ethanol preference showed a significant effect of drinking sessions with a reduction in preference from the 7-hour drinking sessions to the 24-hour drinking sessions. There was a significant main effect of sex ($F(1, 24) = 6.75, p = .016$) where F1 females ($M = 35.80 \pm 9.26\%$) showed a greater preference to ethanol than F1 males ($M = 29.04 \pm 5.85\%$). There was also a significant drinking condition x sex interaction ($F(1, 24) = 6.36, p = .019$) where female ND₁ showed the highest levels of ethanol preference ($M = 41.05 \pm 8.67\%$) and was significantly different from ND₁ males ($p = .007; M = 27.67 \pm 6.56\%$), D₁ males ($p = .039; M = 30.42 \pm 5.16\%$), and D₁ females ($p = .043; M = 30.61 \pm 6.84\%$; Figure 4.8).

F1 g/kg Drinking by day

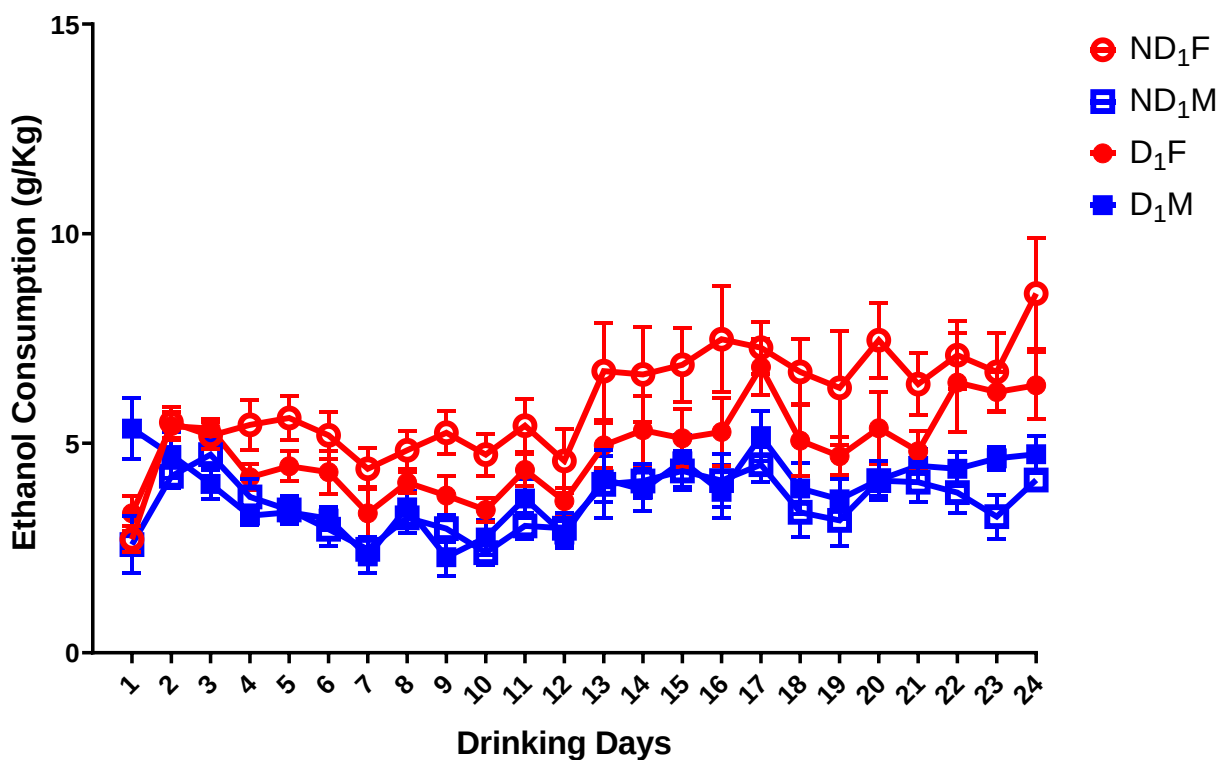


Figure 4.5. Daily g/kg ethanol intake of the F1 cohort. The first 12 sessions had a 7-hour access to ethanol while rats had 24-hour ethanol access for sessions 13-24. Females drank significantly more than males ($F(1, 81.452) = 60.95, p < .001$) and there was a sex x sire condition interaction where the ND₁ females also consumed significantly more than the D₁ females but there was no difference between the males ($F(1, 81.45) = 10.96, p = .001$). Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation.

4.2.3.2. F2

Like the F1 animals, the F2 cohort showed a significant increase ($F(23, 506) = 49.10, p < .001$) in total ethanol intake across the drinking days (Figure 4.6). A similar 2 (sex) x 2 (F0 drinking condition) x 24 (drinking days) linear mixed model analysis looking at g/kg ethanol drinking again showed a significant effect of days ($F(23, 109.30) = 7.86, p < .001$), again indicating a generally stable but slight increase in alcohol consumption per body weight over the duration of the experiment from session 1 ($M = 3.01 \pm 0.79$ g/kg) to 12 ($M = 4.20 \pm 2.01$ g/kg) and then again from session 15 ($M = 4.66 \pm 1.87$ g/kg) to 24 ($M = 5.66 \pm 2.37$ g/kg; Figure 4.7). In addition, there were main effects of sex ($F(1, 49.76) = 24.81, p < .001$) with females consuming higher amounts of ethanol. Unlike in the F1s, there was also a main effect of drinking condition ($F(1, 49.76) = 8.51, p = .005$) with ND₂ rats consuming more ethanol than D₂, this effect was moderated by a significant drinking condition x sex interaction ($F(1, 49.76) = 7.25, p = .010$) where post hoc analysis showed ND₂ females ($M = 6.07 \pm 1.52$ g/kg) consumed significantly higher amounts of g/kg ethanol ($p < .05$ for all) when compared to D₂ females ($M = 4.34 \pm 1.20$ g/kg), D₂ males ($M = 3.76 \pm 1.25$ g/kg) and ND₂ males ($M = 3.83 \pm 0.95$ g/kg) while there was no significant difference between the other 3 conditions. Again, there was no significant day x drinking condition interaction ($F(23, 109.30) = 1.42, p = .118$).

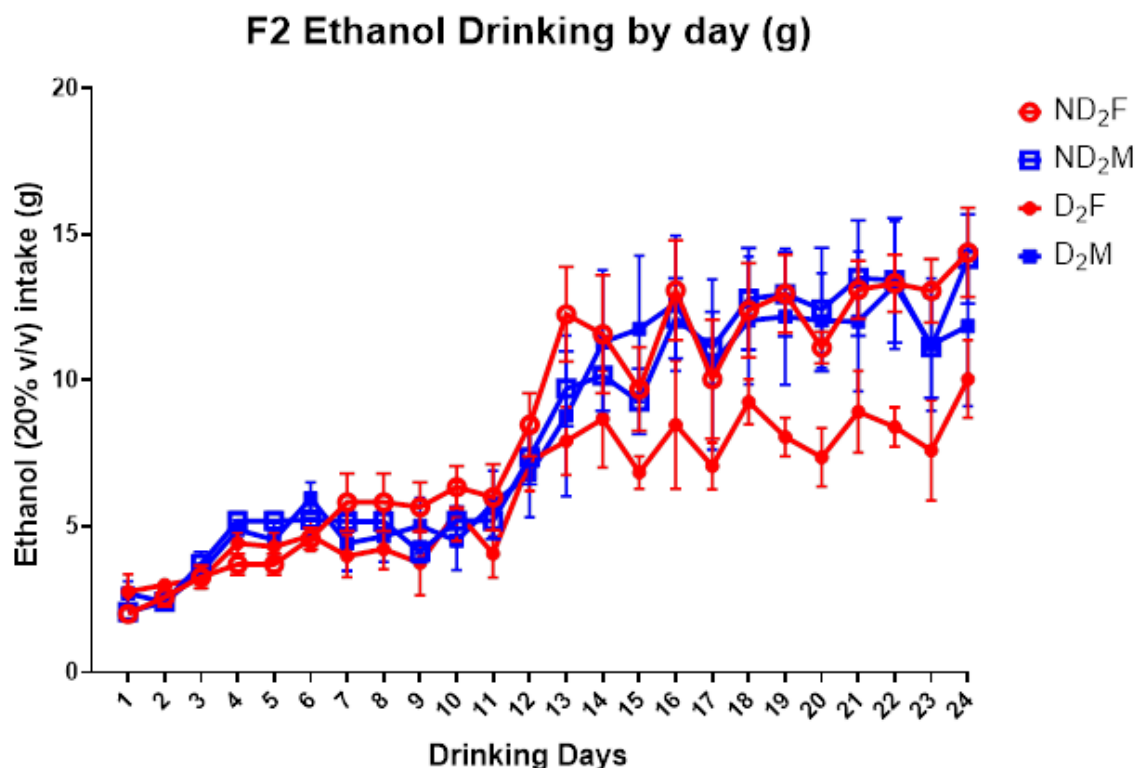


Figure 4.6. Daily ethanol intake of the F2 cohort. There was a clear increase of total ethanol intake across sessions ($F(23, 506) = 49.10, p < .001$). No significant differences between the cohorts were observed here. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation.

Ethanol Preference for the F2 was again looked at in a 2 (sex) x 2 (drinking condition) x 2 (drinking sessions) mixed factorial ANOVA. Unlike the F1s there was no difference in preference between the two sessions ($p = .677$). There was no significant main effect of sex ($p = .113$) or parent ($p = .064$). However, there was again a significant drinking condition x sex interaction ($F(1, 19) = 5.01, p = .037$) and female ND₂ showed the highest ethanol preference ($M = 44.4 \pm 8.27\%$) which was significantly higher than the other 3 conditions ($p < .05$) while there were no significant differences between ND₂ males ($M = 31.4 \pm 7.20\%$), D₂ males ($M = 32.6 \pm 12.1\%$), and D₂ females ($M = 28.8 \pm 5.89$).

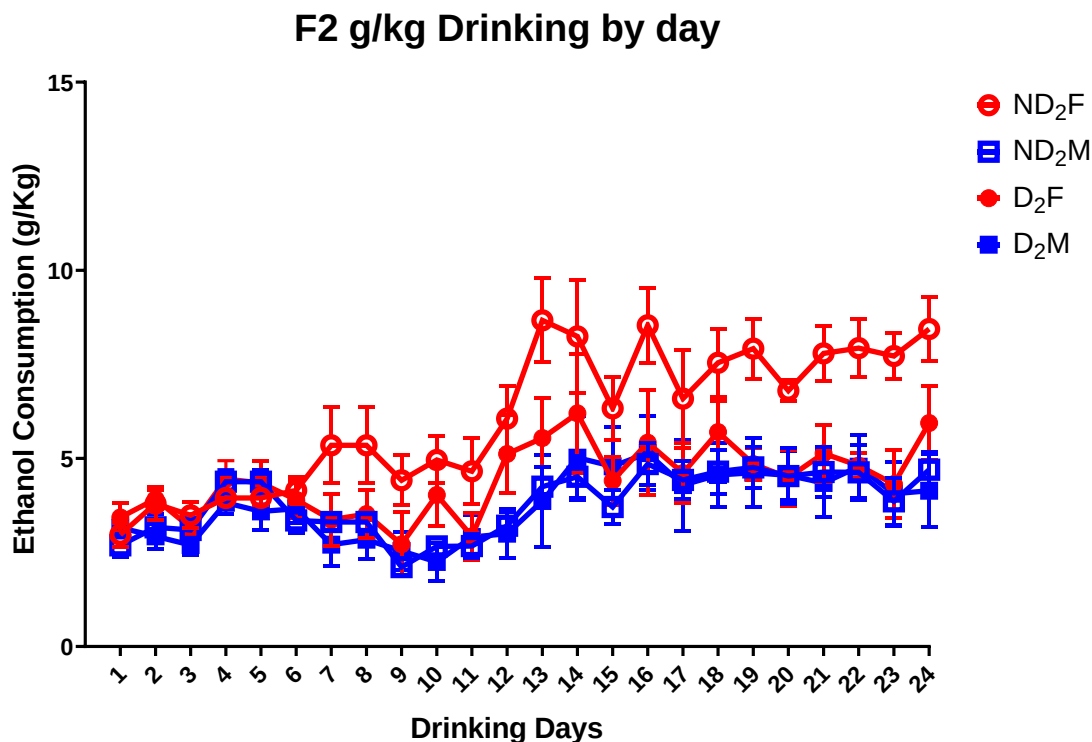


Figure 4.7. Daily g/kg ethanol intake of the F2 cohort. The first 12 sessions had a 7-hour access to ethanol while rats had 24-hour ethanol access for sessions 13-24. Females again, drank significantly more than males ($F(1, 49.76) = 24.81, p < .001$) with the ND2 females consuming significantly more than the other 3 groups ($F(1, 49.76) = 7.25, p = .010$).

4.2.3.3. Crossgenerational comparisons

The F0, F1 and F2 rats were compared on their average g/kg drinking and their ethanol preference. There was a significant difference of average g/kg ethanol drinking ($F(2, 70) = 3.56, p = .034$) with the F0 animals ($M_0 = 3.60$ g/kg, $SD_0 = 1.10$ g/kg) drinking lower on average than the F1 ($M_1 = 4.56$ g/kg, $SD_1 = 1.36$ g/kg) and F2s ($M_2 = 4.43$ g/kg, $SD_2 = 1.37$ g/kg). However, when we compare only the males from the F1 and F2 with the original F0 drinkers we find no significant difference in g/kg ethanol intake ($F(2, 44) = 0.18, p = .837$) which indicates the difference seen above was solely due to females and in particular the higher drinking from both the F1 and F2 ND females. Additionally, there was no significant difference in drinking

preference between the generations ($F(2, 70) = 1.39, p = .257$) with the F0 showing 29.5% preference while the F1 and F2 showed a preference of 32.44% and 33.92% respectively.

The F1 and F2 generations were compared for average g/kg and ethanol preference in a 2 (Sex) x 2 (F0 drinking condition) x 2 (generation) ANOVAs. There was a significant effect of drinking condition where those sired from the ND condition showed a greater g/kg consumption of ethanol ($F(1, 43) = 4.40, p = .042$) and a greater ethanol preference ($F(1, 43) = 6.41, p = .015$) than those sired from D₀ males (Figure 4.8). Again, there was a sex effect where females consumed more ethanol ($F(1, 43) = 27.02, p < .001$) and showed a greater preference to ethanol ($F(1, 43) = 6.86, p = .012$) than males. Importantly, these differences are explained by the significant sex x condition interaction where the ND₁ and ND₂ females consumed greater g/kg ethanol ($F(1, 43) = 6.16, p = .017$) and showed a greater preference to ethanol ($F(1, 43) = 11.99, p = .001$) than all the other groups. Post hoc analysis showed significantly higher g/kg ethanol intake ($p < .015$) and preference ($p < .005$) in the ND females when compared to all other groups while no other conditions were significantly different from each other. No other significant effects were seen.

Lastly, analysis of weights in animals during the rotarod and drinking showed no significant differences between paternal drinking condition, however, the F1 and F2 rats from the D₀ line were on average slightly heavier than the ND₀ line (Appendix A).

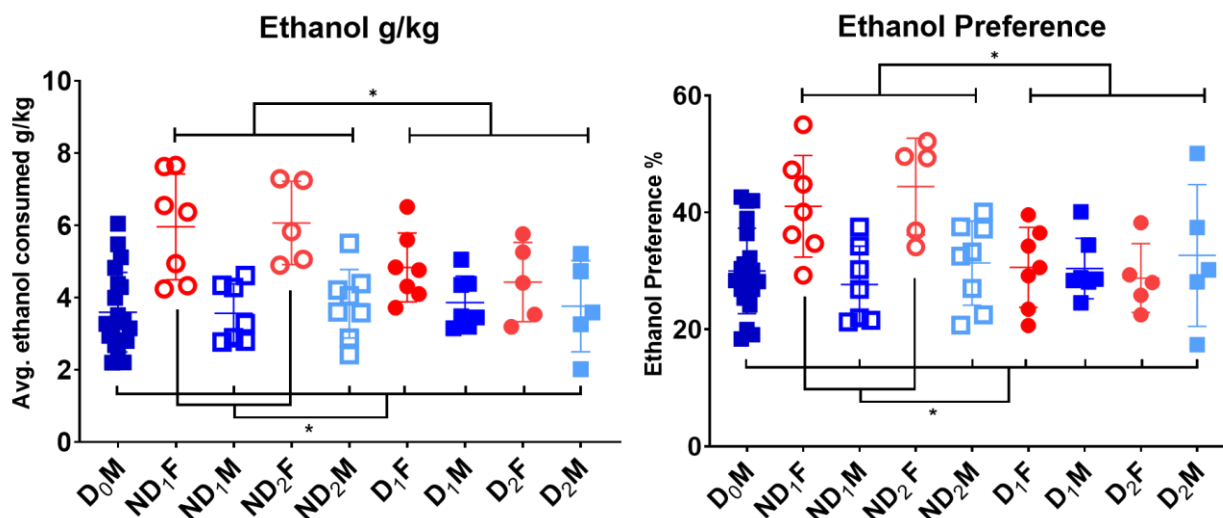


Figure 4.8. Average g/kg ethanol intake and ethanol preference of the F0, F1 and F2 cohorts.

Females on average consumed more ethanol g/kg than the males and showed a greater preference to ethanol. Those sired from the ND condition showed a greater g/kg consumption of ethanol ($F(1, 43) = 4.40, p = .042$) and a greater ethanol preference ($F(1, 43) = 6.41, p = .015$) than those sired from D₀ males. Post Hoc analysis using Tukey's HSD showed ND females had significantly higher ethanol intake ($p < .015$) and preference ($p < .005$) compared to all other groups. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation. * Denotes a significant difference.

4.3. Discussion

The behavioural tests in the F1s showed a significant effect of sire drinking history on ethanol induced motor coordination on the rotarod where D₁ showed smaller decreases in performances and thus reduced sensitivity to the effects of ethanol. There was also a sex x sire drinking interaction with female offspring of drinkers (D₁F) showing reduced drinking. Similar sex dependent effects were also seen in locomotor activity with male offspring of D₀ showing reduced rearing (Table 4.1). In the F2 generation, the same drinking behaviour differences were seen with female D₂ drinkers showing reduced ethanol intake. However, on the rotarod, the

paternal drinking history effects were only seen in the females. Lastly, there were no differences in locomotor activity in the F2s (Table 4.1). These indicate interesting, partly sex-dependent inter- and transgenerational effects of paternal drinking.

Experiment and Measure		Gen.	Effects			Direction of effect
			PPAC	Sex	PPAC*Sex	
Locomotor Activity	Distance Travelled	F1	ns	ns*	ns	F > M ($p = .096$)
		F2	ns	ns	ns	No difference
	Rearing	F1	ns*	Sig.	ns	ND ₁ > D ₁ ($p = .056$), F > M
		F2	ns	ns	ns	No Difference
Accelerating Rotarod	Latency to fall	F1	Sig.	ns*	Sig.	D ₁ > ND ₁ , F > M ($p = .051$)
		F2	Sig.	Sig.	Sig.	F > M, D ₂ F > rest
	% Drop in performance	F1	Sig.	ns	ns	D ₁ > ND ₁
		F2	ns	ns	ns	No Difference
Ethanol Consumption	Ethanol g/Kg consumption	F1	ns*	Sig.	Sig.	D ₁ F < ND ₁ F, F > M, ND > D ($p = .058$)
		F2	Sig.	Sig.	Sig.	F > M, D ₂ F < rest
	Ethanol preference	F1	ns	Sig.	Sig.	F > M, ND ₁ F > rest
		F2	ns*	ns	Sig.	ND ₂ > D ₂ ($p = .064$), ND ₂ F > others

Table 4.1. Summary of Behavioural experiment results. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation. Sig. is significant result, * indicates trending but not significant.

4.3.1. Locomotor activity and Rotarod

In the locomotor activity chamber, rats showed a decrease in distance travelled across the 3 sessions and an increase in rearing behaviour. There were sex differences in the F1s, with

females showing greater levels of activity than males. While the same effect was not seen in F2s, it is important to point out that females did again show greater activity on average. In the rotarod, females performed better than males with longer latencies to fall. Sex differences in rodent motor behaviour have been well documented with studies showing females having higher locomotor activity (Hyde & Jerussi, 1983; Simpson et al., 2012) and performing better than males on motor coordination tasks (Hernandez et al., 2020). These baseline differences must be considered when looking at performance under ethanol because while we see similar performances under the influence of ethanol between the ND₁ males and females, the females show a greater drop from baseline performance. Motor impairment due to ethanol has been reported to affect sexes differently (Hussain et al., 2022; Vetter-O'Hagen et al., 2009) This has been attributed to differences in alcohol metabolism with studies showing lower blood ethanol levels for the same ethanol dose (Mankes et al., 1991) and similar blood ethanol concentrations for greater ethanol intake (Pirino et al., 2022) in females compared to male rats.

While there was no difference in distance travelled, there was a trend of the D₁ rats to show reduced rearing activity when compared to the ND₁. The lack of statistical significance seen here fits well with research which has found no or subtle differences in locomotor activity in offspring of drinkers (Abel, 1993; Wozniak et al., 1991). Another study found differences in activity based on the amounts of ethanol given to the sire and based on rodent strain (Abel, 1989). While other studies report only reduced activity (Abel & Lee, 1988; Abel & Tan, 1988).

There was however a clear difference in rotarod performance based on paternal drinking history. The lower saline latency of the D₁ compared to the ND₁ animals suggests poorer baseline performance. This confirms previous work showing a reduction in baseline balance and

motor coordination in offspring of drinkers (Conner et al., 2020). However, the significantly lower drop in performance under the ethanol dose indicates a reduction in ethanol sensitivity. This is, again, in line with previous research showing a reduction in sensitivity to ethanol in offspring of ethanol sired animals (Finegersh & Homanics, 2014; Hussain et al., 2022; Nieto et al., 2022).

Interestingly, when looking at a transgenerational effect of ethanol, we observed a sex x parent drinking condition interaction with the D₂ females perform significantly better on the rotarod task when compared to all the other animals, while no differences were seen among males. This is the first study we are aware of which has looked at transgenerational effects of PPAC on behaviour and seeing a sex dependent effect a generation removed from paternal drinking gives us new insights into the long-term effects of alcohol.

4.3.2. Ethanol drinking

The F1 and F2 showed similar drinking patterns to the F0. Females also consumed more ethanol than males in both the F1 and F2. This is not surprising as several previous studies has shown that drinking behaviour shows sex-based differences with females consuming greater amounts of ethanol per bodyweight (Hussain et al., 2022; Lancaster et al., 1996), showing higher preferences for ethanol than males (Hussain et al., 2022; Lancaster & Spiegel, 1992; Pirino et al., 2022; Priddy et al., 2017), developing drinking behaviour faster (Melón et al., 2013; Moore, C. F. & Lynch, 2015) and showing different drinking patterns across a session (Juárez & De Tomasi, 1999).

The major difference seen here is the sex dependent interaction in offspring drinking behaviour where only the D₁ and D₂ females showed reduced drinking compared to the ND

offspring. The results seen here are unlike previous studies which found a reduction in male drinking (Finegersh & Homanics, 2014; Rompala et al., 2017) and those which found no differences (Beeler et al., 2019). It is important to mention that these experiments were all done on mice. Whether this indicates a species difference (Ellenbroek & Youn, 2016) or a difference in methodology is currently unclear. However, a reduction in ethanol drinking in female offspring of PPAC has been reported previously by our group (Hussain, et al., 2022) and another recent study (Rathod et al., 2020). In addition, we checked for any influence on the amount of F0 drinking on this effect seen and while there was no significant correlation, data distribution suggests higher drinking in F0s might associate with a greater reduction of drinking in the F1 females and a similar effect in both the F2 sexes, but these would require larger sample sizes to assess (Appendix B). This could mean a dose dependent response to PPAC where greater paternal drinking causes larger behavioural effects in offspring. Such trends have recently been reported in humans (Luan et al., 2022) and are not dissimilar to studies looking at PPAC at high and low doses of ethanol consumption (Abel, 1989). However, we only took the highest drinking F0s for mating which means the variability in ethanol consumption would be compromised for looking at a dose-response relationship.

4.3.3. Crossgenerational differences

The transgenerational effect of lower drinking and reduced ethanol sensitivity in the D₂ females (but not males) is intriguing and largely in line with the transgenerational effects noted in the previous chapter. In addition, we found some interesting other differences between the generations, such as a reduction in distance travelled in the F2 compared to the F1 rats, as well as a superior performance of the F2 on the rotarod. These differences are hard to explain. However,

it is important to keep in mind that the F0 males were isolated for extended periods of time during drinking and while awaiting mating. Given that rats are social animals, it is conceivable that 4 months of isolation (from pre-puberty onwards) induced substantial behavioural changes in the animals. Indeed, research suggests behavioural and sexual changes in animals having gone through extended periods of isolation (Lesscher et al., 2015; Lukkes et al., 2009; Spevak et al., 1973). Furthermore, epigenetic changes due to social isolation have also been identified (Arzate-Mejía et al., 2020; Siuda et al., 2014) which indicate potential fundamental differences between the F0 and the F1 fathers. This is because the F1 fathers were not exposed to the IA2BC drinking paradigm, they were housed in pairs until mating and any epigenetic changes due to the stress of isolation would not have occurred in the F1 fathers.

4.3.4. Mechanisms of PPAC

The results of the present chapter lead to a number of questions, the first being how these transgenerational differences due to paternal drinking are passed on. As none of the foetuses were in direct contact with alcohol, the most logical mechanism for these changes in the offspring is epigenetic alterations in the sperm. Epigenetics refers to the process of altering gene expression without alteration the genetic code. As discussed in the introduction, this can occur through a variety of methods, including DNA methylation and histone modifications. Sperm DNA (Deoxyribonucleic acid) is highly compact, at about 6 times more tightly packed than the average mitotic chromosomes of somatic cells (Kucuk, 2018). This compactness is accomplished by the replacement of histones with protamines, and due to this process, sperm was considered to simply deliver pure paternal DNA. However, recent evidence suggests some sperm DNA remains in its nucleosomal (histone-bound) format, particularly in genes involved in early

embryonic development, which would allow for changes due to DNA methylation and histone modifications to be seen in these cells (Le Blévec et al., 2020). This route is supported by research showing heavy paternal alcohol intake leads to reduced DNA methylation of the regulatory gene *H19* in humans (Ouko et al., 2009) and *Peg 3* and *H19* in mouse sperm (Liang et al., 2014). Another route for carrying epigenetic mechanisms is through the multitude of RNAs (ribonucleic acid) which are also carried by the sperm including non-coding RNAs (ncRNAs) which also carry epigenetic information (Le Blévec et al., 2020; Santiago et al., 2022). Again, there is some evidence showing chronic ethanol exposure alters several small non-coding RNAs (sncRNA) in the sperm (Bedi et al., 2019; Rompala et al., 2018). Lastly, the evidence of seminal fluids playing a mediating role in epigenetic changes and influencing development cannot be discounted (Patlar, 2022)

These studies, while low in number, reinforce the impact of PPAC on offspring development and suggest changes in sperm which might impact offspring development. The other connecting question which remains is what are the driving molecular factors in the F0 fathers due to alcohol drinking which result in these sperm changes? And do the molecular changes seen in the paternal alcohol drinkers directly manifest in the offspring?

4.3.5. Conclusions

The chapter looked at the behavioural changes due to PPAC and whether there were any transgenerational effects. We found clear sex dependent differences in ethanol consumption behaviours with female offspring of drinkers showing reduced ethanol intake. Additionally, we saw a reduction in sensitivity to the motor impairing effects of ethanol in the offspring of drinking rats with this effect being maintained in the F2 females, again showing sexual

dimorphisms. This implicates clear changes in the male germline due to ethanol consumption. The next steps would be to identify the changes occurring due to ethanol consumption, assessing whether the changes persist in the offspring and lastly looking at the mechanisms by which they are epigenetically transmitted to the subsequent generations. This leads to the following chapters which discuss the biological changes seen in the F0 due to drinking, whether any of these changes manifest in the offspring and preliminary results from our lab on the mechanisms involved in the epigenetic communication of PPAC.

Chapter 5: Indications of Molecular and Epigenetic Changes

Previous chapters in this thesis focused on the transgenerational impacts of PPAC on development and behaviour. This chapter aims to investigate the molecular changes induced by PPAC that may underlie the observed behavioural differences, with a specific focus on the liver.

5.0.1. Ethanol, Liver and Mitochondria

Damage to the liver caused by alcohol consumption has been extensively described in both human and animal research. As the liver is the main organ for metabolising ethanol, it is susceptible to both direct and indirect damage (Rocco et al., 2014). Ethanol is metabolised by a series of oxidative reactions in which first alcohol dehydrogenase (ADH) oxidizes ethanol into acetaldehyde. Next, acetaldehyde is metabolised to acetate by aldehyde dehydrogenase before finally the acetate is broken down to water and carbon dioxide. Acetaldehyde is highly reactive and significantly contributes to liver damage by binding with lipids, proteins, and Deoxyribonucleic acid (DNA) to form adducts which impede cellular functions and gene expression (Pohl et al., 2021; Setshedi et al., 2010). In addition, the process generates reactive oxygen species (ROS) which in excess cause damage to multiple cellular organelles and processes, which can ultimately disrupt normal physiology (Auten & Davis, 2009).

Mitochondria are cytoplasmic organelles which are integral to cellular metabolism and are thus involved in multiple cellular processes which include aging, apoptosis, and oxidative phosphorylation (Friedman & Nunnari, 2014). They have their own genome (mtDNA) and multiple copies of mtDNA are present in each cell which form a highly dynamic network (Rafelski, 2013). Defects in mitochondria have been associated with complex disease

presentations like severity and susceptibility to cancers, metabolic disorders, and neurodegenerative disorders (Nunnari & Suomalainen, 2012). Mitochondrial DNA copy number (mtDNA-CN), a measure of mitochondrial DNA (mtDNA) levels per cell, while not a direct measure of mitochondrial function, is correlated with mitochondrial mass, mitochondrial transcription, and respiratory activity (D'Erchia et al., 2015).

5.0.2. Mitochondrial DNA copy number

It is essential to distinguish between mtDNA-CN and other measures of mitochondrial health and function. MtDNA-CN is a measure of the number of mtDNA copies per cell while mitochondrial abundance is measured as the total amount of the organelle in the tissue (Cayci et al., 2012). A study measured mitochondrial abundance with the expression of mitochondrial outer membrane protein TOMM20 in relation to total protein content and found that this did not correlate with mtDNA-CN when looking at the frontal cortex and cerebellums in rats suggesting they are two fundamentally different measures (Thorne, 2023). Equivalent results are reported when looking at blood mtDNA-CN (Picard, 2021). One reason for this is that mtDNA copies vary per mitochondrion based on cell type and pathology (Aryaman et al., 2019) which means an increase in mtDNA-CN may represent an increase in mitochondrial abundance or an increase in DNA copies per mitochondrion. Another reason mtDNA-CN does not perfectly map onto mitochondrial function is due to the compensatory mechanisms of mitochondrial biogenesis which are initiated during mitochondrial dysfunction (Michel et al., 2012). Contrarily, some studies do mention tissue based linear correlations between mtDNA-CN and mitochondrial mass (Castellani et al., 2020; D'Erchia et al., 2015).

Mitochondrial activity is shown to have better correlations with mtDNA-CN, with reduced mtDNA-CN levels being associated with reduced expression of vital complex proteins, altered cellular morphology, and lower respiratory enzyme activity in cell models (Jeng et al., 2008). Thorne (2023) found associations between mtDNA-CN and mitochondrial mRNA expression of electron transport chain (ETC) genes; however, they point out that these were tissue dependent with the association between mtDNA-CN and mt-Nd1 expression in the frontal cortex being stronger than in the cerebellum (Thorne, 2023). Lastly, human tissue samples also show that mtDNA-CN plays a role in determining cytochrome c-oxidase activity (Van den Bogert et al., 1993). Other studies describe the age, sex, and tissue specific differences in mtDNA-CN (Mengel-From et al., 2014; Thorne et al., 2022a; Wai et al., 2010). While the correlations between mitochondrial function and mtDNA-CN are dependent on multiple factors, since we are comparing the same tissue across drinking conditions, it make mtDNA-CN a good proxy for measuring differences in mitochondrial function.

5.0.3. MtDNA-CN in psychiatric disorders

MtDNA-CN appears to be altered across physiological and psychiatric disorders. Reduced mtDNA-CN is associated with chronic kidney and cardiovascular diseases (Ashar et al., 2017; Tin et al., 2016). Additionally, postmortem brain tissue analysis has shown reduced mtDNA-CN is associated with greater cognitive impairment and is linked to Alzheimer's disease (Andrews & Goate, 2020). When looking at psychiatric disorders mtDNA-CN and mitochondrial dysfunction have been associated with autism spectrum disorders (ASD), major depressive disorder (MDD), attention deficit hyperactivity disorder (ADHD) and bipolar disorder (BD) although the directions of the changes in mtDNA-CN are less clear. Studies suggest that these

changes are due to a stress response in oxidative function (Klinedinst & Regenold, 2015; Yoo et al., 2017). Reports on ASD show an elevated mtDNA-CN in children diagnosed with ASD when compared to controls and their unaffected siblings (Chen et al., 2015; Yoo et al., 2017). A study looking at MDD found that mtDNA-CN was negatively correlated with severity of depressive episodes and showed impairments in response to oxidative stress (Czarny et al., 2020). Another study reported raised mtDNA-CN in peripheral blood samples of patients with MDD but decreased copy numbers in patients diagnosed with BD (Tsujii et al., 2019). Lastly, peripheral blood levels of children found higher mtDNA-CN numbers in those diagnosed with ADHD (Ogutlu et al., 2020). While most of these works looked at humans and have used blood samples, rodent research on opiate addiction found reduced mtDNA-CN in the hippocampus of rats and mice, but this effect was mediated by autophagy (Feng et al., 2013). The involvement of mtDNA-CN in a multitude of psychological disorders, its specific nature, and its link to alterations in stress response make it a good target for tissue specific investigation into alcohol use.

5.0.4. Ethanol and Mitochondrial dysfunction

Ethanol metabolism impairs mitochondrial functionality through increased oxidation of mitochondrial proteins and impairs functions like glutathione transport essential for maintaining redox balance (Farfán Labonne et al., 2009; Lluís et al., 2003). The oxidative stress itself sensitizes cells to further oxidative damage (Farfán Labonne et al., 2009). Ethanol consumption is associated with a decrease in mitochondrial protein synthesis and reduction in the rate of ATP synthesis (Cunningham & Bailey, 2001) as well as degradation of mtDNA (Mansouri et al., 1999) and a reduction in mitochondrial biogenesis (Cahill et al., 2002; Nassir & Ibdah, 2014).

This is significant as research shows dysfunction in mitochondrial activity is associated with poorer liver regeneration and even liver failure (Bessone et al., 2019; Zhang, I. W. et al., 2021) supporting the hepatoprotective role for mitochondria in the liver.

The tissue specific nature of mtDNA-CN changes after ethanol exposure is consistent with reports that mtDNA-CN is shown to increase in the heart but reduce in the liver (Marin-Garcia et al., 1995). Studies on alcohol consumption have demonstrated time dependent reductions and later increases in mtDNA levels based on tissue (Mansouri et al., 2001). Additionally, other studies on cardiomyopathy have seen increased mtDNA damage without altering copy number in ethanol exposed rat hearts (Laurent et al., 2014).

Impaired mitochondrial function due to ethanol consumption might also be altered via changes in the levels of proteins involved in mitochondrial biogenesis. Peroxisome proliferator-activated receptor γ coactivator 1 α (*PGC1 α /Pgc1 α*) is considered the master regulator of mitochondrial biogenesis (Ventura-Clapier et al., 2008). Chronic ethanol intake in rodents has been shown to reduce *Pgc1 α* mRNA expression and protein levels (Matyas et al., 2016) in the heart. In hepatic tissue analyses, 15-hour ethanol exposure in rats was shown to reduce gene and protein expression of *Pgc1 α* (Chaung et al., 2008) and the same was seen after a longer 28-day exposure (Lieber et al., 2008).

5.0.5. Ethanol and epigenetics

Our data so far suggests that biochemical (possibly including epigenetic) factors rather than social factors govern the effects of paternal drinking on the offspring as the sire rats were kept under identical housing conditions and were removed prior to the birth of the litters. We thus sought to investigate the biological mechanisms specifically focussing on epigenetic

changes at the level of the DNA. Epigenetic changes refer to the modification of gene expression (gene activity) without a change to the genetic code itself. There are multiple mechanisms through which this can occur, such as histone modification, RNA interference and DNA methylation (Ellenbroek & Youn, 2016). However, as mentioned in Chapter 1, mammalian sperm chromatin has a unique structure with protamines replacing most histones reducing the likelihood of histone modifications being the vehicle for epigenetic inheritance. Next, sperm cells contain small RNAs which are increasingly associated with epigenetic inheritance however the content compared to the RNAs in the maternal RNAs in the oocyte are minimal, leaving their role unclear (Liu, S. & Sharma, 2023). These findings suggest that DNA methylation may be the likely epigenetic mechanism involved in normal development. As mentioned in chapter 1, methylation can also be mediated by changes in seminal fluid proteins although, like the sncRNAs, the mechanisms here are unclear too (Patlar, 2022). Again, most DNA methylation in the sperm is also removed except for a few key loci which makes it the most likely means of epigenetic control. Importantly, there are a substantial number of genes where only the maternal or the paternal copy of the gene is methylated (referred to as genomic imprinted genes). Disruption to the methylation status at known imprinting loci has been shown to have dramatic developmental effects (Monk et al., 2019). Although there is only limited research in this area, there is evidence that ethanol alters DNA methyl transferases (DNMTs), which are the enzymes that methylate cytosine residues in DNA (Bielawski et al., 2002; Liang et al., 2014).

An important gene for epigenetic regulation is methyl-CpG-binding protein 2 (*MeCP2/Mecp2*), the *MECP2* protein recognizes and binds to epigenetic methylated cytosines and recruits various protein complexes that modify epigenetic states to regulate gene expression

(Zimmermann et al., 2015). Early research on *MeCP2* function suggested that *MeCP2* bound to regions of dense CpG dinucleotides or CpG islands and recruited factors involved in transcriptional silencing to nearby genes (Wolffe et al., 1998). However, newer evidence suggests more diverse roles for *MeCP2* such as the direct compaction of chromatin (Nikitina et al., 2007), the control of alternative splicing of pre-mRNA (Young et al., 2005) and modulating expression of transcriptionally active genes by long-range regulation of chromatin structure from methylation sites outside of gene promoters (Yasui et al., 2007). *MeCP2* appears to be integral to development with it showing tissue and region-specific effects and is influenced by and acts upon multiple genetic, epigenetic, and environmental factors (Gonzales & LaSalle, 2010). In addition, *MeCP2* has been implicated in the regulation of specific paternally imprinted genes such as *DLX5*, *UBE3A* (LaSalle, 2007) and the imprinting control region of *H19-IGF2* (Drewell et al., 2002).

MeCP2 is implicated in the development of liver fibrosis and chronic liver disease (Mann, J. et al., 2010; Tao, H. et al., 2011). Other research highlights the role of *MeCP2* in lipid metabolism and gene expression in the liver (Kyle et al., 2016; Yang et al., 2016). While researchers in those studies did not look at alcohol itself, there are clear links between drinking and liver fibrosis (Blomdahl et al., 2021; Cubero et al., 2009). Other studies which have looked at alcohol and *Mecp2* have found an increased ethanol sensitivity and reduced drinking in mice with reduced *Mecp2* expression (Repunte-Canonigo et al., 2014) and an increase in *Mecp2* expression in the hypothalamus of rats which had been subject to foetal alcohol exposure (Gangisetty et al., 2014). This raises the question of whether *MeCP2* expression is influenced by chronic ethanol intake and whether these changes persist after drinking subsides.

5.0.5. Current study

As the focus of this study is paternal alcohol consumption, it is important to note that while mitochondrial DNA itself has shown to mostly be maternally inherited (Sato & Sato, 2013), recent research indicates paternal influences on offspring mitochondrial metabolism including modulation of mitochondrial copy number (Godschalk et al., 2018). This taken with the influence of ethanol on regulation of gene expression suggesting potential pathways to understand the mechanisms involved in the paternal influence of ethanol consumption.

The experiments in this thesis focus on liver energy metabolism and epigenetic analysis of liver DNA, to perform a first assessment of the changes induced by ethanol drinking in the F0. Next, we look at mitochondrial copy number in the F1s to assess any difference between paternal alcohol drinking which may have led to the differential behaviour performances. Lastly, as alcohol liver injury is also shown to have sex dependent effects due to the role of sex hormones (Eagon, 2010; Yokoyama et al., 2005) and differential expressions of genes involved in oxidative stress, inflammation, and metabolic adaptation (Banerjee et al., 2009; Eagon, 2010; Sharma, M. R. et al., 2008; Tadic et al., 2002; Yokoyama et al., 2005), sex differences are also looked by using a set of F0 females.

Quantification of mtDNA-CN is usually measured by qPCR by amplifying a mitochondrial single copy gene and a single copy genomically encoded gene and looking at the ratio of the two. Digital droplet PCR (ddPCR) is a more powerful technique for looking at this although both are applicable (Podlesniy & Trullas, 2017). The genes chosen for mtDNA-CN calculation were mt-Nd1, used as a measure of the mtDNA, and Ndufa9 for the nuclear genome (gDNA) (Thorne et al., 2022a). The expression of *Pgc1a* and *Mecp2* were assessed as *Pgc1a* is

an important co-activator that regulates the genes involved in energy metabolism and acts as a direct link between external stimuli and mitochondrial biogenesis. While *Mecp2* is involved in mediating transcription through influencing binding activity such that it plays a key role in “silencing” genes and is used as a metric for epigenetic changes.

5.1. Methods

5.1.1. Sample collection and storage

Twenty-four F0 males were used: the 9 drinking males (D₀) which sired the D₁ and the 9 non-drinkers (ND₀) which sired the ND₁ (Table 5.1). In addition, 6 drinker males which were not selected for mating were culled and had samples taken at the end of the IA2BC (Chapter 2). F0 male rats used as sires were sacrificed post-mating which was post-natal day (PND) 180. The animals taken immediately post-drinking from F0 were sacrificed at PND 90. As the sires were culled at different ages to the other F0 group they have been spoken as in terms of age at culling with D₀90 (Drinkers), D₀180 (Drinking sires) and ND₀ (non-drinking sires). For the F1 rats, all the rats in the IA2BC paradigm in the F1 generation were used which made 7 animals in each of the four conditions (ND₁F, ND₁M, D₁F, D₁M).

Lastly, as sex differences in mtDNA-CN have been reported (Thorne et al., 2022a) and the F0 animals only comprised of males, we used 3 female Sprague Dawley rats (D₀F) from independent stock litters which underwent the same IA2BC paradigm (described in chapter 2) and had 4 female rats acting as non-drinking controls (ND₀F). This makes the D₀F sample size smaller than suggested based on our power analysis (Chapter 2). Post drinking, on PND 90, these animals were sacrificed and had their tissue samples collected. Rats were sacrificed after the completion of the behavioural experiment they were placed in. This was done through CO₂

asphyxiation before rapid decapitation. Liver samples from animals' part of the IA2BC were collected and analysed here (Other tissue samples collected are mentioned in Chapter 1).

Samples were collected, flash frozen in dry ice and stored at -80 degree Celsius until used.

Generation	Abbreviations	Sex	Condition	Age	N
F0	D ₀ 90 / D ₀ M	Male	Drinker	PND 90	6
	D ₀ 180	Male	Drinker	PND 180	9
	ND ₀ / ND ₀ M	Male	Non-Drinker	PND 180	9
	D ₀ F	Female	Drinker	PND 90	3
	ND ₀ F	Female	Non-Drinker	PND 90	4
F1	D ₁ M	Male	Drinker	PND 90	7
	D ₁ F	Female	Drinker	PND 90	7
	ND ₁ M	Male	Non-Drinker	PND 90	7
	ND ₁ F	Female	Non-Drinker	PND 90	7

Table 5.1. Rats used in molecular experiments. Liver samples collected for mtDNA-CN analysis. The 2 abbreviations for the F0 PND 90 and the non-drinker males are for specification of which rats were used for the sex-differences analysis.

5.1.2. Mitochondrial DNA copy number sample preparation

The study used ddPCR to quantify mtDNA-CN in the liver of the F0 and F1 rats as selfie-ddPCR can measure absolute gene expression without the need for reference genes (Podlesniy & Trullas, 2017). The protocol used was developed by our research group (Thorne et al., 2022a) and used target genes mt-Nd1 as a measure of the mitochondrial genome and Ndufa9 the nuclear genome (Table 5.2). As each diploid cell can have multiple copies of mtDNA while it has only 2 copies of gDNA, mtDNA-CN is calculated as a ratio of mtDNA to gDNA transcripts detected.

Amplicon	Primer sequence		Product size
	Forward (5'–3')	Reverse (5'–3')	
<i>mt-Nd1</i>	TTA ACG TCG AAT ACG CCG CA	AGC TGG TTG AGT ATA ATT CAG GGT	144 bp
<i>Ndufa9</i>	TGA GAT CGA GGA AAC CAA GCC	TAG AGC ACC CTC CTT CAG TAC A	100 bp
<i>Pgc1a</i>	CTG AAG AGG CGA GAG ACA GAA	TGT TCG TTC TGG TCA GGT GC	210 bp
<i>Mecp2</i>	GGG GTA GAA AGC CTG GGA GT	TCA ATG CTG ACG GTT TCC CG	142 bp

Table 5.2. Primer sequences used for assessing mitochondrial DNA copy number and gene expression by ddPCR.

For this, total cellular DNA was isolated with the SingleShot Cell Lysis Kit (Bio-Rad, 1725080; USA). Three milligrams of tissue was homogenized before the addition of 45 µl of cell lysis buffer and 1 µl of proteinase K. Samples were incubated for 10 min at room temperature followed by 5 min at 37°C and 5 min at 75°C to inactivate proteinase k for immediate use. The lysis was diluted to a 1:5 ratio with nuclease free water. Following this, a restriction digest with Hin6I (ThermoFisher Scientific, ER0481; USA) was carried out for optimal DNA partitioning, this enzyme does not cut internal but fragments the amplicon to facilitate partitioning. An enzyme buffer was made, 10 units of the buffer comprised 6.5 units nuclease free water to 1 unit of the provided tango buffer (10x, ThermoFisher Scientific, BY5; USA) and 2.5 units of 10 U/µL Hin6I (Table 5.3). The tango buffer comprised 33 mM Tris-acetate (pH 7.9 at 37°C), 10 mM magnesium acetate, 66 mM potassium acetate, 0.1 mg/ml BSA per unit. The restriction digest sample solution of 10 µl was made by adding 1 µl of the enzyme buffer and 1µl tango

buffer to 8 µl of the diluted sample (Table 5.3). Samples were incubated at 37°C for 1 hr followed by 20 min at 65°C. Samples were then diluted to a final volume of 50 µl (a further 5-fold dilution) for ddPCR. A further 25-fold dilution was done for the ddPCR amplification of mt-Nd1.

Part 1: Cell Lysis		
Component	Concentration	Volume
Lysis solution		~ 50 µl
Homogenised tissue		3 g
Lysis Buffer		45 µl
Proteinase k		1 µl
Diluted lysed sample	5-fold dilution	10 µl
Nuclease-free water	-	8 µl
Lysed solution	-	2 µl
Part 2: Restriction Digest		
Component	Concentration	Volume
Enzyme Buffer		10 µl
Nuclease-free water		6.5 µl
Tango buffer	10x	1 µl
Hin6I	10 U/µL	2.5 µl
Restriction digest solution		10 µl
Diluted lysed sample		8 µl
Enzyme Buffer		1 µl
Tango Buffer	10x	1 µl
Final solution		~ 50 µl
Nuclease-free water		40 µl
Restriction digest sol.		10 µl

Table 5.3. Sample preparation for mitochondrial DNA copy number solutions. Summary of the sample lysis and restriction digest procedure.

5.1.3. Selfie-digital droplet Polymerase Chain Reaction preparation (ddPCR)

Selfie-dPCR measures the amount of DNA and cDNA directly in a cell or tissue lysate that contains native proportions of nucleic acids and is advantageous as it is less sensitive to variance in primer efficiency and nucleic acid purity (Thorne, 2023). It is done in 4 steps, first, isolation of sample DNA and RNA from the sample tissue. Next, gene-specific primer pre-annealing in duplicate. Third, reverse transcription with enzyme in one duplicate (RT +) and without enzyme (RT –) in the other duplicate and lastly, digital PCR and nucleic acid quantification (Figure 5.1).

Reverse transcription quantification of transcript abundance of target genes *Pgc1a* and *Mecp2* by selfie-ddPCR was performed in lines with Podlesniy and Trullas (2017) and Thorne et al., (2022). For this, total cellular DNA and RNA was isolated with the SingleShot Cell Lysis Kit (Bio-Rad, 172-8050; USA). Again, 3 milligram of tissue was homogenized before the addition of 45 µl of cell lysis buffer and 1 µl of proteinase K. Samples were incubated for 10 min at room temperature followed by 10 min at 37°C and 5 min at 75°C. Following lysis, the lysis solution was diluted in a 1:5 ratio with nuclease free water after which 2.25 µl of the diluted lysis sample was added to 9 µl of 5 µM *Pgc1a* and *Mecp2* reverse primer (2.5 µM each, Table 5.4), and samples were incubated in a thermocycler at 70°C for 1 min and brought to 4°C at a slow ramp rate (2° per second) for primer annealing. For reverse transcription a RT mix made of nuclease-free water, 5x RT buffer (250 mM Tris-HCl (pH 8.3 at 25°C), 375 mM KCl, 15 mM MgCl₂, 50 mM DTT; ThermoFisher Scientific, EP0751; USA), 10 µM deoxynucleotides (dNTPs), 40 U/µL RNaseOUT (ThermoFisher Scientific, 10777019; USA) and 0.1M dithiothreitol (DTT) was made for each sample (Table 5.4).

The mixture was split into two 11 μ l mixes and either 0.5 μ l of 200 U/ μ L Maxima H Minus reverse transcriptase (ThermoFisher Scientific, EP0751; USA) or RT buffer was added. Five microliters of the pre-annealed sample was then added to 10 μ l of each mixture to create the RT+/RT- samples. Samples were incubated at 58°C for 30 min, followed by 90°C for 3 min to terminate reverse transcription reactions. For optimal partitioning of DNA into droplets for ddPCR, a restriction digest with HindIII (ThermoFisher Scientific, ER0501; USA) was undertaken prior to ddPCR. A 1:10 dilution of 10 U/ μ L HindIII in the provided R buffer (ThermoFisher Scientific, BR5, USA) was made, where 1x R buffer comprised 10 mM Tris-HCl (pH 8.5), 10 mM MgCl₂, 100 mM KCl and 0.1 mg/mL BSA. Lastly, 1 μ l of this dilution was added to the RT+/RT- reactions. Samples were incubated at 37°C for 1 hr followed by 20 min at 65°C. The prepared sample was then used for ddPCR.

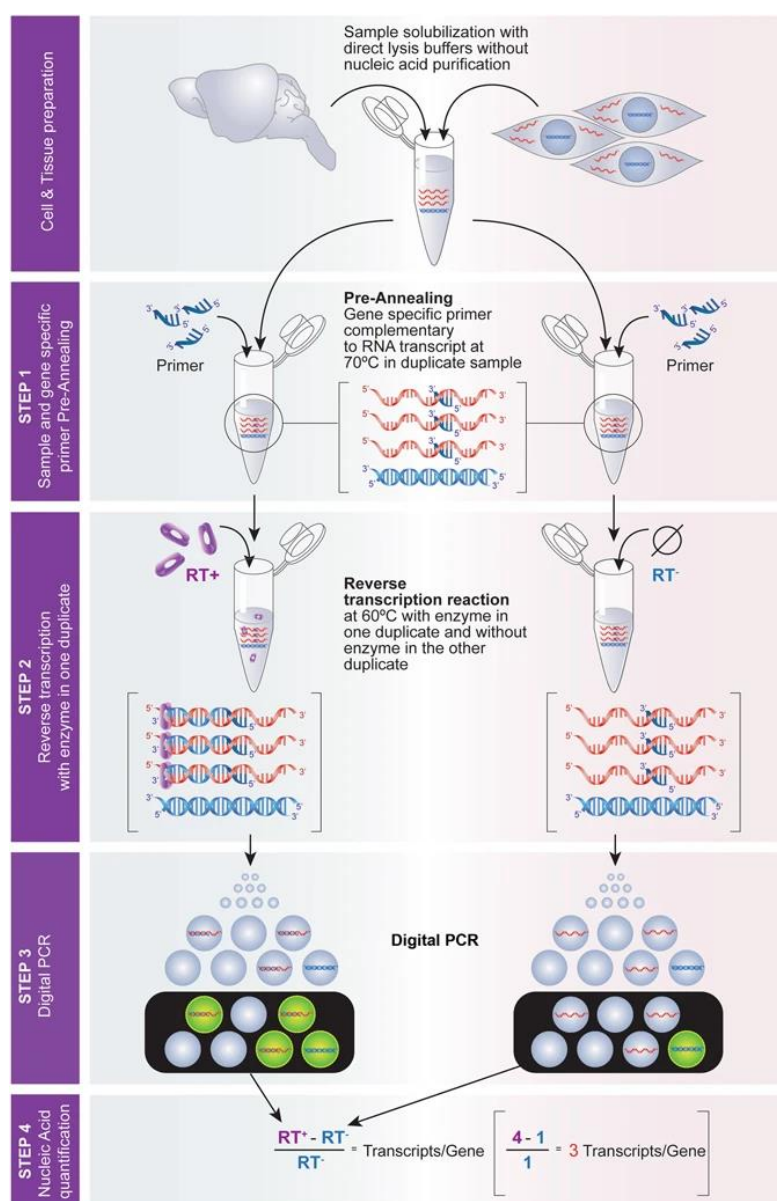


Figure 5.1. Depiction of the Selfie digital droplet PCR method. Selfie-ddPCR to analyze gene transcription in relation to the self-encoding gene (Podlesniy and Trullas, 2017). Selfie-ddPCR can be broken down into 4 steps and allows for the measurement of RNA transcripts in relation to the gene encoding the transcript which allows for absolute values of transcripts per cell (Podlesniy & Trullas, 2017).

Part 1: Cell Lysis		
Component	Concentration	Volume
Lysed solution		~50µl
Homogenised tissue		3 g
Lysis Buffer		45 µl

Proteinase k		1 µl
Diluted lysed sample	5-fold dilution	10 µl
Nuclease-free water	-	8 µl
Lysed solution	-	2 µl
Part 2: Pre-annealing		
Component	Concentration	Volume
Pre-annealed sample		11.25 µl
Target reverse primer mix	5 µM	9 µl
Diluted lysed sample		2.25 µl
Part 3: Reverse Transcription		
Component	Concentration	Volume
RT mix:		23 µl
Nuclease-free water		13.6 µl
RT buffer	5x	4.7 µl
Deoxynucleotides (dNTPs)	10 µM	2.3 µl
Dithiothreitol (DTT)	0.1M	1.2 µl
RNaseOUT	40 U/µL	1.2 µl
RT+		16.5 µl
RT mix		11 µl
Pre-annealed sample		5 µl
Maxima H Minus reverse transcriptase	200 U/µL	0.5 µl
RT-		16.5 µl
RT mix		11.5 µl
Pre-annealed sample		5 µl
RT buffer	5x	.5 µl
Part 4: Restriction Digest		
Component	Concentration	Volume
Restriction digest solution		10 µl
HindIII	10 U/µL	1 µl
R Buffer	1x	9 µl
Final sample solution		17.5 µl
RT+/RT-		16.5 µl

Restriction digest solution		1 μ l
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Table 5.4. Sample preparation for selfie-ddPCR. Solution concentrations and values for the lysis, pre-annealing, reverse transcription and final restriction digest to prepare sample for ddPCR.

5.1.4. Digital droplet PCR method

Selfie-ddPCR used the same steps described by Thorne et al., (2022). One microliter of the final sample was mixed thoroughly with 11 μ l of ddPCR EvaGreen Supermix (Bio-Rad, 186-4034; USA), 9 μ l of nuclease free water and 1.3 μ l of 2.5 μ M forward and reverse primers at room temperature (Table 5.5). To create the droplets, 20 μ l of the ddPCR mix was emulsified in 70 μ l of Droplet Generation Oil (Bio-Rad, 186-4005; USA). No template controls which included the reagents without template were included to confirm the absence of contaminating DNA. Next, 40 μ l of the droplet emulsion was added to a 96-well plate for PCR amplification (C1000 Touch Thermal Cycler with 96-deep well reaction module, Bio-Rad; USA). Amplification setup had a 2°C/s ramp rate with the following cycling: 95°C for 5 min, 95°C for 30 s, 62°C for 1 min for 40 cycles, 4°C for 5 min, and 90°C for 5 min.

QuantaSoft Analysis Pro software (Version 1.0.596, Bio-Rad; USA) Droplet was used for the fluorescence analysis. The fluorescence threshold for positive droplets was adjusted according to the no template controls. The software uses a Poisson distribution, to assess the number of copies per μ L (N) based on the number of positive and negative droplets detected. The mtDNA copy number was the calculated:

$$\text{mtDNA copy number} = [\text{N mtDNA} \times \text{dilution factor}] / [\text{N gDNA}]$$

$$Pgc1\alpha/Mecp2 \text{ transcripts per cell} = [\text{N RT+}] - [\text{N RT-}] / [\text{N RT-}]$$

Part 1: Sample preparation		
Component	Concentration	Volume

ddPCR mix		22.3 μ l
ddPCR EvaGreen Supermix		11 μ l
Forward and reverse primers	2.5 μ M	1.3 μ l
Nuclease-free water		9 μ l
Final sample		1 μ l
Part 2: Droplet generation		
Component	Concentration	Volume
DdPCR mix		20 μ l
Droplet Generation Oil		70 μ l

Table 5.5. Solutions and steps involved in ddPCR. Proportions of samples prepared earlier to the ddPCR mix and primers for optimal ddPCR.

5.1.5. Statistical analysis

All statistical analyses used in this research were conducted using IBM SPSS Statistics 25 (IBM; USA) and GraphPad Prism (version 8.4.1 for Windows, GraphPad Software; USA). Data were analysed by one-way and two-way ANOVAs. To reduce animal to animal variations, trimmed means were used except for in the comparison of sexes in F0 as there were only 7 females in total. The trimmed means remove the same amount (either number or percentage) from both ends of a statistical distribution to eliminate points on the ends to provide a better representation of a data sets central values (Cox, 2013; Tukey, 1962). Like previously, Tukey's HSD test was used as post hoc analysis to assess the significance of differences between pairs of group means. Here we remove the highest and lowest values from each data set where they were used. In addition, Pearsons correlational analyses were used to look at relationships between gene expression, mitochondrial copy numbers and drinking data. Lastly, as there were only 7 F0 females used here, sex and drinking conditions in the F0 were analysed using non-parametric methods. Individual Mann-Whitney U tests were used to compare mtDNA-CN by sex, drinking condition and each sex x drinking condition combination.

5.2. Results

5.2.1. F0 mtDNA-CN in the liver

Comparisons of the mtDNA-CN through a one-way ANOVA presented a significant difference ($F(2, 17) = 6.30, p = .009$). Post hoc analysis showed the ND₀ rats had significantly higher mtDNA-CN than both the D₀90 ($p = .023$) and D₀180 ($p = .021$; Figure 5.2). There were no differences between the two drinking groups. This indicates that chronic ethanol consumption decreased mtDNA-CN in the liver, an effect that persisted for at least 90 days after ethanol drinking was terminated.

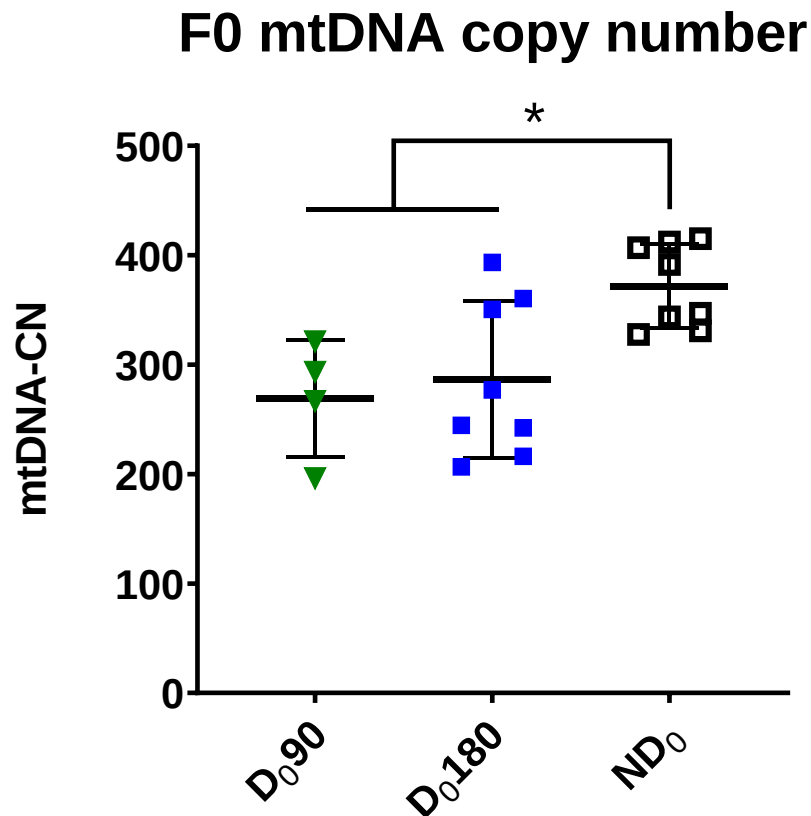


Figure 5.2. MtDNA-CN in the F0 at the end and 90 days after ethanol drinking. Condition dependent changes in mtDNA copy number in the F0 livers are seen due to drinking with reduced mtDNA-CN being seen in D₀90 and these changes are maintained even 3 months after the end of drinking (D₀180), $F(2, 17) = 6.30, p = .009$. Post hoc analysis showed the ND₀ rats had significantly higher mtDNA-CN than both the D₀90 ($p = .023$) and D₀180 ($p = .021$). It is important to note that each of these animals comes from different litters to control for litter effects. Abbreviations: D (drinker), ND (non-drinker) subscripts denote generation * Denotes a significant difference.

5.2.2. Expression of *Pgc1a* and *Mecp2*

Abundance of *Mecp2* and *Pgc1a* RNA transcripts were evaluated using simple one-way ANOVAs. There was a significant difference in *Mecp2* expression ($F(2, 15) = 8.23, p = .004$) with the D₀180 showing a greater expression than the D₀90s ($p = .003$). No significant differences were seen between the ND₀ and either D₀ group. However, group means do suggest the lowest *Mecp2* expression being in D₀90 (Figure 5.3). This suggests increased *Mecp2* activity in rats 90 days post-drinking and suppressed activity during.

There was also a significant difference in *Pgc1a* expression ($F(2, 15) = 7.07, p = .007$). We did not see a difference in expression between the non-drinkers and D₀180 ($p = .653$), but both showed significantly higher expression than the D₀90 ($p < .027$). These, like in *Mecp2*, indicate a reduced expression of *Pgc1a* immediately after chronic ethanol consumption but show a recovery around 90 days after.

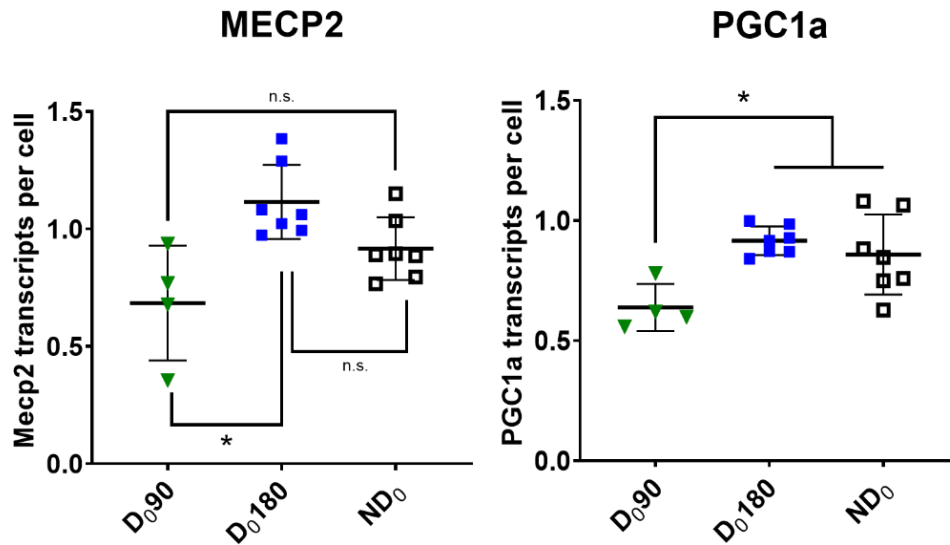


Figure 5.3. mRNA expression of *Mecp2* and *Pgc1a* in the F0 male livers. Male F0 drinkers immediately after (D₀90) and three months post drinking (D₀180) compared to non-drinker (ND₀) animals. Condition based differences in expression are seen with animals who had just completed the drinking paradigm (D₀90) showing the lowest expression of both *Mecp2* and *Pgc1a*. Drinkers who were kept 3 months post drinking (D₀180) show the highest expression of *Mecp2* ($p = .108$) and similar levels of *Pgc1a* ($p = .653$) compared to ND₀. No significant difference is seen between D₀90 and ND₀ in *Mecp2* ($p = .110$) but there is a clear difference in *Pgc1a* expression ($p = .027$). Lastly, D₀180 is significantly higher than D₀90 in both *Mecp2* ($p = .003$) and *Pgc1a* ($p = .006$). * Denote significant differences.

5.2.3. Associations of expression of *Mecp2* & *Pgc1a* to mtDNA-CN

Correlations between liver mitochondrial copy number and target gene expression showed fascinating trends. Firstly, a significant positive correlation ($r = .89$, $p = .019$) was seen between mtDNA-CN and *Mecp2* expression in the D₀90 where lower copy number was associated with lower levels of *Mecp2* expression (Figure 5.4). This effect was less pronounced in the D₀180 and completely absent in the ND₀. In the *Pgc1a*, we saw no significant correlations but there were trends which should be highlighted. Just like *Mecp2*, the ND₀ saw no correlation

in *Pgc1a* expression and mtDNA-CN. There were positive correlation trends in the D₀180 ($r = .64, p = .089$) where *Pgc1a* expression was associated with an increase in mtDNA-CN and to a lower degree in the D₀90 group ($r = .63, p = .178$). Taken together, these suggest that ethanol drinking affects *Pgc1a* and *Mecp2* expression and that these reductions are associated with lower mtDNA-CN levels.

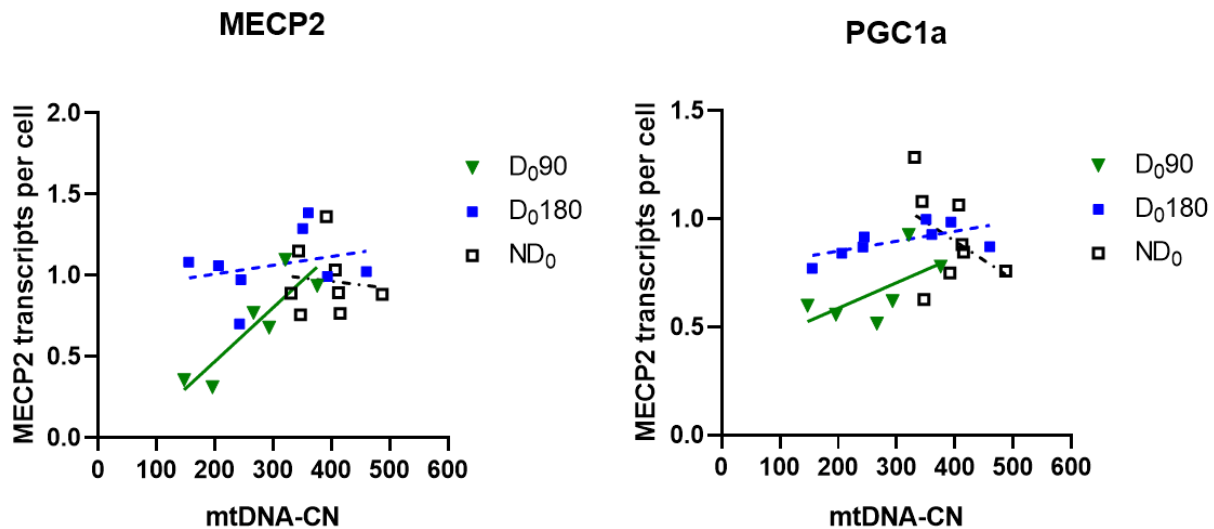


Figure 5.4. Expression of *Mecp2* and *Pgc1a* to the mtDNA-CN in the liver of male F0 drinker (PND90 and PND180) and non-drinker (PND180) animals. Condition based differences are seen with animals who had just completed the drinking paradigm showing the clearest positive association of mtDNA-CN with *Mecp2* and showing that trend in *Pgc1a*. The positive association of *Pgc1a* with mtDNA-CN is still seen in the D₀180 drinkers ($p = .089$). No association is seen in the ND₀ animals. The rats show clear clustering based on condition particularly in *Mecp2*. A significant positive correlation ($r = .89, p = .019$) was seen between mtDNA-CN and *Mecp2* expression in the D₀90 where lower copy number was associated with lower levels of *Mecp2* expression.

5.2.4. Sex differences in mtDNA-CN

As mentioned previously, sex differences have been reported in mtDNA-CN in rats. To assess these, we compared our non-drinking (ND₀M) and immediately post drinking males

(D₀M) to matched females (ND₀F and D₀F) using Mann-Whitney U tests. There were no effects of sex ($U = 46.00, p = .535$) but there were significant differences in mtDNA-CN based on drinking condition ($U = 8.00, p < .001$) where non-drinkers had higher mtDNA-CN ($M = 408.90 \pm 86.86$) than drinking animals ($M = 252.10 \pm 83.84$), again confirming the effects of ethanol on liver mitochondria. Interestingly, the individual comparisons show an interaction between sex and drinking condition on mtDNA-CN. The ND₀ females ($M = 496.80 \pm 85.63$) on average had the highest mtDNA-CN which was significantly higher than the ND₀ males ($M = 373.70 \pm 60.29$; $U = 5.00, p = .036$) and the D₀M ($M = 266.60 \pm 83.37$; $U = 8.00, p = .003$). The ND₀F were also nearly significantly different from the D₀F ($M = 223.2 \pm 94.09$; $U = 0.00, p = .057$) but this was impacted by the small sample size. Next, ND₀M had significantly higher mtDNA-CN than D₀F ($U = 1.00, p = .014$) and D₀M ($U = 7.00, p = .011$). Lastly, there was no difference between the drinking males and females ($U = 12.00, p = .548$). These suggest sex differences in base levels of mitochondrial copy numbers in the livers in rats and potentially female rats being more sensitive to the effects of ethanol on mtDNA-CN, but this cannot be confirmed due to the small sample size.

F0 mtDNA-CN sex difference

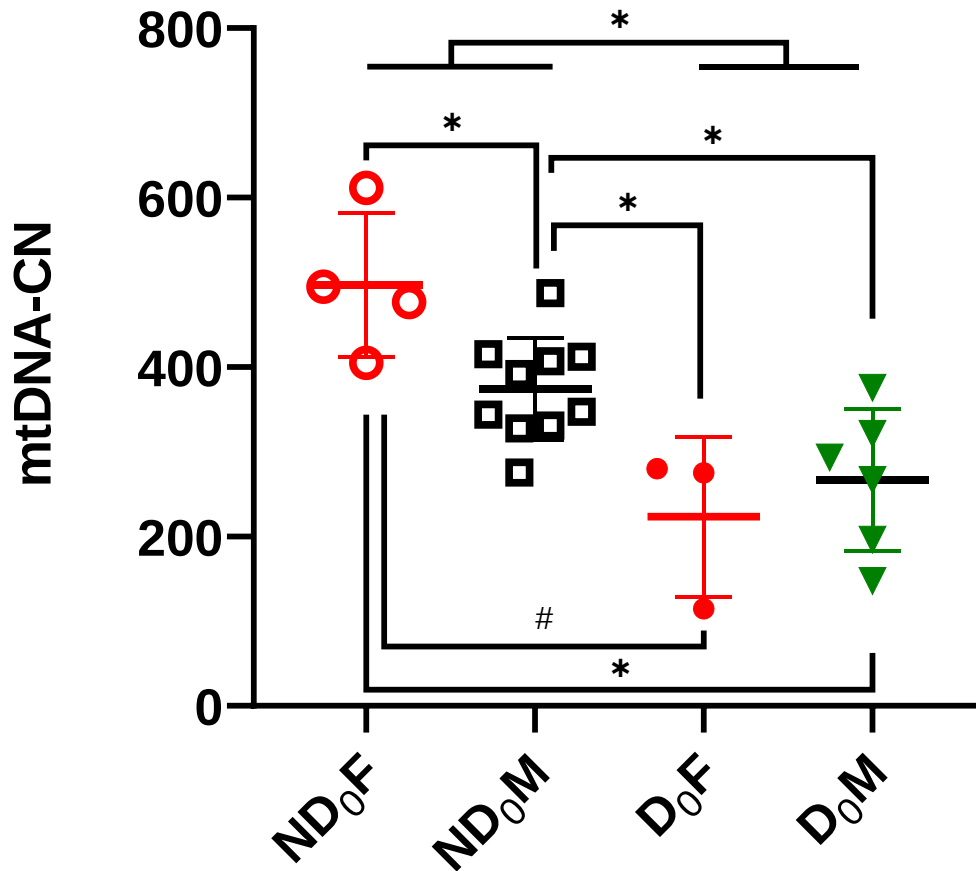


Figure 5.5. MtDNA-CN comparing sex differences in the F0 animals. There were significant differences in mtDNA-CN based on drinking condition ($U = 8.00$, $p < .001$) where non-drinkers had higher mtDNA-CN than drinking animals. Individual comparisons indicate an interaction between sex and drinking condition on mtDNA-CN. The ND₀ females had significantly higher mtDNA-CN than the ND₀ ($U = 5.00$, $p = .036$) and D₀ ($U = 8.00$, $p = .003$) males. The ND₀F were also nearly significantly different from the D₀F ($U = 0.00$, $p = .057$) but this was impacted by the small sample size. ND₀M had significantly higher mtDNA-CN than D₀F ($U = 1.00$, $p = .014$) and D₀M ($U = 7.00$, $p = .011$). Lastly, there was no difference between the drinking males and females ($U = 12.00$, $p = .548$). Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation * Denote significant difference, # denotes near significant ($p = .057$).

5.2.5. Association of ethanol drinking to mtDNA-CN

Mitochondrial copy number in the livers were compared to the drinking behaviours in the F0 to assess whether the amount of drinking is related to mtDNA-CN. To do this mtDNA-CN, ethanol preference and total ethanol drinking were converted to z-scores and charted. While statistical analysis did not show significant effects, the z-score distributions are non-random (Figure 5.6). As we have seen, ethanol drinking reduces mtDNA-CN and within those which have consumed ethanol, increased drinking or preference is not associated with increased mtDNA-CN. However, it is not clear whether lower rates of ethanol consumption among the D₀ rats is associated with lower reductions in mtDNA-CN. Additionally, the greater negative association of higher ethanol drinking and preference to lower mtDNA-CN is seen in the D₀180 rats not the D₀90 rats.

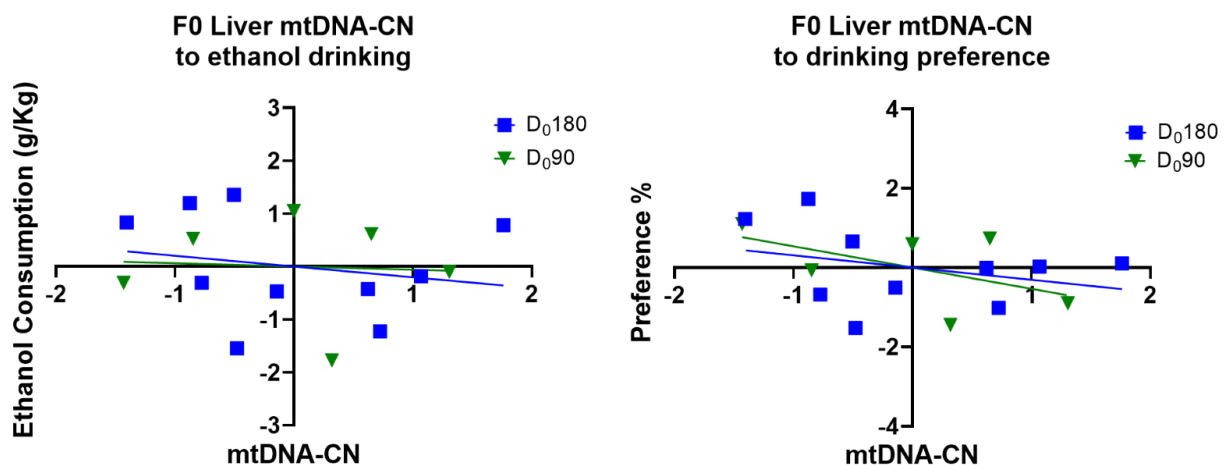


Figure 5.6. Association of the z-scores of alcohol preference (%) and ethanol consumption (g/kg) to liver mtDNA-CN of the male F0 drinkers (PND90 and PND180). No significant correlations are seen, however, if there was no association between mtDNA-CN and drinking we would see an even scattering of points through all 4 quadrants. Here, drinking does not increase mtDNA-CN (seen through no points in the top right quadrants) but its impact on reduction of mtDNA-CN is harder to ascertain.

5.2.6. F1 mtDNA-CN

Mitochondrial copy number was investigated in the F1 animals which were part of the IA2BC. As a reminder, all animals here consumed ethanol and the comparisons were using a two-way ANOVA looking at sex and paternal condition. There was a significant main effect of paternal drinking condition ($F(1, 16) = 5.73, p = .029$) with offspring of drinkers (D_1) having lower mtDNA-CN than the ND_1 . There were no effects of sex nor any interactions (Figure 5.7). These indicate father drinking interacts with offspring ethanol consumption to cause an increased reduction in mtDNA-CN in offspring. These might suggest either an initial lower mtDNA-CN in offspring of drinking fathers or a greater loss of mtDNA-CN due to ethanol.

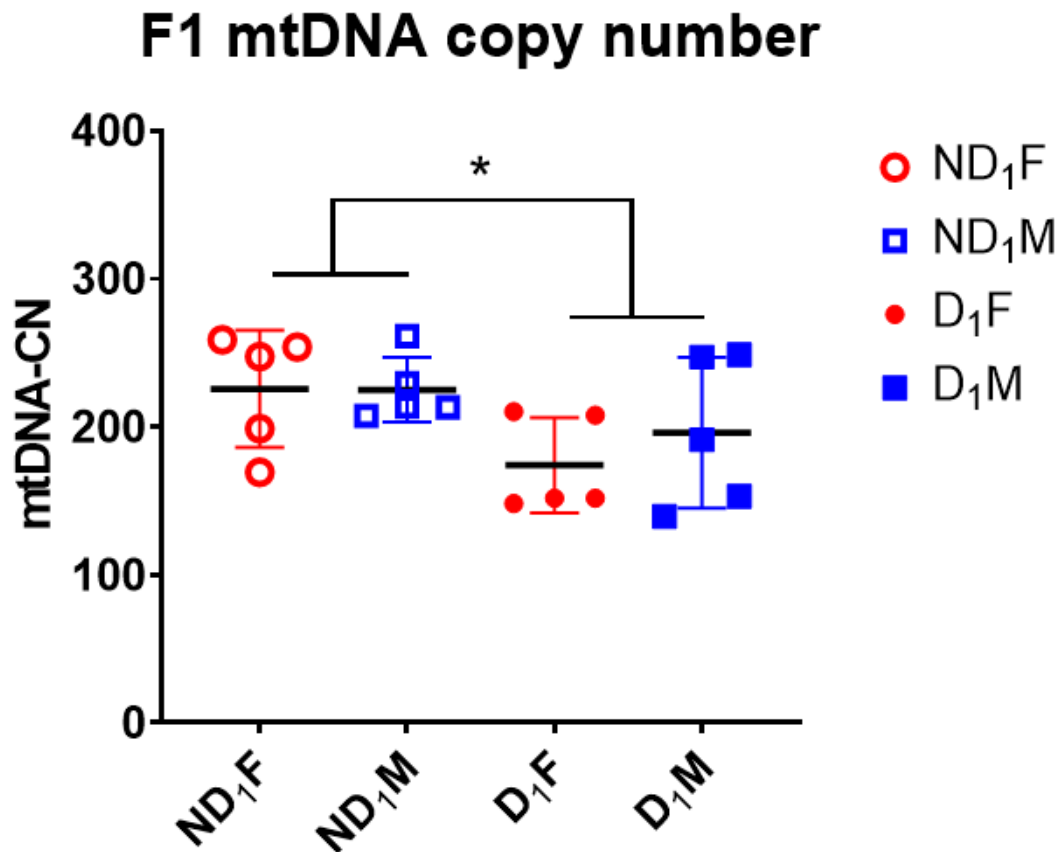


Figure 5.7. mtDNA-CN in the liver of F1 animals. Condition based differences are seen with animals who had drinking sires showing lower mt-DNA-CN ($F(1, 16) = 5.73, p = .029$). There were no sex differences. Abbreviations: parent conditions D (drinker) and ND (non-drinker), F (female), M (male), subscripts denote generation * Denote significant differences.

As there were differences in offspring of D₀ when compared to ND₀ in mtDNA-CN, we assessed whether the mtDNA-CN in the D₀ sires were associated with the level of mtDNA-CN in the offspring. For this we calculated z-scores of the mtDNA-CN for both the F0 and F1 generations and found that there were no significant correlations. Interestingly, there may be a sexually dimorphic effect of ethanol drinking on offspring mtDNA-CN with female and male

offspring showing different trends (Figure 5.8), although the statistical power is too low at this moment to draw any firm conclusions.

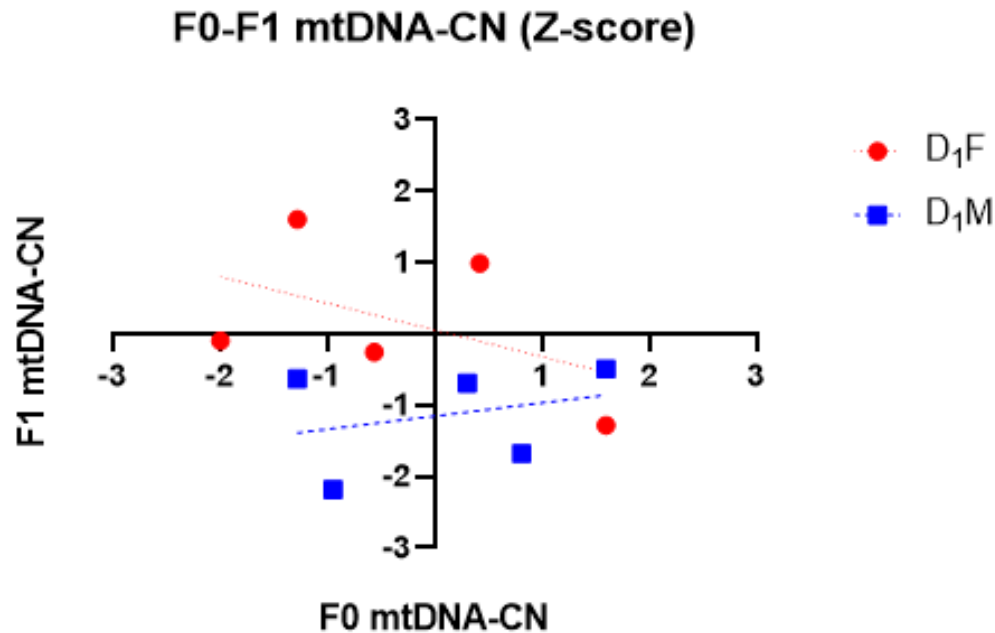


Figure 5.8. Z-score comparisons of mtDNA-CN in F0 drinkers to F1. There were no significant associations of parent mtDNA-CN to offspring mtDNA-CN z-scores. However, the uneven spread in each quadrant of the figure does indicate a relationship between parent mtDNA-CN and offspring mtDNA-CN particularly in the D₁M. The association also appears to be sexually dimorphic. Abbreviations: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation

5.2.7. Crossgenerational comparisons

Lastly, we compared the mtDNA-CN between the F0 and the F1 rats with a 2 x 2 x 2 ANOVA looking at generation, sex, and drinking/parent condition. There was a significant generation effect ($F(1, 43) = 48.87, p < .001$) with the F0 having a higher mtDNA-CN. This was mainly due to F1s not having a group which did not drink alcohol. There was also a main effect of condition with ND rats having higher mtDNA-CN ($F(1, 43) = 35.78, p < .001$). Importantly,

there was a generation x sire condition interaction ($F(1, 43) = 16.88, p < .001$), a sex x condition interaction ($F(1, 43) = 5.78, p = .021$) and sex x generation x sire condition interaction ($F(1, 43) = 4.03, p = .051$). Post-hoc analyses show where ND₀ have a significantly higher mtDNA-CN when compared to D₀ and the female ND₀ have the highest mtDNA-CN. The ND₁ have higher mtDNA-CN than the D₁ but the sex effects seen in the F0 are not seen in the F1 livers. It is important to note that the ND₁ rats were also exposed to ethanol meaning the only rats not exposed to alcohol were the ND₀ which means the D₀ and the ND₁ rats are similar in terms of parental drinking history. Here we see no differences between D₀ and ND₁ in terms of mtDNA-CN.

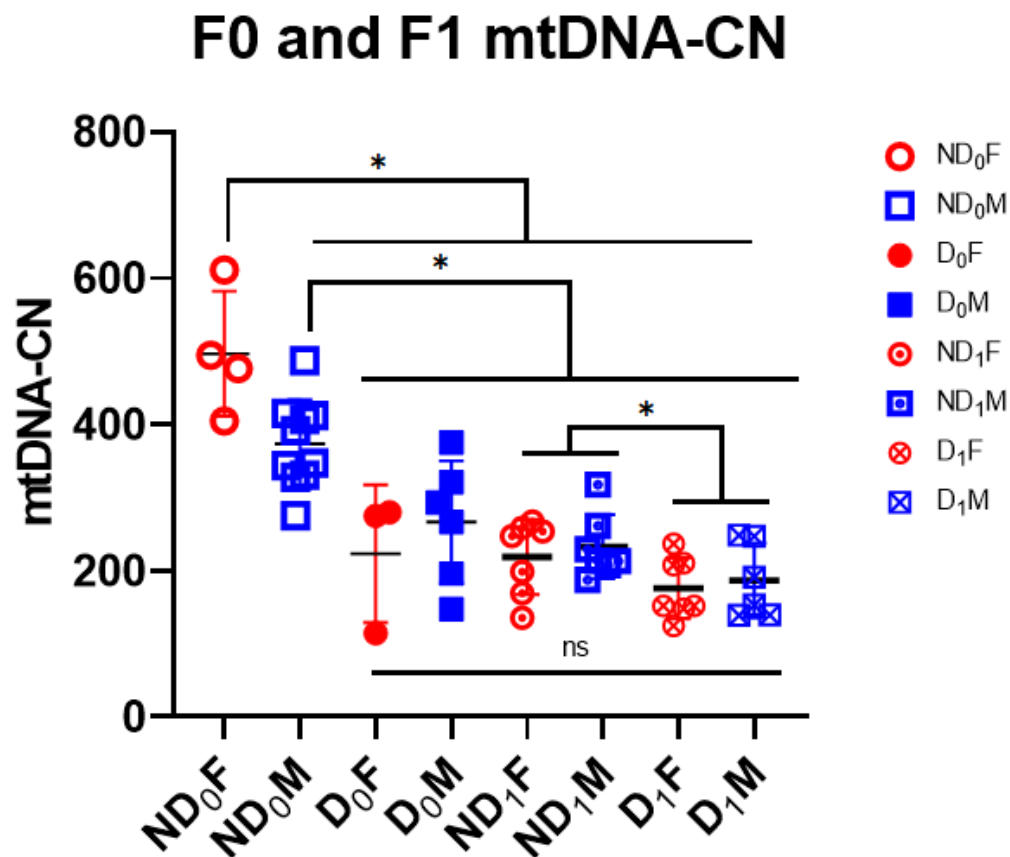


Figure 5.9. Comparisons of the F0 and F1 liver mtDNA-CN. Significant generation ($F(1, 43) = 48.87, p < .001$), drinking condition ($F(1, 43) = 35.78, p < .001$), generation x drinking condition ($F(1, 43) = 16.88, p < .001$) and generation x sex effects ($F(1, 43) = 5.78, p = .021$) were seen. The non-drinking F0 females (ND₀F) had the highest mtDNA-CN followed by the F0 males (ND₀M). The lowest mtDNA-CN is seen in the F1 drinkers (D₁). The F1 non-drinkers and the F0 drinkers show no differences in mtDNA-CN as all of them had consumed ethanol ($p = .905$). This shows that ethanol exposure significantly reduces mtDNA-CN, the effects of ethanol are the same on the D₀ and ND₁, and offspring of D₀ show further reductions in mtDNA-CN under the influence of ethanol. Abbreviations: F0/sire condition D (drinker), ND (non-drinker); F (female), M (male), subscripts denote generation * Denotes significance

5.3. Discussion

The results indicated a clear effect of ethanol drinking on mitochondrial copy number, *Pgc1a* and *Mecp2* expression with changes in mtDNA-CN persisting even 3 months after the end of drinking. Ethanol is associated with lower expression of both *Pgc1a* and *Mecp2* after chronic alcohol drinking as well as lower mtDNA-CN. The lower mtDNA-CN persists for at least 3 months after drinking, however, *Pgc1a* expression returns to a similar level to the ND₀ while *Mecp2* shows a higher level of expression than controls. These data are in line with the reported role of alcohol in liver damage and regeneration and suggest that the effects may impact future generations via epigenetic mechanisms.

5.3.1. Liver mitochondria function

The reduced mtDNA-CN in the F0 drinkers further supports the ability of the intermittent access paradigm for studying ethanol drinking behaviours and biological effects. Most studies looking at ethanol induced mitochondrial damage have used experimenter-applied ethanol administration (Han, D. et al., 2017; Marin-Garcia et al., 1995). We now show that the IA2BC paradigm, which leads to substantial ethanol intake as shown in the previous chapters, also induces clear mitochondrial alterations.

The low mtDNA-CN observed even in the D₀180 rats show that chronic ethanol consumption has long lasting effects on the regenerative properties of the liver. On the other hand, the expression of *Pgc1a* was only reduced immediately after ethanol consumption, returning to normal levels 90 days after. This suggests a return of the liver to a normal mitochondrial biogenesis rate. While mitogenesis in the liver might recover, we do see reduced mtDNA-CN in rats even 3 months post-drinking. The study also showed clear associations between copy number and *Pgc1a* expression in the drinking animals where the higher expression was associated with higher mtDNA-CN. The association between the two makes sense as the lower mtDNA-CN would be associated with lower mitochondrial biogenesis as seen through low *PGC1a* expression. No such correlation between the two was seen in the controls which again indicates the influence of ethanol on mediating this relationship and replicates the effects seen in prior research linking ethanol drinking to changes in liver mitochondrial function (Albano, 2006; Han, D. et al., 2017; Morio et al., 2021). The similar *Pgc1a* expression rates between the D₀180 and the ND₀ but the ND₀ having higher mtDNA-CN suggest no increase in biogenesis to counteract the effects of chronic alcohol consumption but rather *Pgc1a* no longer being suppressed as it was during drinking. Liver regeneration has previously shown to be inhibited by chronic ethanol administration (Dippold et al., 2013; Duguay et al., 1982) and this delay could be linked to mtDNA-CN number due to the significant role they play in the regeneration process (Li, Z. et al., 2021). Liver regeneration mainly occurs through the proliferation of mature hepatocytes (Michalopoulos, 2017), if these hepatocytes have seen a reduction in mtDNA-CN, as our data suggests, then the proliferated hepatocytes would also have lower mtDNA-CN despite their markers of biogenesis returning to control levels.

The study does show that mtDNA-CN is affected by drinking in ways which are non-random. There is no evidence of an increase in mtDNA-CN with high ethanol consumption in the liver but the existence of an association between higher drinking and reduced mtDNA-CN could not be elucidated here. We did not find any correlation between mtDNA-CN and the amount of ethanol consumed or preference, however, this could be due to limitations in the sample selection procedure. The 6 animals taken as the D₀90 were from the lower half of the drinking animals while all the higher drinkers had about another 3 months post drinking before being taken. This delay in the D₀180 could have affected their mtDNA-CN, they were the highest drinkers and could have had lower mtDNA-CN at PND90 than what was measured at PND180. Additionally, the *Pgc1a* expression of the D₀180 was higher than the D₀90 which suggests an increase in biogenesis from the levels post-drinking. This is backed up by time dependent changes in mtDNA levels post drinking in different tissues (Mansouri et al., 2001). Therefore, a better selection such as having a mix of the highest and lowest drinkers being selected for mating as well as for the D₀90 might have shed more light on the levels of drinking to mtDNA-CN associations.

The study also looked at sex differences in the F0 by using a small cohort of female rats treated in matched ways to the male F0 animals. We again found an effect of drinking with reduced mtDNA-CN in the drinking females compared to the non-drinkers. Interestingly, there was a sex-condition interaction where the ND₀ females had the highest mtDNA-CN. Earlier research has shown sex dependent differences in mtDNA-CN in specific brain regions (Thorne et al., 2022a) which makes the differences seen here understandable. While there were no differences in the drinking males and females in terms of mtDNA-CN, the females did on

average have a lower mtDNA-CN and the lack of significance may just be down to the small sample size. Overall, this still suggests a greater impact of on mtDNA-CN in females due to ethanol. Additionally, as females consistently consume more ethanol per bodyweight (Chapter 4) the greater damage could be due to higher ethanol exposure which would mean a cyclic effect of females being predisposed to higher ethanol drinking which in turn leads to a higher likelihood of mitochondrial damage.

5.3.2. Epigenetic changes and transgenerational effects

Mecp2 expression was suppressed during alcohol drinking but appeared to be over expressed after drinking ended. Unlike *Pgc1a*, the greatest expression was seen in the D₀180 suggesting an increase in *Mecp2* activity in the months following the cessation of alcohol consumption. The greater expression in the D₀180 3 months post-drinking matches findings which note increased *Mecp2* protein and mRNA expression in the prefrontal cortex of rats after exposure to ethanol (Cui et al., 2020) and is similar to the increased activity seen in the hypothalamus of rodents which underwent prenatal ethanol exposure (Gangisetty et al., 2014). Additionally, serum samples in humans have also shown a down regulation in microRNAs leading to a suppression of *MeCP2* in patients with alcohol use disorder (Aşçıbaşı et al., 2023). Our study provides evidence of differential *Mecp2* expression in rat livers based on ethanol drinking condition, whether there is a link between *MeCP2* changes and mtDNA-CN is less clear. *Mecp2* appears to express differently in recent drinkers with expression being positively associated with mtDNA-CN in D₀90 and this trend reducing in the D₀180. When *Pgc1a* expression returns to normal in the D₀180s we see an over expression of *Mecp2*. This overcompensation can have major effects as *MeCP2* influences a wide number of genes

(Zimmermann et al., 2015). For our study, these might imply a functional link between methylation status and mtDNA-CN with the suppression of *Mecp2* response and reduced mitochondrial activity during drinking leading to altered methylation status of genes associated with liver function and therefore mtDNA-CN. This could also explain the differences in F1 mtDNA-CN seen in this study. Conversely, these changes may correlate but not be linked to each other with *MeCP2* expression being altered by changes in microRNA expression (Aşçıbaşı et al., 2023) while the damage to mtDNA-CN would be through oxidative stress related damage (Farfán Labonne et al., 2009). We did not look at *Mecp2* in the F1 animals which would have helped identify whether *Mecp2* plays a regulatory role in mtDNA-CN. Future studies would aim to address this to better understand the role of methylation status on mtDNA-CN and liver function across generations.

A look at the mtDNA-CN of the F1s shows that the intergenerational changes reported in the offspring in terms of behaviour are also biological. We saw that the F1 offspring of drinkers (D₁) had lower mtDNA-CN than those of the non-drinkers which suggests an epigenetic effect of drinking on offspring liver mitochondria metabolism. Previous research on paternal stress liver damage caused through benzo[a]pyrene found reduced mtDNA-CN in the F1 offspring liver (Godschalk et al., 2018). Similarly, we show that metabolic stress to the liver induced through chronic ethanol exposure which effects the F0 animals also causes epigenetic changes which reduce mtDNA-CN in the F1 which indicate liver mitochondria are susceptible to epigenetic regulation through yet unknown mechanisms.

5.3.3. Limitations

We did not see a difference in sexes in terms of the D₁ rats, but a major reason for this could be that all F1 rats assessed here underwent ethanol drinking, which means the lack of difference in the ND₁ sexes is like what was seen in the D₀ rats. Future analysis should look at copy number without alcohol drinking to check for changes based solely from paternal drinking. We also do see some associations between the mtDNA-CN of the parents and their offspring but again the F1 going through drinking as well as the low sample size hinder the ability to clearly look at this. Nevertheless, the impacts of *MeCP2* on paternally imprinted genes (Drewell et al., 2002; LaSalle, 2007) coupled with its over expression in the D₀ and reduced mtDNA-CN in the D₁ indicates a role of paternal drinking in offspring behaviours and that the changes in behaviour seen might be due to the metabolic stress and epigenetic changes brought about by ethanol.

5.3.4. Conclusions

The difference in mtDNA-CN between drinker offspring and non-drinker offspring correlates well with the behavioural differences seen earlier. This suggests ethanol drinking in the F0 causes liver damage and changes in liver regeneration seen through lower mtDNA-CN and reduced *Pgc1a* expression. This comes along with possible epigenetic changes as seen in differences in *Mecp2* expression which is important in regulating methyl binding. These changes in the liver mtDNA-CN may confer a protective advantage on subsequent generations, particularly in females as seen in the reduced drinking and better locomotor activity performance under ethanol as seen in the previous chapters. The F1 behaviour is likely a balance between the reward circuits in the brain responding to the euphoric aspects of ethanol consumption balanced by acetaldehyde toxicity due to differences in liver metabolism. The following chapters bring

together the results of this thesis with other research being conducted by our lab as well as discussing the wider implications of our findings.

Chapter 6: General Discussion

6.1 Thesis Overview

This thesis investigated the intergenerational and transgenerational effects of paternal alcohol drinking in a rat model with a focus on developmental and behavioural changes. In addition, we investigated changes brought about in fathers due to ethanol drinking. The study found clear behavioural differences in the offspring of drinkers which were transgenerationally inherited through the male germline (Table 6.1). In addition, some of these changes were sex specific. Lastly, we investigated the livers of the F0 rats and found ethanol drinking males had decreased mitochondrial DNA copy number (mtDNA-CN), a marker of mitochondrial activity, altered expression of *Pgc1a*, a marker of mitogenesis, and altered expression of *Mecp2*, which is involved in DNA methylation (Table 6.1). Differences in mtDNA-CN were seen in the F1 generation suggesting the behavioural changes seen may be a result of the biological changes brought about by paternal drinking.

The thesis demonstrates that paternal preconceptual alcohol consumption causes behavioural and biological changes in the offspring and these behavioural changes persist for another generation suggesting the effects of PPAC are truly transgenerational. This study stands apart in a few aspects, first, we use a free choice intermittent ethanol access paradigm which tries to better mimic human patterns of ethanol consumption (Spoelder et al., 2015; Beeler et al., 2019; Hussain et al., 2022) than most paternal ethanol studies which have used forced ethanol exposure (Finegersh & Homanics, 2014, Rompala et al., 2017, Romapala et al., 2018; Nieto et al., 2022). This additionally allows us to avoid the stress associated with forced feeding paradigms while also allowing us to look at potential dose dependent effects of PPAC. Next, the

males were kept for a full spermatogenesis cycle before mating which has been done by few studies (Neito et al., 2022; Hussain et al., 2022) and allows us to understand the long-term effect of PPAC on the germline. Lastly, the study found true transgenerational effects of PPAC on behaviour which, to our knowledge, have not been done before. While studies have indicated potential effects (Abel & Lee, 1988, Cambiasso et al., 2022) or looked at developmental signs like birthweights and litter effects (Stockard & Papanicolaou, 1916 a, b) none have looked at behaviours one generation removed from chronic paternal alcohol use. In this regard, the thesis provides evidence supporting the impacts of PPAC while providing new insights into its transgenerational effects. Expanding on the key findings from this thesis, we will place them in context of other work in our lab and the field of preconceptual alcohol consumption.

Experiment	Generation	Effects			Description
		PPAC	Sex	PPAC*Sex interaction	
Developmental milestones	F1	Yes	N/A	N/A	Sig. delay in pinnae detachment, mean delays in other milestones, increased litter variance and pup deaths in D ₁
	F2	Yes	N/A	N/A	Sig. delay in coat growth, eye opening and pup deaths, mean delays in other milestones and increased litter variance in D ₂
Surface righting	F1	No	No	Yes	At PND2, D ₁ males achieved surface righting faster than other groups.
	F2	No	No	No	No differences
Locomotor Activity	F1	No*	Yes	No	The D ₁ M reared less than the other conditions, Females show greater locomotor activity than males.
	F2	No	No	No	No differences

Accelerating Rotarod	F1	Yes	No*	Yes	ND ₁ rats perform better than D ₁ under saline but D ₁ perform better when injected 1g/kg ethanol. Females perform better than males on average.
	F2	Yes	Yes	Yes	The differences are seen due to D ₂ F performing better than the other groups
Ethanol Consumption	F1	No*	Yes	Yes	The D ₁ F showed reduced drinking compared to the ND ₁ F. Females drank more than males
	F2	Yes	Yes	Yes	The D ₂ F showed reduced drinking compared to the ND ₂ F. Females drank more than males
MtDNA-CN	F1	Yes	No	No	D ₁ rats had a lower abundance of mtDNA-CN than the ND ₁ rats after ethanol intake.
Experiment	Generation	Effects			Description
		Ethanol exposure	Sex	Ethanol*Sex interaction	
MtDNA-CN	F0	Yes	No*	Yes	D rats had lower mtDNA-CN than ND. ND ₀ F had the highest mtDNA-CN and D ₀ F had the lowest.
MtDNA-CN	F0	Yes	N/A	N/A	D ₀ 90 and D ₀ 180 both show lower mtDNA-CN than the ND ₀
Pgc1a	F0	Yes	N/A	N/A	D ₀ 90 has lower Pgc1a expression than ND ₀ and D ₀ 180.
Mecp2	F0	Yes	N/A	N/A	D ₀ 180 shows higher Mecp2 expression than D ₀ 90 and ND ₀

Table 6.1. Summary of key findings from the study. * next to a “No” denotes trends which did not reach significance and N/A denote not applicable. Abbreviations: F0/sire condition: D (drinker), ND (non-drinker), F (female), M (male), subscripts denote generation of the rats

6.2 Sexually Dimorphic Effects of Ethanol

Even before assessing PPAC, sex differences in rodent motor behaviour have been well documented with studies showing females having higher locomotor activity (Hyde & Jerussi, 1983; Simpson et al., 2012) and performing better than males on motor coordination tasks (Hernandez et al., 2020). Likewise, several other studies report sex-dependent differences in drinking behaviours with females consuming higher amounts of ethanol per bodyweight (Hussain et al., 2022; Lancaster et al., 1996; Randall, P. A. et al., 2017), showing greater preferences for ethanol (Lancaster & Spiegel, 1992; Pirino et al., 2022; Priddy et al., 2017) and developing drinking behaviour faster than males (Melón et al., 2013; Moore, C. F. & Lynch, 2015). Our study demonstrates clear sex differences in certain behaviours; females show higher locomotor activity in terms of distance travelled and rearing when compared to males and perform better in the rotarod task. When looking at the acute effects of ethanol, ND₁ females performed better than males on the rotarod task when injected with saline but showed a greater drop in performance when injected with ethanol, additionally, the ND₁ females showed a greater preference for ethanol and had a higher g/kg ethanol intake than their matched males. These are consistent with studies showing sex-dependent motor impairments due to ethanol (Hussain et al., 2022; Vetter-O'Hagen et al., 2009).

One of the reasons for this dimorphism could be the difference in alcohol metabolism with rodent studies showing females having similar blood ethanol concentrations despite greater ethanol intake (Pirino et al., 2022; Randall, P. A. et al., 2017) compared to males. This is explained by females showing higher maximum blood ethanol concentration peaks and faster elimination rates (Crippens et al., 1999; Thomasson, 1995). These are mirrored in humans with

women showing greater clearance of ethanol per lean body mass (Kwo et al., 1998) but having significantly lower lean body mass than males in addition to significantly slower gastric alcohol metabolism with lesser alcohol dehydrogenase (ADH) activity compared to men (Baraona et al., 2001). Another study showed that blood alcohol was similar between men and women for intravenous injections, but women had higher blood concentrations when alcohol was taken orally suggesting the role of lower gastric ADH and not liver activity for the enhanced vulnerability of women to acute and chronic alcohol complications (Frezza et al., 1990). We also conducted preliminary work on Aldehyde dehydrogenase (ALDH) activity in the F0 livers and found lower ALDH activity in females (Appendix C) suggesting a higher build-up of acetaldehyde in the liver. Low ALDH activity causes an excess of acetaldehyde which leads to higher oxidative stress and damage (Ohta et al., 2004). The adverse effects brought about by acetaldehyde build up act to prevent further alcohol intake (Quertemont & Didone, 2006). However, we found key sex-based differences in liver function with the mtDNA-CN in the F0 livers in the females being higher than the males. This is not just specific to the liver with mtDNA-CN and mitochondrial respiratory complex gene expression in the frontal cortex and cerebellum being sexually dimorphic (Thorne et al., 2022a). This increased mitochondrial function could be part of the mechanisms which compensate for the lower ADH/ALDH activity and lead to the faster ethanol elimination rates we see in females.

Lastly, females and males are also shown to have different neurobiology in response to alcohol (Flores-Bonilla & Richardson, 2020). For example, dopamine release patterns in response to ethanol in the NAc is sex dependent with females showing significant increases in extracellular dopamine at lower ethanol doses and a return to baseline levels at high doses

(Blanchard et al., 1993). In contrast, males show slight increases in extracellular dopamine at low doses of ethanol but decrease in dopamine levels at high doses (Blanchard et al., 1993). Females therefore require lower ethanol doses than males for reinforcement (Hauser et al., 2019) and estradiol appears to enhance the rewarding effects of alcohol (Hilderbrand & Lasek, 2018). Taken together with the metabolic differences, these different mechanisms combine to influence the higher ethanol drinking rate we see in females.

While we have mentioned baseline differences and differences due to ethanol exposure between males and females, it is important to note we also found sexually dimorphic behaviours in offspring of PPAC. In our study, these effects were seen in early surface righting, post-weaning locomotor activity, ethanol induced motor coordination performance and ethanol drinking (Chapters 3 and 4). Additionally, these sex effects in rotarod performance and ethanol drinking were transgenerational as they were seen in the F2 generation (Chapter 4) and we discuss these below.

6.3 Impacts of PPAC on Birth and Early Development

Multiple studies have shown the impact of PatEE on litter characteristics and early development. Although we did not find significant differences in average litter size, we did find a larger number of pre-weaning deaths (deaths in the first 3 weeks) in drinker offspring compared to non-drinkers which is consistent with some reports (Abel, 1989; Rathod et al., 2020; Stockard & Papanicolaou, 1916b). We failed to see any differences in sex ratios between the ethanol exposure conditions which was contrary to some studies (Abel, 1995; Cicero et al., 1990; Emanuele et al., 2001; Stockard & Papanicolaou, 1916b). While we did not measure litter weights at birth, we found no differences in offspring weights on PND 2 and PND 4 which adds

to the already mixed reporting which is seen around this. Studies of litter weights due to PatEE are mixed with some studies reporting the reduction of weights in ethanol exposed offspring (Bielawski & Abel, 1997; Bielawski et al., 2002; Ceccanti et al., 2016; Chang et al., 2019; Chang et al., 2017; Ledig et al., 1998; Meek et al., 2007) and others reporting no significant differences (Abel & Lee, 1988; Liang et al., 2014; Randall, C. L. et al., 1982; Rompala et al., 2016). One reason for this could be that studies find increased incidents of runts in drinker offspring litters despite overall litter weights being the same (Abel, 1993; Abel, 1995; Asimes, 2018; Rathod et al., 2020). These may play a role in the mixed consensus on birthweights depending on how the pup weights are measured because a litter which has more runts can also have a few larger pups which make up the average litter weight.

Additionally, smaller litters tend to have larger pups due to an abundance of maternal resources which can also affect such calculations. Although this study attempted to minimize these by reducing litters to a maximum size, we did have litters with fewer pups which were heavier than those from the larger litters (Chapter 3). The study also measured only one male and female pup from each litter at the timepoints as a representative for the litter, this did not allow us to assess the number of runts per litter, however, the measures we did use still indicate an influence of PPAC on early litter development. We did look at litter size based on drinking and while there were no significant effects, the sires with the highest ethanol intakes had either the largest or the smallest litters suggesting the increased variance in litter sizes might have to do with amounts of ethanol consumed (Appendix D). These suggest a need to further investigate the role paternal drinking plays in embryonic development.

Early developmental milestones like eye opening and pinnae detachment were reached later in the D₁ when compared to the ND₁ rats indicating deficits in neurodevelopment mirroring those reports found by others (Meek et al., 2007). These early developmental deficits are a good measure of neocortical development and the delay seen fits well with research showing abnormal gene expression patterns in the neocortex of offspring of ethanol exposed male mice (Bottom, 2021; Conner et al., 2020). However, these studies either use acute ethanol administration or use chronic ethanol exposure until the time of mating. One study which had a 2-week break between ethanol exposure and mating did not find differences in offspring birth weights and early development but did see behavioural differences in adulthood (Wozniak et al., 1991). Our study shows that early developmental changes due to PPAC can be seen even if the ethanol exposure is stopped about 8-weeks prior to mating. Future research on tissue samples collected from PND 2 and PND 4 pups will hopefully elucidate other early developmental differences seen due to PPAC.

6.4 Differences in Behavioural Phenotypes linked to PPAC

6.4.1 PPAC and locomotor activity and coordination

Most PatEE research has focused on differences in behaviours in adulthood. Here we found behavioural changes in the offspring of drinkers in tests ranging from early development to adolescence and into adulthood. The surface righting reflex showed a sexually dimorphic difference with the male D₁ pups at PND 2 having a faster surface righting reflex compared to the ND₁ which disappeared at PND 4 (Chapter 3). At PND 24, D₁ males showed lower rates of rearing in the locomotor activity test (Chapter 4). Apart from the differences in activity based on sex, motor coordination in the F1 showed the D₁ rats performed poorer than the ND₁ on the

rotarod at saline indicating reduction in baseline motor coordination. The earlier surface righting coupled with poorer rotarod performance is in line with research which found that pups born to sires who had stopped alcohol consumption prior to mating showed earlier attainment of surface righting than controls but also performed worse on balance and motor coordination tasks (Conner et al., 2020; Jamerson et al., 2004). This is thought to be linked to thickening of cortical layers found in pups of ethanol exposed rats (Jamerson et al., 2004)

Multiple studies have found PPAC playing a role in altered locomotor activity at different ages of offspring development (Abel, 1993; Ledig et al., 1998). While our findings replicate those of Beeler et al., (2019) who also found no difference in total distance, we did see reduced rearing in ethanol sired male offspring. Some studies have seen increased locomotor activity in offspring of paternal ethanol exposed rats and have linked it ADHD like hyperactivity behaviours (Conner et al., 2020; Kim, P. et al., 2014). There is some clinical evidence to support this with children of alcoholic fathers showing higher incidence rates of ADHD (Han, J. et al., 2015; Knopik et al., 2009) . However, both animal and human studies show no clear unified phenotype of PPAC with one rat study showing hypoactivity in offspring of ethanol sired rats (Abel, 1989) and a human twin study stating associations between paternal drinking and offspring ADHD depended on broad definitions of ADHD symptoms (Knopik et al., 2009). While the effects of PatEE on locomotor activity have been inconsistent, multiple studies which have found no differences in open field tasks have seen the effects of PatEE on tests like radial maze for spatial discrimination, rotarod for motor coordination, elevated plus maze for anxiety like and the forced swim test for depression like behaviours (Finegersh & Homanics, 2014; Nieto et al., 2022; Wozniak et al., 1991).

6.4.2. Sex x PPAC interactions in depression and anxiety behaviours

In addition to work mentioned in the thesis, our lab looked at behaviours related to anxiety and depression in the F1 offspring through the affective disorders test (ADT, Appendix E). The ADT is an adapted version of the successive alleys test (SAT) which provides a measure of anxiety and integrates a food reward which allows for the measurement of anticipatory pleasure (Blackburne, 2021). Analysis so far has shown no differences between the D₁ and ND₁ in terms of anxiety behaviours but there are sex x condition interactions in measured anticipatory behaviour with D₁ females showing higher rearing, indicating more pronounced anticipation than ND₁ females and D₁ males showing lower rearing than ND₁ males (Appendix E), indicating sexually dimorphic effects of PPAC on anticipatory (depression-related) behaviour. Taken together, the project adds to the multitude of research suggesting offspring of ethanol exposed sires are more likely to develop behavioural deficits in adulthood (Abel & Lee, 1988; Beeler et al., 2019; Ledig et al., 1998; Rompala et al., 2017).

Studies in humans found that paternal AUD was associated with increased rates of anxiety and depression symptoms particularly in the male offspring (Mahesh et al., 2021). While our lab found no differences in anxiety like behaviours in the ADT, D₁ males showed a reduction in rearing behaviours which is taken as an indication of reduced anticipatory behaviour, a symptom of depression. This is in line with other rodent studies that reported effects of PatEE on offspring depression although the differences were also seen in the female offspring (Kim, P. et al., 2014; Liang et al., 2014). On the other hand, the relationship of PatEE and anxiety disorders appears to be complex with some studies reporting no differences in anxiety-like behaviours (Conner et al., 2020; Nieto et al., 2022; Rompala et al., 2017) while others seeing a reduction of

anxiety-like behaviours in male offspring of PPAC in open field tests (Beeler et al., 2019; Koabel et al., 2021). Again, multiple factors, including sex, strain, treatment type, length, dosage and study design may have contributed to these divergent results.

6.4.3 PPAC causes differences in ethanol sensitivity

One of the strongest effects of this study was the impact of PPAC on offspring ethanol sensitivity. The offspring of drinkers showed a blunted response of ethanol on motor coordination compared to offspring of non-drinking sires. This is in line with previous studies which have seen a reduction in ethanol sensitivity in offspring of ethanol sired animals on the rotarod (Finegersh & Homanics, 2014; Hussain et al., 2022). These findings align with human studies which suggest reduced sensitivity to alcohol seen in sons and daughters of drinkers which may contribute to an enhanced risk of alcoholism (Eng et al., 2005; Schuckit, 1988). Not all studies report similar rotarod results, one found no differences in rotarod performance but observed quicker recovery of ethanol sired male offspring from acute ethanol injections when compared to controls (Rathod et al., 2020). Other studies also show sex-based differences in ethanol sensitivity with no difference in performance of female offspring from controls (Beeler et al., 2019; Nelson et al., 1988; Nieto et al., 2022; Rompala et al., 2017).

The PPAC again showed sexually dimorphic effects in tests of ethanol consumption (Chapter 4). We found female offspring of drinkers showed reduced ethanol consumption and reduced preference when compared to offspring of non-drinkers. However, these effects were not seen in the males. The opposite is seen in mice studies where male offspring of alcohol-exposed sires that show reduced alcohol intake (Finegersh & Homanics, 2014; Rompala et al., 2017). Again, only male offspring of ethanol exposed sires showed a decreased preference to the

rewarding effects of ethanol at high doses in conditioned place aversion and context induced relapse tasks (Campbell et al., 2018; Ceccanti et al., 2016). However, like our study, some did see ethanol-sired females having reduced drinking compared to controls (Hussain et al., 2022; Rathod et al., 2020).

The differences seen here may be attributed to a few factors. First, the various animals and strains used along with the different lengths of paternal exposures and ages at which the drinking and rotarod tests were carried out can all influence outcomes. Example, the above-mentioned studies showing reduced ethanol consumption in males only were done in mice. Second, and more interestingly, there appear to be clear sexually dimorphic effects in ethanol sensitivity due to PPAC interacting with the existing pre-established differences in ethanol metabolizing pathways and reward mechanisms between males and females. The reduced sensitivity to the motor impairing effects of ethanol was seen in both male and female D₁ rats but the reduced drinking only in females may indicate certain pathways may be affected in both sexes while other changes due to PPAC might be sex specific.

The reduced ethanol intake of the D₁ females could instead be an indication of an enhancement of the rewarding properties of ethanol. This is because females have already shown heightened response to ethanol in dopamine pathways due to ethanol when compared to male (Blanchard et al., 1993; Blanchard & Glick, 2002). In addition, serotonin pathways have also been linked to reward and addiction (Liu, Z. et al., 2020). Serotonin knockout models of ethanol show sexually dimorphic interactions with ethanol with females with reduced SERT activity consuming significantly more ethanol than the wild-type SERT while this effect not being seen in the males (Hussain et al., 2022). Additionally, SERT appears to modulate mtDNA-CN and

adenosine triphosphate production in a sex dependent manner in the frontal cortex but not in the cerebellum of rats with reduced SERT male rats showing higher expressions of Complex I subunit mt- Nd1 but reduced SERT females showing a reduction in expression (Thorne et al., 2022a). Taken together, PPAC might affect these reward pathways differently in the frontal cortex leading to lower ethanol intake for achievement of the same reward in females and therefore reduced drinking. At the same time, the similar effects seen in F1 ethanol induced motor coordination in both sexes may be due to PPAC having similar impacts on the cerebellum, which is a key region for motor coordination (Manto et al., 2012), in both sexes.

6.5 Behavioural and Developmental Effects of PPAC are Transgenerational

One of the most unique aspects of this thesis was identifying true transgenerational effects of PPAC. The study showed that D₂ litters having slower development, larger litter size variances and greater pup death in early development (Chapter 3). These initial changes were followed by differences in effects on motor coordination induced by ethanol and ethanol drinking which indicated the ethanol sensitivity differences due to D₀ drinking persist past the F1 generation. Interestingly, these differences are sex specific with the male D₂ rats performing no different to the controls on all tests where sex was looked at as a variable. In addition to the greater variance in litter sizes, we see an association of F₀ drinking with F₂ litter sizes where higher drinking D₀ had larger D₂ litters (Appendix D). These differences support previous work which indicates the persistence of litter effects and early development (Stockard & Papanicolaou, 1916a; Stockard & Papanicolaou, 1916b) but to the best of our knowledge, no study has looked at the transgenerational effects of preconceptual paternal ethanol consumption. Govorko et al (2012) have shown that changes due to FASD can be passed down through the male germline

and human studies have seen grandmothers of children with partial foetal alcohol syndrome having histories of alcohol abuse, suggesting transgenerational effects of FASDs (Kvigne et al., 2008). While research on paternal transgenerational effects due to alcohol is scarce, there has been a substantial amount of work looking at the transgenerational effects of paternal drug use in nicotine, opiates and cocaine (Goldberg & Gould, 2019). For example, one study on heroin exposure in Sprague Dawley sires saw both F1 and F2 male offspring had increased anxiety-like behaviours as well as increased aggression (Farah Naquiah et al., 2016). Another study on paternal cocaine exposure also found sensitivity differences in the F2 generations and saw an increase in cocaine intake behaviour (Le et al., 2017). However, both these studies only focussed on male offspring which hides any potential sex differences like the ones we saw in our study. This is important as the studies which have looked at the transgenerational impact of paternal drug exposure on both sexes and have found sexually dimorphic changes in the F1 and F2 generations (Goldberg et al., 2021; McCarthy, D. M. et al., 2018; Vyssotski, 2011). Human studies have long identified trends of transgenerational epigenetic inheritance which means specific rodent studies are required to identify the underlying mechanisms especially since reports in transgenerational studies on diet have found persistence of behavioural phenotype changes despite the epigenetic markers being wiped out (Bottom et al., 2022; Gapp et al., 2014; Radford et al., 2014).

6.6 Ethanol Consumption Impacts on the F0 and F1 liver

6.6.1 Mitochondrial function and regeneration

The reduced mtDNA-CN in F0 drinkers immediately after the end of drinking and even 3 months after the final drinking day shows the impact of ethanol on liver function (Figure 5.1).

This is in line with studies which show alcohol reduces the regenerative properties of the liver (Larosche et al., 2010; Mansouri et al., 1999). We saw a return of *Pgc1a* expression levels similar to that of the non-drinkers in rats which had stopped drinking 3 months prior, which indicated a return to non-drinker levels of mitochondrial biogenesis (Chapter 5).

The two processes however clustered based on drinking condition, where lower *Pgc1a* expression was associated with lower mtDNA-CN in drinking animals but not in the non-drinkers again indicating the influence of ethanol on mediating liver mitochondrial function (Albano, 2006; Han, D. et al., 2017; Morio et al., 2021). However, we do not see a return to non-drinker levels of mtDNA-CN 3 months later which suggests a lag between mitochondrial biogenesis and return normal mtDNA-CN which could be part of the slower liver regeneration mentioned above. One hypothesis for this could be based on two points. First, chronic ethanol exposure causes a reduction in mtDNA-CN in the liver as seen by our data and next, liver regeneration is dependent on proliferation of mature hepatocytes (Michalopoulos, 2017). If these hepatocytes have seen a reduction in mtDNA-CN due to ethanol, then the proliferated hepatocytes in regeneration would also have lower mtDNA-CN despite *Pgc1a* expression returning to control levels. The lower mtDNA-CN would slow the regeneration process and it would take a full turnover of hepatocytes over a much longer period of time to return to control mtDNA-CN levels.

6.6.2 Methylation changes and intergenerational inheritance

The other key finding was the suppression of *Mecp2* expression in D₀90 followed by an overexpression in the D₀180. While *MeCP2* does not directly measure DNA methylation, its key role in the methylation process suggests alterations in global methylation patterns in the liver due

to ethanol. This lines up with other areas also showing altered *MeCP2* expression after ethanol exposure and changes to this regulation after treatment for AUD (Aşçıbaşı et al., 2023; Cui et al., 2020; Krasteva et al., 2020). The changes in *Mecp2* expression raise questions about the long-term alterations of global methylation patterns and how these might play a role in inheritance. We have already seen differences in behavioural phenotypes based on lengths of ethanol exposure and time waited before mating. This would require future focus as altered *MeCP2* expression has been reported in studies looking at intergenerational effects with it being altered in the frontal cortex and striatum of offspring of ethanol exposed rats (Kim, P. et al., 2014) and it being shown to mediate the transgenerational transmission of chronic pain (Tao, W. et al., 2020).

Clear differences in the mtDNA-CN of the D₁ rats after ethanol consumption were also observed, suggesting epigenetic inheritance of mtDNA-CN in the liver due to PPAC. Our study did not look at *MeCP2* expression in the F1 generation, in contrast, Kim et al. (2014) found *MeCP2* changes in the offspring but did not look at it in the sires. This makes it hard to ascertain whether the changes seen in mitochondrial function are regulated through methylation status or through other mechanisms like ROS damage. The differences seen in the F1 mtDNA-CN may indicate a role for methylation, while the immediate decrease in mtDNA-CN and Pgc1a expression can be due to metabolic challenges, the difference in mtDNA-CN in the F1s suggests epigenetic regulation.

Continuing with the sexually dimorphic effects seen in PPAC, we also see opposite trends of mtDNA-CN in the offspring when compared to the D₀ mtDNA-CN. Higher sire mtDNA-CN is associated with higher mtDNA-CN in the male offspring, but the opposite trend is seen in the female offspring with higher mtDNA-CN in the sires linked to lower F1 mtDNA-CN.

This becomes interesting when looking at drinking behaviour where we saw trends of higher drinking in F0s showing greater reduction of drinking in the F1 females. So not only is ethanol reducing mtDNA-CN in the F0, but we can also see indications of these being linked to offspring drinking behaviours where D₁ females showed an overall reduction of drinking. It is important to stress that we only saw trends, and these were not significant associations. What these do indicate, are potential dose-response effects of the amount of ethanol consumed by fathers in the offspring. Such trends have recently been reported in humans (Luan et al., 2022) and are not dissimilar to studies looking at PPAC at high and low doses of ethanol consumption (Abel, 1989). While there are likely multiple factors involved in mediating this relationship, in hindsight the thesis would have selected a wider range of drinking as opposed to the top 10 as this would have allowed for greater variance which may have helped in assessing dose-response effects.

One other interesting element we see from the thesis is that the epigenetic influence of paternal stress to the liver through alcohol is related to hepatic mtDNA-CN reductions in the offspring. Similarly, one previous study on paternal stress liver damage caused through benzo[a]pyrene found reduced mtDNA-CN in the F1 offspring liver (Godschalk et al., 2018). This suggests liver mitochondria are susceptible to regulation through an unknown epigenetic mechanism and raises wider questions on whether any genotoxic stress induced liver damage has an epigenetic component of inheritance. For example, as hepatic infections such as hepatitis B are known to cause damage to mitochondrial DNA in the liver (Hossain et al., 2020) would the offspring of individuals having suffered from it inherit changes in liver function? While we could not find research which directly answers that question, there is evidence to show that male rats

with a history of liver damage transmit epigenetic suppressive adaptation of the fibrogenic component of wound-healing through the male germline (Zeybel et al., 2012) and another study which showed environmental stress can lead to alterations in hepatic glucose metabolism in offspring (Wu et al., 2016). Taken together, these suggest stress acting upon the liver in fathers can pass down through the male germline in both adaptive and maladaptive ways.

6.7 Mechanisms of PPAC

This study, along with others, has shown that alcohol causes changes in those directly exposed to it. Some of these include changes in epigenetic modifications in the liver, blood, and brain regions as well as alterations in reward, HPA and cognitive pathways (Ajoolah et al., 2023; Gangisetty et al., 2019; Krasteva et al., 2020). While the mechanisms are still unclear, the behavioural phenotypes reported in the F1 and F2 generations indicate these changes are more likely to be passed down to the offspring through epigenetic changes rather than genetic mutations. This is because our study found differences between the F1 and F2 in a range of tests with some differences being exacerbated while others no longer being present. This would suggest the absence of clear, consistent point mutations at DNA mutation hotspots. This is further backed up by the variances in results seen when looking at other animal models of paternal exposure (Chapter 1).

In addition to the heterogeneity in outcomes seen, studies have also suggested that methylation levels in some regions due to alcohol cease to exist after a particular period making understanding the duration of these changes important (Wieting et al., 2019). Nevertheless, as we have demonstrated, the effects of PPAC can occur in offspring born to sires who were not

exposed to alcohol for 8 weeks before mating and the next step is to identify the changes which support epigenetic inheritance through the male germline.

6.7.1 Changes in sperm epigenetics

This thesis did not look for epigenetic alterations in the F0 sperm. However, sperm samples of the mated sires were collected and are being analysed by our lab. Preliminary tests focused on the methylation status of CTCF binding sites in the H19/Igf2 imprinting control region and have found methylation differences in 1 of 4 investigated CTCF sites (Davie, unpublished). However, analysis of the whole region by bisulphite sequencing would be required to identify which CpG sequences are differentially methylated. This backs up the body of work reporting methylation changes in imprinted genes in the sperm due to ethanol (Knezovich & Ramsay, 2012; Liang et al., 2014). The lab seeing differences in only 1 site could be for a few reasons. First, spermatozoa collection was only done after the end of mating which meant a further 3 weeks had passed from the last exposure to alcohol, given the additional time from drinking there is a likelihood that the sperm assessed are not like the ones which were part of fertilizing the F1. Second, we could simply be targeting the wrong regions, some studies have not found any methylation changes in this region (Chang et al., 2017; Knezovich & Ramsay, 2012) and another study shows how the CTCF binding sites are dispensable for imprinting establishment in the male germlines (Szabó et al., 2004). Lastly, it may be due to the techniques used when assessing methylation changes. However, the lab has only just begun work in this area and further assessment of the sperm methylation might bring different results as is seen through sperm methylation changes being identified in, DNMTs, *MeCP2*, *BDNF* and *POMC* among others (Bielawski et al., 2002; Finegersh & Homanics, 2014; Govorko et al., 2012).

As mentioned, DNA methylation changes are not the only way epigenetic inheritance is transferred with alcohol affecting sperm. Recent research has identified changes in histone methylation levels in the sperm due to ethanol (Bedi et al., 2022; Cambiasso et al., 2022). Cambiasso et al. (2022) found these changes also in the germline of the male offspring of alcohol drinkers which might explain how behavioural phenotypes can be passed down transgenerationally. Novel next generation sequencing techniques to assess chromatin accessibility in sperm and has revealed unique euchromatic landscapes which are reflected in the embryo (Jung et al., 2017). As techniques improve and the understanding of sperm chromatin profiles increase, we should be better able to examine the effects of alcohol on these epigenetic mechanisms.

Lastly, the work done on sperm and seminal fluid RNA modifications has found crucial roles played by them in epigenetic regulation. Altered miR10a which is suggested to function as a posttranscriptional regulator for *BDNF* (Jiajie et al., 2017), altered tRNA-Gly-GCC which is associated with controlling retrotransposon-regulated gene networks in the fertilized oocyte (Sharma, U. et al., 2016) and altered ratio of tRNA-derived small RNAs to Piwi-interacting RNAs in sperm which is associated with late term foetal growth restriction (Bedi et al., 2022). Overall, the identified changes across all epigenetic mechanisms seem to play a vital role in prenatal development.

6.7.2 The effect of PPAC on fetoplacental growth

One other phenomenon which has been noted is the impact of PatEE on female fertility and gestation. A study found an increased number of male foetuses in dams pregnant from ethanol exposed sires, however, there were no sex-ratio or litter size differences at birth (Abel,

1995). Abel (1995) also saw a reduction in foetal weights but did not see these weight differences in newborn pups. Heavy PPAC is shown to increase the risk of early pregnancy loss and stillbirths in humans (Henriksen et al., 2004). Other studies report alterations in gestation lengths and increase in placental weights in dams of alcohol sired litters (Chang et al., 2019; Klassen & Persaud, 1976; Meek et al., 2007). While the study here did not look at gestational days, all animals for each generation were mated on the same day and the differences in days of litter birth showed that dams mated with ethanol drinking D₀ and ethanol sired D₁ rats on an average gave birth later than dams mated with ND rats (Appendix F). Ethanol is known to affect sperm density, sperm motility and damage sperm DNA reducing viability which would affect conception and could be one explanation for the delays seen (Dare et al., 2002; Finelli et al., 2021). However, we cannot rule out the possibility of ethanol induced epigenetic changes in the sperm affecting placental function and leading to alterations in the offspring development. While research on the effects of PPAC on the placental function is scarce, studies show paternal drinking is associated with altered transcription of placental mitochondrial genes and placental histology (Thomas et al., 2021; Thomas et al., 2022). There are also reported changes in placental expression of imprinted genes *Ascl2*, *Igf2*, *H19*, *Peg3* and *Slc22a18* (Thomas et al., 2022). This is in line with works which show the effects of paternal influences on placental health and offspring development particularly through altered expression of imprinted genes (Bhadsavle & Golding, 2022).

6.8 Implications

6.8.1 Sex differences

The thesis reiterates the need for the inclusion of both sexes in any neuroscience research, while many of the studies on paternal ethanol exposure mentioned here have looked at sex differences this has historically not been the case (Beery & Zucker, 2011; Thorne et al., 2022b). Human and animal research have shown sex differences in ethanol response with females having higher alcohol preference to ethanol and higher intake of alcohol per body weight (Hussain et al., 2022; Lancaster et al., 1996; Pirino et al., 2022; Priddy et al., 2017). Females acquire ethanol self-administration faster and show a greater preference for ethanol than males (Melón et al., 2013; Moore, C. F. & Lynch, 2015; Torres et al., 2014) and show altered withdrawal symptoms (Marsland et al., 2021; Varlinskaya & Spear, 2004). Women also show a higher risk of alcohol related metabolic damage (Ward & Coutelle, 2003) which is mirrored in the greater reduction in mtDNA-CN after ethanol seen here. Other reports further highlight the differences due to alcohol such as males showing more severe reductions in cortical thickness and reduced myelinated fibre density in the prefrontal cortex, whereas females show more robust decreases in neurogenesis in the hippocampus in response to alcohol (Flores-Bonilla & Richardson, 2020). Several other health consequences are greater in women with AUD as they experience higher risks of developing cancers, alcohol-related liver injury, and cardiovascular disease compared to men with AUD despite comparable levels of drinking (Mann, K. et al., 2005; White, 2020). Binge drinking in women is associated with a higher risk of disease related mortality when compared to men (Schwarzinger et al., 2018).

The sex differences in behaviours and outcomes could be modulated by a variety of functions. First, there are differences in ethanol metabolism with men showing significantly higher ADH and ALDH enzyme activity which results in a much higher bioavailability of

ethanol in women and represents an underlying pharmacokinetic reason for why men tend to cope better with alcohol than women (Farkouh et al., 2020). Differences in reward pathways have been observed with sex chromosome complement independently impacting gene expression of prodynorphin, the endogenous ligand for the kappa opioid receptor, which has been implicated in regulation of dopamine signalling and in alcohol reward in the striatum (Barker & Taylor, 2019). In addition, dopamine pathway differences in the ventral tegmental area (VTA) have been implicated in sex specific cocaine reward. Lastly, genetic sex differences have been observed in BDNF-related genes as well as serotonin pathways which are both implicated in addictions. Other modulating mechanisms could be, differences in amygdala related stress response and hypothalamus driven social behaviour (Flores-Bonilla & Richardson, 2020). Lastly, oestrogen and other endocrine hormones could play a role as studies show ovariectomized females having lower levels of baseline ethanol intake compared to intact females with the effect being reversed with oestradiol replacement (Ford et al., 2002; Ford et al., 2004). While the above works do not focus on intergenerational impacts, these baseline differences need to be taken into consideration when designing future research on PPAC as most of the changes noticed in the offspring were also sexually dimorphic and can be factors as to why we either see protective or detrimental changes in the offspring.

6.8.2 Alcohol metabolism in the liver as a driver for behaviour

One of the key findings of the project so far has been the differences in mtDNA-CN across generations. We did not see as steep a drop in mtDNA-CN due to ethanol in the F1 as we did in the F0 females. This is coupled with reduced sensitivity to the motor impairing effects in females in both the F1 and F2 might indicate a change in metabolism of ethanol due to PPAC.

Based on our findings here and as females show slower alcohol clearance (Soldin & Mattison, 2009), the greater drop in performance in the non-drinkers when injected with ethanol compared to the males makes sense. These effects not being seen in the offspring of drinkers might imply changes in alcohol metabolism rates which also could explain the drop in ethanol consumption in the female offspring. However, we should note a previous study did not find any changes in ethanol in the F1 offspring due to paternal preconceptual ethanol exposure based on blood samples taken 60 and 240 minutes after injection (Finegersh & Homanics, 2014). It is important to consider that the mentioned study was done in mice and while both rats and mice have high rates of ethanol metabolism there are significant factors which could influence metabolism rates, including ages, differences seen between strains within species and differences between rats and mice themselves (Desroches et al., 1995; Eriksson, 1973; Kelly et al., 1987). Additionally, Finegersh and Homanics (2014) looked at clearance 60 minutes and 240 minutes after injection while the differences we see occur in the first 30 minutes. Taken together metabolism might still play a role in the differences seen.

As hypothesized earlier, the reduction in liver function would alter the compensatory mechanisms involved in ethanol clearance in females and as females already show reduced ADH/ALDH activity compared to males, the lower clearance rate of ethanol would lead to ethanol consumption and preference rates dropping due to increased toxicity. These might protect the future generations from alcoholic liver injury and AUD. This fits well the work indicating paternal liver damage is seen to reduce liver function in the offspring (Godschalk et al., 2018; Wu et al., 2016). Male offspring did not show the same drops in ethanol consumption which may be due to no changes in ADH/ALDH clearance being inherited although this would

need to be studied. In addition, epigenetic changes seen in the parent drinkers do not match the changes seen in the offspring (Chang et al., 2017; Knezovich & Ramsay, 2012; Liang et al., 2014) this might imply the changes brought about by PPAC might affect offspring through other unidentified metabolic changes and not just alterations in cortical pathways which has been the focus of most studies.

6.8.3 Sex Differences in epigenetic modifications

Lastly, since previous research has shown that prenatal alcohol exposure can pass down alterations through the male germline (Sarkar, 2016) it would have been interesting to see whether the paternal drinking effects would pass down the female D₁ germline as they were the only ones which showed differences in ethanol drinking behaviour. There is some evidence to suggest there are sex-specific differences in the development of epigenetic marks. In each generation, the epigenome undergoes two major waves of reprogramming, one after fertilization and the other one during germline development (Reik, 2007). One proposed theory is that in male (XY) and female (XX) pre-implantation embryos, sex chromosomes establish sexually dimorphic interactions with the autosomes even before overt differences appear (Engel, 2018). Later, sex hormones act as modulators of epigenetic mechanisms and are shown to be involved in regulation of recruitment of epigenetic modifiers to target genes (Sandovici et al., 2022).

The next phase of epigenetic reprogramming is, during germline development, evidence shows that there are sex-specific signatures during histone remodelling leading to different epigenomic landscapes in males and females (Huang et al., 2021). There are further descriptions of the development of sperm-specific and oocyte specific genomes which can explain why certain changes are only passed through paternal or maternal germlines (Pembrey et al., 2006;

van den Berg & Pinger, 2016) and passed differentially to male and female offspring (Andres et al., 2015; Pembrey et al., 2006; Savva et al., 2021). Acquisition of DNA methylation occurs at different timepoints in germline development. In the female germline, this process takes place after birth while DNA methylation is acquired before birth in the male germline while both appear to be carried out by different mechanisms (Shirane & Lorincz, 2023; Xu et al., 2022).

The different windows and mechanisms provide additional routes for sex-specific effects on the intergenerational programming. This along with the interactions of epigenetic marks with the environment has the potential to induce alterations dependent on parental sex. Therefore, alcohol consumption could potentially be exacerbating sex-related differences in the epigenome contributing to the sexual dimorphism we see.

6.8.4 Implications for humans

When broadening the work on early outcomes of PPAC to human research, PPAC is associated with reports of higher risk of congenital heart disease and ventricle malformations (Steinberger et al., 2002; Zhang, S. et al., 2020) and risk of childhood acute lymphoblastic leukaemia and risk of childhood malignancy and brain tumours (Milne et al., 2013). Early research reports FASDs features in children born to alcoholic fathers (Lemoine et al., 2003) while others indicate low birthweights and differences in anogenital distance (Xia et al., 2018). To counter these, there are reports which have found no effect of paternal drinking on birth weights, cognitive ability, or early motor development in infants (Hutchinson et al., 2019; McCormack et al., 2018; Passaro et al., 1998; Roeleveld et al., 1992). A systematic review highlights the bias and high rates of methodological limitations in human studies of PPAC which might be influencing the results seen (McBride & Johnson, 2016). The mixed results may also

indicate the subtlety in phenotypes due to PPAC and an increased focus in this field coupled with improvement in techniques will help uncover the nuances which may have been missed until now.

These studies indicate the vital role parental environment plays in offspring outcomes. The effects seen in this thesis demonstrate clear effects of paternal alcohol on offspring and the sexual dimorphic nature of it. The sexual dimorphism of outcomes further stresses the need for research to buck the trends of sex bias in neuroscience and biomedical research (Beery & Zucker, 2011). These also imply a father's role in their child's development is more than just the genes passed down and the social interactions after birth. With epigenetic changes in imprinted genes leading to a host of problems in future generations (Bajrami & Spiroski, 2016). Future research in this area would lead to a better understanding of the mechanisms underlying this epigenetic inheritance in humans and how to counter it. At the same time, these results indicate the need to update the general awareness about the effects of alcohol on both parents on future generations to assist in things like family planning.

6.9 Limitations and Future Directions

While the wider project is still underway, the thesis has provided a good start for the understanding of transgenerational impacts of PPAC. However, along with its successes and interesting prospects, there are a few methodological limitations which the thesis has revealed.

6.9.1 Intermittent ethanol access and isolation

The first issue appears with the intermittent access 2 bottle choice drinking paradigm used for ethanol developing ethanol consumption in the F0. The method by design required

isolation of animals for 8-weeks of drinking. Rats are social animals, and like humans, social isolation increases anxiety and depression-like behaviours in rats (Kinley et al., 2021; Kokare et al., 2010; Novoa et al., 2021). Isolating rats during adolescence is associated with depression symptoms which do not resolve when animals are reunited later (Shetty & Sadananda, 2017). Importantly, isolation in rats during adolescence increases alcohol consumption (Lesscher et al., 2015). As such, isolating rats during adolescence for the drinking paradigm for extended periods might not purely be measuring changes due to ethanol intake. A recent study by our lab supports this, as a modified version of the IA2BC was used where animals were kept together on non-drinking days (Bradley, unpublished). The results showed reduced drinking in the paired housing cohort compared to the isolated group suggesting an effect of isolation on drinking behaviour. This becomes increasingly important in epigenetic research as social isolation is shown to cause epigenetic changes (Li, M. et al., 2016) which might influence the effects seen across generations. Maternal social isolation is shown to impact the blood plasma metabolome even up to the F3 offspring (Heynen et al., 2022). It is important to keep in mind that in the present thesis both controls and drinker F0 males were isolated for a prolonged period, and thus potential isolation stress induced changes would be seen in both groups. It is also possible that the isolation stress interacted with the effects induced by high ethanol consumption. Importantly, our lab's work on isolated vs paired drinking has shown that those in the paired condition still show elevated levels of drinking which means the effects seen in this thesis might still persist. Future research looking into paternal alcohol consumption through drinking could use this updated protocol to account for the effects of social isolation on the epigenome.

6.9.2 Assessment of dose-response relationship of paternal ethanol exposure

Another reason for the use of the IA2BC was for looking at a dose response relationship between ethanol drinking in the fathers and the tested outcomes. However, we selected the highest drinking rats for mating and results have shown trends but no clear relationships between F0 ethanol consumption and measures such as F0 mtDNA-CN and F1 ethanol consumption. While trends are a good start, it is important to see significant effects to make inferences about the dose-response effects of PPAC. It would have been better to either use a wider distribution among the drinking animals, although this would increase the risk of not finding any significant effects. As all animals did not mate successfully, using a larger sample size would have mitigated some of the sample effects we saw as we moved down the generations.

6.9.3 Use of adolescent rats at time of first ethanol exposure

One thing which needs to be addressed is that the F0 rats were adolescents at the start of the drinking paradigm. Apart from the earlier mentioned intake differences between adults and adolescents, ethanol exposure in adolescence appears to have long lasting effects after the discontinuation of exposure (Broadwater & Spear, 2013; Evrard et al., 2006). This could mean the influence of age at ethanol exposure in sires may impact cross generational inheritance. There is already evidence which indicates sperm motility and seminal fluid composition changes with age (Mukhopadhyay et al., 2010; Nieschlag et al., 1982) but there seems to be no impact on DNA fragmentation, chromosome aberrations or fertility (Brahem et al., 2011; Carlsen et al., 2005). Previous research on paternal ethanol exposure (as described in Chapter 1) has been heterogenous in sire exposure ages and there is no clear evidence of it being a factor. However, ethanol exposure length, ethanol dose, animal strain and duration between final exposure and mating appear to play more of a role in the differences seen. Separate research focussing on the

specific differences in intergenerational effects due to age of sire at ethanol exposure should be conducted to elucidate these differences, but this was beyond the scope of the current project.

6.9.4 Dependence on proxy measures

The molecular analysis in this thesis heavily relies on proxy measurements for different mechanisms. As mentioned earlier, mtDNA-CN does not have a 1:1 relationship with mitochondrial function or mitochondrial mass (Picard, 2021; Thorne, 2023; Van den Bogert et al., 1993) and while it has been shown to be a useful measure when comparing the same tissue sample across treatment conditions it does not provide the entire picture. The same can be said about for the use of *Mecp2* expression as an indicator of the methylation status of DNA (Kyle et al., 2016; Yang et al., 2016), as it gives us an idea of potential methylation changes but does not directly measure DNA methylation status, nor the sites of these methylation changes. A genome wide methylation analysis would be a better way to fully assess potential epigenetic changes due to ethanol.

Another issue mentioned before is with the sample collection in the offspring. Liver samples were only collected in the F1 after they had gone through the drinking paradigm which means all the F1 results are affected by ethanol. This prevents us from assessing valuable information in terms of differences in baseline liver function between the F0 and F1. At the same time, the study did not look at *Mecp2* and *Pgc1a* expression the F1 livers and although affected by drinking it would have been interesting to see whether the expression status was different in the offspring. One of the main reasons for this error was the focus on brain regions at the start of the project, as such brain tissue from animals have been collected which are unaffected by ethanol drinking and will be analysed in the future. However, the changes seen in other organs

like the liver appear to be important in understanding the behavioural phenotypes seen in the subsequent generations and as such future work focusing on these metabolic mechanisms and PPAC is necessary.

6.9.5 Assessment of male and female germlines for inheritance

One aspect which has not been spoken about in detail so far is how the maternal germline of PPAC offspring and maternal preconceptual alcohol consumption (MPAC) affect future generations. Our study found differences due to paternal drinking being passed down through the male germline and would be interesting to see whether the effects of PPAC would have also been seen through mating the F1 females. While this was not done, our lab has looked at preconceptual maternal ethanol drinking before with the results showing no differences between controls and MPAC offspring (unpublished data). This could however be due to the small sample size used and future work on MPAC is needed. Available research suggests maternal and paternal alcohol exposure can cause separate effects. While the offspring of adult males exposed to ethanol present with reductions in the junctional zone and increases in the labyrinth layer (Thomas et al., 2022), maternal alcohol exposures promote an expansion of the junctional zone suggesting differences in placental adaptation (Gårdebjer, 2015). Rodent studies have found offspring of preception ethanol exposed females have found increased locomotor activity, increased impulsive behaviour, impaired spatial learning memory and an increase in brain expression of norepinephrine transporter and impaired stress hormone response (Choi et al., 2010; Jabbar et al., 2016). However, reports are mixed in humans with heavy maternal alcohol use before pregnancy being shown to affect vocal development in infants (McDonald & Watson, 2020) but a large-scale study finding no impact of MPAC (McCarthy, F. P. et al., 2013)

Lastly, genome-wide DNA methylation patterns in the hypothalamus of offspring who had either one or both parents being exposed to ethanol preconception and found methylation patterns differed whether it was the mother or father who was exposed (Asimes et al., 2017). While some of these are like the effects seen by paternal alcohol consumption, others affect distinct aspects of behaviour and thus MPAC requires attention independent of prenatal and paternal ethanol exposure.

Transgenerational PPAC inheritance in our study appears to only impact females which makes it imperative to understand the mechanisms involved. It gives rise to questions such as whether the changes seen can pass down the female germline as the F2 males do not present different phenotypes making it unlikely the male germline will pass on any effects to a F3 generation. This is further complicated by evidence showing that epigenetic changes detected in the sperm do not show the same phenotype in the offspring (Gapp et al., 2014; Radford et al., 2014). Another question revolves around whether these mechanisms are protective or detrimental to the female offspring. A starting point to uncover all these questions is to understand the changes occurring in the F0 due to ethanol drinking and working our way through the generations. It would be interesting to follow both maternal and paternal germlines of PPAC and MPAC to see how long their effects last. The study also raises questions about how much time post-abstinence it would take for these changes induced by alcohol consumption to reverse in rats and what the equivalent would be in humans.

With regards to the actual sperm analysis, we allowed one spermatogenesis cycle before mating the F0 animals post ethanol exposure, while other studies have used shorter timeframes. It would be interesting to assess how long these changes are seen for post drinking and assess the

impact of chronic ethanol exposure on sperm progenitor cells and time for the impacts to reverse. While many studies so far have shown epigenetic changes in the sperm, the effect sizes of significant changes in methylation at individual cytosines are often small (e.g., ~10 to 20%) and as such question their significance (Shea et al., 2015). New techniques for loci specific analysis and better understanding of chromatin and non-coding RNAs in the sperm will all be necessary to understand the missing variance in epigenetic heritability.

6.10 Conclusions

This thesis forms the first part of a wider project to understand the transgenerational effects of paternal preconceptual alcohol consumption. The behavioural experiments show clear differences in early development and ethanol sensitivity between offspring of drinkers and non-drinkers. The differences seen persist through the next generation showing clear transgenerational heritability of male alcohol use. Paternal alcohol use also appears to affect offspring in a sexually dimorphic manner and might act as a protective mechanism, particularly in the females. Lastly, we show the effects of alcohol on liver function and suggest that changes in liver due to alcohol consumption are heritable and might drive some of the behavioural differences seen. Given the conflation between inter and transgenerational effects in alcohol research, our findings manage to clarify the differences seen and provide another perspective on the mechanisms driving these changes.

To end, we will revisit the quote at the start of this thesis which asked us to talk about births due to alcohol. While the original intention of the quote may not have been to go into a 150-page doctoral thesis worth of analysis about the subject, us and all the researchers we have

cited here can agree that we should not only talk but aim to further understand all aspects about births due to alcohol. Especially since many of us are born because of it.

Appendix A: Weights During Ethanol Consumption

Analysis of weights in animals during drinking showed no significant differences between paternal drinking condition, however, the F1 and F2 rats from the D₀ line were on average slightly heavier than the ND₀ line.

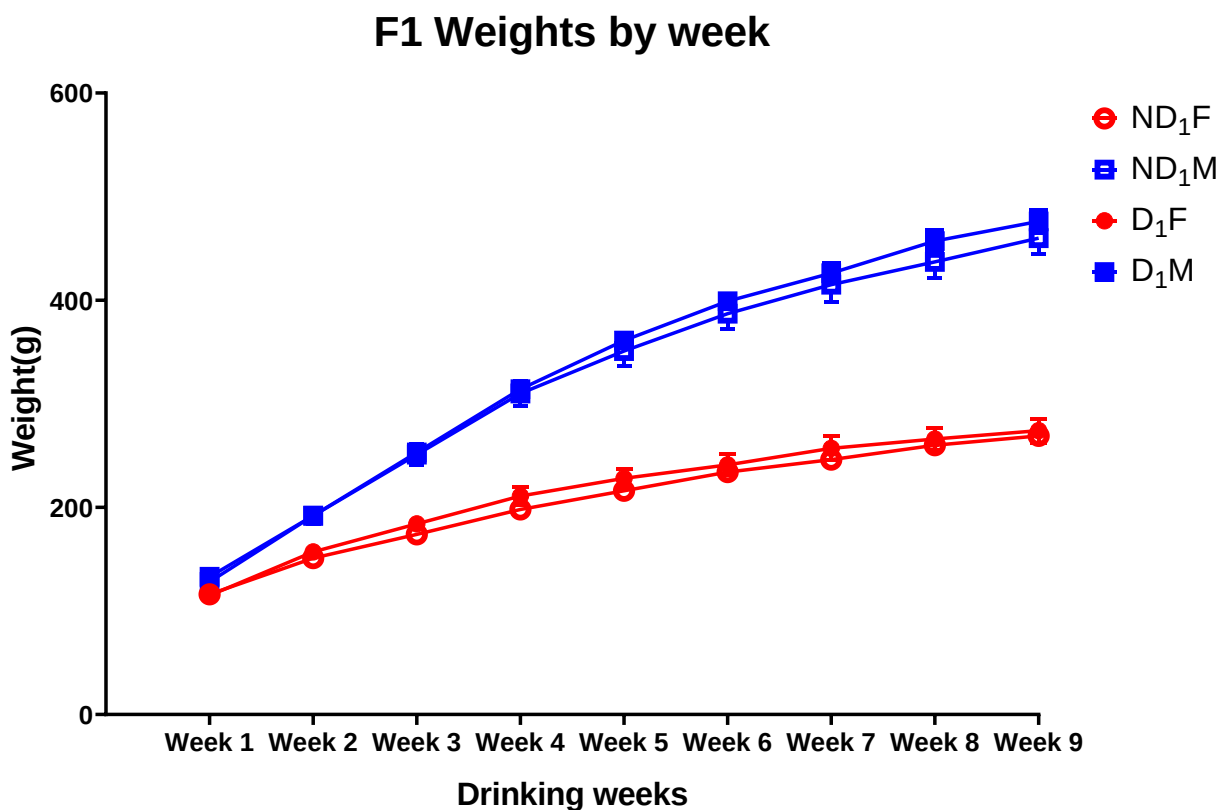


Figure A.1. Weights across drinking weeks F1. Rats started ethanol intake PND 30-35 and carried on for 24 sessions. Weights were measured once a week and animals gained weight overtime ($F(8, 192) = 1893.32, p < .001$). The males weighed more than females ($F(1, 24) = 197.77, p < .001$) and the D₁ offspring appeared to slightly heavier on average as drinking progressed but this was not significant ($F(1, 24) = 0.81, p = .378$).

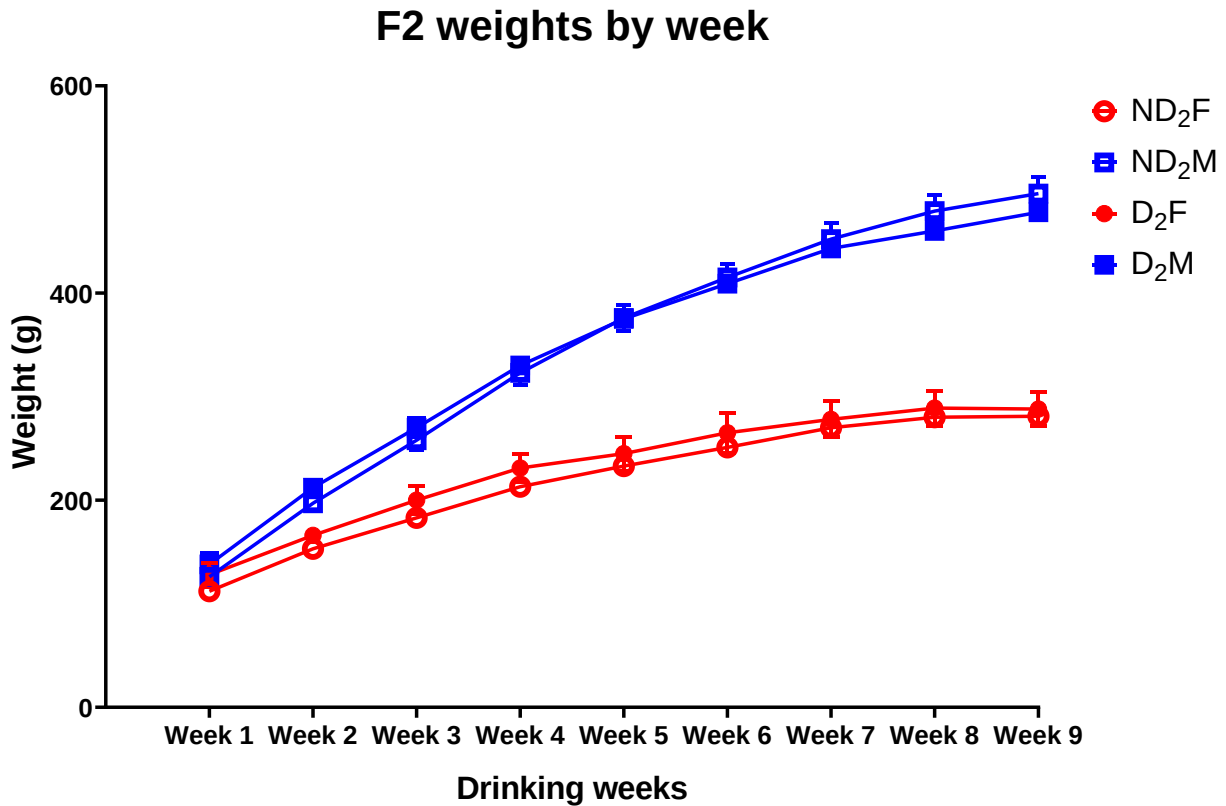


Figure A.2. Weights across drinking weeks F2. Rats started ethanol intake PND 30-35 and carried on for 24 sessions. Weights were measured once a week and there was a significant gain in weight across time ($F(8, 152) = 1504.61, p < .001$). The males weighed more than females ($F(1, 19) = 114.09, p < .001$) and there was a sire condition $\times$ week interaction ($F(8, 152) = 3.15, p = .003$) where the D₁ offspring appeared to slightly heavier on average on certain weeks but there was no main effect of sire drinking condition ($F(1, 19) = 0.29, p = .596$).

Appendix B: Parent Drinking Compared to Offspring

Correlations looking at D_0 drinking compared to D_1 and D_2 offspring. While no correlation was significant the data distributions suggest higher F_0 drinking corresponding to lower drinking in the F_1 females and the F_2 males. But these relationships are unclear and require further research.

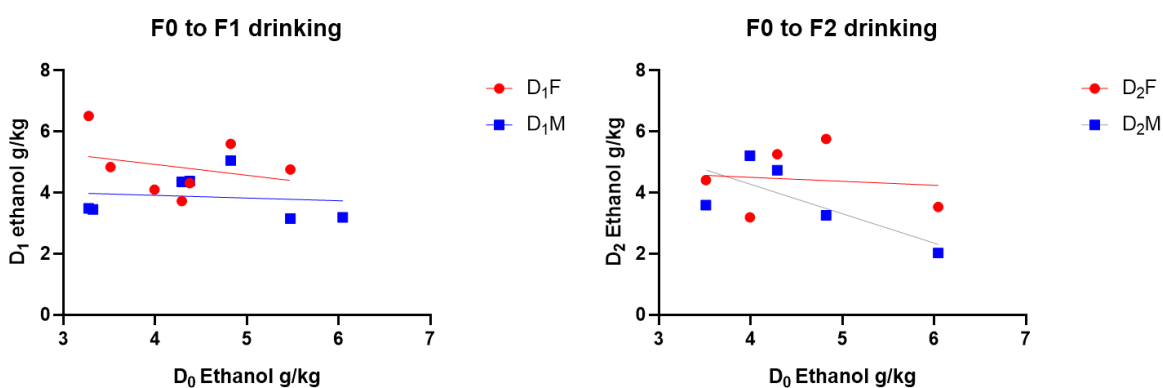


Figure B.1. Parent to offspring drinking. There are no clear associations between parent and offspring drinking in the females. Higher drinking in the D_0 might be associated with lower drinking in the D_1 females and D_2 males, but this needs further research.

Appendix C: ALDH activity

We conducted preliminary work on Aldehyde dehydrogenase (ALDH) activity in the F0 livers. There was significantly lower ALDH activity in females ($F = 10.24$, $p = .008$), suggesting a higher buildup of acetaldehyde in the liver which indicates a difference in oxidative stress and damage seen. This could explain the greater drop in mtDNA-CN in females.

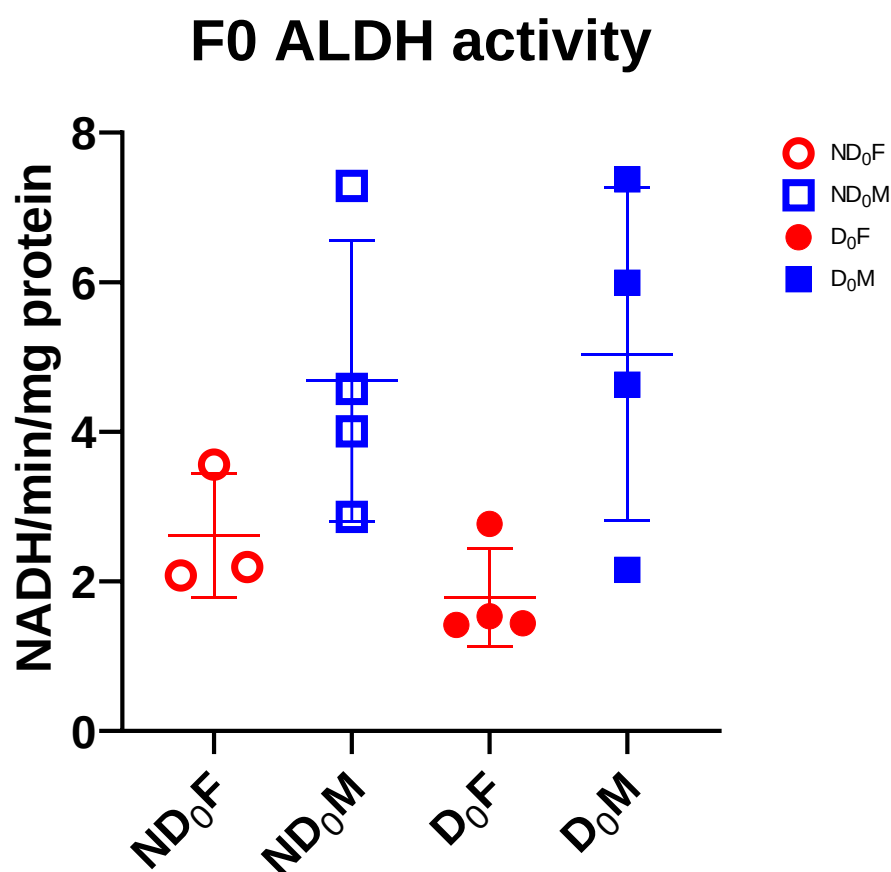


Figure C.1. Aldehyde dehydrogenase activity in the F0. Males show higher rates of acetaldehyde dehydrogenase activity which could play a role in the behavioural differences seen in drinking between males and females.

Appendix D: F0 Drinking to litter sizes

Comparison of F0 offspring to F1 litter size based on drinking show no significant effects ($r = .31, p = .377$). However, the sires with the highest ethanol intakes had either the largest or the smallest litter sizes which suggests the increase in variance seen comes from excess ethanol intake (Figure D.1). In, addition, all pup deaths in the first 3 weeks came from litters with the highest drinking sires. The F2 litters had a significant correlation of higher litters with F0 drinking ($r = .85, p = .024$) the larger litters had more pup deaths too suggest associations with D_0 drinking and F2 litters (Figure D.2).

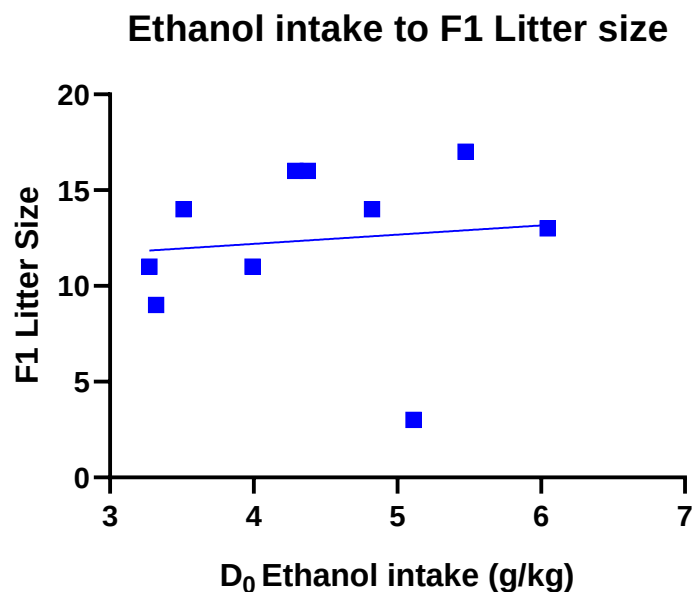


Figure D.1. Comparison of ethanol intake to F1 litter size. As drinking increases, so does the litter variance and likelihood of pup death. No significant correlation between litter size and ethanol intake.

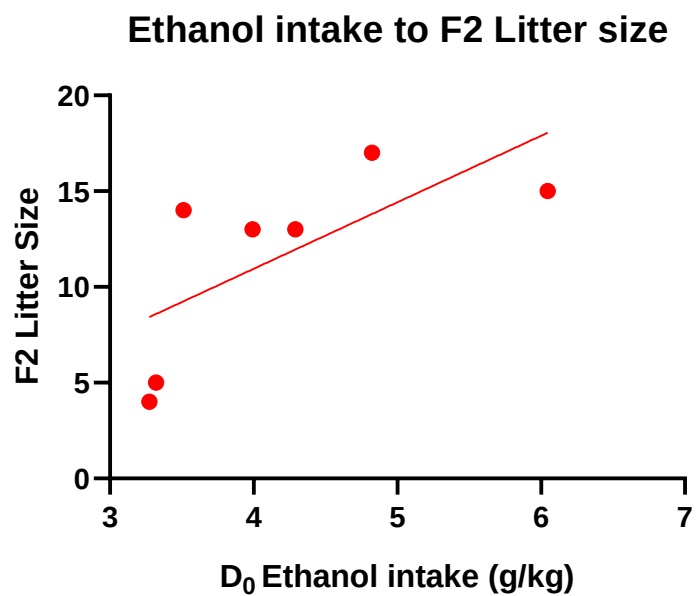


Figure D.2. Comparison of ethanol intake to F2 litter size. As drinking increases, so does the litter variance and likelihood of pup death at a significant rate ($r = .85$, $p = .024$)

Appendix E: Affective Disorders Test

Our lab looked at behaviours related to anxiety and depression in the F1 offspring through the affective disorders test (ADT). The ADT is an adapted version of the successive alleys test (SAT) with the integration of an anticipation box and food which provide measures of anticipation (Blackburne, 2021). Animals are maintained at 90% their free feeding weight for the duration of the experiment. The test has a three-day habituation period where rats are put in the anticipation box for a period of 10 minutes and recorded for movements and ultrasonic vocalisations (USV). On testing days, rats begin with the anticipation box and are then placed on the SAT with a food reward at the end of each alley. The anticipation for the food reward should be seen in increases in box activity and is measured by USV calls and rearing behaviour while scores were given for how far down the alley rats went to claim their food reward with higher SAT scores indicate lower anxiety.

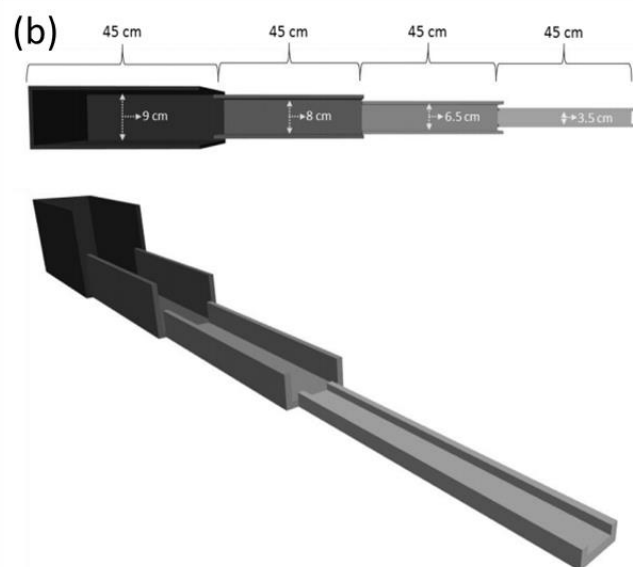


Figure E.1 Apparatus for Affective Disorders Test. (a) The anticipatory box (b) successive alleys apparatus. (Blackburne, 2021)

Analysis of the SAT score for anxiety has shown a sex effect ($F = 11.80, p = .002$) but no differences between the D₁ and ND₁ in terms of anxiety behaviours (Figure E.2). There are sex x condition interactions ($F = 6.69, p = .015$) in measured anticipatory behaviour with D₁ females showing higher rearing, indicating more pronounced anticipation than ND₁ females and D₁ males showing lower rearing than ND₁ males (Figure E.3), indicating sexually dimorphic effects of PPAC on anticipatory (depression-related) behaviour.

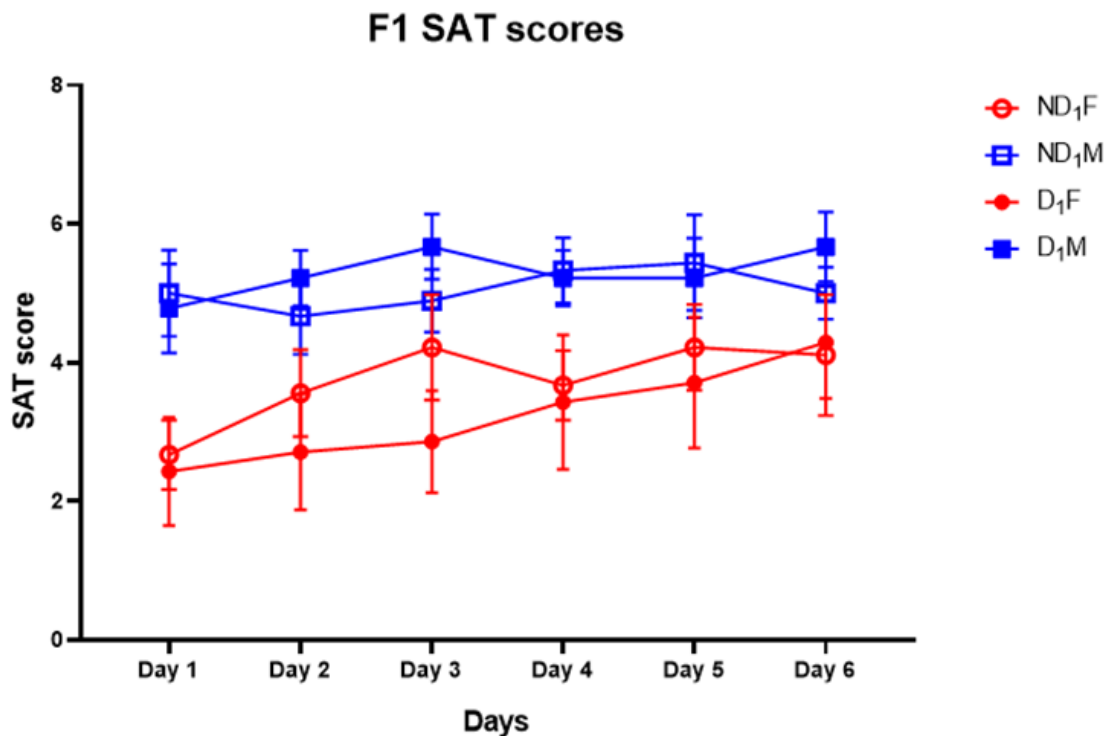


Figure E.2. Successive Alley Scores for Anxiety. There were no differences between drinker (D₁) and non-drinker (ND₁) offspring but males showed less anxiety behaviours by consistently travelling further down the alleys.

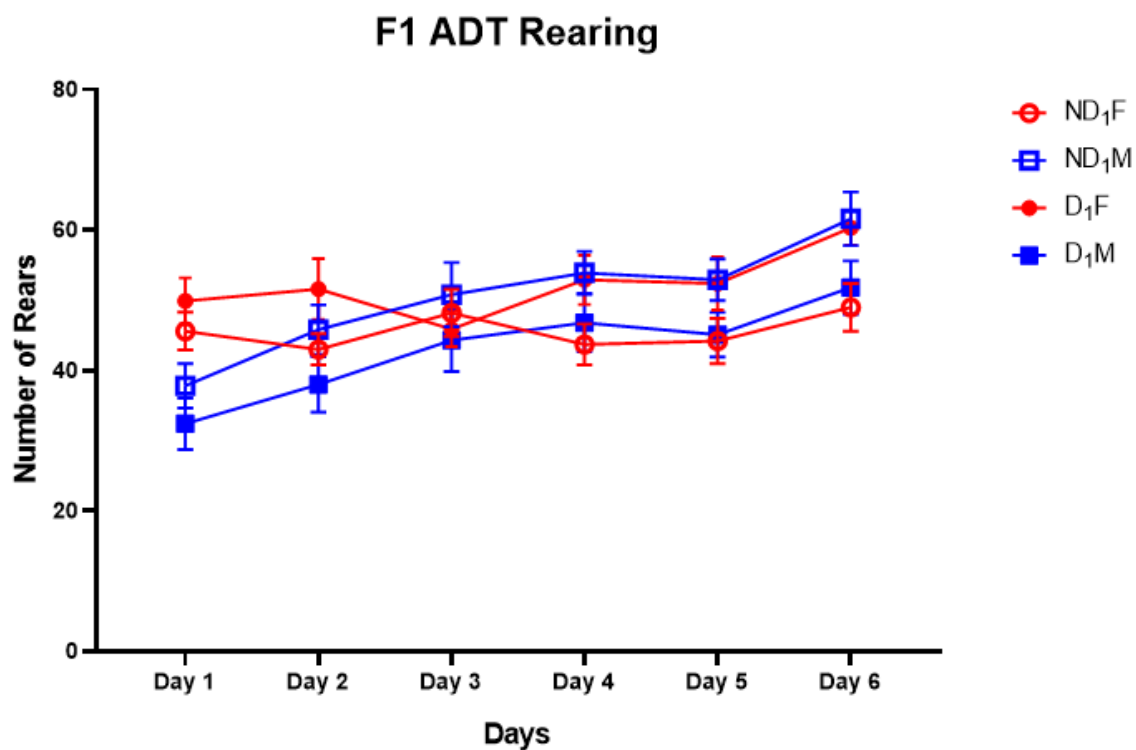


Figure E.3. Rearing scores for anticipation on ADT. There are sex x condition interactions in measured anticipatory behaviour with female offspring of ethanol exposed rats (D₁F) showing higher rearing, indicating pronounced anticipation when compared to the female offspring of non-drinkers (ND₁F) and D₁ males showing lower rearing than ND₁ males.

Appendix F: Mate Pairing to Litter Birth

While not looking at gestational days, the differences in days of litter birth showed that dams mated with ethanol drinking D_0 and ethanol sired D_1 rats on an average gave birth later than dams mated with ND rats. The ND_1 ($M = 25.00 \pm 2.67$ days) and ND_2 ($M = 26.29 \pm 0.49$ days) were generally being born earlier than the drinkers D_1 ($M = 26.60 \pm 4.67$ days) and D_2 ($M = 28.29 \pm 3.95$ days) and while this may not be due to changes in gestation, it does indicate some differences between the two conditions with regards to mating which should be investigated.

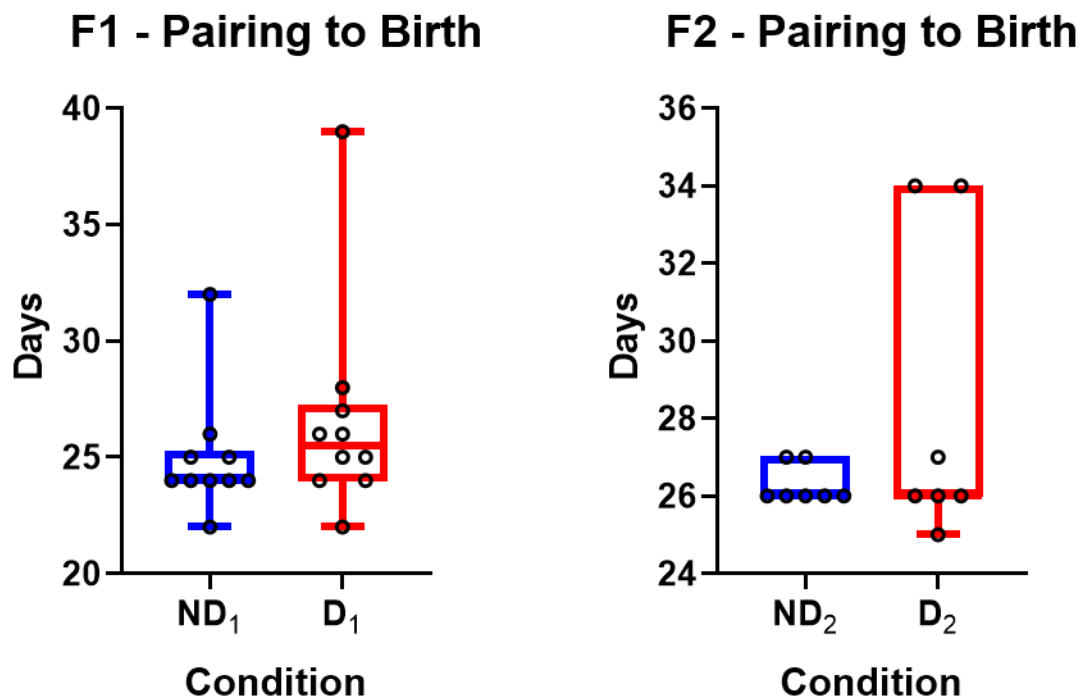


Figure F.1. Time between being paired for mating and litter birth. Mating occurs on average quicker in the non-drinkers with the ND_1 ($M = 25.00 \pm 2.67$ days) and ND_2 ($M = 26.29 \pm 0.49$ days) generally being born earlier than the drinkers D_1 ($M = 26.60 \pm 4.67$ days) and D_2 ($M = 28.29 \pm 3.95$ days) suggesting differences in mating behaviour, maternal gestation or both in the ethanol exposed animals and their germline.

References

- Abel, E. (1989). Paternal and Maternal Alcohol Consumption: Effects on Offspring in Two Strains of Rats. *Alcoholism, Clinical and Experimental Research*, 13(4), 533-541.
10.1111/j.1530-0277.1989.tb00373.x
- Abel, E. (1991a). Alarm substance emitted by rats in the forced-swim test is a low volatile pheromone. *Physiology & Behavior*, 50(4), 723-727. 10.1016/0031-9384(91)90009-D
- Abel, E. (1991b). Paternal alcohol consumption affects grooming response in rat offspring. *Alcohol (Fayetteville, N.Y.)*, 8(1), 21-23. 10.1016/0741-8329(91)91168-2
- Abel, E. (1993). Paternal alcohol exposure and hyperactivity in rat offspring: Effects of amphetamine. *Neurotoxicology and Teratology*, 15(6), 445-449. 10.1016/0892-0362(93)90063-T
- Abel, E. (1995). An update on incidence of FAS: FAS is not an equal opportunity birth defect. *Neurotoxicology and Teratology*, 17(4), 437-443. 10.1016/0892-0362(95)00005-C
- Abel, E. (2004). Paternal contribution to fetal alcohol syndrome. *Addiction Biology*, 9(2), 127-133. 10.1080/13556210410001716980
- Abel, E., & Lee, J. (1988). Paternal Alcohol Exposure Affects Offspring Behavior but not Body or Organ Weights in Mice. *Alcoholism, Clinical and Experimental Research*, 12(3), 349-355. 10.1111/j.1530-0277.1988.tb00205.x

- Abel, E., & Tan, S. E. (1988). Effects of paternal alcohol consumption on pregnancy outcome in rats. *Neurotoxicology and Teratology*, 10(3), 187-192. 10.1016/0892-0362(88)90016-5
- Ajoolabady, A., Kaplowitz, N., Lebeaupin, C., Kroemer, G., Kaufman, R. J., Malhi, H., & Ren, J. (2023). Endoplasmic reticulum stress in liver diseases. *Hepatology (Baltimore, Md.)*, 77(2), 619-639. 10.1002/hep.32562
- Albano, E. (2006). Alcohol, oxidative stress and free radical damage. *Proceedings of the Nutrition Society*, 65(3), 278-290. 10.1079/PNS2006496
- Alvarez Sedó, C., Bilinski, M., Lorenzi, D., Uriondo, H., Noblía, F., Longobucco, V., Lagar, E. V., & Nodar, F. (2017). Effect of sperm DNA fragmentation on embryo development: clinical and biological aspects. *JBRA Assisted Reproduction*, 21(4), 343-350. 10.5935/1518-0557.20170061
- Alvarez-Fernandez, A., Borziak, K., McDonald, G. C., Dorus, S., & Pizzari, T. (2019). Female novelty and male status dynamically modulate ejaculate expenditure and seminal fluid proteome over successive matings in red junglefowl. *Scientific Reports*, 9(1), 5852. 10.1038/s41598-019-41336-5
- Andres, A., Hull, H. R., Shankar, K., Casey, P. H., Cleves, M. A., & Badger, T. M. (2015). Longitudinal body composition of children born to mothers with normal weight, overweight, and obesity. *Obesity*, 23(6), 1252-1258.
- Andrews, S. J., & Goate, A. M. (2020). Mitochondrial DNA copy number is associated with cognitive impairment. *Alzheimer's & Dementia*, 16(S5)10.1002/alz.047543

- Arifin, W. N., & Zahiruddin, W. M. (2017). Sample Size Calculation in Animal Studies Using Resource Equation Approach. *The Malaysian Journal of Medical Sciences : MJMS*, 24(5), 101-105. 10.21315/mjms2017.24.5.11
- Armstrong, E. M. (2017). Making Sense of Advice About Drinking During Pregnancy: Does Evidence Even Matter? *The Journal of Perinatal Education*, 26(2), 65-69. 10.1891/1058-1243.26.2.65
- Aronson, H., & Gilbert, A. (1963). Preadolescent Sons of Male Alcoholics: An Experimental Study of Personality Patterning. *Archives of General Psychiatry*, 8(3), 235-241. 10.1001/archpsyc.1963.01720090023002
- Aryaman, J., Johnston, I. G., & Jones, N. S. (2019). Mitochondrial Heterogeneity. *Frontiers in Genetics*, 9, 718. 10.3389/fgene.2018.00718
- Arzate-Mejía, R. G., Lottenbach, Z., Schindler, V., Jawaidd, A., & Mansuy, I. M. (2020). Long-Term Impact of Social Isolation and Molecular Underpinnings. *Frontiers in Genetics*, 11, 589621. 10.3389/fgene.2020.589621
- Aşçıbaşı, K., Deveci, A., Özyılmaz, B., Erdoğan, K. M., & Çakıroğlu, E. (2023). Evaluation of microRNA expression levels during the craving period of patients diagnosed with severe alcohol use disorder. *Journal of Substance Use, ahead-of-print(ahead-of-print)*, 1-6. 10.1080/14659891.2023.2217267

- Åsenius, F., Danson, A. F., & Marzi, S. J. (2020). DNA methylation in human sperm: a systematic review. *Human Reproduction Update*, 26(6), 841-873.
10.1093/humupd/dmaa025
- Ashar, F. N., Zhang, Y., Longchamps, R. J., Lane, J., Moes, A., Grove, M. L., Mychaleckyj, J. C., Taylor, K. D., Coresh, J., Rotter, J. I., Boerwinkle, E., Pankratz, N., Guallar, E., & Arking, D. E. (2017). Association of Mitochondrial DNA Copy Number With Cardiovascular Disease. *JAMA Cardiology*, 2(11), 1247-1255.
10.1001/jamacardio.2017.3683
- Asimes, A. (2018). *Mechanisms and Consequences of Epigenetic Inheritance following Parental Preconception Alcohol Exposure* Available from ProQuest One Academic Eastern Edition
<https://search.proquest.com/docview/2047585233>
- Asimes, A., Torcaso, A., Pinceti, E., Kim, C. K., Zeleznik-Le, N. J., & Pak, T. R. (2017). Adolescent binge-pattern alcohol exposure alters genome-wide DNA methylation patterns in the hypothalamus of alcohol-naïve male offspring. *Alcohol (Fayetteville, N.Y.)*, 60, 179-189. 10.1016/j.alcohol.2016.10.010
- Australian Bureau of Statistics. (2021). *Alcohol consumption*. (). Aus: ABS.
<https://www.abs.gov.au/statistics/health/health-conditions-and-risks/alcohol-consumption/2020-21>
- Auten, R. L., & Davis, J. M. (2009). Oxygen Toxicity and Reactive Oxygen Species: The Devil Is in the Details. *Pediatric Research*, 66(2), 121-127. 10.1203/PDR.0b013e3181a9eafb

- Bajrami, E., & Spiroski, M. (2016). Genomic Imprinting. *Open Access Macedonian Journal of Medical Sciences*, 4(1), 181-184. 10.3889/oamjms.2016.028
- Balhorn, R. (2007). The protamine family of sperm nuclear proteins. *Genome Biology*, 8(9), 227. 10.1186/gb-2007-8-9-227
- Banerjee, A., Rose, R., Johnson, G. A., Burghardt, R. C., & Ramaiah, S. K. (2009). The Influence of Estrogen on Hepatobiliary Osteopontin (SPP1) Expression in a Female Rodent Model of Alcoholic Steatohepatitis. *Toxicologic Pathology*, 37(4), 492-501. 10.1177/0192623309335633
- Baraona, E., Abittan, C. S., Dohmen, K., Moretti, M., Pozzato, G., Chayes, Z. W., Schaefer, C., & Lieber, C. S. (2001). Gender Differences in Pharmacokinetics of Alcohol. *Alcoholism, Clinical and Experimental Research*, 25(4), 502-507. 10.1111/j.1530-0277.2001.tb02242.x
- Barbier, E., Tapocik, J. D., Juergens, N., Pitcairn, C., Borich, A., Schank, J. R., Sun, H., Schuebel, K., Zhou, Z., Yuan, Q., Vendruscolo, L. F., Goldman, D., & Heilig, M. (2015). DNA Methylation in the Medial Prefrontal Cortex Regulates Alcohol-Induced Behavior and Plasticity. *The Journal of Neuroscience*, 35(15), 6153-6164. 10.1523/JNEUROSCI.4571-14.2015
- Barker, J. M., & Taylor, J. R. (2019). Sex differences in incentive motivation and the relationship to the development and maintenance of alcohol use disorders. *Physiology & Behavior*, 203, 91-99.

- Barkur, R. R., & Bairy, L. K. (2015). Evaluation of passive avoidance learning and spatial memory in rats exposed to low levels of lead during specific periods of early brain development. *International Journal of Occupational Medicine and Environmental Health*, 28(3), 533-544. 10.13075/ijomeh.1896.00283
- Barney, T. M., Vore, A. S., & Deak, T. (2022). Acute Ethanol Challenge Differentially Regulates Expression of Growth Factors and miRNA Expression Profile of Whole Tissue of the Dorsal Hippocampus. *Frontiers in Neuroscience*, 16, 884197. 10.3389/fnins.2022.884197
- Bedi, Y., Chang, R. C., Gibbs, R., Clement, T. M., & Golding, M. C. (2019). Alterations in sperm-inherited noncoding RNAs associate with late-term fetal growth restriction induced by preconception paternal alcohol use. *Reproductive Toxicology (Elmsford, N.Y.)*, 87, 11-20. 10.1016/j.reprotox.2019.04.006
- Bedi, Y., Wang, H., Thomas, K. N., Basel, A., Prunier, J., Robert, C., & Golding, M. C. (2022). Alcohol induced increases in sperm Histone H3 lysine 4 trimethylation correlate with increased placental CTCF occupancy and altered developmental programming. *Scientific Reports*, 12(1), 8839. 10.1038/s41598-022-12188-3
- Beeler, E., Nobile, Z. L., & Homanics, G. E. (2019). Paternal Preconception Every-Other-Day Ethanol Drinking Alters Behavior and Ethanol Consumption in Offspring. *Brain Sciences*, 9(3), 56. 10.3390/brainsci9030056

Beery, A. K., & Zucker, I. (2011). Sex bias in neuroscience and biomedical research.

Neuroscience & Biobehavioral Reviews, 35(3), 565-572.

Bell, C. G., Finer, S., Lindgren, C. M., Wilson, G. A., Rakyan, V. K., Teschendorff, A. E., Akan, P., Stupka, E., Down, T. A., Prokopenko, I., Morison, I. M., Mill, J., Pidsley, R., Deloukas, P., Frayling, T. M., Hattersley, A. T., McCarthy, M. I., Beck, S., & Hitman, G. A. (2010).

Integrated Genetic and Epigenetic Analysis Identifies Haplotype-Specific Methylation in the FTO Type 2 Diabetes and Obesity Susceptibility Locus. *PLoS ONE*, 5(11), e14040.

10.1371/journal.pone.0014040

Bell, R. L., Rodd, Z. A., Lumeng, L., Murphy, J. M., & McBride, W. J. (2006). The alcohol-preferring P rat and animal models of excessive alcohol drinking. *Addiction Biology*, 11(3),

270-288. 10.1111/j.1369-1600.2005.00029.x

Bessone, F., Razori, M. V., & Roma, M. G. (2019). Molecular pathways of nonalcoholic fatty liver disease development and progression. *Cellular and Molecular Life Sciences : CMLS*,

76(1), 99-128. 10.1007/s00018-018-2947-0

Bhadsavle, S. S., & Golding, M. C. (2022). Paternal epigenetic influences on placental health and their impacts on offspring development and disease. *Frontiers in Genetics*, 13, 1068408.

Bielawski, D. M., & Abel, E. L. (1997). Acute treatment of paternal alcohol exposure produces malformations in offspring. *Alcohol (Fayetteville, N.Y.)*, 14(4), 397-401. 10.1016/S0741-

8329(97)87951-7

- Bielawski, D. M., Zaher, F. M., Svinarich, D. M., & Abel, E. L. (2002). Paternal Alcohol Exposure Affects Sperm Cytosine Methyltransferase Messenger RNA Levels. *Alcoholism, Clinical and Experimental Research*, 26(3), 347-351. 10.1111/j.1530-0277.2002.tb02544.x
- Biermann, T., Reulbach, U., Lenz, B., Frieling, H., Muschler, M., Hillemacher, T., Kornhuber, J., & Bleich, S. (2009). N-methyl-D-aspartate 2b receptor subtype (NR2B) promoter methylation in patients during alcohol withdrawal. *Journal of Neural Transmission (Vienna, Austria : 1996)*, 116(5), 615-622. 10.1007/s00702-009-0212-2
- Billah, M. M., Khatiwada, S., Morris, M. J., & Maloney, C. A. (2022). Effects of paternal overnutrition and interventions on future generations. *International Journal of Obesity*, 46(5), 901-917. 10.1038/s41366-021-01042-7
- Binder, N. K., Sheedy, J. R., Hannan, N. J., & Gardner, D. K. (2015). Male obesity is associated with changed spermatozoa Cox4i1 mRNA level and altered seminal vesicle fluid composition in a mouse model. *Molecular Human Reproduction*, 21(5), 424-434. 10.1093/molehr/gav010
- Blackburne, K. (2021). *The Affective Disorder Test: a new test to assess depression and anxiety-like behaviour in rats*
- Blanchard, B. A., & Glick, S. D. (2002). Sex differences in mesolimbic dopamine responses to ethanol and relationship to ethanol intake in rats. *Recent Developments in Alcoholism: Alcoholism and Women* (pp. 231-241). Springer.

Blanchard, B. A., Steindorf, S., Wang, S., & Glick, S. D. (1993). Sex Differences in Ethanol-induced Dopamine Release in Nucleus Accumbens and in Ethanol Consumption in Rats. *Alcoholism, Clinical and Experimental Research*, 17(5), 968-973. 10.1111/j.1530-0277.1993.tb05650.x

Blaseg, E., Von Wald, T., & Hansen, K. A. (2022). Vitamin D levels and human sperm DNA fragmentation: a prospective, cohort study. *Basic and Clinical Andrology*, 32(1), 1-14. 10.1186/s12610-022-00166-8

Blomdahl, J., Nasr, P., Ekstedt, M., & Kechagias, S. (2021). Moderate alcohol consumption is associated with advanced fibrosis in non-alcoholic fatty liver disease and shows a synergistic effect with type 2 diabetes mellitus. *Metabolism, Clinical and Experimental*, 115, 154439. 10.1016/j.metabol.2020.154439

Boccaletto, P., Stefaniak, F., Ray, A., Cappannini, A., Mukherjee, S., Purta, E., Kurkowska, M., Shirvanizadeh, N., Destefanis, E., Groza, P., Avşar, G., Romitelli, A., Pir, P., Dassi, E., Conticello, S. G., Aguilo, F., & Bujnicki, J. M. (2022). MODOMICS: a database of RNA modification pathways. 2021 update. *Nucleic Acids Research*, 50(D1), D231-D235. 10.1093/nar/gkab1083

Bohnsack, J. P., Teppen, T., Kyzar, E. J., Dzitoyeva, S., & Pandey, S. C. (2019). The lncRNA BDNF-AS is an epigenetic regulator in the human amygdala in early onset alcohol use disorders. *Translational Psychiatry*, 9(1), 34. 10.1038/s41398-019-0367-z

- Bolland, K. A., Bolland, J. M., Tomek, S., Devereaux, R. S., Mrug, S., & Wimberly, J. C. (2016). Trajectories of Adolescent Alcohol Use by Gender and Early Initiation Status. *Youth & Society*, 48(1), 3-32. 10.1177/0044118X13475639
- Bönsch, D., Lenz, B., Fiszer, R., Frieling, H., Kornhuber, J., & Bleich, S. (2006). Lowered DNA methyltransferase (DNMT-3b) mRNA expression is associated with genomic DNA hypermethylation in patients with chronic alcoholism. *Journal of Neural Transmission*, 113(9), 1299-1304. 10.1007/s00702-005-0413-2
- Bönsch, D., Lenz, B., Reulbach, U., Kornhuber, J., & Bleich, S. (2004). Homocysteine associated genomic DNA hypermethylation in patients with chronic alcoholism. *Journal of Neural Transmission*, 111(12), 1611-1616. 10.1007/s00702-004-0232-x
- Borges, E., Zanetti, B. F., Setti, A. S., Braga, D. P. d. A. F., Provenza, R. R., & Iaconelli, A. (2019). Sperm DNA fragmentation is correlated with poor embryo development, lower implantation rate, and higher miscarriage rate in reproductive cycles of non–male factor infertility. *Fertility and Sterility*, 112(3), 483-490. 10.1016/j.fertnstert.2019.04.029
- Bottom, R. T. (2021). *The Effects of Direct and Indirect Ethanol Exposure on Brain and Behavioral Development in Mice* (Ph.D.). Available from ProQuest One Academic. (2562836986). <https://www.proquest.com/dissertations-theses/effects-direct-indirect-ethanol-exposure-on-brain/docview/2562836986/se-2?accountid=14782>
https://tewaharoa.victoria.ac.nz/openurl/64VUW_INST/64VUW_INST:VUWNUI??url_ver=Z39.88-2004&rft_val_fmt=info:ofi/fmt:kev:mtx:dissertation&genre=dissertations&sid=ProQ:ProQu

[est+Dissertations+%2526+Theses+Global&atitle=&title=The+Effects+of+Direct+and+Indirect+Ethanol+Exposure+on+Brain+and+Behavioral+Development+in+Mice&issn=&date=2021-01-01&volume=&issue=&spage=&au=Bottom%252C+Riley+Thomas&isbn=9798534694994&jtitle=&btitle=&rft_id=info:eric/&rft_id=info:doi/](https://www.eric.ed.gov/fulltext/Dissertations/2526/Theses/Global&atitle=&title=The+Effects+of+Direct+and+Indirect+Ethanol+Exposure+on+Brain+and+Behavioral+Development+in+Mice&issn=&date=2021-01-01&volume=&issue=&spage=&au=Bottom%252C+Riley+Thomas&isbn=9798534694994&jtitle=&btitle=&rft_id=info:eric/&rft_id=info:doi/)

- Bottom, R. T., Kozanian, O. O., Rohac, D. J., Erickson, M. A., & Huffman, K. J. (2022). Transgenerational Effects of Prenatal Ethanol Exposure in Prepubescent Mice. *Frontiers in Cell and Developmental Biology*, 10, 812429. 10.3389/fcell.2022.812429
- Bradley, C. (unpublished). *Investigating Social Housing: the effects of isolation on rodent ethanol consumption*. Unpublished manuscript.
- Brahem, S., Mehdi, M., Elghezal, H., & Saad, A. (2011). The effects of male aging on semen quality, sperm DNA fragmentation and chromosomal abnormalities in an infertile population. *Journal of Assisted Reproduction and Genetics*, 28(5), 425-432. 10.1007/s10815-011-9537-5
- Broadwater, M., & Spear, L. P. (2013). Consequences of ethanol exposure on cued and contextual fear conditioning and extinction differ depending on timing of exposure during adolescence or adulthood. *Behavioural Brain Research*, 256, 10-19. 10.1016/j.bbr.2013.08.013

- Bromfield, J. J. (2018). Review: The potential of seminal fluid mediated paternal–maternal communication to optimise pregnancy success. *Animal*, 12, s104-s109.
10.1017/S1751731118000083
- Burchert, S., Stammel, N., & Knaevelsrud, C. (2017). Transgenerational trauma in a post-conflict setting: Effects on offspring PTSS/PTSD and offspring vulnerability in Cambodian Families. *Psychiatry Research*, 254, 151-157. 10.1016/j.psychres.2017.04.033
- Bygren, L. O., Kaati, G., & Edvinsson, S. (2001). Longevity determined by paternal ancestors' nutrition during their slow growth period. *Acta Biotheoretica*, 49(1), 53-59.
10.1023/A:1010241825519
- Cahill, A., Cunningham, C. C., Adachi, M., Ishii, H., Bailey, S. M., Fromenty, B., & Davies, A. (2002). Effects of Alcohol and Oxidative Stress on Liver Pathology: The Role of the Mitochondrion. *Alcoholism, Clinical and Experimental Research*, 26(6), 907-915.
10.1111/j.1530-0277.2002.tb02621.x
- Calhoun, F., & Warren, K. (2007). Fetal alcohol syndrome: Historical perspectives. *Neuroscience and Biobehavioral Reviews*, 31(2), 168-171. 10.1016/j.neubiorev.2006.06.023
- Cambiasso, M. Y., Gotfryd, L., Stinson, M. G., Birolo, S., Salamone, G., Romanato, M., Calvo, J. C., & Fontana, V. A. (2022). Paternal alcohol consumption has intergenerational consequences in male offspring. *Journal of Assisted Reproduction and Genetics*, 39(2), 441-459. 10.1007/s10815-021-02373-0

- Campbell, E. J., Flanagan, J. P. M., Marchant, N. J., & Lawrence, A. J. (2018). Reduced alcohol-seeking in male offspring of sires exposed to alcohol self-administration followed by punishment-imposed abstinence. *Pharmacology Research & Perspectives*, 6(2), e00384-n/a. 10.1002/prp2.384
- Carlsen, E., Swan, S. H., Petersen, J. H., & Skakkebaek, N. E. (2005). Longitudinal changes in semen parameters in young Danish men from the Copenhagen area. *Human Reproduction (Oxford, England)*, 20(4), 942-949. 10.1093/humrep/deh704
- Carnicella, S., Ron, D., & Barak, S. (2014). Intermittent ethanol access schedule in rats as a preclinical model of alcohol abuse. *Alcohol (Fayetteville, N.Y.)*, 48(3), 243-252. 10.1016/j.alcohol.2014.01.006
- Castellani, C. A., Longchamps, R. J., Sun, J., Guallar, E., & Arking, D. E. (2020). Thinking outside the nucleus: Mitochondrial DNA copy number in health and disease. *Mitochondrion*, 53, 214-223. 10.1016/j.mito.2020.06.004
- Cayci, T., Kurt, Y. G., Akgul, E. O., & Kurt, B. (2012). Does mtDNA copy number mean mitochondrial abundance? *Journal of Assisted Reproduction and Genetics*, 29(8), 855. 10.1007/s10815-012-9803-1
- Ceccanti, M., Coccurello, R., Carito, V., Ciafrè, S., Ferraguti, G., Giacobazzo, G., Mancinelli, R., Tirassa, P., Chaldakov, G. N., Pascale, E., Ceccanti, M., Codazzo, C., & Fiore, M. (2016). Paternal alcohol exposure in mice alters brain NGF and BDNF and increases

ethanol-elicited preference in male offspring. *Addiction Biology*, 21(4), 776-787.

10.1111/adb.12255

Centers for Disease Control and Prevention. (2023, October,). *Alcohol Use During Pregnancy*.

cdc.gov. <https://www.cdc.gov/ncbddd/fasd/alcohol-use.html>

Chang, R. C., Skiles, W. M., Chronister, S. S., Wang, H., Sutton, G. I., Bedi, Y. S., Snyder, M.,

Long, C. R., & Golding, M. C. (2017). DNA methylation-independent growth restriction and altered developmental programming in a mouse model of preconception male alcohol exposure. *Epigenetics*, 12(10), 841-853. 10.1080/15592294.2017.1363952

Chang, R. C., Wang, H., Bedi, Y., & Golding, M. C. (2019). Preconception paternal alcohol

exposure exerts sex-specific effects on offspring growth and long-term metabolic programming. *Epigenetics & Chromatin*, 12(1), 9. 10.1186/s13072-019-0254-0

Charan, J., & Kantharia, N. D. (2013). How to calculate sample size in animal studies? *Journal*

of Pharmacology & Pharmacotherapeutics, 4(4), 303-306. 10.4103/0976-500X.119726

Chastain, L. G., Franklin, T., Gangisetty, O., Cabrera, M. A., Mukherjee, S., Shrivastava, P.,

Jabbar, S., & Sarkar, D. K. (2019). Early life alcohol exposure primes hypothalamic microglia to later-life hypersensitivity to immune stress: possible epigenetic mechanism.

Neuropsychopharmacology : Official Publication of the American College of

Neuropsychopharmacology, 44(9), 1579-1588. 10.1038/s41386-019-0326-7

- Chastain, L. G., & Sarkar, D. K. (2017). Alcohol effects on the epigenome in the germline: Role in the inheritance of alcohol-related pathology. *Alcohol (Fayetteville, N.Y.)*, 60, 53-66. 10.1016/j.alcohol.2016.12.007
- Chaung, W. W., Jacob, A., Ji, Y., & Wang, P. (2008). *Suppression of PGC-1 α by Ethanol: Implications of Its Role in Alcohol Induced Liver Injury*. e-Century Publishing Corporation.
- Chen, S., Li, Z., He, Y., Zhang, F., Li, H., Liao, Y., Wei, Z., Wan, G., Xiang, X., Hu, M., Xia, K., Chen, X., & Tang, J. (2015). Elevated mitochondrial DNA copy number in peripheral blood cells is associated with childhood autism. *BMC Psychiatry*, 15(1), 50. 10.1186/s12888-015-0432-y
- Choi, I., Kim, P., Kim, H. J., Cheong, J. H., & Shin, C. Y. (2010). Effects of preconceptional ethanol consumption on ADHD-like symptoms in Sprague–Dawley rat offsprings. *International Journal of Developmental Neuroscience*, 28(8), 675. 10.1016/j.ijdevneu.2010.07.099
- Ciafrè, S., Carito, V., Ferraguti, G., Greco, A., Chaldakov, G. N., Fiore, M., & Ceccanti, M. (2019). How alcohol drinking affects our genes: an epigenetic point of view. *Biochemistry and Cell Biology*, 97(4), 345-356. 10.1139/bcb-2018-0248
- Cicero, T. J., Adams, M. L., O'connor, L., Nock, B., Meyer, E. R., & Wozniak, D. (1990). Influence of chronic alcohol administration on representative indices of puberty and sexual maturation in male rats and the development of their progeny. *Journal of Pharmacology and Experimental Therapeutics*, 255(2), 707-715.

- Conner, K. E., Bottom, R. T., & Huffman, K. J. (2020). The Impact of Paternal Alcohol Consumption on Offspring Brain and Behavioral Development. *Alcoholism, Clinical and Experimental Research*, 44(1), 125-140. 10.1111/acer.14245
- Correa, M., Arizzi, M. N., Betz, A., Mingote, S., & Salamone, J. D. (2003). Open field locomotor effects in rats after intraventricular injections of ethanol and the ethanol metabolites acetaldehyde and acetate. *Brain Research Bulletin*, 62(3), 197-202. 10.1016/j.brainresbull.2003.09.013
- Cox, N. (2013). TRIMMEAN: Stata module for trimmed means as descriptive or inferential statistics. <https://econpapers.repec.org/software/bocbocode/S457593.htm>
- Crean, A. J., Kopps, A. M., & Bonduriansky, R. (2014). Revisiting telegony: offspring inherit an acquired characteristic of their mother's previous mate. *Ecology Letters*, 17(12), 1545-1552. <https://doi.org/10.1111/ele.12373>
- Crippens, D., White, M. L., George, M. A., Jaworski, J. N., Brunner, L. J., Lancaster, F. E., & Gonzales, R. A. (1999). Gender Differences in Blood Levels, But Not Brain Levels, of Ethanol in Rats. *Alcoholism, Clinical and Experimental Research*, 23(3), 414-420. 10.1111/j.1530-0277.1999.tb04131.x
- Crossin, R., Cleland, L., Wilkins, C., Rychert, M., Adamson, S., Potiki, T., Pomerleau, A. C., MacDonald, B., Faletanoai, D., & Hutton, F. (2023). The New Zealand drug harms ranking study: A multi-criteria decision analysis. *Journal of Psychopharmacology*, , 02698811231182012.

Cubero, F. J., Urtasun, R., & Nieto, N. (2009). Alcohol and Liver Fibrosis. *Seminars in Liver Disease*, 29(2), 211-221. 10.1055/s-0029-1214376

Cui, H. Z., Sun, M. Z., Wang, R. Z., Li, C. Y., Huang, Y. X., Huang, Q. J., & Qiao, X. M. (2020). DNA Methylation in the Medial Prefrontal Cortex Regulates Alcohol-Related Behavior in Rats. *Yíchuán*, 42(1), 112-125. 10.16288/j.ycz.19-261

Cunningham, C. C., & Bailey, S. M. (2001). Ethanol Consumption and Liver Mitochondria Function. *Biological Signals and Receptors*, 10(3-4), 271-282. 10.1159/000046892

Czarny, P., Wigner, P., Strycharz, J., Swiderska, E., Synowiec, E., Szatkowska, M., Sliwinska, A., Talarowska, M., Szemraj, J., Su, K., Maes, M., Sliwinski, T., & Galecki, P. (2020). Mitochondrial DNA copy number, damage, repair and degradation in depressive disorder. *The World Journal of Biological Psychiatry*, 21(2), 91-101. 10.1080/15622975.2019.1588993

Dare, W. N., Noronha, C. C., Kusemiju, O. T., & Okanlawon, O. A. (2002). The effect of ethanol on spermatogenesis and fertility in male Sprague-Dawley rats pretreated with acetylsalicylic acid. *The Nigerian Postgraduate Medical Journal*, 9(4), 194-198.

Dasgupta, A. (2016). *Alcohol, Drugs, Genes and the Clinical Laboratory*. Elsevier Science.

Davie, M. (unpublished). *Impacts of ethanol consumption on sperm DNA methylation*. Unpublished manuscript.

- Dawson, D. A., Goldstein, R. B., Patricia Chou, S., June Ruan, W., & Grant, B. F. (2008). Age at First Drink and the First Incidence of Adult-Onset DSM-IV Alcohol Use Disorders. *Alcoholism: Clinical and Experimental Research*, 32(12), 2149-2160. 10.1111/j.1530-0277.2008.00806.x
- D'Erchia, A. M., Atlante, A., Gadaleta, G., Pavesi, G., Chiara, M., De Virgilio, C., Manzari, C., Mastropasqua, F., Prazzoli, G. M., Picardi, E., Gissi, C., Horner, D., Reyes, A., Sbisà, E., Tullo, A., & Pesole, G. (2015). Tissue-specific mtDNA abundance from exome data and its correlation with mitochondrial transcription, mass and respiratory activity. *Mitochondrion*, 20, 13-21. 10.1016/j.mito.2014.10.005
- Desroches, D., Orevillo, C., & Verina, D. (1995). Sex- and strain-related differences in first-pass alcohol metabolism in mice. *Alcohol*, 12(3), 221-226. 10.1016/0741-8329(94)00098-X
- Dippold, R. P., Vadigepalli, R., Gonye, G. E., Patra, B., & Hoek, J. B. (2013). Chronic Ethanol Feeding Alters miRNA Expression Dynamics During Liver Regeneration. *Alcoholism, Clinical and Experimental Research*, 37(s1), E59-E69. 10.1111/j.1530-0277.2012.01852.x
- Doremus, T. L., Brunell, S. C., Rajendran, P., & Spear, L. P. (2005). Factors influencing elevated ethanol consumption in adolescent relative to adult rats. *Alcoholism, Clinical and Experimental Research*, 29(10), 1796-1808. 10.1097/01.alc.0000183007.65998.aa
- Drewell, R. A., Goddard, C. J., Thomas, J. O., & Surani, M. A. (2002). Methylation-dependent silencing at the H19 imprinting control region by MeCP2. *Nucleic Acids Research*, 30(5), 1139-1144. 10.1093/nar/30.5.1139

- Duguay, L., Coutu, D., Hetu, C., & Joly, J. G. (1982). Inhibition of liver regeneration by chronic alcohol administration. *Gut*, 23(1), 8-13. 10.1136/gut.23.1.8
- Dugué, P., Wilson, R., Lehne, B., Jayasekara, H., Wang, X., Jung, C., Joo, J. E., Makalic, E., Schmidt, D. F., Baglietto, L., Severi, G., Gieger, C., Ladwig, K., Peters, A., Kooner, J. S., Southey, M. C., English, D. R., Waldenberger, M., Chambers, J. C., . . . Milne, R. L. (2021). Alcohol consumption is associated with widespread changes in blood DNA methylation: Analysis of cross-sectional and longitudinal data. *Addiction Biology*, 26(1), e12855-n/a. 10.1111/adb.12855
- Dyr, W., & Kostowski, W. (2008). Warsaw high-preferring (WHP) and Warsaw low-preferring (WLP) lines of rats selectively bred for high and low voluntary ethanol intake: Preliminary phenotypic characterization. *Alcohol (Fayetteville, N.Y.)*, 42(3), 161-170. 10.1016/j.alcohol.2008.01.008
- Eagon, P. K. (2010). Alcoholic liver injury: Influence of gender and hormones. *World Journal of Gastroenterology : WJG*, 16(11), 1377-1384. 10.3748/wjg.v16.i11.1377
- Easey, K. E., & Sharp, G. C. (2021). The impact of paternal alcohol, tobacco, caffeine use and physical activity on offspring mental health: A systematic review and meta-analysis. *Reproductive Health*, 18(1)10.1186/s12978-021-01266-w
- Ellenbroek, B., Derks, N., & Park, H. (2005). Early maternal deprivation retards neurodevelopment in Wistar rats. *Stress (Amsterdam, Netherlands)*, 8(4), 247-257. 10.1080/10253890500404634

- Ellenbroek, B., & Youn, J. U. (2016). *Gene-environment interactions in psychiatry: nature, nurture, neuroscience*. Academic Press.
- Emanuele, N. V., LaPaglia, N., Steiner, J., Colantoni, A., Thiel, D. H., & Emanuele, M. A. (2001). Peripubertal Paternal EtOH Exposure: Testicular Oxidative Injury, Fecundity, and Offspring. *Endocrine*, 14(2), 213-220. 10.1385/ENDO:14:2:213
- Eng, M. Y., Schuckit, M. A., & Smith, T. L. (2005). The level of response to alcohol in daughters of alcoholics and controls. *Drug and Alcohol Dependence*, 79(1), 83-93. 10.1016/j.drugalcdep.2005.01.002
- Engel, N. (2018). Sex differences in early embryogenesis: Inter-chromosomal regulation sets the stage for sex-biased gene networks: The dialogue between the sex chromosomes and autosomes imposes sexual identity soon after fertilization. *BioEssays*, 40(9), 1800073.
- Eriksson, C. J. P. (1973). Ethanol and acetaldehyde metabolism in rat strains genetically selected for their ethanol preference. *Biochemical Pharmacology*, 22(18), 2283-2292. 10.1016/0006-2952(73)90009-9
- Ervin, C. S., Little, R. E., Streissguth, A. P., & Beck, D. E. (1984). Alcoholic Fathering and Its Relation to Child's Intellectual Development: A Pilot Investigation. *Alcoholism, Clinical and Experimental Research*, 8(4), 362-365. 10.1111/j.1530-0277.1984.tb05681.x
- Evrard, S. G., Duhalde-Vega, M., Tagliaferro, P., Mirochnic, S., Caltana, L. R., & Brusco, A. (2006). A low chronic ethanol exposure induces morphological changes in the adolescent rat

brain that are not fully recovered even after a long abstinence: An immunohistochemical study. *Experimental Neurology*, 200(2), 438-459. 10.1016/j.expneurol.2006.03.001

Farah Naquiah, M. Z., James, R. J., Suratman, S., Lee, L. S., Mohd Hafidz, M. I., Salleh, M. Z., & Teh, L. K. (2016). Transgenerational effects of paternal heroin addiction on anxiety and aggression behavior in male offspring. *Behavioral and Brain Functions*, 12(1), 23. 10.1186/s12993-016-0107-y

Farfán Labonne, B. E., Gutiérrez, M., Gómez-Quiroz, L. E., Konigsberg Fainstein, M., Bucio, L., Souza, V., Flores, O., Ortiz, V., Hernández, E., Kershenobich, D., & Gutiérrez-Ruiz, M. C. (2009). Acetaldehyde-induced mitochondrial dysfunction sensitizes hepatocytes to oxidative damage. *Cell Biology and Toxicology*, 25(6), 599-609. 10.1007/s10565-008-9115-5

Farkouh, A., Riedl, T., Gottardi, R., Czejka, M., & Kautzky-Willer, A. (2020). Sex-related differences in pharmacokinetics and pharmacodynamics of frequently prescribed drugs: a review of the literature. *Advances in Therapy*, 37, 644-655.

Feinberg, A. P. (2007). Phenotypic plasticity and the epigenetics of human disease. *Nature*, 447(7143), 433-440. 10.1038/nature05919

Feng, Y., Jia, Y., Su, L., Wang, D., Lv, L., Xu, L., & Yao, Y. (2013). Decreased mitochondrial DNA copy number in the hippocampus and peripheral blood during opiate addiction is mediated by autophagy and can be salvaged by melatonin. *Autophagy*, 9(9), 1395-1406. 10.4161/auto.25468

Ferraguti, G., Codazzo, C., Petrella, C., Coccurello, R., Ceccanti, M., & Fiore, M. (2020).

Brainstem expression of SLC6A4, HTR2C, NGF, BDNF, TRKANGF, TRKBBDNF and P75NTR following paternal alcohol exposure in the male mouse. *Bio-Medical Reviews (Varna)*, 31, 75. 10.14748/bmr.v31.7707

Festing, M. F. W., & Altman, D. G. (2002). Guidelines for the Design and Statistical Analysis of Experiments Using Laboratory Animals. *ILAR Journal*, 43(4), 244-258.

10.1093/ilar.43.4.244

Finegersh, A., & Homanics, G. E. (2014). Paternal Alcohol Exposure Reduces Alcohol Drinking and Increases Behavioral Sensitivity to Alcohol Selectively in Male Offspring. *PLoS ONE*, 9(6), e99078. 10.1371/journal.pone.0099078

Finegersh, A., Rompala, G. R., Martin, D. I. K., & Homanics, G. E. (2015). Drinking beyond a lifetime: New and emerging insights into paternal alcohol exposure on subsequent generations. *Alcohol (Fayetteville, N.Y.)*, 49(5), 461-470. 10.1016/j.alcohol.2015.02.008

Finelli, R., Mottola, F., & Agarwal, A. (2021). Impact of alcohol consumption on male fertility potential: a narrative review. *International Journal of Environmental Research and Public Health*, 19(1), 328.

Fischer, A. G., & Ullsperger, M. (2017). An update on the role of serotonin and its interplay with dopamine for reward. *Frontiers in Human Neuroscience*, 11, 484. 10.3389/fnhum.2017.00484

- Flores-Bonilla, A., & Richardson, H. N. (2020). Sex Differences in the Neurobiology of Alcohol Use Disorder. *Alcohol Research*, 40(2), 4. 10.35946/arc.v40.2.04
- Foley, K. A., Ossenkopp, K., Kavaliers, M., & Macfabe, D. F. (2014). Pre- and Neonatal Exposure to Lipopolysaccharide or the Enteric Metabolite, Propionic Acid, Alters Development and Behavior in Adolescent Rats in a Sexually Dimorphic Manner. *PLoS ONE*, 9(1), e87072. 10.1371/journal.pone.0087072
- Ford, M. M., Eldridge, J. C., & Samson, H. H. (2002). Microanalysis of ethanol self-administration: estrous cycle phase-related changes in consumption patterns. *Alcoholism: Clinical and Experimental Research*, 26(5), 635-643.
- Ford, M. M., Eldridge, J. C., & Samson, H. H. (2004). Determination of an estradiol dose-response relationship in the modulation of ethanol intake. *Alcoholism: Clinical and Experimental Research*, 28(1), 20-28.
- Frezza, M., di Padova, C., Pozzato, G., Terpin, M., Baraona, E., & Lieber, C. S. (1990). High blood alcohol levels in women. The role of decreased gastric alcohol dehydrogenase activity and first-pass metabolism. *The New England Journal of Medicine*, 322(2), 95-99. 10.1056/NEJM199001113220205
- Friedman, J. R., & Nunnari, J. (2014). Mitochondrial form and function. *Nature (London)*, 505(7483), 335-343. 10.1038/nature12985
- Frye, G. D., & Breese, G. R. (1981). An evaluation of the locomotor stimulating action of ethanol in rats and mice. *Psychopharmacology*, 75(4), 372-379. 10.1007/BF00435856

Gangisetty, O., Bekdash, R., Maglakelidze, G., & Sarkar, D. K. (2014). Fetal Alcohol Exposure Alters Proopiomelanocortin Gene Expression and Hypothalamic-Pituitary-Adrenal Axis Function via Increasing MeCP2 Expression in the Hypothalamus. *PLoS ONE*, 9(11), e113228. 10.1371/journal.pone.0113228

Gangisetty, O., Sinha, R., & Sarkar, D. K. (2019). Hypermethylation of Proopiomelanocortin and Period 2 Genes in Blood Are Associated with Greater Subjective and Behavioral Motivation for Alcohol in Humans. *Alcoholism, Clinical and Experimental Research*, 43(2), 212-220. 10.1111/acer.13932

Gapp, K., von Ziegler, L., Tweedie-Cullen, R. Y., & Mansuy, I. M. (2014). Early life epigenetic programming and transmission of stress-induced traits in mammals: how and when can environmental factors influence traits and their transgenerational inheritance? *BioEssays*, 36(5), 491-502. 10.1002/bies.201300116

Garaycochea, J. I., Crossan, G. P., Langevin, F., Mulderrig, L., Louzada, S., Yang, F., Guilbaud, G., Park, N., Roerink, S., Nik-Zainal, S., Stratton, M. R., & Patel, K. J. (2018). Alcohol and endogenous aldehydes damage chromosomes and mutate stem cells. *Nature*, 553(7687), 171-177. 10.1038/nature25154

Gårdebjer, E. M. (2015). *Metabolic effects following periconceptional alcohol exposure in rat adult offspring: modulations by a postnatal western diet*
<https://trove.nla.gov.au/work/213241007>

- Geisler, H. C., Westerga, J., & Gramsbergen, A. (1993). Development of posture in the rat. *Acta Neurobiologiae Experimentalis*, 53(4), 517-523.
<https://www.ncbi.nlm.nih.gov/pubmed/8109261>
- Godschalk, R., Remels, A., Hoogendoorn, C., van Benthem, J., Luijten, M., Duale, N., Brunborg, G., Olsen, A., Bouwman, F. G., Munnia, A., Peluso, M., Mariman, E., & van Schooten, F. J. (2018). Paternal Exposure to Environmental Chemical Stress Affects Male Offspring's Hepatic Mitochondria. *Toxicological Sciences*, 162(1), 241-250. 10.1093/toxsci/kfx246
- Goldberg, L. R., Zeid, D., Kutlu, M. G., Cole, R. D., Lallai, V., Sebastian, A., Albert, I., Fowler, C. D., Parikh, V., & Gould, T. J. (2021). Paternal nicotine enhances fear memory, reduces nicotine administration, and alters hippocampal genetic and neural function in offspring. *Addiction Biology*, 26(1), e12859. 10.1111/adb.12859
- Goldberg, L. R., & Gould, T. J. (2019). Multigenerational and transgenerational effects of paternal exposure to drugs of abuse on behavioral and neural function. *The European Journal of Neuroscience*, 50(3), 2453-2466. 10.1111/ejn.14060
- Gonzales, M. L., & LaSalle, J. M. (2010). The Role of MeCP2 in Brain Development and Neurodevelopmental Disorders. *Current Psychiatry Reports*, 12(2), 127-134.
 10.1007/s11920-010-0097-7
- Govorko, D., Bekdash, R. A., Zhang, C., & Sarkar, D. K. (2012). Male Germline Transmits Fetal Alcohol Adverse Effect on Hypothalamic Proopiomelanocortin Gene Across Generations. *Biological Psychiatry* (1969), 72(5), 378-388. 10.1016/j.biopsych.2012.04.006

- Greeson, K. W., Crow, K. M. S., Edenfield, R. C., & Easley, 4., Charles A. (2023). Inheritance of paternal lifestyles and exposures through sperm DNA methylation. *Nature Reviews. Urology*, 20(6), 356-370. 10.1038/s41585-022-00708-9
- Han, D., Johnson, H. S., Rao, M. P., Martin, G., Sancheti, H., Silkwood, K. H., Decker, C. W., Nguyen, K. T., Casian, J. G., Cadenas, E., & Kaplowitz, N. (2017). Mitochondrial remodeling in the liver following chronic alcohol feeding to rats. *Free Radical Biology & Medicine*, 102, 100-110. 10.1016/j.freeradbiomed.2016.11.020
- Han, J., Kwon, H., Ha, M., Paik, K., Lim, M., Gyu Lee, S., Yoo, S., & Kim, E. (2015). The effects of prenatal exposure to alcohol and environmental tobacco smoke on risk for ADHD: A large population-based study. *Psychiatry Research*, 225(1), 164-168. 10.1016/j.psychres.2014.11.009
- Harkness, L. L. (1993). Transgenerational Transmission of War-Related Trauma. *International Handbook of Traumatic Stress Syndromes* (pp. 635-643). Springer US. 10.1007/978-1-4615-2820-3_53
- Hauser, S. R., Deehan, G. A., Knight, C. P., Waeiss, R. A., Truitt, W. A., Johnson, P. L., Bell, R. L., McBride, W. J., & Rodd, Z. A. (2019). Conditioned stimuli affect ethanol-seeking by female alcohol-preferring (P) rats: the role of repeated-deprivations, cue-pretreatment, and cue-temporal intervals. *Psychopharmacology*, 236(9), 2835-2846. 10.1007/s00213-019-05264-6

- Heijmans, B. T., Tobi, E. W., Stein, A. D., Putter, H., Blauw, G. J., Susser, E. S., Slagboom, P. E., & Lumey, L. H. (2008). Persistent epigenetic differences associated with prenatal exposure to famine in humans. *Proceedings of the National Academy of Sciences - PNAS*, 105(44), 17046-17049. 10.1073/pnas.0806560105
- Henriksen, T. B., Hjollund, N. H., Jensen, T. K., Bonde, J. P., Andersson, A., Kolstad, H., Ernst, E., Giwercman, A., Skakkebaek, N. E., & Olsen, J. (2004). Alcohol Consumption at the Time of Conception and Spontaneous Abortion. *American Journal of Epidemiology*, 160(7), 661-667. 10.1093/aje/kwh259
- Hernandez, A. R., Truckenbrod, L. M., Campos, K. T., Williams, S. A., & Burke, S. N. (2020). Sex Differences in Age-Related Impairments Vary Across Cognitive and Physical Assessments in Rats. *Behavioral Neuroscience*, 134(2), 69-81. 10.1037/bne0000352
- Heynen, J. P., Paxman, E. J., Sanghavi, P., McCreary, J. K., Montana, T., & Metz, G. A. S. (2022). Trans- and Multigenerational Maternal Social Isolation Stress Programs the Blood Plasma Metabolome in the F3 Generation. *Metabolites*, 12(7), 572. 10.3390/metabo12070572
- Hilderbrand, E. R., & Lasek, A. W. (2018). Estradiol enhances ethanol reward in female mice through activation of ER α and ER β . *Hormones and Behavior*, 98, 159-164. 10.1016/j.yhbeh.2018.01.001

- Hill, S. Y., Rumpala, G., Homanics, G. E., & Zezza, N. (2017). Cross-generational effects of alcohol dependence in humans on HRAS and TP53 methylation in offspring. *Epigenomics*, 9(9), 1189-1203. 10.2217/epi-2017-0052
- Hollander, J., McNivens, M., Pautassi, R. M., & Nizhnikov, M. E. (2019). Offspring of male rats exposed to binge alcohol exhibit heightened ethanol intake at infancy and alterations in T-maze performance. *Alcohol (Fayetteville, N.Y.)*, 76, 65-71. 10.1016/j.alcohol.2018.07.013
- Holoch, D., & Moazed, D. (2015). RNA-mediated epigenetic regulation of gene expression. *Nature Reviews. Genetics*, 16(2), 71-84. 10.1038/nrg3863
- Hossain, M. G., Akter, S., Ohsaki, E., & Ueda, K. (2020). Impact of the Interaction of Hepatitis B Virus with Mitochondria and Associated Proteins. *Viruses*, 12(2), 175. 10.3390/v12020175
- Hottenga, J. J., Ligthart, L., Mbarek, H., & Boomsma, D. I. (2018). Transancestral GWAS of alcohol dependence reveals common genetic underpinnings with psychiatric disorders. *Nature Neuroscience*, 21(12), 1656-1669. 10.1038/s41593-018-0275-1
- Howe, S. W. (1983). *Effects of early handling on developing auditory mechanisms in the rat as assessed by the auditory brainstem response in adulthood* Available from ComDisDome <https://search.proquest.com/docview/85140437>
- Huang, T., Wang, Y., Vazquez-Ferrer, E., Theofel, I., Requena, C. E., Hanna, C. W., Kelsey, G., & Hajkova, P. (2021). Sex-specific chromatin remodelling safeguards transcription in germ cells. *Nature*, 600(7890), 737-742.

- Hussain, S., Lesscher, H. M. D., Day, D. J., Ellenbroek, B. A., AISS Behaviour Neuroscience, & dASS, B. (2022). Genetics and epigenetics: paternal adolescent ethanol consumption in serotonin transporter knock-out rats and offspring sensitivity to ethanol. *Psychopharmacology*, 239(10), 3145-3159. 10.1007/s00213-022-06195-5
- Hutchinson, D., Youssef, G. J., McCormack, C., Wilson, J., Allsop, S., Najman, J., Elliott, E., Burns, L., Jacobs, S., Honan, I., Rossen, L., Fiedler, H., Teague, S., Ryan, J., Olsson, C. A., & Mattick, R. P. (2019). Prenatal alcohol exposure and infant gross motor development: a prospective cohort study. *BMC Pediatrics*, 19(1), 149. 10.1186/s12887-019-1516-5
- Hwa, L. S., Chu, A., Levinson, S. A., Kayyali, T. M., DeBold, J. F., & Miczek, K. A. (2011). Persistent Escalation of Alcohol Drinking in C57BL/6J Mice With Intermittent Access to 20% Ethanol. *Alcoholism, Clinical and Experimental Research*, 35(11), 1938-1947. 10.1111/j.1530-0277.2011.01545.x
- Hyde, J. F., & Jerussi, T. P. (1983). Sexual dimorphism in rats with respect to locomotor activity and circling behavior. *Pharmacology, Biochemistry and Behavior*, 18(5), 725-729. 10.1016/0091-3057(83)90014-X
- Ishikawa, H., Ishikawa, T., Yamamoto, H., Fukao, A., & Yokoyama, K. (2007). Genotoxic effects of alcohol in human peripheral lymphocytes modulated by ADH1B and ALDH2 gene polymorphisms. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis*, 615(1), 134-142. 10.1016/j.mrfmmm.2006.11.026

- Jabbar, S., Chastain, L. G., Gangisetty, O., Cabrera, M. A., Sochacki, K., & Sarkar, D. K. (2016). Preconception Alcohol Increases Offspring Vulnerability to Stress. *Neuropsychopharmacology (New York, N.Y.)*, 41(11), 2782-2793. 10.1038/npp.2016.92
- Jamerson, P. A., Wulser, M. J., & Kimler, B. F. (2004). Neurobehavioral effects in rat pups whose sires were exposed to alcohol. *Brain Research. Developmental Brain Research*, 149(2), 103-111. 10.1016/j.devbrainres.2003.12.010
- Jana, K., Jana, N., De, D. K., & Guha, S. K. (2010). Ethanol induces mouse spermatogenic cell apoptosis in vivo through over-expression of Fas/Fas-L, p53, and caspase-3 along with cytochrome c translocation and glutathione depletion. *Molecular Reproduction and Development*, 77(9), 820-833. 10.1002/mrd.21227
- Jeng, J., Yeh, T., Lee, J., Lin, S., Fong, T., & Hsieh, R. (2008). Maintenance of mitochondrial DNA copy number and expression are essential for preservation of mitochondrial function and cell growth. *Journal of Cellular Biochemistry*, 103(2), 347-357. 10.1002/jcb.21625
- Jiajie, T., Yanzhou, Y., Hoi-Hung, A. C., Zi-Jiang, C., & Wai-Yee, C. (2017). Conserved miR-10 family represses proliferation and induces apoptosis in ovarian granulosa cells. *Scientific Reports*, 7(1), 41304. 10.1038/srep41304
- Jiménez, J. A., & Zylka, M. J. (2021). Controlling litter effects to enhance rigor and reproducibility with rodent models of neurodevelopmental disorders. *Journal of Neurodevelopmental Disorders*, 13(1), 2. 10.1186/s11689-020-09353-y

Jones, B. J., & Roberts, D. J. (1968). The quantitative measurement of motor inco-ordination in naive mice using an accelerating rotarod. *Journal of Pharmacy and Pharmacology*, 20(4), 302-304. 10.1111/j.2042-7158.1968.tb09743.x

Joo, K., Kwon, Y., Myung, S., & Kim, T. (2012). The Effects of Smoking and Alcohol Intake on Sperm Quality: Light and Transmission Electron Microscopy Findings. *Journal of International Medical Research*, 40(6), 2327-2335. 10.1177/030006051204000631

Juárez, J., & De Tomasi, E. B. (1999). Sex Differences in Alcohol Drinking Patterns During Forced and Voluntary Consumption in Rats. *Alcohol (Fayetteville, N.Y.)*, 19(1), 15-22. 10.1016/S0741-8329(99)00010-5

Jung, Y. H., Sauria, M. E. G., Lyu, X., Cheema, M. S., Ausio, J., Taylor, J., & Corces, V. G. (2017). Chromatin States in Mouse Sperm Correlate with Embryonic and Adult Regulatory Landscapes. *Cell Reports*, 18(6), 1366-1382. 10.1016/j.celrep.2017.01.034

Kalueff, A. V., Stewart, A. M., Song, C., Berridge, K. C., Graybiel, A. M., & Fentress, J. C. (2016). Neurobiology of rodent self-grooming and its value for translational neuroscience. *Nature Reviews Neuroscience*, 17(1), 45-59.

Kasimanickam, V., Buhr, M., & Kasimanickam, R. (2019). Patterns of expression of sperm and seminal plasma microRNAs in boar semen. *Theriogenology*, 125, 87-92. 10.1016/j.theriogenology.2018.10.021

- Kaufman, J., Yang, B., Douglas-Palumberi, H., Crouse-Artus, M., Lipschitz, D., Krystal, J. H., & Gelernter, J. (2007). Genetic and environmental predictors of early alcohol use. *Biological Psychiatry*, 61(11), 1228-1234.
- Kayani, M. A., & Parry, J. M. (2010). The in vitro genotoxicity of ethanol and acetaldehyde. *Toxicology in Vitro*, 24(1), 56-60. 10.1016/j.tiv.2009.09.003
- Kekäläinen, J., Jokiniemi, A., Janhunen, M., & Huuskonen, H. (2020). Offspring phenotype is shaped by the nonsperm fraction of semen. *Journal of Evolutionary Biology*, 33(5), 584-594. 10.1111/jeb.13592
- Kelly, S. J., Bonthius, D. J., & West, J. R. (1987). Developmental Changes in Alcohol Pharmacokinetics in Rats. *Alcoholism: Clinical and Experimental Research*, 11(3), 281-286. 10.1111/j.1530-0277.1987.tb01308.x
- Kiefer, S. W., Bice, P. J., & Badia-Elder, N. (1994). Alterations in Taste Reactivity to Alcohol in Rats Given Continuous Alcohol Access Followed by Abstinence. *Alcoholism, Clinical and Experimental Research*, 18(3), 555-559. 10.1111/j.1530-0277.1994.tb00909.x
- Kim, J., & Shukla, S. (2006). Acute in Vivo Effect of Ethanol (Binge Drinking) on Histone H3 Modifications in Rat Tissues. *Alcohol and Alcoholism (Oxford)*, 41(2), 126-132. 10.1093/alcalc/agh248
- Kim, P., Choi, C. S., Park, J. H., Joo, S. H., Kim, S. Y., Ko, H. M., Kim, K. C., Jeon, S. J., Park, S. H., Han, S., Ryu, J. H., Cheong, J. H., Han, J. Y., Ko, K. N., & Shin, C. Y. (2014). Chronic exposure to ethanol of male mice before mating produces attention deficit

hyperactivity disorder-like phenotype along with epigenetic dysregulation of dopamine transporter expression in mouse offspring. *Journal of Neuroscience Research*, 92(5), 658-670. 10.1002/jnr.23275

Kinley, B. L., Kyne, R. F., Lawton-Stone, T. S., Walker, D. M., & Paul, M. J. (2021). Long-term consequences of peri-adolescent social isolation on social preference, anxiety-like behaviour, and vasopressin neural circuitry of male and female rats. *The European Journal of Neuroscience*, 54(11), 7790-7804. 10.1111/ejn.15520

Klassen, R. W., & Persaud, T. V. N. (1976). Experimental studies on the influence of male alcoholism on pregnancy and progeny. *Experimentelle Pathologie*, 12(1), 38-45. 10.1016/S0014-4908(76)80031-2

Klinedinst, N. J., & Regenold, W. T. (2015). A mitochondrial bioenergetic basis of depression. *Journal of Bioenergetics and Biomembranes*, 47(1-2), 155-171. 10.1007/s10863-014-9584-6

Knezovich, J. G., & Ramsay, M. (2012). The Effect of Preconception Paternal Alcohol Exposure on Epigenetic Remodeling of the H19 and Rasgrf1 Imprinting Control Regions in Mouse Offspring. *Frontiers in Genetics*, 3, 10. 10.3389/fgene.2012.00010

Knopik, V. S., Jacob, T., Haber, J. R., Swenson, L. P., & Howell, D. N. (2009). Paternal Alcoholism and Offspring ADHD Problems: A Children of Twins Design. *Twin Research and Human Genetics*, 12(1), 53-62. 10.1375/twin.12.1.53

Koabel, J., McNivens, M., McKee, P., Pautassi, R., Bordner, K., & Nizhnikov, M. (2021). The offspring of alcohol-exposed sires exhibit heightened ethanol intake and behavioral alterations in the elevated plus maze. *Alcohol (Fayetteville, N.Y.)*, 92, 65-72.
10.1016/j.alcohol.2021.01.009

Kokare, D. M., Dandekar, M. P., Singru, P. S., Gupta, G. L., & Subhedar, N. K. (2010). Involvement of α -MSH in the social isolation induced anxiety- and depression-like behaviors in rat. *Neuropharmacology*, 58(7), 1009-1018.
10.1016/j.neuropharm.2010.01.006

Kokare, D. M., Kyzar, E. J., Zhang, H., Sakharkar, A. J., & Pandey, S. C. (2017). Adolescent Alcohol Exposure-Induced Changes in Alpha-Melanocyte Stimulating Hormone and Neuropeptide Y Pathways via Histone Acetylation in the Brain During Adulthood. *International Journal of Neuropsychopharmacology*, 20(9), 758-768. 10.1093/ijnp/pyx041

Kotova, N., Vare, D., Schultz, N., Gradecka Meesters, D., Stepnik, M., Grawé, J., Helleday, T., & Jenssen, D. (2013). Genotoxicity of alcohol is linked to DNA replication-associated damage and homologous recombination repair. *Carcinogenesis*, 34(2), 325-330.
10.1093/carcin/bgs340

Krasteva, M., Koycheva, Y., Taseva, T., & Simeonova, S. (2020). Changes in the Expression of DNA Methylation Related Genes in Leukocytes of Persons with Alcohol and Drug Dependence. *Acta Medica Bulgarica*, 47(4), 11-17. 10.2478/amb-2020-0039

Kreuz, S., & Fischle, W. (2016). Oxidative stress signaling to chromatin in health and disease.

Epigenomics, 8(6), 843-862. 10.2217/epi-2016-0002

Kruman, I. I., & Fowler, A. (2014). Impaired one carbon metabolism and DNA methylation in

alcohol toxicity. *Journal of Neurochemistry*, 129(5), 770-780. 10.1111/jnc.12677

Kucuk, N. (2018). Sperm DNA and detection of DNA fragmentations in sperm. *Turkish Journal*

of Urology, 44(1), 1-5. 10.5152/tud.2018.49321

Kvigne, V. L., Leonardson, G. R., Borzelleca, J., & Welty, T. K. (2008). Characteristics of

Grandmothers who have Grandchildren with Fetal Alcohol Syndrome or Incomplete Fetal

Alcohol Syndrome. *Maternal and Child Health Journal*, 12(6), 760-765. 10.1007/s10995-

007-0308-y

Kwo, P. Y., Ramchandani, V. A., O'Connor, S., Amann, D., Carr, L. G., Sandrasegaran, K.,

Kopecky, K. K., & Li, T. (1998). Gender differences in alcohol metabolism: Relationship to

liver volume and effect of adjusting for body mass. *Gastroenterology (New York, N.Y.*

1943), 115(6), 1552-1557. 10.1016/S0016-5085(98)70035-6

Kyle, S. M., Saha, P. K., Brown, H. M., Chan, L. C., & Justice, M. J. (2016). MeCP2 co-

ordinates liver lipid metabolism with the NCoR1/HDAC3 corepressor complex. *Human*

Molecular Genetics, 25(14), 3029-3041. 10.1093/hmg/ddw156

Kyzar, E. J., Zhang, H., & Pandey, S. C. (2019). Adolescent Alcohol Exposure Epigenetically

Suppresses Amygdala Arc Enhancer RNA Expression to Confer Adult Anxiety

- Susceptibility. *Biological Psychiatry* (1969), 85(11), 904-914.
10.1016/j.biopsych.2018.12.021
- Kyzar, E. J., Zhang, H., Sakharkar, A. J., & Pandey, S. C. (2017). Adolescent alcohol exposure alters lysine demethylase 1 (LSD1) expression and histone methylation in the amygdala during adulthood. *Addiction Biology*, 22(5), 1191-1204.
- Lanata, C. M., Chung, S. A., & Criswell, L. A. (2018). DNA methylation 101: what is important to know about DNA methylation and its role in SLE risk and disease heterogeneity. *Lupus Science & Medicine*, 5(1), e000285. 10.1136/lupus-2018-000285
- Lancaster, F. E., Brown, T. D., Coker, K. L., Elliott, J. A., & Wren, S. B. (1996). Sex Differences in Alcohol Preference and Drinking Patterns Emerge during the Early Postpubertal Period in Sprague-Dawley Rats. *Alcoholism, Clinical and Experimental Research*, 20(6), 1043-1049. 10.1111/j.1530-0277.1996.tb01945.x
- Lancaster, F. E., & Spiegel, K. S. (1992). Sex differences in pattern of drinking. *Alcohol (Fayetteville, N.Y.)*, 9(5), 415-420. 10.1016/0741-8329(92)90041-8
- Lara, N. L. M., Santos, I. C., Costa, G. M. J., Cordeiro-Junior, D. A., Almeida, A. C. G., Madureira, A. P., Zanini, M. S., & França, L. R. (2016). Duration of spermatogenesis and daily sperm production in the rodent *Proechimys guyannensis*. *Zygote (Cambridge, England)*, 24(5), 783-793. 10.1017/S0967199416000137
- Larosche, I., Lettéron, P., Berson, A., Fromenty, B., Huang, T., Moreau, R., Pessayre, D., & Mansouri, A. (2010). Hepatic Mitochondrial DNA Depletion after an Alcohol Binge in

- Mice: Probable Role of Peroxynitrite and Modulation by Manganese Superoxide Dismutase. *The Journal of Pharmacology and Experimental Therapeutics*, 332(3), 886-897.
10.1124/jpet.109.160879
- LaSalle, J. M. (2007). The Odyssey of MeCP2 and Parental Imprinting. *Epigenetics*, 2(1), 5-10.
10.4161/epi.2.1.3697
- Laurent, D., Mathew, J. E., Mitry, M., Taft, M., Force, A., & Edwards, J. G. (2014). Chronic Ethanol Consumption Increases Myocardial Mitochondrial DNA Mutations: A Potential Contribution by Mitochondrial Topoisomerases. *Alcohol and Alcoholism (Oxford)*, 49(4), 381-389. 10.1093/alcalc/agu029
- Le Blévec, E., Muroňová, J., Ray, P. F., & Arnoult, C. (2020). Paternal epigenetics: Mammalian sperm provide much more than DNA at fertilization. *Molecular and Cellular Endocrinology*, 518, 110964. 10.1016/j.mce.2020.110964
- Le, Q., Yan, B., Yu, X., Li, Y., Song, H., Zhu, H., Hou, W., Ma, D., Wu, F., Zhou, Y., & Ma, L. (2017). Drug-seeking motivation level in male rats determines offspring susceptibility or resistance to cocaine-seeking behaviour. *Nature Communications*, 8(1), 15527.
10.1038/ncomms15527
- Ledig, M., Simler, S., Ciesielski, L., & Mandel, P. (1993). Alcohol exposure before pregnancy: effect on GABA levels and turnover in rat offspring. *Alcohol and Alcoholism (Oxford)*, 28(2), 175-179. 10.1093/oxfordjournals.alcalc.a045358

Ledig, M., Misslin, R., Vogel, E., Holownia, A., Copin, J. C., & Tholey, G. (1998). Paternal alcohol exposure: developmental and behavioral effects on the offspring of rats.

Neuropharmacology, 37(1), 57-66. 10.1016/S0028-3908(97)00185-8

Lee, H. J., Ryu, J. S., Choi, N. Y., Park, Y. S., Kim, Y. I., Han, D. W., Ko, K. S., Shin, C. Y., Hwng, H. S., Kang, K. S., & Ko, K. (2013). Transgenerational effects of paternal alcohol exposure in mouse offspring. *Animal Cells and Systems*, 17(6), 429-434.

10.1080/19768354.2013.865675

Lee, M., & Williams, D. I. (1977). Stimulation Retards Teeth Eruption and Eye Opening in the Rat. *Perceptual and Motor Skills*, 44(3), 974. 10.2466/pms.1977.44.3.974

Leichter, J. (1986). Effect of paternal alcohol ingestion on fetal growth in rats. *Growth; (United States)*, 50:2 <https://www.osti.gov/biblio/6774691>

Lemoine, P., Harousseau, H., Borteyru, J. P., & Menuet, J. C. (2003). Children of Alcoholic Parents—Observed Anomalies: Discussion of 127 Cases. *Therapeutic Drug Monitoring*, 25(2), 132-136. 10.1097/00007691-200304000-00002

Lesscher, H. M. B., Spoelder, M., Rotte, M. D., Janssen, M. J., Hesselings, P., Lozeman-Van't Klooster, J. G., Baars, A. M., Vanderschuren, L. J. M. J., Sub Neurobiologie van geag, ASS E&C1, & Emotion and Cognition. (2015). Early social isolation augments alcohol consumption in rats. *Behavioural Pharmacology*, 26(7 Spec No), 673-680.

10.1097/FBP.0000000000000165

- Li, M., Du, W., Shao, F., & Wang, W. (2016). Cognitive dysfunction and epigenetic alterations of the BDNF gene are induced by social isolation during early adolescence. *Behavioural Brain Research*, 313, 177-183. 10.1016/j.bbr.2016.07.025
- Li, Z., Liang, Y., Ying, H., Chen, M., He, X., Wang, Y., Tong, Y., & Cai, X. (2021). Mitochondrial dysfunction attenuates rapid regeneration in livers with toxin-induced fibrosis. *Annals of Translational Medicine*, 9(7), 527. 10.21037/atm-20-4639
- Liang, F., Diao, L., Jiang, N., Zhang, J., Wang, H., Zhou, W., Huang, G., & Ma, D. (2015). Chronic exposure to ethanol in male mice may be associated with hearing loss in offspring. *Asian Journal of Andrology*, 17(6), 985-990. 10.4103/1008-682X.160267
- Liang, F., Diao, L., Liu, J., Jiang, N., Zhang, J., Wang, H., Zhou, W., Huang, G., & Ma, D. (2014). Paternal ethanol exposure and behavioral abnormalities in offspring: Associated alterations in imprinted gene methylation. *Neuropharmacology*, 81, 126-133. 10.1016/j.neuropharm.2014.01.025
- Lieber, C. S., Leo, M. A., Wang, X., & DeCarli, L. M. (2008). Effect of chronic alcohol consumption on Hepatic SIRT1 and PGC-1 α in rats. *Biochemical and Biophysical Research Communications*, 370(1), 44-48. 10.1016/j.bbrc.2008.03.005
- Liljequist, S., & Karcz-Kubicha, M. (1993). Genetic aspects on the effects of ethanol and central stimulants on locomotor activity and brain dopamine metabolism in mice. *Alcohol and Alcoholism (Oxford, Oxfordshire) Supplement.*, 2, 457-461.
<https://www.ncbi.nlm.nih.gov/pubmed/7748339>

- Liu, N., & Pan, T. (2015). RNA epigenetics. *Translational Research : The Journal of Laboratory and Clinical Medicine*, 165(1), 28-35. 10.1016/j.trsl.2014.04.003
- Liu, S., & Sharma, U. (2023). Sperm RNA Payload: Implications for Intergenerational Epigenetic Inheritance. *International Journal of Molecular Sciences*, 24(6), 5889. 10.3390/ijms24065889
- Liu, Z., Lin, R., & Luo, M. (2020). Reward Contributions to Serotonergic Functions. *Annual Review of Neuroscience*, 43(1), 141-162. 10.1146/annurev-neuro-093019-112252
- Lluis, J. M., Colell, A., García-Ruiz, C., Kaplowitz, N., & Fernández-Checa, J. C. (2003). Acetaldehyde impairs mitochondrial glutathione transport in HepG2 cells through endoplasmic reticulum stress. *Gastroenterology (New York, N.Y. 1943)*, 124(3), 708-724. 10.1053/gast.2003.50089
- Lohoff, F. W., Roy, A., Jung, J., Longley, M., Rosoff, D. B., Luo, A., O'Connell, E., Sorcher, J. L., Sun, H., Schwandt, M., Hodgkinson, C. A., Goldman, D., Momenan, R., McIntosh, A. M., Adams, M. J., Walker, R. M., Evans, K. L., Porteous, D., Smith, A. K., . . . Kaminsky, Z. A. (2021). Epigenome-wide association study and multi-tissue replication of individuals with alcohol use disorder: evidence for abnormal glucocorticoid signaling pathway gene regulation. *Molecular Psychiatry*, 26(6), 2224-2237. 10.1038/s41380-020-0734-4
- Luan, M., Zhang, X., Fang, G., Liang, H., Yang, F., Song, X., Chen, Y., Yuan, W., & Miao, M. (2022). Preconceptional paternal alcohol consumption and the risk of child behavioral

problems: a prospective cohort study. *Scientific Reports*, 12(1), 1508. 10.1038/s41598-022-05611-2

Lukkes, J. L., Mokin, M. V., Scholl, J. L., & Forster, G. L. (2009). Adult rats exposed to early-life social isolation exhibit increased anxiety and conditioned fear behavior, and altered hormonal stress responses. *Hormones and Behavior*, 55(1), 248-256.
10.1016/j.yhbeh.2008.10.014

Maeda, K., Yamada, H., Munetsuna, E., Fujii, R., Yamazaki, M., Ando, Y., Mizuno, G., Ishikawa, H., Ohashi, K., Tsuboi, Y., Hattori, Y., Ishihara, Y., Hashimoto, S., Hamajima, N., & Suzuki, K. (2022). Association of drinking behaviors with TXNIP DNA methylation levels in leukocytes among the general Japanese population. *The American Journal of Drug and Alcohol Abuse*, 48(3), 302-310. 10.1080/00952990.2022.2037137

Mahesh, P. K. B., Leung, J., Senananayake, S., Kumarapeli, V., Ferdinando, R., Jayawickrama, W. i. U., Lakmini, W. D., Perera, C., Thasleem, S., & McLaughlin, D. (2021). Effects of paternal alcoholism on the psycho-behavioural outcomes of the offspring: a systematic review and meta-analysis. *Journal of the College of Community Physicians of Sri Lanka*, 27(3), 442-461. 10.4038/jccpsl.v27i3.8398

Mankes, R. F., Glick, S. D., Van der Hoeven, T., & LeFevre, R. (1991). Alcohol Preference and Hepatic Alcohol Dehydrogenase Activity in Adult Long-Evans Rats is Affected by Intrauterine Sibling Contiguity. *Alcoholism, Clinical and Experimental Research*, 15(1), 80-85. 10.1111/j.1530-0277.1991.tb00521.x

Mann, J., Chu, D. C. K., Maxwell, A., Oakley, F., Zhu, N., Tsukamoto, H., & Mann, D. A.

(2010). MeCP2 Controls an Epigenetic Pathway That Promotes Myofibroblast Transdifferentiation and Fibrosis. *Gastroenterology (New York, N.Y. 1943)*, 138(2), 705-714.e4. 10.1053/j.gastro.2009.10.002

Mann, K., Ackermann, K., Croissant, B., Mundle, G., Nakovics, H., & Diehl, A. (2005).

Neuroimaging of gender differences in alcohol dependence: are women more vulnerable? *Alcoholism: Clinical and Experimental Research*, 29(5), 896-901.

Mansouri, A., Demeilliers, C., Amsellem, S., Pessayre, D., & Fromenty, B. (2001). Acute

Ethanol Administration Oxidatively Damages and Depletes Mitochondrial DNA in Mouse Liver, Brain, Heart, and Skeletal Muscles: Protective Effects of Antioxidants. *The Journal of Pharmacology and Experimental Therapeutics*, 298(2), 737-743.

<http://jpet.aspetjournals.org/content/298/2/737.abstract>

Mansouri, A., Gaou, I., de Kerguenec, C., Amsellem, S., Haouzi, D., Berson, A., Moreau ‡, A.,

Feldmann ‡, G., Lettéron, P., Pessayre, D., & Fromenty, B. (1999). An alcoholic binge causes massive degradation of hepatic mitochondrial DNA in mice. *Gastroenterology (New York, N.Y. 1943)*, 117(1), 181-190. 10.1016/S0016-5085(99)70566-4

Manto, M., Bower, J. M., Conforto, A. B., Delgado-García, J. M., da Guarda, S. N. F., Gerwig,

M., Habas, C., Hagura, N., Ivry, R. B., Mariën, P., Molinari, M., Naito, E., Nowak, D. A., Oulad Ben Taib, N., Pelisson, D., Tesche, C. D., Tilikete, C., & Timmann, D. (2012).

Consensus Paper: Roles of the Cerebellum in Motor Control—The Diversity of Ideas on

- Cerebellar Involvement in Movement. *Cerebellum (London, England)*, 11(2), 457-487.
10.1007/s12311-011-0331-9
- Manzardo, A. M., Henkhaus, R. S., & Butler, M. G. (2012). Global DNA promoter methylation in frontal cortex of alcoholics and controls. *Gene*, 498(1), 5-12. 10.1016/j.gene.2012.01.096
- Marin-Garcia, J., Ananthakrishnan, R., & Goldenthal, M. J. (1995). Heart Mitochondria Response to Alcohol Is Different than Brain and Liver. *Alcoholism, Clinical and Experimental Research*, 19(6), 1463-1466. 10.1111/j.1530-0277.1995.tb01008.x
- Marsland, P., Parrella, A., Vore, A. S., Barney, T. M., Varlinskaya, E. I., & Deak, T. (2021). Male, but not female, Sprague Dawley rats display enhanced fear learning following acute ethanol withdrawal (hangover). *Pharmacology Biochemistry and Behavior*, 208, 173229.
- Matyas, C., Varga, Z. V., Mukhopadhyay, P., Paloczi, J., Lajtos, T., Erdelyi, K., Nemeth, B. T., Nan, M., Hasko, G., Gao, B., & Pacher, P. (2016). Chronic plus binge ethanol feeding induces myocardial oxidative stress, mitochondrial and cardiovascular dysfunction, and steatosis. *American Journal of Physiology. Heart and Circulatory Physiology*, 310(11), H1658-H1670. 10.1152/ajpheart.00214.2016
- McBride, N., & Johnson, S. (2016). Fathers' Role in Alcohol-Exposed Pregnancies: Systematic Review of Human Studies. *American Journal of Preventive Medicine*, 51(2), 240-248.
10.1016/j.amepre.2016.02.009
- McCarthy, D. M., Morgan, T. J., Jr., Lowe, S. E., Williamson, M. J., Spencer, T. J., Biederman, J., & Bhide, P. G. (2018). Nicotine exposure of male mice produces behavioral impairment

in multiple generations of descendants. *PLOS Biology*, 16(10), e2006497.

<https://doi.org/10.1371/journal.pbio.2006497>

McCarthy, F. P., O'Keeffe, L. M., Khashan, A. S., North, R. A., Poston, L., McCowan, L. M. E., Baker, P. N., Dekker, G. A., Roberts, C. T., Walker, J. J., & Kenny, L. C. (2013).

Association between maternal alcohol consumption in early pregnancy and pregnancy outcomes. *Obstetrics and Gynecology*, 122(4), 830-837. 10.1097/AOG.0b013e3182a6b226

McCormack, C., Hutchinson, D., Burns, L., Youssef, G., Wilson, J., Elliott, E., Allsop, S.,

Najman, J., Jacobs, S., Rossen, L., Olsson, C., & Mattick, R. (2018). Maternal and partner prenatal alcohol use and infant cognitive development. *Drug and Alcohol Dependence*, 185, 330-338. 10.1016/j.drugalcdep.2017.12.038

McDonald, B. W., & Watson, P. E. (2020). Maternal alcohol intakes before and during

pregnancy: Impact on the mother and infant outcome to 18 months. *Nordisk Alkohol- & Narkotikatidskrift : NAT*, 37(2), 153-171. 10.1177/1455072520905404

McPherson, N. O., Owens, J. A., Fullston, T., & Lane, M. (2015). Preconception diet or exercise

intervention in obese fathers normalizes sperm microRNA profile and metabolic syndrome in female offspring. *American Journal of Physiology: Endocrinology and Metabolism*, 308(9), E805-E821. 10.1152/ajpendo.00013.2015

McQueen, D. B., Zhang, J., & Robins, J. C. (2019). Sperm DNA fragmentation and recurrent

pregnancy loss: a systematic review and meta-analysis. *Fertility and Sterility*, 112(1), 54-60.e3. 10.1016/j.fertnstert.2019.03.003

- Meek, L. R., Myren, K., Sturm, J., & Burau, D. (2007). Acute paternal alcohol use affects offspring development and adult behavior. *Physiology & Behavior*, 91(1), 154-160. 10.1016/j.physbeh.2007.02.004
- Meisch, R. A., & Thompson, T. (1973). Ethanol as a reinforcer: Effects of fixed-ratio size and food deprivation. *Psychopharmacologia*, 28(2), 171-183. 10.1007/BF00421402
- Melón, L. C., Wray, K. N., Moore, E. M., & Boehm, S. L. (2013). Sex and age differences in heavy binge drinking and its effects on alcohol responsivity following abstinence. *Pharmacology, Biochemistry and Behavior*, 104, 177-187. 10.1016/j.pbb.2013.01.005
- Meng, W., Sjöholm, L. K., Kononenko, O., Tay, N., Zhang, D., Sarkisyan, D., Geske, J. R., Ing, A., Qiu, W., Watanabe, H., Almamoun, R., Frieling, H., Bleich, S., Cui, D., Biernacka, J. M., Mayfield, R. D., Dang, Y., Karpyak, V. M., Schumann, G., . . . Liu, Y. (2021). Genotype-dependent epigenetic regulation of DLGAP2 in alcohol use and dependence. *Molecular Psychiatry*, 26(8), 4367-4382. 10.1038/s41380-019-0588-9
- Mengel-From, J., Thinggaard, M., Dalgård, C., Kyvik, K. O., Christensen, K., & Christiansen, L. (2014). Mitochondrial DNA copy number in peripheral blood cells declines with age and is associated with general health among elderly. *Human Genetics*, 133(9), 1149-1159. 10.1007/s00439-014-1458-9
- Merrill, S. M., Moore, S. R., Gladish, N., Giesbrecht, G. F., Dewey, D., Konwar, C., MacIssac, J. L., Kobor, M. S., & Letourneau, N. L. (2021). Paternal adverse childhood experiences:

- Associations with infant DNA methylation. *Developmental Psychobiology*, 63(6), e22174-n/a. 10.1002/dev.22174
- Mesquita, A. R., Pêgo, J. M., Summavielle, T., Maciel, P., Almeida, O. F. X., & Sousa, N. (2007). Neurodevelopment milestone abnormalities in rats exposed to stress in early life. *Neuroscience*, 147(4), 1022-1033. 10.1016/j.neuroscience.2007.04.007
- Michalopoulos, G. K. (2017). Hepatostat: Liver regeneration and normal liver tissue maintenance. *Hepatology (Baltimore, Md.)*, 65(4), 1384-1392. 10.1002/hep.28988
- Michel, S., Wanet, A., De Pauw, A., Rommelaere, G., Arnould, T., & Renard, P. (2012). Crosstalk between mitochondrial (dys)function and mitochondrial abundance. *Journal of Cellular Physiology*, 227(6), 2297-2310. 10.1002/jcp.23021
- Milne, E., Greenop, K. R., Scott, R. J., de Klerk, N. H., Bower, C., Ashton, L. J., Heath, J. A., & Armstrong, B. K. (2013). Parental alcohol consumption and risk of childhood acute lymphoblastic leukemia and brain tumors. *Cancer Causes & Control*, 24(2), 391-402. 10.1007/s10552-012-0125-5
- Ministry of Health. (2022). *Annual Data Explorer 2021/22: New Zealand Health Survey*. (). NZ: Ministry of Health. <https://minhealthnz.shinyapps.io/nz-health-survey-2021-22-annual-data-explorer/>
- Monk, D., Mackay, D. J. G., Eggermann, T., Maher, E. R., & Riccio, A. (2019). Genomic imprinting disorders: lessons on how genome, epigenome and environment interact. *Nature Reviews. Genetics*, 20(4), 235-248. 10.1038/s41576-018-0092-0

- Montesinos, J., Pascual, M., Rodríguez-Arias, M., Miñarro, J., & Guerri, C. (2016). Involvement of TLR4 in the long-term epigenetic changes, rewarding and anxiety effects induced by intermittent ethanol treatment in adolescence. *Brain, Behavior, and Immunity*, 53, 159-171. 10.1016/j.bbi.2015.12.006
- Moore, C. F., & Lynch, W. J. (2015). Alcohol preferring (P) rats as a model for examining sex differences in alcohol use disorder and its treatment. *Pharmacology, Biochemistry and Behavior*, 132, 1-9. 10.1016/j.pbb.2015.02.014
- Moore, L. D., Le, T., & Fan, G. (2013). DNA Methylation and Its Basic Function. *Neuropsychopharmacology (New York, N.Y.)*, 38(1), 23-38. 10.1038/npp.2012.112
- Morio, B., Panthu, B., Bassot, A., & Rieusset, J. (2021). Role of mitochondria in liver metabolic health and diseases. *Cell Calcium (Edinburgh)*, 94, 102336. 10.1016/j.ceca.2020.102336
- Mouzon, B., Chaytow, H., Crynen, G., Bachmeier, C., Stewart, J., Mullan, M., Stewart, W., & Crawford, F. (2012). Repetitive mild traumatic brain injury in a mouse model produces learning and memory deficits accompanied by histological changes. *Journal of Neurotrauma*, 29(18), 2761-2773. 10.1089/neu.2012.2498
- Mukhopadhyay, D., Varghese, A. C., Pal, M., Banerjee, S. K., Bhattacharyya, A. K., Sharma, R. K., & Agarwal, A. (2010). Semen quality and age-specific changes: a study between two decades on 3,729 male partners of couples with normal sperm count and attending an andrology laboratory for infertility-related problems in an Indian city. *Fertility and Sterility*, 93(7), 2247-2254. 10.1016/j.fertnstert.2009.01.135

- Muschler, M. A. N., Hillemacher, T., Kraus, C., Kornhuber, J., Bleich, S., & Frieling, H. (2010). DNA methylation of the POMC gene promoter is associated with craving in alcohol dependence. *Journal of Neural Transmission*, 117(4), 513-519. 10.1007/s00702-010-0378-7
- Myers, R. D., Robinson, D. E., West, M. W., Biggs, T. A. G., & Mcmillen, B. A. (1998). Genetics of Alcoholism: Rapid Development of a New High-Ethanol-Preferring (HEP) Strain of Female and Male Rats. *Alcohol (Fayetteville, N.Y.)*, 16(4), 343-357. 10.1016/S0741-8329(98)00031-7
- Nassir, F., & Ibdah, J. A. (2014). Role of mitochondria in alcoholic liver disease. *World Journal of Gastroenterology : WJG*, 20(9), 2136-2142. 10.3748/wjg.v20.i9.2136
- National Health Service. (2023, 23 October,). *How can I improve my chances of becoming a dad?* nhs.uk. <https://www.nhs.uk/common-health-questions/mens-health/how-can-i-improve-my-chances-of-becoming-a-dad/>
- Nelson, B. K., Brightwell, W. S., MacKenzie-Taylor, D. R., Burg, J. R., & Massari, V. J. (1988). Neurochemical, but not behavioral, deviations in the offspring of rats following prenatal or paternal inhalation exposure to ethanol. *Neurotoxicology and Teratology*, 10(1), 15-22. 10.1016/0892-0362(88)90062-1
- Nieschlag, E., Lammers, U., Freischem, C. W., Langer, K., & Wickings, E. J. (1982). Reproductive functions in young fathers and grandfathers. *The Journal of Clinical Endocrinology and Metabolism*, 55(4), 676-681. 10.1210/jcem-55-4-676

- Nieto, S. J., Harding, M. J., Nielsen, D. A., & Kosten, T. A. (2022). Paternal alcohol exposure has task- and sex-dependent behavioral effect in offspring. *Alcoholism, Clinical and Experimental Research*, 46(12), 2191-2202. 10.1111/acer.14964
- Nieto, S. J., & Kosten, T. A. (2023). Paternal alcohol exposure attenuates maintenance and reinstated operant responding for alcohol in the offspring of rats. *Alcohol, Clinical & Experimental Research*, 10.1111/acer.15136
- Nikitina, T., Shi, X., Ghosh, R. P., Horowitz-Scherer, R. A., Hansen, J. C., & Woodcock, C. L. (2007). Multiple Modes of Interaction between the Methylated DNA Binding Protein MeCP2 and Chromatin. *Molecular and Cellular Biology*, 27(3), 864-877. 10.1128/MCB.01593-06
- Niu, Y., Zhao, X., Wu, Y., Li, M., Wang, X., & Yang, Y. (2013). N6-methyl-adenosine (m6A) in RNA: An Old Modification with A Novel Epigenetic Function. *Genomics, Proteomics & Bioinformatics*, 11(1), 8-17. 10.1016/j.gpb.2012.12.002
- Novoa, J., Rivero, C. J., Pérez-Cardona, E. U., Freire-Arvelo, J. A., Zegers, J., Yarur, H. E., Santiago-Marerro, I. G., Agosto-Rivera, J. L., González-Pérez, J. L., Gysling, K., & Segarra, A. C. (2021). Social isolation of adolescent male rats increases anxiety and K⁺-induced dopamine release in the nucleus accumbens: Role of CRF-R1. *The European Journal of Neuroscience*, 54(3), 4888-4905. 10.1111/ejn.15345
- Nunnari, J., & Suomalainen, A. (2012). Mitochondria: In Sickness and in Health. *Cell*, 148(6), 1145-1159. 10.1016/j.cell.2012.02.035

- Obe, G., & Anderson, D. (1987). Genetic effects of ethanol. *Mutation Research/Reviews in Genetic Toxicology*, 186(3), 177-200. 10.1016/0165-1110(87)90003-0
- Ogutlu, H., Esin, I. S., Erdem, H. B., Tatar, A., & Dursun, O. B. (2020). Mitochondrial DNA copy number is associated with attention deficit hyperactivity disorder. *Psychiatria Danubina*, 32(2), 168-175. 10.24869/psyd.2020.168
- Ohta, S., Ohsawa, I., Kamino, K., Ando, F., & Shimokata, H. (2004). Mitochondrial ALDH2 Deficiency as an Oxidative Stress. *Mitochondrial Pathogenesis* (pp. 36-44). Springer Berlin Heidelberg. 10.1007/978-3-662-41088-2_4
- Olfson, E., Edenberg, H. J., Nurnberger Jr, J., Agrawal, A., Bucholz, K. K., Almasy, L. A., Chorlian, D., Dick, D. M., Hesselbrock, V. M., & Kramer, J. R. (2014). An ADH 1B Variant and Peer Drinking in Progression to Adolescent Drinking Milestones: Evidence of a Gene-by-Environment Interaction. *Alcoholism: Clinical and Experimental Research*, 38(10), 2541-2549.
- ONS. (2022). *Drug misuse in England and Wales: year ending June 2022*. (). ONS website: Office for National Statistics.
<https://www.ons.gov.uk/peoplepopulationandcommunity/crimeandjustice/articles/drugmisuseinenglandandwales/yearendingjune2022>
- Oppermann, U. (2013). Why is epigenetics important in understanding the pathogenesis of inflammatory musculoskeletal diseases? *Arthritis Research & Therapy*, 15(2), 209. 10.1186/ar4186

- Orford, J., & Velleman, R. (1990). Offspring of parents with drinking problems: drinking and drug-taking as young adults. *British Journal of Addiction*, 85(Jun 90), 779-794.
<https://search.proquest.com/docview/57759673>
- Otero, N. K. H., Thomas, J. D., Saski, C. A., Xia, X., & Kelly, S. J. (2012). Choline Supplementation and DNA Methylation in the Hippocampus and Prefrontal Cortex of Rats Exposed to Alcohol During Development. *Alcoholism, Clinical and Experimental Research*, 36(10), 1701-1709. 10.1111/j.1530-0277.2012.01784.x
- Ouko, L. A., Shantikumar, K., Knezovich, J., Haycock, P., Schnugh, D. J., & Ramsay, M. (2009). Effect of Alcohol Consumption on CpG Methylation in the Differentially Methylated Regions of H19 and IG-DMR in Male Gametes—Implications for Fetal Alcohol Spectrum Disorders. *Alcoholism, Clinical and Experimental Research*, 33(9), 1615-1627. 10.1111/j.1530-0277.2009.00993.x
- Overstreet, D. H., Knapp, D. J., & Breese, G. R. (2004). Similar anxiety-like responses in male and female rats exposed to repeated withdrawals from ethanol. *Pharmacology, Biochemistry and Behavior*, 78(3), 459-464. 10.1016/j.pbb.2004.04.018
- Ozkaragoz, T., Satz, P., & Noble, E. P. (1997). Neuropsychological functioning in sons of active alcoholic, recovering alcoholic, and social drinking fathers. *Alcohol (Fayetteville, N.Y.)*, 14(1), 31-37. 10.1016/S0741-8329(96)00084-5
- Painter, R. C., Rooij, S. R. d., Bossuyt, P. M., Simmers, T. A., Osmond, C., Barker, D. J., Bleker, O. P., & Roseboom, T. J. (2006). Early onset of coronary artery disease after prenatal

- exposure to the Dutch famine. *The American Journal of Clinical Nutrition*, 84(2), 322-327.
10.1093/ajcn/84.2.322
- Pandey, S. C., Sakharkar, A. J., Tang, L., & Zhang, H. (2015). Potential role of adolescent alcohol exposure-induced amygdaloid histone modifications in anxiety and alcohol intake during adulthood. *Neurobiology of Disease*, 82, 607-619. 10.1016/j.nbd.2015.03.019
- Pandey, S. C., Ugale, R., Zhang, H., Tang, L., & Prakash, A. (2008). Brain Chromatin Remodeling: A Novel Mechanism of Alcoholism. *The Journal of Neuroscience*, 28(14), 3729-3737. 10.1523/JNEUROSCI.5731-07.2008
- Passaro, K. T., Little, R. E., Savitz, D. A., & Noss, J. (1998). Effect of paternal alcohol consumption before conception on infant birth weight. *Teratology (Philadelphia)*, 57(6), 294-301. 10.1002/(SICI)1096-9926(199806)57:63.0.CO;2-X
- Patlar, B. (2022). On the Role of Seminal Fluid Protein and Nucleic Acid Content in Paternal Epigenetic Inheritance. *International Journal of Molecular Sciences*, 23(23), 14533. doi: 10.3390/ijms232314533. 10.3390/ijms232314533
- Patlar, B., Weber, M., & Ramm, S. A. (2019). Genetic and environmental variation in transcriptional expression of seminal fluid proteins. *Heredity*, 122(5), 595-611.
10.1038/s41437-018-0160-4
- Pembrey, M. E., Bygren, L. O., Kaati, G., Edvinsson, S., Northstone, K., Sjöström, M., & Golding, J. (2006). Sex-specific, male-line transgenerational responses in humans. *European Journal of Human Genetics : EJHG*, 14(2), 159-166. 10.1038/sj.ejhg.5201538

- Peng, H., Shi, J., Zhang, Y., Zhang, H., Liao, S., Li, W., Lei, L., Han, C., Ning, L., Cao, Y., Zhou, Q., Chen, Q., & Duan, E. (2012). A novel class of tRNA-derived small RNAs extremely enriched in mature mouse sperm. *Cell Research*, 22(11), 1609-1612. 10.1038/cr.2012.141
- Perez, M. F., & Lehner, B. (2019). Intergenerational and transgenerational epigenetic inheritance in animals. *Nature Cell Biology*, 21(2), 143-151. 10.1038/s41556-018-0242-9
- Philibert, R., Dogan, M., Noel, A., Miller, S., Krukow, B., Papworth, E., Cowley, J., Knudsen, A., Beach, S. R. H., & Black, D. (2018). Genome-wide and digital polymerase chain reaction epigenetic assessments of alcohol consumption. *American Journal of Medical Genetics. Part B, Neuropsychiatric Genetics*, 177(5), 479-488. 10.1002/ajmg.b.32636
- Phillips, B. J., & Jenkinson, P. (2001). Is ethanol genotoxic? A review of the published data. *Mutagenesis*, 16(2), 91-101. 10.1093/mutage/16.2.91
- Picard, M. (2021). Blood mitochondrial DNA copy number: What are we counting? *Mitochondrion*, 60, 1-11. 10.1016/j.mito.2021.06.010
- Pirino, B. E., Martin, C. R., Carpenter, B. A., Curtis, G. R., Curran-Alfaro, C. M., Samels, S. B., Barker, J. M., Karkhanis, A. N., & Barson, J. R. (2022). Sex-related differences in pattern of ethanol drinking under the intermittent-access model and its impact on exploratory and anxiety-like behavior in Long-Evans rats. *Alcoholism, Clinical and Experimental Research*, 46(7), 1282-1293. 10.1111/acer.14853

- Podlesniy, P., & Trullas, R. (2017). Absolute measurement of gene transcripts with Selfie-digital PCR. *Scientific Reports*, 7(1), 8328-15. 10.1038/s41598-017-08270-w
- Pohl, K., Moodley, P., & Dhanda, A. D. (2021). Alcohol's Impact on the Gut and Liver. *Nutrients*, 13(9), 3170. 10.3390/nu13093170
- Ponomarev, I., Wang, S., Zhang, L., Harris, R. A., & Mayfield, R. D. (2012). Gene Coexpression Networks in Human Brain Identify Epigenetic Modifications in Alcohol Dependence. *The Journal of Neuroscience*, 32(5), 1884-1897. 10.1523/JNEUROSCI.3136-11.2012
- Poon, H. K., Lee, K. H., Wong, C. L., O, W. S., & Chow, P. H. (2009). Absence of paternal accessory sex gland secretions disturbs epigenetic reprogramming and expression of Igf2 and Dlk1 in golden hamster embryos. *Theriogenology*, 71(9), 1367-1380. 10.1016/j.theriogenology.2008.12.016
- Priddy, B. M., Carmack, S. A., Thomas, L. C., Vendruscolo, J. C. M., Koob, G. F., & Vendruscolo, L. F. (2017). Sex, strain, and estrous cycle influences on alcohol drinking in rats. *Pharmacology, Biochemistry and Behavior*, 152, 61-67. 10.1016/j.pbb.2016.08.001
- Przybycien-Szymanska, M. M., Rao, Y. S., Prins, S. A., & Pak, T. R. (2014). Parental Binge Alcohol Abuse Alters F1 Generation Hypothalamic Gene Expression in the Absence of Direct Fetal Alcohol Exposure. *PLoS ONE*, 9(2), e89320. 10.1371/journal.pone.0089320
- Quertemont, E., & Didone, V. (2006). Role of acetaldehyde in mediating the pharmacological and behavioral effects of alcohol. *Alcohol Research and Health: The Journal of the National*

Institute on Alcohol Abuse and Alcoholism, 29(4), 258-265.

<https://www.ncbi.nlm.nih.gov/pubmed/17718404>

- Radford, E. J., Ito, M., Shi, H., Corish, J. A., Yamazawa, K., Isganaitis, E., Seisenberger, S., Hore, T. A., Reik, W., Erkek, S., Peters, A. H. F. M., Patti, M., & Ferguson-Smith, A. C. (2014). In utero undernourishment perturbs the adult sperm methylome and intergenerational metabolism. *Science (American Association for the Advancement of Science)*, 345(6198), 785. 10.1126/science.1255903
- Rafelski, S. M. (2013). Mitochondrial network morphology: building an integrative, geometrical view. *BMC Biology*, 11(1), 71. 10.1186/1741-7007-11-71
- Rahimipour, M., Talebi, A. R., Anvari, M., Sarcheshmeh, A. A., & Omid, M. (2013). Effects of different doses of ethanol on sperm parameters, chromatin structure and apoptosis in adult mice. *European Journal of Obstetrics & Gynecology and Reproductive Biology*, 170(2), 423-428. 10.1016/j.ejogrb.2013.06.038
- Ramm, S. A. (2020). Seminal fluid and accessory male investment in sperm competition. *Philosophical Transactions of the Royal Society B*, 375(1813), 20200068. 10.1098/rstb.2020.0068
- Randall, C. L., Burling, T. A., Lochry, E. A., & Sutker, P. B. (1982). *The effect of paternal alcohol consumption on fetal development in mice*. Elsevier Ireland Ltd. 10.1016/0376-8716(82)90028-X

- Randall, C. L., Carpenter, J. A., Lester, D., & Friedman, H. J. (1975). Ethanol-induced mouse strain differences in locomotor activity. *Pharmacology, Biochemistry and Behavior*, 3(3), 533-535. 10.1016/0091-3057(75)90069-6
- Randall, P. A., Stewart, R. T., & Besheer, J. (2017). Sex differences in alcohol self-administration and relapse-like behavior in Long-Evans rats. *Pharmacology, Biochemistry and Behavior*, 156, 1-9. 10.1016/j.pbb.2017.03.005
- Rathod, R. S., Ferguson, C., Seth, A., Baratta, A. M., Plasil, S. L., & Homanics, G. E. (2020). Effects of Paternal Preconception Vapor Alcohol Exposure Paradigms on Behavioral Responses in Offspring. *Brain Sciences*, 10(9), 658. 10.3390/brainsci10090658
- Reik, W. (2007). Stability and flexibility of epigenetic gene regulation in mammalian development. *Nature*, 447(7143), 425-432.
- Reilly, J. N., McLaughlin, E. A., Stanger, S. J., Anderson, A. L., Hutcheon, K., Church, K., Mihalas, B. P., Tyagi, S., Holt, J. E., Eamens, A. L., & Nixon, B. (2016). Characterisation of mouse epididymosomes reveals a complex profile of microRNAs and a potential mechanism for modification of the sperm epigenome. *Scientific Reports*, 6(1), 31794. 10.1038/srep31794
- Repunte-Canonigo, V., Chen, J., Lefebvre, C., Kawamura, T., Kreifeldt, M., Basson, O., Roberts, A. J., & Sanna, P. P. (2014). MeCP2 regulates ethanol sensitivity and intake. *Addiction Biology*, 19(5), 791-799. 10.1111/adb.12047

- Ricci, E., Al Beitawi, S., Cipriani, S., Candiani, M., Chiaffarino, F., Viganò, P., Noli, S., & Parazzini, F. (2017). Semen quality and alcohol intake: a systematic review and meta-analysis. *Reproductive Biomedicine Online*, 34(1), 38-47. 10.1016/j.rbmo.2016.09.012
- Rocco, A., Compare, D., Angrisani, D., Sanduzzi Zamparelli, M., & Nardone, G. (2014). Alcoholic disease: Liver and beyond. *World Journal of Gastroenterology : WJG*, 20(40), 14652-14659. 10.3748/wjg.v20.i40.14652
- Rodgers, A. B., Morgan, C. P., Leu, N. A., & Bale, T. L. (2015). Transgenerational epigenetic programming via sperm microRNA recapitulates effects of paternal stress. *Proceedings of the National Academy of Sciences - PNAS*, 112(44), 13699-13704. 10.1073/pnas.1508347112
- Rodgers, A. B., Morgan, C. P., Bronson, S. L., Revello, S., & Bale, T. L. (2013). Paternal Stress Exposure Alters Sperm MicroRNA Content and Reprograms Offspring HPA Stress Axis Regulation. *The Journal of Neuroscience*, 33(21), 9003-9012. 10.1523/JNEUROSCI.0914-13.2013
- Roeleveld, N., Vingerhoets, E., Zielhuis, G. A., & Gabreëls, F. (1992). Mental retardation associated with parental smoking and alcohol consumption before, during, and after pregnancy. *Preventive Medicine*, 21(1), 110-119. 10.1016/0091-7435(92)90010-F
- Rompala, G. R., & Homanics, G. E. (2019). Intergenerational Effects of Alcohol: A Review of Paternal Preconception Ethanol Exposure Studies and Epigenetic Mechanisms in the Male

Germline. *Alcoholism, Clinical and Experimental Research*, 43(6), 1032-1045.

10.1111/acer.14029

Rompala, G. R., Finegersh, A., & Homanics, G. E. (2016). Paternal preconception ethanol exposure blunts hypothalamic-pituitary-adrenal axis responsivity and stress-induced excessive fluid intake in male mice. *Alcohol (Fayetteville, N.Y.)*, 53, 19-25.

10.1016/j.alcohol.2016.03.006

Rompala, G. R., Finegersh, A., Slater, M., & Homanics, G. E. (2017). Paternal preconception alcohol exposure imparts intergenerational alcohol-related behaviors to male offspring on a pure C57BL/6J background. *Alcohol (Fayetteville, N.Y.)*, 60, 169-177.

10.1016/j.alcohol.2016.11.001

Rompala, G. R., Mounier, A., Wolfe, C. M., Lin, Q., Lefterov, I., & Homanics, G. E. (2018). Heavy Chronic Intermittent Ethanol Exposure Alters Small Noncoding RNAs in Mouse Sperm and Epididymosomes. *Frontiers in Genetics*, 9, 32. 10.3389/fgene.2018.00032

Roseboom, T. J., Painter, R. C., van Abeelen, A. F. M., Veenendaal, M. V. E., & de Rooij, S. R. (2011). Hungry in the womb: What are the consequences? Lessons from the Dutch famine. *Maturitas*, 70(2), 141-145. 10.1016/j.maturitas.2011.06.017

Rossetto, I. M. U., Cagnon, V. H. A., Lizarte, F. S. N., Tirapelli, L. F., Tirapelli, D. P. C., Arantes, R. M. S., Chuffa, L. G. A., Martinez, F. E., & Martinez, M. (2019). Ethanol and caffeine consumption modulates the expression of miRNAs in the cerebellum and plasma of UChB rats. *Life Sciences (1973)*, 229, 180-186. 10.1016/j.lfs.2019.05.016

- Russo, S. J., Mazei-Robison, M. S., Ables, J. L., & Nestler, E. J. (2009). Neurotrophic factors and structural plasticity in addiction. *Neuropharmacology*, 56(Suppl 1), 73-82.
10.1016/j.neuropharm.2008.06.059
- Rustay, N. R., Wahlsten, D., & Crabbe, J. C. (2003). Influence of task parameters on rotarod performance and sensitivity to ethanol in mice. *Behavioural Brain Research*, 141(2), 237-249. 10.1016/S0166-4328(02)00376-5
- Sabino, V., Kwak, J., Rice, K. C., & Cottone, P. (2013). Pharmacological Characterization of the 20% Alcohol Intermittent Access Model in Sardinian Alcohol-Preferring Rats: A Model of Binge-Like Drinking. *Alcoholism, Clinical and Experimental Research*, 37(4), 635-643.
10.1111/acer.12008
- Sakharkar, A. J., Kyzar, E. J., Gavin, D. P., Zhang, H., Chen, Y., Krishnan, H. R., Grayson, D. R., & Pandey, S. C. (2019). Altered amygdala DNA methylation mechanisms after adolescent alcohol exposure contribute to adult anxiety and alcohol drinking. *Neuropharmacology*, 157, 107679. 10.1016/j.neuropharm.2019.107679
- Samson, H. H. (1986). Initiation of Ethanol Reinforcement using a Sucrose-Substitution Procedure in Food- and Water-Sated Rats. *Alcoholism, Clinical and Experimental Research*, 10(4), 436-442. 10.1111/j.1530-0277.1986.tb05120.x
- Sandovici, I., Fernandez-Twinn, D. S., Hufnagel, A., Constância, M., & Ozanne, S. E. (2022). Sex differences in the intergenerational inheritance of metabolic traits. *Nature Metabolism*, 4(5), 507-523. 10.1038/s42255-022-00570-4

- Santiago, E., Moreno, D. F., & Acar, M. (2022). Phenotypic plasticity as a facilitator of microbial evolution. *Environmental Epigenetics*, 8(1), dvac020. 10.1093/eep/dvac020
- Sarkar, D. K. (2016). Male germline transmits fetal alcohol epigenetic marks for multiple generations: a review. *Addiction Biology*, 21(1), 23-34. 10.1111/adb.12186
- Sato, M., & Sato, K. (2013). Maternal inheritance of mitochondrial DNA by diverse mechanisms to eliminate paternal mitochondrial DNA. *Biochimica Et Biophysica Acta (BBA) - Molecular Cell Research*, 1833(8), 1979-1984. 10.1016/j.bbamcr.2013.03.010
- Savva, C., Helguero, L. A., González-Granillo, M., Couto, D., Melo, T., Li, X., Angelin, B., Domingues, M. R., Kutter, C., & Korach-André, M. (2021). Obese mother offspring have hepatic lipidic modulation that contributes to sex-dependent metabolic adaptation later in life. *Communications Biology*, 4(1), 14.
- Schandler, S., Antick, J., Cohen, M., Brannock, J., & Caine, K. (1988). Visuospatial learning in elementary school children with and without a family history of alcoholism. *Journal of Studies on Alcohol*, 49(1988), 538-545. <https://search.proquest.com/docview/57766427>
- Schjenken, J. E., & Robertson, S. A. (2020). The Female Response to Seminal Fluid. *Physiological Reviews*, 100(3), 1077-1117. 10.1152/physrev.00013.2018
- Scholte, R. H. J., Poelen, E. A. P., Willemsen, G., Boomsma, D. I., & Engels, R. C. M. E. (2008). Relative risks of adolescent and young adult alcohol use: The role of drinking fathers, mothers, siblings, and friends. *Addictive Behaviors*, 33(1), 1-14. 10.1016/j.addbeh.2007.04.015

- Schuckit, M. A. (1988). Reactions to Alcohol in Sons of Alcoholics and Controls. *Alcoholism, Clinical and Experimental Research*, 12(4), 465-470. 10.1111/j.1530-0277.1988.tb00228.x
- Schwarzinger, M., Thiébaud, S. P., Baillot, S., Mallet, V., & Rehm, J. (2018). Alcohol use disorders and associated chronic disease—a national retrospective cohort study from France. *BMC Public Health*, 18(1), 1-9.
- Setshedi, M., Wands, J. R., & Monte, S. M. d. l. (2010). Acetaldehyde Adducts in Alcoholic Liver Disease. *Oxidative Medicine and Cellular Longevity*, 3(3), 178-185. 10.4161/oxim.3.3.12288
- Sharma, M. R., Polavarapu, R., Roseman, D., Patel, V., Eaton, E., Kishor, P. B. K., & Nanji, A. A. (2008). Increased severity of alcoholic liver injury in female versus male rats: A microarray analysis. *Experimental and Molecular Pathology*, 84(1), 46-58. 10.1016/j.yexmp.2007.10.001
- Sharma, U. (2019). Paternal Contributions to Offspring Health: Role of Sperm Small RNAs in Intergenerational Transmission of Epigenetic Information. *Frontiers in Cell and Developmental Biology*, 7, 215. 10.3389/FCELL.2019.00215
- Sharma, U., Conine, C. C., Shea, J. M., Boskovic, A., Derr, A. G., Bing, X. Y., Belleannee, C., Kucukural, A., Serra, R. W., Sun, F., Song, L., Carone, B. R., Ricci, E. P., Li, X. Z., Fauquier, L., Moore, M. J., Sullivan, R., Mello, C. C., Garber, M., & Rando, O. J. (2016). Biogenesis and function of tRNA fragments during sperm maturation and fertilization in

- mammals. *Science (American Association for the Advancement of Science)*, 351(6271), 391-396. 10.1126/science.aad6780
- Shea, J., Serra, R., Carone, B., Shulha, H., Kucukural, A., Ziller, M., Vallaster, M., Gu, H., Tapper, A., Gardner, P., Meissner, A., Garber, M., & Rando, O. (2015). Genetic and Epigenetic Variation, but Not Diet, Shape the Sperm Methylome. *Developmental Cell*, 35(6), 750-758. 10.1016/j.devcel.2015.11.024
- Shen, H. M., Chia, S. E., & Ong, C. N. (1999). Evaluation of oxidative DNA damage in human sperm and its association with male infertility. *Journal of Andrology*, 20(6), 718-723. 10.1002/j.1939-4640.1999.tb03376.x
- Shetty, R. A., & Sadananda, M. (2017). Immediate and delayed anxiety- and depression-like profiles in the adolescent Wistar-Kyoto rat model of endogenous depression following postweaning social isolation. *Behavioural Brain Research*, 320, 323-332. 10.1016/j.bbr.2016.12.030
- Shirane, K., & Lorincz, M. (2023). Epigenetic Mechanisms Governing Female and Male Germline Development in Mammals. *Sexual Development*, 16(5-6), 1-23.
- Simmons, L. W., & Lovegrove, M. (2019). Nongenetic paternal effects via seminal fluid. *Evolution Letters*, 3(4), 403-411. <https://doi.org/10.1002/evl3.124>
- Simms, J. A., Steensland, P., Medina, B., Abernathy, K. E., Chandler, L. J., Wise, R., & Bartlett, S. E. (2008). Intermittent Access to 20% Ethanol Induces High Ethanol Consumption in

- Long-Evans and Wistar Rats. *Alcoholism, Clinical and Experimental Research*, 32(10), 1816-1823. 10.1111/j.1530-0277.2008.00753.x
- Simpson, J., Ryan, C., Curley, A., Mulcaire, J., & Kelly, J. P. (2012). Sex differences in baseline and drug-induced behavioural responses in classical behavioural tests. *Progress in Neuro-Psychopharmacology & Biological Psychiatry*, 37(2), 227-236.
10.1016/j.pnpbp.2012.02.004
- Siuda, D., Wu, Z., Chen, Y., Guo, L., Linke, M., Zechner, U., Xia, N., Reifenberg, G., Kleinert, H., Forstermann, U., & Li, H. (2014). Social isolation-induced epigenetic changes in midbrain of adult mice. *Journal of Physiology and Pharmacology : An Official Journal of the Polish Physiological Society*, 65(2), 247.
<https://www.ncbi.nlm.nih.gov/pubmed/24781734>
- Skinner, M. K. (2016). Epigenetic transgenerational inheritance. *Nature Reviews. Endocrinology*, 12(2), 68-70. 10.1038/nrendo.2015.206
- Smirnov, K., & Sitnikova, E. (2019). Developmental milestones and behavior of infant rats: The role of sensory input from whiskers. *Behavioural Brain Research*, 374, 112143.
10.1016/j.bbr.2019.112143
- Soldin, O. P., & Mattison, D. R. (2009). Sex Differences in Pharmacokinetics and Pharmacodynamics. *Clinical Pharmacokinetics*, 48(3), 143-157. 10.2165/00003088-200948030-00001

- Spear, L. P. (2016). Alcohol Consumption in Adolescence: a Translational Perspective. *Current Addiction Reports*, 3(1), 50-61. 10.1007/s40429-016-0088-9
- Spevak, A. M., Quadagno, D. M., Knoeppel, D., & Poggio, J. P. (1973). The effects of isolation on sexual and social behavior in the rat. *Behavioral Biology*, 8(1), 63-73. 10.1016/S0091-6773(73)80007-0
- Spoelder, M., Dourojeanni, J. P. F., Git, K. C. G. d., Baars, A. M., Lesscher, H. M., & Vanderschuren, L. (2017). Individual differences in voluntary alcohol intake in rats: relationship with impulsivity, decision making and Pavlovian conditioned approach. *Psychopharmacology*, 234(14), 2177-2196. 10.1007/s00213-017-4617-6
- Spoelder, M., Hesseling, P., Baars, A. M., Lozeman-van 't Klooster, J. G., Rotte, M. D., Vanderschuren, L. J. M. J., & Lesscher, H. M. B. (2015). Individual Variation in Alcohol Intake Predicts Reinforcement, Motivation, and Compulsive Alcohol Use in Rats. *Alcoholism, Clinical and Experimental Research*, 39(12), 2427-2437. 10.1111/acer.12891
- Steinberger, E. K., Ferencz, C., & Loffredo, C. A. (2002). Infants with single ventricle: A population-based epidemiological study. *Teratology (Philadelphia)*, 65(3), 106-115. 10.1002/tera.10017
- Stockard, C. R., & Papanicolaou, G. (1916a). A Further Analysis of the Hereditary Transmission of Degeneracy and Deformities by the Descendants of Alcoholized Mammals. *The American Naturalist*, 50(590), 65-88. 10.1086/279525

- Stockard, C. R., & Papanicolaou, G. (1916b). A Further Analysis of the Hereditary Transmission of Degeneracy and Deformities by the Descendants of Alcoholized Mammals. II. *The American Naturalist*, 50(591), 144-177. 10.1086/279531
- Szabó, P. E., Tang, S. E., Silva, F. J., Tsark, W. M. K., & Mann, J. R. (2004). Role of CTCF Binding Sites in the Igf2/H19 Imprinting Control Region. *Molecular and Cellular Biology*, 24(11), 4791-4800. 10.1128/MCB.24.11.4791-4800.2004
- Tadic, S. D., Elm, M. S., Li, H., Van Londen, G. J., Subbotin, V. M., Whitcomb, D. C., & Eagon, P. K. (2002). Sex differences in hepatic gene expression in a rat model of ethanol-induced liver injury. *Journal of Applied Physiology*, 93(3), 1057-1068. 10.1152/japplphysiol.00568.2001
- Tang, Y., Banan, A., Forsyth, C. B., Fields, J. Z., Lau, C. K., Zhang, L. J., & Keshavarzian, A. (2008). Effect of Alcohol on miR-212 Expression in Intestinal Epithelial Cells and Its Potential Role in Alcoholic Liver Disease. *Alcoholism, Clinical and Experimental Research*, 32(2), 355-364. 10.1111/j.1530-0277.2007.00584.x
- Tao, H., Huang, C., Yang, J., Ma, T., Bian, E., Zhang, L., Lv, X., Jin, Y., & Li, J. (2011). MeCP2 controls the expression of RASAL1 in the hepatic fibrosis in rats. *Toxicology (Amsterdam)*, 290(2), 327-333. 10.1016/j.tox.2011.10.011
- Tao, W., Chen, C., Wang, Y., Zhou, W., Jin, Y., Mao, Y., Wang, H., Wang, L., Xie, W., Zhang, X., Li, J., Li, J., Li, X., Tang, Z., Zhou, C., Pan, Z. Z., & Zhang, Z. (2020). MeCP2 mediates

transgenerational transmission of chronic pain. *Progress in Neurobiology*, 189, 101790.

10.1016/j.pneurobio.2020.101790

Terracina, S., Ferraguti, G., Tarani, L., Messina, M. P., Lucarelli, M., Vitali, M., De Persis, S.,

Greco, A., Minni, A., Polimeni, A., Ceccanti, M., Petrella, C., & Fiore, M. (2022).

Transgenerational Abnormalities Induced by Paternal Preconceptual Alcohol Drinking:

Findings from Humans and Animal Models. *Current Neuropharmacology*, 20(6), 1158-

1173. 10.2174/1570159X19666211101111430

The American College of Obstetricians and Gynecologists. (2023, February,). *Alcohol and*

Pregnancy. acog.org. [https://www.acog.org/womens-health/infographics/alcohol-and-](https://www.acog.org/womens-health/infographics/alcohol-and-pregnancy)

[pregnancy](https://www.acog.org/womens-health/infographics/alcohol-and-pregnancy)

Thomas, K. N., Zimmer, K. N., Basel, A., Roach, A. N., Mehta, N. A., Thomas, K. R., Dotson,

L. J., Bedi, Y. S., & Golding, M. C. (2022). Paternal alcohol exposures program

intergenerational hormetic effects on offspring fetoplacental growth. *Frontiers in Cell and*

Developmental Biology, 10, 930375.

Thomas, K. N., Zimmer, K. N., Roach, A. N., Basel, A., Mehta, N. A., Bedi, Y. S., & Golding,

M. C. (2021). Maternal background alters the penetrance of growth phenotypes and sex-

specific placental adaptation of offspring sired by alcohol-exposed males. *The FASEB*

Journal, 35(12)

Thomasson, H. R. (1995). Gender differences in alcohol metabolism. Physiological responses to

ethanol. *Recent Developments in Alcoholism : An Official Publication of the American*

- Medical Society on Alcoholism, the Research Society on Alcoholism, and the National Council on Alcoholism*, 12, 163-179. <https://www.ncbi.nlm.nih.gov/pubmed/7624539>
- Thor, S., Hemmingsson, T., Danielsson, A., & Landberg, J. (2022). Fathers' alcohol consumption and risk of substance-related disorders in offspring. *Drug and Alcohol Dependence*, 233, 109354. 10.1016/j.drugalcdep.2022.109354
- Thorne, B. (2023). *Mitochondrial abundance and function in the serotonin transporter knockout model: a sexually dimorphic association* <https://doi.org/10.26686/wgtn.22714660>
- Thorne, B., Ellenbroek, B. A., & Day, D. J. (2022a). The serotonin reuptake transporter modulates mitochondrial copy number and mitochondrial respiratory complex gene expression in the frontal cortex and cerebellum in a sexually dimorphic manner. *Journal of Neuroscience Research*, 100(3), 869-879. 10.1002/jnr.25010
- Thorne, B., Ellenbroek, B. A., & Day, D. J. (2022b). Sex bias in the serotonin transporter knockout model: Implications for neuropsychiatric disorder research. *Neuroscience & Biobehavioral Reviews*, 134, 104547.
- Tin, A., Grams, M. E., Ashar, F. N., Lane, J. A., Rosenberg, A. Z., Grove, M. L., Boerwinkle, E., Selvin, E., Coresh, J., Pankratz, N., & Arking, D. E. (2016). Association between Mitochondrial DNA Copy Number in Peripheral Blood and Incident CKD in the Atherosclerosis Risk in Communities Study. *Journal of the American Society of Nephrology*, 27(8), 2467-2473. 10.1681/ASN.2015060661

Torres, O. V., Walker, E. M., Beas, B. S., & O'Dell, L. E. (2014). Female rats display enhanced rewarding effects of ethanol that are hormone dependent. *Alcoholism: Clinical and Experimental Research*, 38(1), 108-115.

Trautman, A. (2023). Effects of Alcohol Use on Sperm Chromatin Structure. *South Dakota Medicine : The Journal of the South Dakota State Medical Association*, 76(9), 408.
<http://europepmc.org/abstract/MED/37738493>

Tsuji, N., Otsuka, I., Okazaki, S., Yanagi, M., Numata, S., Yamaki, N., Kawakubo, Y., Shirakawa, O., & Hishimoto, A. (2019). Mitochondrial DNA Copy Number Raises the Potential of Left Frontopolar Hemodynamic Response as a Diagnostic Marker for Distinguishing Bipolar Disorder From Major Depressive Disorder. *Frontiers in Psychiatry*, 10, 312. 10.3389/fpsy.2019.00312

Tukey, J. W. (1962). The Future of Data Analysis. *The Annals of Mathematical Statistics*, 33(1), 1-67. <http://www.jstor.org/stable/2237638>

Tweedie-Cullen, R. Y., Brunner, A. M., Grossmann, J., Mohanna, S., Sichau, D., Nanni, P., Panse, C., & Mansuy, I. M. (2012). Identification of Combinatorial Patterns of Post-Translational Modifications on Individual Histones in the Mouse Brain. *PloS One*, 7(5), e36980. 10.1371/journal.pone.0036980

Uzbay, I. T., & Wallis, C. J. (1999). Lack of Tolerance to Ethanol-Induced Motor Impairment on Accelerod Performance in Rats. *Pharmacology, Biochemistry and Behavior*, 63(4), 607-611. 10.1016/S0091-3057(99)00025-8

- van Amsterdam, J., Nutt, D., Phillips, L., & van den Brink, W. (2015). European rating of drug harms. *Journal of Psychopharmacology (Oxford)*, 29(6), 655-660.
10.1177/0269881115581980
- van Amsterdam, J., Ramaekers, J. G., Verkes, R., Kuypers, K. P., Goudriaan, A. E., & van den Brink, W. (2020). Alcohol-and drug-related public violence in Europe. *European Journal of Criminology*, 17(6), 806-825.
- van den Berg, G. J., & Pinger, P. R. (2016). Transgenerational effects of childhood conditions on third generation health and education outcomes. *Economics & Human Biology*, 23, 103-120.
- Van den Bogert, C., De Vries, H., Holtrop, M., Muus, P., Dekker, H. L., Van Galen, M. J. M., Bolhuis, P. A., & Taanman, J. (1993). Regulation of the expression of mitochondrial proteins: relationship between mtDNA copy number and cytochrome- c oxidase activity in human cells and tissues. *Biochimica Et Biophysica Acta*, 1144(2), 177-183. 10.1016/0005-2728(93)90170-K
- van der Zwaluw, C. S., Engels, R. C., Vermulst, A. A., Rose, R. J., Verkes, R. J., Buitelaar, J., Franke, B., & Scholte, R. H. (2010). A serotonin transporter polymorphism (5-HTTLPR) predicts the development of adolescent alcohol use. *Drug and Alcohol Dependence*, 112(1-2), 134-139.
- van Haaster, L. H., & de rooij, D. G. (1993). Spermatogenesis is accelerated in the immature Djungarian and Chinese hamster and rat. *Biology of Reproduction*, 49(6), 1229-1235.
10.1095/biolreprod49.6.1229

- Varlinskaya, E. I., & Spear, L. P. (2004). Acute ethanol withdrawal (hangover) and social behavior in adolescent and adult male and female Sprague-Dawley rats. *Alcoholism: Clinical and Experimental Research*, 28(1), 40-50.
- Veenendaal, M. V. E., Painter, R. C., de Rooij, S. R., Bossuyt, P. M. M., van der Post, J. A. M., Gluckman, P. D., Hanson, M. A., & Roseboom, T. J. (2013). Transgenerational effects of prenatal exposure to the 1944-45 Dutch famine. *BJOG : An International Journal of Obstetrics and Gynaecology*, 120(5), 548-553. 10.1111/1471-0528.12136
- Ventura-Clapier, R., Garnier, A., & Veksler, V. (2008). Transcriptional control of mitochondrial biogenesis: the central role of PGC-1 α . *Cardiovascular Research*, 79(2), 208-217. 10.1093/cvr/cvn098
- Verhulst, B., Neale, M. C., & Kendler, K. S. (2015). The heritability of alcohol use disorders: a meta-analysis of twin and adoption studies. *Psychological Medicine*, 45(5), 1061-1072. 10.1017/S0033291714002165
- Vetter, C. S., Doremus-Fitzwater, T. L., & Spear, L. P. (2007). Time course of elevated ethanol intake in adolescent relative to adult rats under continuous, voluntary-access conditions. *Alcoholism, Clinical and Experimental Research*, 31(7), 1159-1168. 10.1111/j.1530-0277.2007.00417.x
- Vetter-O'Hagen, C., Varlinskaya, E., & Spear, L. (2009). Sex Differences in Ethanol Intake and Sensitivity to Aversive Effects during Adolescence and Adulthood. *Alcohol and Alcoholism (Oxford, Oxfordshire)*, 44(6), 547-554. 10.1093/alcalc/agp048

- Vojtech, L., Woo, S., Hughes, S., Levy, C., Ballweber, L., Sauteraud, R. P., Strobl, J., Westerberg, K., Gottardo, R., Tewari, M., & Hladik, F. (2014). Exosomes in human semen carry a distinctive repertoire of small non-coding RNAs with potential regulatory functions. *Nucleic Acids Research*, 42(11), 7290-7304. 10.1093/nar/gku347
- Voordeckers, K., Colding, C., Grasso, L., Pardo, B., Hoes, L., Kominek, J., Gielens, K., Dekoster, K., Gordon, J., Van der Zande, E., Bircham, P., Swings, T., Michiels, J., Van Loo, P., Nuyts, S., Pasero, P., Lisby, M., & Verstrepen, K. J. (2020). Ethanol exposure increases mutation rate through error-prone polymerases. *Nature Communications*, 11(1), 3664-3. 10.1038/s41467-020-17447-3
- Vyssotski, D. L. (2011). Transgenerational epigenetic compensation and natural selection. *Evolocus*, , 1-6. <http://evolocus.com/evolocus/v1/evolocus-01-001-c.pdf>
- Wai, T., AO, A., XIAOYUN ZHANG, CYR, D., DUFORT, D., & SHOUBRIDGE, E. A. (2010). The Role of Mitochondrial DNA Copy Number in Mammalian Fertility. *Biology of Reproduction*, 83(1), 52-62. 10.1095/biolreprod.109.080887
- Wang, X., Feng, J., Xue, Y., Guan, Z., Zhang, D., Liu, Z., Gong, Z., Wang, Q., Huang, J., Tang, C., Zou, T., & Yin, P. (2016). Structural basis of N6-adenosine methylation by the METTL3–METTL14 complex. *Nature (London)*, 534(7608), 575-578. 10.1038/nature18298

- Wang, Z., & Goonewardene, L. A. (2004). use of MIXED models in the analysis of animal experiments with repeated measures data. *Canadian Journal of Animal Science*, 84(1), 1-11. 10.4141/a03-123
- Ward, R. J., & Coutelle, C. H. (2003). Women and alcohol susceptibility: could differences in alcohol metabolism predispose women to alcohol-related diseases? *Archives of Women's Mental Health*, 6, 231-238.
- Wayner, M. J., & Greenberg, I. (1972). Effects of hypothalamic stimulation, acclimation and periodic withdrawal on ethanol consumption. *Physiology & Behavior*, 9(5), 737-740. 10.1016/0031-9384(72)90043-1
- Wei, J., Huang, K., Yang, C., & Kang, C. (2017). Non-coding RNAs as regulators in epigenetics. *Oncology Reports*, 37(1), 3-9. 10.3892/or.2016.5236
- Weinstock, M., Fride, E., & Hertzberg, R. (1988). Prenatal stress effects on functional development of the offspring. *Progress in Brain Research* (pp. 319-331). Elsevier Science & Technology. 10.1016/S0079-6123(08)60513-0
- Westman, J., Wahlbeck, K., Laursen, T. M., Gissler, M., Nordentoft, M., Hällgren, J., Arffman, M., & Ösby, U. (2015). Mortality and life expectancy of people with alcohol use disorder in Denmark, Finland and Sweden. *Acta Psychiatrica Scandinavica*, 131(4), 297-306.
- White, A. M. (2020). Gender differences in the epidemiology of alcohol use and related harms in the United States. *Alcohol Research: Current Reviews*, 40(2)

WHO. (2022). *Alcohol*. (). <https://www.who.int/news-room/fact-sheets/detail/alcohol>

WHO. (2023). *World health statistics 2023: monitoring health for the SDGs, Sustainable Development Goal*. World Health Organization.

Wieting, J., Rhein, M., Hillemacher, T., Lichtinghagen, R., Hoppe, V., Müschen, L., Glahn, A., Frieling, H., Bleich, S., & Muschler, M. (2019). DNA Methylation of the Leptin Gene Promoter is Altered by Chronic Alcohol Exposure in an Animal Model for Alcohol Dependence. *European Addiction Research*, 25(2), 49-55. 10.1159/000496111

Windham, G. C., Fenster, L., Hopkins, B., & Swan, S. H. (1995). The Association of Moderate Maternal and Paternal Alcohol Consumption with Birthweight and Gestational Age. *Epidemiology (Cambridge, Mass.)*, 6(6), 591-597. 10.1097/00001648-199511000-00005

Wise, R. A. (1973). Voluntary ethanol intake in rats following exposure to ethanol on various schedules. *Psychopharmacologia*, 29(3), 203-210. 10.1007/BF00414034

Wolffe, A. P., Jones, P. L., Jan Veenstra, G. C., Wade, P. A., Vermaak, D., Kass, S. U., Landsberger, N., & Strouboulis, J. (1998). Methylated DNA and MeCP2 recruit histone deacetylase to repress transcription. *Nature Genetics*, 19(2), 187-191. 10.1038/561

Wolffgramm, J., & Heyne, A. (1995). *From controlled drug intake to loss of control: the irreversible development of drug addiction in the rat*. Elsevier B.V. 10.1016/0166-4328(95)00131-C

- Wozniak, D. F., Cicero, T. J., Kettinger, L. I., & Meyer, E. R. (1991). Paternal alcohol consumption in the rat impairs spatial learning performance in male offspring. *Psychopharmacologia*, 105(2), 289-302. 10.1007/BF02244324
- Wu, L., Lu, Y., Jiao, Y., Liu, B., Li, S., Li, Y., Xing, F., Chen, D., Liu, X., Zhao, J., Xiong, X., Gu, Y., Lu, J., Chen, X., & Li, X. (2016). Paternal Psychological Stress Reprograms Hepatic Gluconeogenesis in Offspring. *Cell Metabolism*, 23(4), 735-743. 10.1016/j.cmet.2016.01.014
- Xia, R., Jin, L., Li, D., Liang, H., Yang, F., Chen, J., Yuan, W., & Miao, M. (2018). Association Between Paternal Alcohol Consumption Before Conception and Anogenital Distance of Offspring. *Alcoholism, Clinical and Experimental Research*, 42(4), 735-742. 10.1111/acer.13595
- Xu, R., Li, S., Wu, Q., Li, C., Jiang, M., Guo, L., Chen, M., Yang, L., Dong, X., & Wang, H. (2022). Stage-specific H3K9me3 occupancy ensures retrotransposon silencing in human pre-implantation embryos. *Cell Stem Cell*, 29(7), 1051-1066. e8.
- Yang, J., Liu, L., Tao, H., Hu, W., Shi, P., Deng, Z., & Li, J. (2016). MeCP2 silencing of LncRNA H19 controls hepatic stellate cell proliferation by targeting IGF1R. *Toxicology (Amsterdam)*, 359-360, 39-46. 10.1016/j.tox.2016.06.016
- Yasui, D. H., Peddada, S., Bieda, M. C., Vallero, R. O., Hogart, A., Nagarajan, R. P., Thatcher, K. N., Farnham, P. J., & LaSalle, J. M. (2007). Integrated epigenomic analyses of neuronal

- MeCP2 reveal a role for long-range interaction with active genes. *Proceedings of the National Academy of Sciences - PNAS*, 104(49), 19416-19421. 10.1073/pnas.0707442104
- Yin, X., Anwar, A., Wang, Y., Hu, H., Liang, G., & Zhang, C. (2022). Paternal environmental exposure-induced spermatozoal small noncoding RNA alteration mediates the intergenerational epigenetic inheritance of multiple diseases. *Frontiers of Medicine*, 16(2), 176-184. 10.1007/s11684-021-0885-y
- Yokoyama, Y., Nimura, Y., Nagino, M., Bland, K. I., & Chaudry, I. H. (2005). Current Understanding of Gender Dimorphism in Hepatic Pathophysiology1. *The Journal of Surgical Research*, 128(1), 147-156. 10.1016/j.jss.2005.04.017
- Yoo, H. J., Park, M., & Kim, S. A. (2017). Difference in mitochondrial DNA copy number in peripheral blood cells between probands with autism spectrum disorders and their unaffected siblings. *The World Journal of Biological Psychiatry*, 18(2), 151-156. 10.1080/15622975.2016.1234069
- You, M., Liang, X., Ajmo, J. M., & Ness, G. C. (2008). Involvement of mammalian sirtuin 1 in the action of ethanol in the liver. *American Journal of Physiology - Gastrointestinal and Liver Physiology*, 294(4), 892-898. 10.1152/ajpgi.00575.2007
- Young, J., Hong, E., Castle, J., & Crespo-Barreto, J. (2005). Inaugural Article by a Recently Elected Academy Member: Inaugural Article: Regulation of RNA splicing by the methylation-dependent transcriptional repressor methyl-CpG binding protein 2. *Proceedings*

of the National Academy of Sciences - PNAS, 102(49), 17751.

<https://search.proquest.com/docview/201365826>

Young-Wolff, K. C., Enoch, M., & Prescott, C. A. (2011). The influence of gene–environment interactions on alcohol consumption and alcohol use disorders: A comprehensive review.

Clinical Psychology Review, 31(5), 800-816. 10.1016/j.cpr.2011.03.005

Youssef, N. A., Lockwood, L., Su, S., Hao, G., & Rutten, B. P. F. (2018). The Effects of Trauma, with or without PTSD, on the Transgenerational DNA Methylation Alterations in

Human Offsprings. *Brain Sciences*, 8(5), 83. 10.3390/brainsci8050083

Zeybel, M., Hardy, T., Wong, Y., Mathers, J., Fox, C., Gackowska, A., Oakley, F., Burt, A.,

Wilson, C., Anstee, Q., Barter, M., Masson, S., Elsharkawy, A., Mann, D., & Mann, J.

(2012). Multigenerational epigenetic adaptation of the hepatic wound-healing response.

Nature Medicine, 18(10), 1592. 10.1038/nm1012-1592c

Zhang, I. W., López-Vicario, C., Duran-Güell, M., & Clària, J. (2021). Mitochondrial

Dysfunction in Advanced Liver Disease: Emerging Concepts. *Frontiers in Molecular*

Biosciences, 8, 772174. 10.3389/fmolb.2021.772174

Zhang, S., Wang, L., Yang, T., Chen, L., Zhao, L., Wang, T., Chen, L., Ye, Z., Zheng, Z., & Qin,

J. (2020). *Parental alcohol consumption and the risk of congenital heart diseases in*

offspring: An updated systematic review and meta-analysis. SAGE Publications.

10.1177/2047487319874530

Zimmermann, C. A., Hoffmann, A., Raabe, F., & Spengler, D. (2015). Role of Mecp2 in Experience-Dependent Epigenetic Programming. *Genes*, 6(1), 60-86.
10.3390/genes6010060